GEOMETRY · COMMON CORE HIGH SCHOOL MATH

Question Collection

Main • Backward Branches • Forward 1 • Forward 2 (later courses)

63 question sets • 1880 questions

CONTENTS

Set 1	**HSG-CO.A.1**	Precise definitions of angle, circle, and line segment	6
Set 2	**HSG-CO.A.1**	Precise definitions of perpendicular lines and parallel lines	43
Set 3	**HSG-CO.A.2**	Transformations as functions: inputs, outputs, and preimages	74
Set 4	**HSG-CO.A.2**	Compare transformations that preserve distance and angle with those that do not	111
Set 5	**HSG-CO.A.3**	Rotations and reflections that carry a rectangle, parallelogram, or trapezoid onto it	142
Set 6	**HSG-CO.A.3**	Rotations and reflections that carry a regular polygon onto itself	179
Set 7	**HSG-CO.A.4**	Definitions of reflections and translations	216
Set 8	**HSG-CO.A.4**	Definition of a rotation	253
Set 9	**HSG-CO.A.5**	Draw the image of a figure under a rotation, reflection, or translation	290
Set 10	**HSG-CO.A.5**	Specify a sequence of transformations that carries one figure onto another	327
Set 11	**HSG-CO.B.6**	Predict the effect of a rigid motion on a figure	364
Set 12	**HSG-CO.B.6**	Decide whether two figures are congruent using rigid motions	395
Set 13	**HSG-CO.B.7**	Congruent triangles: a rigid motion exactly when all corresponding parts are congr	426
Set 14	**HSG-CO.B.8**	Explain SAS congruence with rigid motions	463
Set 15	**HSG-CO.B.8**	Explain ASA congruence with rigid motions	500
Set 16	**HSG-CO.B.8**	Explain SSS congruence with rigid motions	537
Set 17	**HSG-CO.C.9**	Prove that vertical angles are congruent	574

CONTENTS (continued)

Set 18	**HSG-CO.C.9**	Prove the angle theorems for parallel lines cut by a transversal	611
Set 19	**HSG-CO.C.9**	Prove the perpendicular bisector theorem and its converse	648
Set 20	**HSG-CO.C.10**	Prove that the angles of a triangle add to 180°	685
Set 21	**HSG-CO.C.10**	Prove that the base angles of an isosceles triangle are congruent	722
Set 22	**HSG-CO.C.10**	Prove the midsegment theorem	759
Set 23	**HSG-CO.C.10**	Prove that the medians of a triangle meet at one point	802
Set 24	**HSG-CO.C.11**	Prove that opposite sides and opposite angles of a parallelogram are congruent	845
Set 25	**HSG-CO.C.11**	Prove that the diagonals of a parallelogram bisect each other	882
Set 26	**HSG-CO.C.11**	Prove the converses: conditions that make a quadrilateral a parallelogram	919
Set 27	**HSG-CO.C.11**	Prove that rectangles are parallelograms with congruent diagonals	956
Set 28	**HSG-CO.D.12**	Construct a copy of a segment	993
Set 29	**HSG-CO.D.12**	Construct a copy of an angle	1030
Set 30	**HSG-CO.D.12**	Construct the perpendicular bisector of a segment (bisect a segment)	1067
Set 31	**HSG-CO.D.12**	Construct the bisector of an angle	1104
Set 32	**HSG-CO.D.12**	Construct a perpendicular to a line through a point on or off the line	1141
Set 33	**HSG-CO.D.12**	Construct a line parallel to a given line through a point not on it	1178
Set 34	**HSG-CO.D.13**	Construct an equilateral triangle inscribed in a circle	1215

CONTENTS (continued)

Set 35	**HSG-CO.D.13**	Construct a square inscribed in a circle	1252
Set 36	**HSG-CO.D.13**	Construct a regular hexagon inscribed in a circle	1289
Set 37	**HSG-SRT.A.1.a**	Verify that a dilation maps a line to a parallel line (or to itself through the center)	1326
Set 38	**HSG-SRT.A.1.b**	Verify that a dilation multiplies lengths by the scale factor	1363
Set 39	**HSG-SRT.A.2**	Decide whether two figures are similar using similarity transformations	1400
Set 40	**HSG-SRT.A.2**	Explain similarity of triangles as equal angles and proportional sides	1437
Set 41	**HSG-SRT.A.3**	Establish the AA criterion with similarity transformations	1474
Set 42	**HSG-SRT.B.4**	Prove the triangle proportionality theorem	1511
Set 43	**HSG-SRT.B.4**	Prove the converse of the triangle proportionality theorem	1548
Set 44	**HSG-SRT.B.4**	Prove the Pythagorean Theorem using similar triangles	1585
Set 45	**HSG-SRT.B.5**	Use similarity criteria to solve problems in geometric figures	1622
Set 46	**HSG-SRT.B.5**	Use congruence criteria to prove relationships in figures	1659
Set 47	**HSG-SRT.B.5**	Use similarity criteria to prove relationships in figures	1696
Set 48	**HSG-SRT.C.6**	Understand trigonometric ratios as properties of an angle, by similarity	1733
Set 49	**HSG-SRT.C.7**	Explain and use the relationship between the sine and cosine of complementary	1770
Set 50	**HSG-SRT.C.8**	Find unknown sides of right triangles with trigonometric ratios in applied problems	1807
Set 51	**HSG-SRT.C.8**	Find unknown angles with inverse trigonometric ratios in applied problems	1844

CONTENTS (continued)

Set 52	**HSG-SRT.C.8**	Solve right triangles in applied problems with trigonometry and the Pythagorean T	1881
Set 53	**HSG-C.A.1**	Prove that all circles are similar	1918
Set 54	**HSG-C.A.2**	Relate inscribed angles and central angles	1955
Set 55	**HSG-C.A.2**	Inscribed angles on a diameter are right angles	1992
Set 56	**HSG-C.A.2**	A radius is perpendicular to the tangent at the point of tangency	2029
Set 57	**HSG-C.A.2**	Relate circumscribed angles and central angles	2066
Set 58	**HSG-C.A.2**	Relationships among radii and chords	2103
Set 59	**HSG-C.A.3**	Construct the inscribed circle of a triangle	2140
Set 60	**HSG-C.A.3**	Construct the circumscribed circle of a triangle	2177
Set 61	**HSG-C.A.3**	Prove that opposite angles of an inscribed quadrilateral are supplementary	2214
Set 62	**HSG-C.B.5**	Derive arc length from similarity and define radian measure	2251
Set 63	**HSG-C.B.5**	Derive and use the formula for the area of a sector	2288

Precise definitions of angle, circle, and line segment

MAIN	Geometry main question	**HSG-CO.A.1**
BACKWARD 1	Identify points, lines, rays, segments, and angles	**4.G.A.1**
BACKWARD 2	See an angle as two rays with a common endpoint, measured by…	**4.MD.C.5.a**
BACKWARD 3	Use the center, radius, and diameter of a circle	**7.G.B.4**
FORWARD 1	Test proposed definitions with counterexamples	**HSG-CO.A.1**
FORWARD 2	Write precise definitions of new figures from earlier definitions	**HSG-CO.A.1**

SET 1 • HSG-CO.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Write a precise definition of a circle. Use only the words point, plane, and distance, and words built from them.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q1/5

Points A and B lie on line ℓ. Which is a precise definition of segment AB?

A. Every point of line ℓ

B. Point A and every point of line ℓ on the same side of A as B

C. Points A and B together with every point C of line ℓ for which AC + CB = AB

D. Every point of the plane that is closer to A than to B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q2/5

Write a precise definition of an angle. Name its vertex and its sides in your definition.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q3/5

Mia writes: "A circle is all the points that are 5 cm from point O."

What must she add to make the definition precise? Explain why.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q4/5

True or false?

An angle with its vertex at the center of a circle that cuts off $\frac{1}{360}$ of the circle measures 1°, whatever the radius of the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q5/5

SET 1 • HSG-CO.A.1

BACKWARD 1

Identify points, lines, rays, segments, and angles • Grade 4 • Geometry • 4.G.A.1

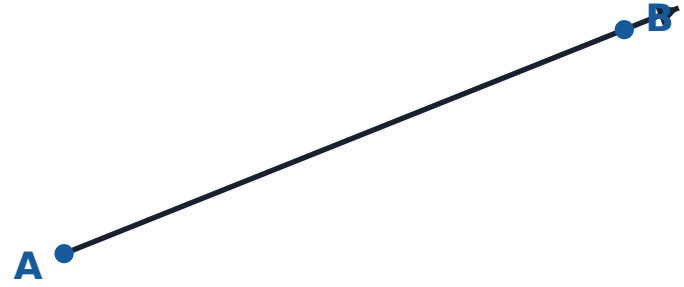
What is shown in the figure?

A. Ray BA

B. Line AB

C. Ray AB

D. Segment AB



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 1 • BACKWARD 1 • Q1/5

How many endpoints does a line segment have? How many does a ray have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 1 • BACKWARD 1 • Q2/5

True or false?

A line segment has exactly two endpoints.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 1 • BACKWARD 1 • Q3/5

Which figure goes on without end in exactly one direction?

A. A line

B. A point

C. A line segment

D. A ray

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 1 • BACKWARD 1 • Q4/5

True or false?

Ray AB and ray BA are the same ray.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 1 • BACKWARD 1 • Q5/5

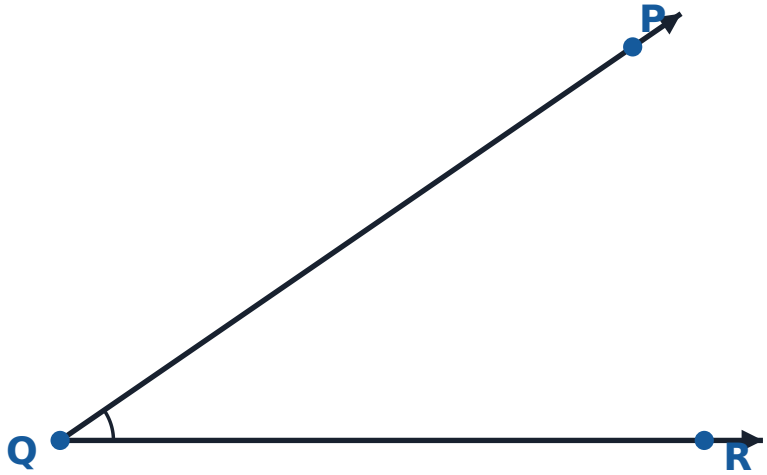
SET 1 • HSG-CO.A.1

BACKWARD 2

See an angle as two rays with a common endpoint, measured by an arc • Grade 4 • Measurement & Data • 4.MD.C.5.a

Name the vertex and the two sides of ∠PQR.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 1 • BACKWARD 2 • Q1/5

An angle has its vertex at the center of a circle and cuts off $\frac{1}{4}$ of the circle. What is its measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 1 • BACKWARD 2 • Q2/5

An angle measures 60°. What fraction of a circle centered at its vertex does it cut off?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 1 • BACKWARD 2 • Q3/5

True or false?

The measure of an angle depends on how long its sides are drawn.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 1 • BACKWARD 2 • Q4/5

True or false?

A one-degree angle turns through $\frac{1}{360}$ of a full circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 1 • BACKWARD 2 • Q5/5

SET 1 • HSG-CO.A.1

BACKWARD 3

Use the center, radius, and diameter of a circle • Grade 7 • Geometry • 7.G.B.4

A circle has center O and radius 7 cm. Point P is in the plane of the circle, 7 cm from O. Is P on the circle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q1/5

A circle has a diameter of 18 in. What is its radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q2/5

A circle has center C and radius 4. Point D is in the same plane, 3 units from C. Where is D?

A. Inside the circle

B. Outside the circle

C. On the circle

D. There is not enough information

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q3/5

True or false?

Every radius of a circle has the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q4/5

True or false?

A point 10 units from the center of a circle of radius 5 lies on the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q5/5

SET 1 • HSG-CO.A.1

FORWARD 1

Test proposed definitions with counterexamples • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Proposed definition: "A circle is a set of points that are all the same distance from a point O."

Give a set that fits this wording but is NOT a circle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q1/5

Proposed definition: "Point M is the midpoint of segment AB if AM = MB."

Give a counterexample, and repair the definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q2/5

Which set fits the description "all points 2 units from point O" but is NOT a circle?

A. A sphere of radius 2 centered at O

B. A circle of radius 2 centered at O in a plane

C. A disk of radius 2

D. A segment of length 2 from O

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q3/5

True or false?

"Two lines are parallel if they never meet" is a complete definition, even in three-dimensional space.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q4/5

Proposed definition: "A diameter is a segment that passes through the center of a circle."

Give a counterexample, and repair the definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q5/5

SET 1 • HSG-CO.A.1

FORWARD 2

Write precise definitions of new figures from earlier definitions • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Write a precise definition of ray AB. Use the word "between".

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q1/5

Write a precise definition of the midpoint of segment AB.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q2/5

Segment PQ is a diameter of a circle with center O. Write a precise definition of a semicircle with endpoints P and Q.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q3/5

Which is a precise definition of the bisector of ∠ABC?

A. A ray from B that lies outside ∠ABC

B. Ray BD, where D is in the interior of ∠ABC and m∠ABD = m∠DBC

C. The segment that joins A and C

D. Any line through B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q4/5

True or false?

The definition "the midpoint of AB is a point M with AM = MB" is precise enough, because every such point lies on segment AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q5/5

Precise definitions of perpendicular lines and parallel lines

MAIN	Geometry main question	**HSG-CO.A.1**
BACKWARD 1	Recognize perpendicular and parallel lines and right angles	**4.G.A.1**
BACKWARD 2	Find the other angles when two lines cross	**7.G.B.5**
FORWARD 1	Prove consequences of the definition of perpendicular lines	**HSG-CO.C.9**
FORWARD 2	Reason about three lines in a plane and in space	**HSG-CO.C.9**

SET 2 • HSG-CO.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Write a precise definition of parallel lines.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q1/5

Write a precise definition of perpendicular lines.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q2/5

Lines ℓ and m do not intersect. Which additional fact makes them parallel?

A. They lie in the same plane

B. They both look horizontal in a drawing

C. They are drawn the same length

D. They are both perpendicular to one segment in space

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q3/5

True or false?

If two lines meet and form one right angle, they form four right angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

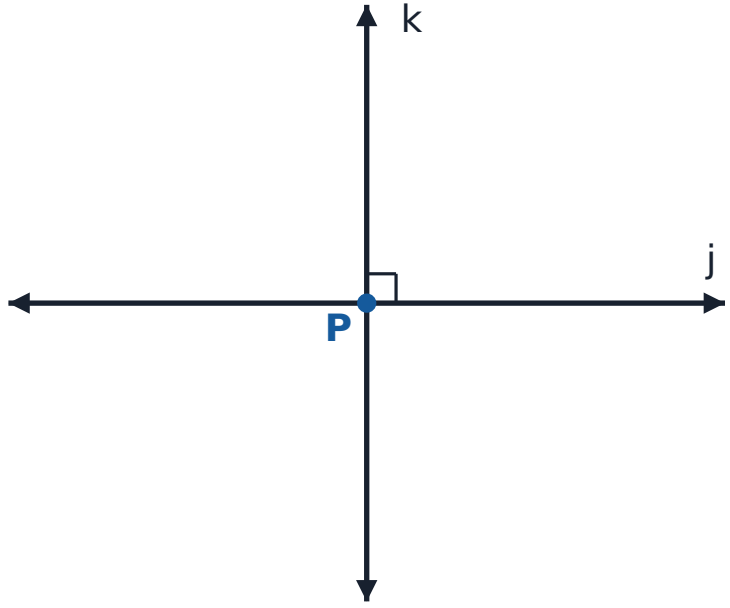
ID S2-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q4/5

Lines j and k meet at point P, and one of the angles they form is marked as a right angle.

Are j and k perpendicular? Use the definition to explain.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q5/5

SET 2 • HSG-CO.A.1

BACKWARD 1

Recognize perpendicular and parallel lines and right angles • Grade 4 • Geometry • 4.G.A.1

True or false?

Lines that meet at a 90° angle are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q1/5

True or false?

Any two lines that cross are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q2/5

Two perpendicular lines cross. How many right angles do they form?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q3/5

What is true of the opposite sides of a rectangle?

A. They are parallel

B. They are perpendicular

C. They meet at one point

D. They are neither parallel nor perpendicular

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q4/5

Which pair of streets is most like perpendicular lines?

A. Two streets that cross at a square corner

B. Two straight streets that never meet

C. Two streets that curve around a park

D. Two streets that meet at a sharp point

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q5/5

SET 2 • HSG-CO.A.1

BACKWARD 2

Find the other angles when two lines cross • Grade 7 • Geometry • 7.G.B.5

Two lines cross. One of the angles measures 90°. Find the other three angles.

Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q1/5

Two lines cross. One of the angles measures 70°. Find the other three angles.

Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q2/5

True or false?

Vertical angles formed by two crossing lines are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q3/5

Two angles form a linear pair. One measures 90°. What does the other measure?

A. 90°

B. 180°

C. 45°

D. 270°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q4/5

True or false?

Two lines cross and one angle measures 100°. The angle next to it also measures 100°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • HSG-CO.A.1

FORWARD 1

Prove consequences of the definition of perpendicular lines • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines ℓ and m are perpendicular and meet at point P. Prove that all four angles at P are right angles.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q1/5

Two lines meet at point P and form a linear pair of congruent angles. Prove that the lines are perpendicular.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q2/5

Lines a and b intersect, and one of the angles they form measures (3x + 15)°. For what value of x are a and b perpendicular?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q3/5

True or false?

If two intersecting lines form a pair of vertical angles that are supplementary, the lines are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q4/5

Lines p and q meet at point O. Which statement is enough to prove that p ⊥ q?

A. Two of the angles at O are vertical angles

B. Two adjacent angles at O add to 180°

C. Two of the angles at O are acute

D. One of the four angles at O measures 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q5/5

SET 2 • HSG-CO.A.1

FORWARD 2

Reason about three lines in a plane and in space • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

In a plane, lines a and b are both perpendicular to line t. Prove that a and b are parallel.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q1/5

True or false?

In three-dimensional space, two lines that are each perpendicular to the same line must be parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q2/5

In a plane, line a is parallel to line b, and line t is perpendicular to line a. What is true about t and b?

A. t and b do not meet

B. t is parallel to b

C. The angle between t and b cannot be known

D. t is perpendicular to b

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q3/5

In a coordinate plane, line t is the x-axis. Line a is perpendicular to t at (0, 0), and line b is perpendicular to t and passes through (5, 2). Write an equation of line b.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q4/5

True or false?

In a plane, through a point not on a line there is exactly one line parallel to the given line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q5/5

Transformations as functions: inputs, outputs, and preimages

MAIN	Geometry main question	**HSG-CO.A.2**
BACKWARD 1	Evaluate a rule for a given input	**8.F.A.1**
BACKWARD 2	Apply coordinate rules for translations, reflections, and rotations	**8.G.A.3**
BACKWARD 3	Read and locate ordered pairs in all four quadrants	**6.NS.C.6.c**
FORWARD 1	Compose two transformations and write one rule	**HSG-CO.A.2**
FORWARD 2	Find the inverse of a transformation	**HSG-CO.A.2**

SET 3 • HSG-CO.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.2

The transformation T is defined by T(x, y) = (x + 4, y − 3). Find T(2, 5).

T(2, 5) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q1/5

The transformation R is defined by $R(x, y) = (-y, x)$. Find $R(3, -1)$.

$R(3, -1) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q2/5

The transformation F takes each point (x, y) to (x, −y). The output point is F(P) = (4, 7). Find the input point P.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q3/5

Every point of the plane is moved 5 units right and 2 units up. Which function describes this transformation?

A. $T(x, y) = (x - 5, y - 2)$

B. $T(x, y) = (x + 5, y + 2)$

C. $T(x, y) = (5x, 2y)$

D. $T(x, y) = (x + 2, y + 5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q4/5

True or false?

Under the transformation D(x, y) = (2x, 2y), the point (0, 0) is its own output.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q5/5

SET 3 • HSG-CO.A.2

BACKWARD 1

Evaluate a rule for a given input • Grade 8 • Functions • 8.F.A.1

f(x) = 3x − 2. Find f(4).

f(4) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q1/5

A rule takes a number, doubles it, and adds 1. What is the output for the input 6?

Output: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q2/5

g(x) = x + 5. Which input gives the output 12?

Input: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q3

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q3/5

True or false?

A function assigns exactly one output to each input.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q4/5

True or false?

The rule "the input 3 gives the outputs 4 and 5" describes a function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q5

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q5/5

SET 3 • HSG-CO.A.2

BACKWARD 2

Apply coordinate rules for translations, reflections, and rotations • Grade 8 • Geometry • 8.G.A.3

Translate the point (2, −1) 3 units left and 4 units up. Where does it go?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q1/5

Reflect the point (5, 2) across the y-axis.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q2/5

Rotate the point (4, 1) by 180° about the origin.

A. (1, 4)

B. (−1, 4)

C. (−4, −1)

D. (4, −1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q3/5

True or false?

Reflecting the point (3, 6) across the x-axis gives (3, −6).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q4/5

True or false?

Rotating the point (0, 5) by 90° counterclockwise about the origin gives (5, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q5

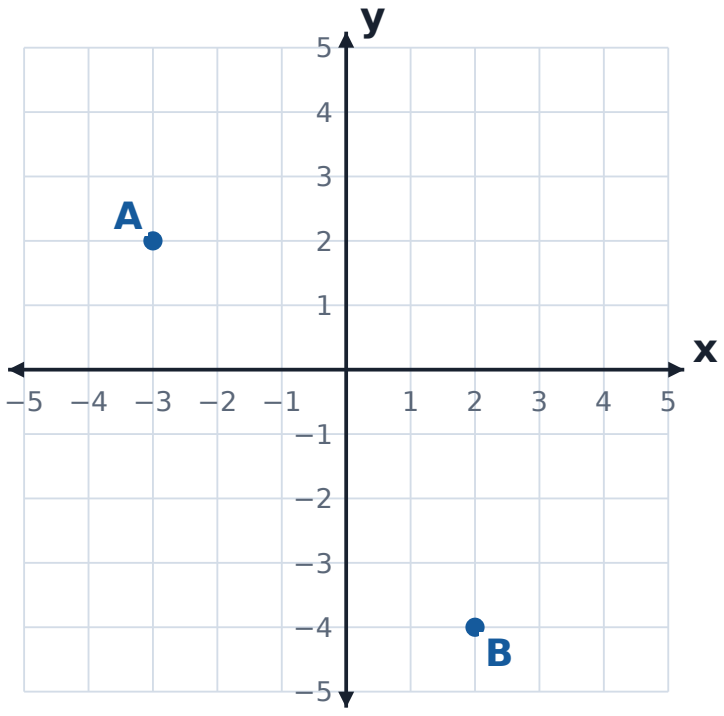
Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q5/5

SET 3 • HSG-CO.A.2

BACKWARD 3

Read and locate ordered pairs in all four quadrants • Grade 6 • The Number System • 6.NS.C.6.c

What are the coordinates of point A?



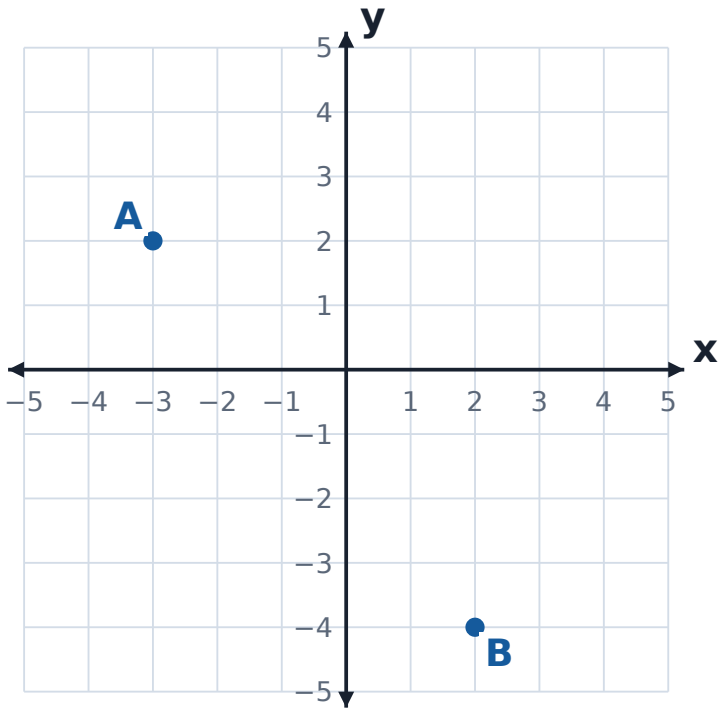
A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q1/5

What are the coordinates of point B?



B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q2/5

Which point is in Quadrant III?

A. $(-2, -5)$

B. $(-2, 5)$

C. $(2, 5)$

D. $(2, -5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q3/5

True or false?

The point (0, −3) lies on the y-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q4/5

True or false?

The point $(-2, 5)$ is 2 units to the right of the y-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q5/5

SET 3 • HSG-CO.A.2

FORWARD 1

Compose two transformations and write one rule • High School — Geometry • Geometry: Congruence • HSG-CO.A.2

$T(x, y) = (x + 2, y)$ and $R(x, y) = (-x, y)$. Find $R(T(1, 4))$.

$R(T(1, 4)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q1/5

$T(x, y) = (x + 2, y)$ and $R(x, y) = (-x, y)$. Write a single rule for $R(T(x, y))$.

$R(T(x, y)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q2/5

$T(x, y) = (x + 2, y)$ and $R(x, y) = (-x, y)$. Find $T(R(1, 4))$.

$T(R(1, 4)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q3/5

True or false?

For any two transformations, doing them in either order gives the same result.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q4/5

$T(x, y) = (x + 3, y - 1)$ and $S(x, y) = (x - 1, y + 5)$. Which rule is $S(T(x, y))$?

A. $(x - 3, y - 5)$

B. $(x + 2, y + 4)$

C. $(x + 2, y - 4)$

D. $(x + 4, y - 6)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q5/5

SET 3 • HSG-CO.A.2

FORWARD 2

Find the inverse of a transformation • High School — Geometry • Geometry: Congruence • HSG-CO.A.2

Find a rule for the inverse of $T(x, y) = (x - 6, y + 2)$.

Inverse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q1/5

$R(x, y) = (-y, x)$ is a 90° counterclockwise rotation about the origin. Find a rule for its inverse.

Inverse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q2/5

True or false?

Every reflection is its own inverse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q3/5

D(x, y) = (3x, 3y). Which rule undoes D?

A. $(x - 3, y - 3)$

B. $(3y, 3x)$

C. $(\frac{x}{3}, \frac{y}{3})$

D. $(-3x, -3y)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q4/5

True or false?

The transformation P(x, y) = (x, 0), which moves every point straight up or down to the x-axis, has an inverse transformation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q5/5

Compare transformations that preserve distance and angle with those that do not

MAIN	Geometry main question	**HSG-CO.A.2**
BACKWARD 1	Find the distance between two points	**8.G.B.8**
BACKWARD 2	Know that rigid motions keep lengths and angle measures	**8.G.A.1**
FORWARD 1	Classify transformations as rigid motions, similarity transformatio...	**HSG-SRT.A.2**
FORWARD 2	Prove with variables that a transformation preserves or scales ev...	**HSG-SRT.A.2**

SET 4 • HSG-CO.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.2

Which transformation does NOT preserve distance?

A. $R(x, y) = (-x, y)$

B. $T(x, y) = (x + 4, y - 1)$

C. $H(x, y) = (2x, y)$

D. $Q(x, y) = (-y, x)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q1/5

H(x, y) = (2x, y). Points A(0, 0) and B(3, 4) are 5 units apart.

Find H(A), H(B), and the distance between them. Does H preserve distance?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q2/5

The translation $T(x, y) = (x - 2, y + 5)$ maps $A(1, 1)$ and $B(4, 5)$ to A' and B'. Show that $A'B' = AB$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q3/5

True or false?

The transformation V(x, y) = (x, 3y) preserves the measure of every angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q4/5

S(x, y) = (3x, 3y). Does S preserve distance? Does S preserve angle measure? Explain each.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q5/5

SET 4 • HSG-CO.A.2

BACKWARD 1

Find the distance between two points • Grade 8 • Geometry • 8.G.B.8

Find the distance between (1, 2) and (4, 6).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q1/5

Find the distance between $(-2, 3)$ and $(4, 3)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q2/5

Find the distance between (0, 0) and (5, 12).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q3/5

True or false?

The distance from (0, 0) to (1, 1) is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q4/5

Find the distance between $(2, -1)$ and $(5, 3)$.

A. 5

B. 25

C. $\sqrt{7}$

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q5/5

SET 4 • HSG-CO.A.2

BACKWARD 2

Know that rigid motions keep lengths and angle measures • Grade 8 • Geometry • 8.G.A.1

True or false?

A rotation keeps every segment the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q1/5

True or false?

A reflection can change the measure of an angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q2/5

A triangle is translated. What changes?

A. Its angle measures

B. The positions of its vertices

C. Its side lengths

D. Its area

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q3/5

A segment 7 cm long is reflected across a line. How long is its image?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q4/5

A 35° angle is rotated 90° about a point. What is the measure of its image?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q5/5

SET 4 • HSG-CO.A.2

FORWARD 1

Classify transformations as rigid motions, similarity transformations, or neither • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Which transformation preserves angle measure but not distance?

A. $(x, y) \rightarrow (4x, 4y)$

B. $(x, y) \rightarrow (4x, y)$

C. $(x, y) \rightarrow (x + 4, y)$

D. $(x, y) \rightarrow (x, y + 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q1/5

Classify $(x, y) \to (-2x, -2y)$ as a rigid motion, a similarity transformation that is not rigid, or neither. Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q2/5

Classify $(x, y) \to (x + y, y)$ as a rigid motion, a similarity transformation that is not rigid, or neither. Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q3/5

True or false?

Every rigid motion is a similarity transformation with scale factor 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q4/5

True or false?

A transformation that multiplies every distance by the same number k > 0 always changes angle measures.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q5/5

SET 4 • HSG-CO.A.2

FORWARD 2

Prove with variables that a transformation preserves or scales every distance • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Prove that $T(x, y) = (x + a, y + b)$ preserves the distance between ANY two points (x_1, y_1) and (x_2, y_2).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q1/5

Prove that D(x, y) = (3x, 3y) multiplies the distance between ANY two points by 3.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q2/5

$H(x, y) = (2x, y)$. The distance between the images of (x_1, y_1) and (x_2, y_2) is $\sqrt{(4(x_2 - x_1)^2 + (y_2 - y_1)^2)}$.

For which pairs of points is the distance unchanged?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q3/5

True or false?

One example in which a distance stays the same is enough to prove that a transformation preserves every distance.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q4/5

True or false?

One example in which a distance changes is enough to prove that a transformation does not preserve distance.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q5/5

Rotations and reflections that carry a rectangle, parallelogram, or trapezoid onto itself

MAIN	Geometry main question	**HSG-CO.A.3**
BACKWARD 1	Identify lines of symmetry of figures	**4.G.A.3**
BACKWARD 2	Know the side and angle properties of parallelograms, rectangles,…	**5.G.B.3**
BACKWARD 3	Rotate points by 90° and 180° about the origin	**8.G.A.3**
FORWARD 1	Use a symmetry to deduce congruent parts of the figure	**HSG-CO.B.6**
FORWARD 2	Combine symmetries and identify a figure from its symmetries	**HSG-CO.A.5**

SET 5 • HSG-CO.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

Rectangle ABCD is not a square. Describe every rotation (with angle greater than 0° and at most 360°) and every reflection that carries the rectangle onto itself.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q1/5

Parallelogram PQRS has no right angles, and its sides are not all the same length. Which transformation carries it onto itself?

A. A 90° rotation about the point where its diagonals meet

B. A reflection across diagonal PR

C. A reflection across the line through the midpoints of PQ and RS

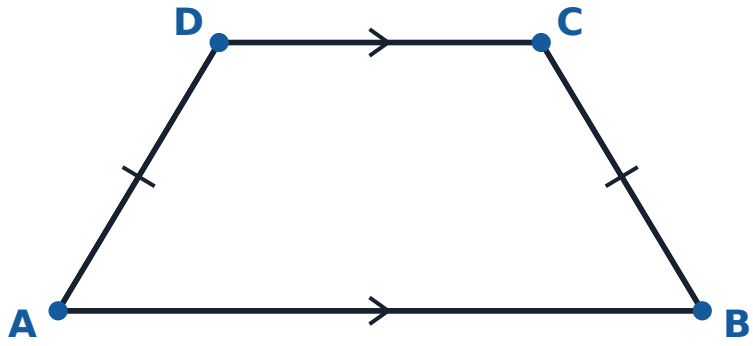
D. A 180° rotation about the point where its diagonals meet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q2/5

Trapezoid ABCD is isosceles, with bases AB and DC and legs AD = BC. Describe every rotation and reflection that carries it onto itself.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q3/5

True or false?

A 90° rotation about its center carries a rectangle that is not a square onto itself.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q4/5

Trapezoid EFGH has EF ∥ HG, EF = 4, HG = 8, and legs FG = 5 and EH = 3. Which rotations and reflections carry it onto itself?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q5/5

SET 5 • HSG-CO.A.3

BACKWARD 1

Identify lines of symmetry of figures • Grade 4 • Geometry • 4.G.A.3

How many lines of symmetry does a square have?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q1

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q1/5

How many lines of symmetry does a rectangle that is not a square have?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q2/5

True or false?

Each diagonal of a rectangle that is not a square is a line of symmetry of the rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q3

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q3/5

Which figure has exactly one line of symmetry?

A. A square

B. A circle

C. An isosceles triangle that is not equilateral

D. A scalene triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q4

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q4/5

True or false?

An isosceles trapezoid has a line of symmetry.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • HSG-CO.A.3

BACKWARD 2

Know the side and angle properties of parallelograms, rectangles, and trapezoids • Grade 5 • Geometry • 5.G.B.3

True or false?

In every parallelogram, the diagonals have equal lengths.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q1/5

True or false?

Every rectangle is a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q2

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q2/5

Which property do all rectangles have?

A. Diagonals that are perpendicular

B. Four sides of equal length

C. Four right angles

D. Exactly one pair of parallel sides

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q3

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q3/5

A parallelogram has one angle of 70°. Find its other three angles.

Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q4/5

An isosceles trapezoid has a 65° angle at one end of its longer base. What is the angle at the other end of that base?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q5/5

SET 5 • HSG-CO.A.3

BACKWARD 3

Rotate points by 90° and 180° about the origin • Grade 8 • Geometry • 8.G.A.3

Rotate the point (3, 1) by 180° about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q1/5

Rotate the point (2, 0) by 90° counterclockwise about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q2/5

True or false?

A 180° rotation about the origin sends (x, y) to (−x, −y).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q3/5

Rotate the point $(-4, 2)$ by 90° counterclockwise about the origin.

A. $(-2, -4)$

B. $(4, -2)$

C. $(-4, -2)$

D. $(2, 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q4/5

True or false?

A rotation of 360° about a point moves every point to a new position.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q5/5

SET 5 • HSG-CO.A.3

FORWARD 1

Use a symmetry to deduce congruent parts of the figure • High School — Geometry • Geometry: Congruence • HSG-CO.B.6

Isosceles trapezoid ABCD (AB ∥ DC) is carried onto itself by the reflection across the line through the midpoints of AB and DC. Which vertex does A go to? Use this to name a pair of congruent angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q1/5

A 180° rotation about point O carries parallelogram PQRS onto itself, sending P to R and Q to S. Name two pairs of congruent sides that follow.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q2/5

True or false?

Because a reflection carries an isosceles trapezoid onto itself, its two diagonals are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q3/5

A reflection carries rectangle ABCD onto itself and swaps vertices A and B (and C and D). Which equality follows?

A. AD = BC

B. AB = BC

C. AC = AB

D. AD = AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q4/5

True or false?

If a 180° rotation about a point carries a quadrilateral onto itself, the quadrilateral must be a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q5/5

SET 5 • HSG-CO.A.3

FORWARD 2

Combine symmetries and identify a figure from its symmetries • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Rectangle ABCD (not a square) has center O. Reflect it across the line through the midpoints of AB and CD, then across the line through the midpoints of BC and DA. Which single transformation has the same result?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q1/5

A quadrilateral is carried onto itself by exactly two transformations: the 360° rotation and one reflection across a line through two of its vertices. Which could it be?

A. A parallelogram that is not a rectangle or a rhombus

B. A kite that is not a rhombus

C. A square

D. A rectangle that is not a square

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q2/5

True or false?

A quadrilateral that is carried onto itself by a 90° rotation about a point must be a square.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q3/5

A quadrilateral is carried onto itself by a 180° rotation but by no reflection. What is the most specific name for it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q4/5

True or false?

Doing the same reflection twice gives the same result as the 360° rotation (the identity).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q5/5

Rotations and reflections that carry a regular polygon onto itself

MAIN	Geometry main question	**HSG-CO.A.3**
BACKWARD 1	Find the angle of a fraction of a full turn	**4.MD.C.5.b**
BACKWARD 2	Recognize regular polygons	**5.G.B.3**
BACKWARD 3	Find lines of symmetry of regular polygons	**4.G.A.3**
FORWARD 1	Work backward from a symmetry to the regular polygon	**HSG-CO.A.3**
FORWARD 2	Combine symmetries of a regular polygon	**HSG-CO.A.5**

SET 6 • HSG-CO.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

Describe every rotation about its center, with angle greater than 0° and at most 360°, that carries a regular pentagon onto itself.

Rotations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q1/5

How many reflections carry a regular hexagon onto itself? Describe their lines.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q2/5

Which rotation about its center carries a regular octagon onto itself?

A. 135°

B. 100°

C. 150°

D. 60°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q3/5

True or false?

Each line of symmetry of a regular nonagon (9 sides) passes through one vertex and the midpoint of the opposite side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q4/5

Describe every reflection that carries an equilateral triangle onto itself.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q5/5

SET 6 • HSG-CO.A.3

BACKWARD 1

Find the angle of a fraction of a full turn • Grade 4 • Measurement & Data • 4.MD.C.5.b

How many degrees are in $\frac{1}{5}$ of a full turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q1/5

How many degrees are in $\frac{1}{8}$ of a full turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q2/5

How many degrees are in $\frac{3}{6}$ of a full turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q3/5

True or false?

$\frac{1}{12}$ of a full turn is 30°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q4/5

True or false?

$\frac{1}{9}$ of a full turn is 36°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q5/5

SET 6 • HSG-CO.A.3

BACKWARD 2

Recognize regular polygons • Grade 5 • Geometry • 5.G.B.3

Which is a regular polygon?

A. A right triangle

B. An equilateral triangle

C. A rhombus that is not a square

D. A rectangle that is not a square

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q1/5

True or false?

A polygon with all sides equal must be regular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q2/5

How many sides does a regular decagon have?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q3/5

What is the name of the regular polygon with 4 sides?

Name: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q4/5

True or false?

A regular hexagon has 6 equal angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q5/5

SET 6 • HSG-CO.A.3

BACKWARD 3

Find lines of symmetry of regular polygons • Grade 4 • Geometry • 4.G.A.3

How many lines of symmetry does an equilateral triangle have?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q1

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q1/5

How many lines of symmetry does a regular pentagon have?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q2

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q2/5

True or false?

Every line of symmetry of a regular polygon passes through its center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q3

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q3/5

A line of symmetry of a square passes through

A. two adjacent vertices

B. two opposite vertices, or the midpoints of two opposite sides

C. exactly one vertex and no midpoint

D. no point of the square

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q4

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q4/5

True or false?

A regular hexagon has exactly 3 lines of symmetry.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q5

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q5/5

SET 6 • HSG-CO.A.3

FORWARD 1

Work backward from a symmetry to the regular polygon • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

The smallest positive rotation that carries a regular polygon onto itself is 24°. How many sides does it have?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q1/5

A regular polygon has 20 lines of symmetry. What is the smallest positive rotation that carries it onto itself?

Rotation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q2/5

True or false?

A regular polygon can have 50° as its smallest positive rotation onto itself.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q3/5

Which can be the smallest positive rotation that carries a regular polygon onto itself?

A. 35°

B. 70°

C. 80°

D. 40°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q4/5

Counting the 360° rotation, how many rotations and reflections in all carry a regular 12-gon onto itself?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q5/5

SET 6 • HSG-CO.A.3

FORWARD 2

Combine symmetries of a regular polygon • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Regular hexagon ABCDEF is labeled counterclockwise. It is rotated 120° counterclockwise about its center. Which vertex position does A move to?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q1/5

Regular hexagon ABCDEF is labeled counterclockwise. Rotate it 60° counterclockwise about its center, then reflect it across line AD (through the original positions of A and D). Which original vertex position does vertex B end at?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q2/5

True or false?

For a regular pentagon, a reflection followed by a different reflection that carries it onto itself is a rotation that carries it onto itself.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q3/5

Square ABCD is labeled counterclockwise. Reflect it across line AC, then rotate it 90° counterclockwise about its center. Which original vertex position does vertex B end at?

A. The position of A

B. The position of C

C. The position of D

D. The position of B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q4/5

Two successive 72° rotations of a regular pentagon about its center equal which single transformation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q5/5

Definitions of reflections and translations

MAIN	Geometry main question	**HSG-CO.A.4**
BACKWARD 1	Find the midpoint of a horizontal or vertical segment	**6.NS.C.8**
BACKWARD 2	Recognize perpendicular and parallel segments on a grid	**4.G.A.1**
BACKWARD 3	Reflect and translate points with coordinate rules	**8.G.A.3**
FORWARD 1	Find lines of reflection and images in slanted, non-axis cases	**HSG-CO.A.4**
FORWARD 2	Prove from the definitions that reflections and translations keep l…	**HSG-CO.A.4**

SET 7 • HSG-CO.A.4

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

Complete a precise definition. A reflection across line ℓ maps each point P not on ℓ to the point P′ such that _____. Each point on ℓ maps to _____.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q1/5

Points A and B are different. Write a precise definition of the translation that maps A to B.

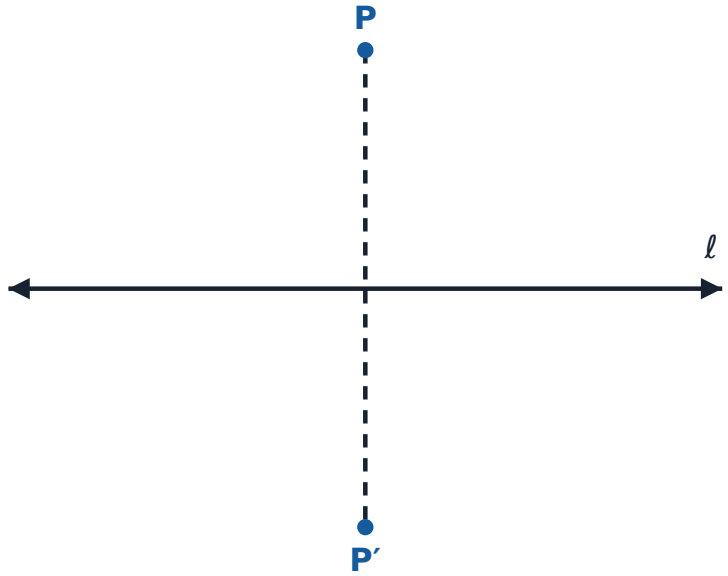
Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q2/5

Point P′ is the image of point P under a reflection across line ℓ, and P is not on ℓ. Which must be true?



A. ℓ is perpendicular to PP′ and passes through its midpoint

B. ℓ passes through P

C. PP′ is parallel to ℓ

D. P and P′ are on the same side of ℓ

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q3/5

True or false?

Under a translation, all the segments that join points to their images are parallel (or on one line) and have the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q4/5

A reflection maps P(1, 2) to P′(5, 2). Use the definition to find the line of reflection.

Line: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q5/5

SET 7 • HSG-CO.A.4

BACKWARD 1

Find the midpoint of a horizontal or vertical segment • Grade 6 • The Number System • 6.NS.C.8

Find the midpoint of the segment from (2, 4) to (8, 4).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q1/5

Find the midpoint of the segment from (1, 1) to (1, 9).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q2/5

On a number line, what number is halfway between −3 and 7?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q3/5

True or false?

The midpoint of the segment from (−4, 0) to (6, 0) is (1, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q4/5

True or false?

The midpoint of a segment is closer to one endpoint than to the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q5/5

SET 7 • HSG-CO.A.4

BACKWARD 2

Recognize perpendicular and parallel segments on a grid • Grade 4 • Geometry • 4.G.A.1

True or false?

A horizontal segment and a vertical segment are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q1/5

True or false?

Any two segments of the same length are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q2/5

Which pair of segments is parallel?

A. From (0, 0) to (3, 1) and from (0, 2) to (3, 3)

B. From (1, 1) to (2, 3) and from (1, 1) to (3, 2)

C. From (0, 0) to (3, 1) and from (0, 0) to (1, 3)

D. From (0, 0) to (4, 0) and from (0, 0) to (0, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q3/5

A segment goes from (2, 1) to (2, 6). Describe a segment through (5, 3) that is perpendicular to it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q4/5

Which describes two lines that are both perpendicular to the x-axis?

A. They are parallel

B. They are perpendicular to each other

C. They meet at the origin

D. They are the same line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q5/5

SET 7 • HSG-CO.A.4

BACKWARD 3

Reflect and translate points with coordinate rules • Grade 8 • Geometry • 8.G.A.3

Reflect the point (4, −3) across the x-axis.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q1/5

Translate the point $(-1, 5)$ 6 units right and 2 units down.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q2/5

Reflect the point (2, 7) across the line y = x.

A. (7, 2)

B. (−7, −2)

C. (−2, 7)

D. (2, −7)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q3/5

True or false?

Reflecting the point $(-6, 1)$ across the y-axis gives $(6, 1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q4/5

True or false?

Translating (3, 3) by 2 units left gives (3, 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q5/5

SET 7 • HSG-CO.A.4

FORWARD 1

Find lines of reflection and images in slanted, non-axis cases • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

A reflection maps P(0, 0) to P′(4, 2). Find an equation of the line of reflection.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q1/5

A translation maps A(2, −1) to A′(−3, 4). Where does it map B(0, 6)?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q2/5

True or false?

A reflection maps P to P′ and Q to Q′, where P and Q are not on the line of reflection. Then PP′ is parallel to QQ′ (or on the same line).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q3/5

A reflection maps A(1, 3) to A′(3, 1). Is the line of reflection y = x? Use the definition to decide.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q4/5

A reflection maps P(−1, 4) to P′(5, 4). Which point is on the line of reflection?

A. (−1, 0)

B. (2, −3)

C. (5, 0)

D. (4, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q5/5

SET 7 • HSG-CO.A.4

FORWARD 2

Prove from the definitions that reflections and translations keep lengths • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

Points P and Q are on the same side of line ℓ, and line PQ is not perpendicular to ℓ. A reflection across ℓ maps them to P′ and Q′. Let PP′ meet ℓ at M and QQ′ meet ℓ at N. Prove that PQ = P′Q′.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q1/5

A translation maps P to P′ and Q to Q′, where Q is not on line PP′. Prove that PQ = P′Q′.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q2/5

True or false?

A translation maps P to P′ and Q to Q′. Then PP′Q′Q is always a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q3/5

True or false?

Under a reflection, a point and its image are always the same distance from the line of reflection.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q4/5

In a proof that a reflection across ℓ keeps the length of segment PQ, which fact is used first?

A. ℓ bisects $\angle PQP'$

B. ℓ is parallel to PQ

C. ℓ is the perpendicular bisector of PP' and of QQ'

D. ℓ passes through P

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q5/5

Definition of a rotation

MAIN	Geometry main question	**HSG-CO.A.4**
BACKWARD 1	Describe turns in degrees, clockwise and counterclockwise	**4.MD.C.5.b**
BACKWARD 2	Points on a circle are the same distance from its center	**7.G.B.4**
BACKWARD 3	Rotate points 90°, 180°, and 270° about the origin	**8.G.A.3**
FORWARD 1	Rotate about a center that is not the origin	**HSG-CO.A.4**
FORWARD 2	Find the center and angle of a rotation from points and their imag..	**HSG-CO.A.4**

SET 8 • HSG-CO.A.4

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

Complete a precise definition. A counterclockwise rotation about point O through angle θ maps each point P ≠ O to the point P′ such that _____ and _____. Point O maps to _____.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q1/5

A rotation about point O maps point P to P′ (P ≠ P′). Which must be true?

A. O is on the perpendicular bisector of PP′

B. O is the midpoint of PP′

C. OP′ = 2 · OP

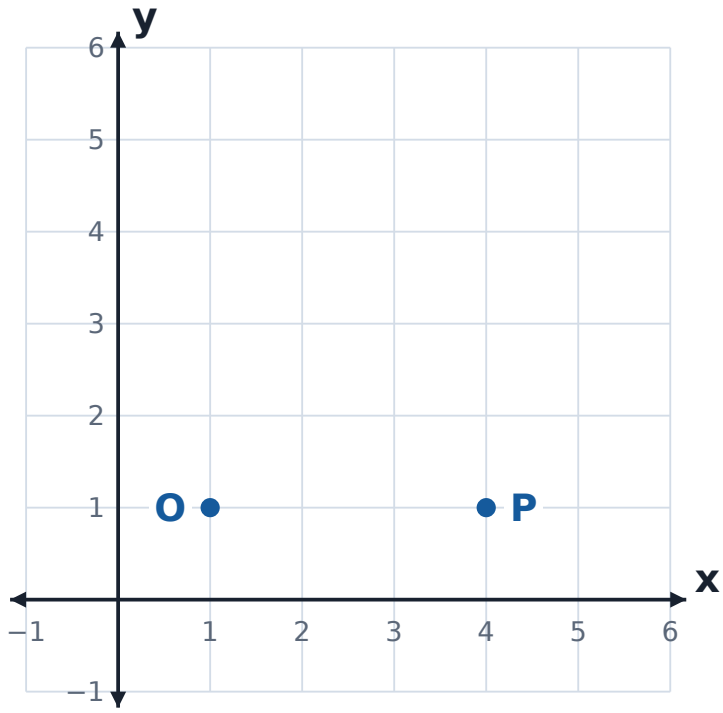
D. PP′ is perpendicular to OP

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q2/5

A rotation of 90° counterclockwise about O(1, 1) maps P(4, 1) to P′. Use the definition to find P′.



P′ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q3/5

True or false?

Under a rotation about point O, each point P moves along the circle with center O that passes through P.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q4/5

A rotation maps P(2, 0) to P′(0, 2). Explain why its center must lie on the line y = x.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q5/5

SET 8 • HSG-CO.A.4

BACKWARD 1

Describe turns in degrees, clockwise and counterclockwise • Grade 4 • Measurement & Data • 4.MD.C.5.b

A ray points right. It turns 90° counterclockwise. Which way does it point now?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q1/5

A ray points up. It turns 180°. Which way does it point now?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q2/5

True or false?

A clockwise turn of 270° ends in the same direction as a counterclockwise turn of 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q3/5

A ray points up. It turns 90° clockwise. Which way does it point now?

A. Down

B. Left

C. Right

D. Up

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q4/5

How many degrees are in $\frac{3}{4}$ of a full turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q5/5

SET 8 • HSG-CO.A.4

BACKWARD 2

Points on a circle are the same distance from its center • Grade 7 • Geometry • 7.G.B.4

A circle has center O and radius 6. Point P is on the circle. What is OP?

OP = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q1/5

True or false?

If OA = OB, then A and B lie on one circle with center O.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q2/5

Which point is on the circle with center (0, 0) and radius 5?

A. (4, 4)

B. (3, 4)

C. (2, 3)

D. (5, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q3/5

A circle has center O and diameter 10 cm. How far is every point of the circle from O?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q4/5

True or false?

Two points the same distance from O are always the same point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q5/5

SET 8 • HSG-CO.A.4

BACKWARD 3

Rotate points 90°, 180°, and 270° about the origin • Grade 8 • Geometry • 8.G.A.3

Rotate (5, 2) by 90° counterclockwise about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q1/5

Rotate $(-3, 4)$ by $180°$ about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q2/5

Rotate (1, 6) by 90° clockwise about the origin.

A. (−6, 1)

B. (6, −1)

C. (1, −6)

D. (−1, −6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q3/5

True or false?

Rotating (2, 3) by 270° counterclockwise about the origin gives (3, −2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q4/5

True or false?

Rotating (4, 0) by 90° counterclockwise about the origin gives (0, −4).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q5/5

SET 8 • HSG-CO.A.4

FORWARD 1

Rotate about a center that is not the origin • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

Rotate P(5, 4) by 90° clockwise about C(2, 3).

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q1/5

Rotate P(1, 5) by 180° about C(−1, 2).

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q2/5

Rotate P(3, −2) by 90° counterclockwise about C(0, −2).

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q3/5

Rotate P(1, 4) by 270° counterclockwise about C(1, 1).

A. (4, 1)

B. (4, 4)

C. (1, −2)

D. (−2, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q4/5

True or false?

Rotating P(4, 0) by 90° counterclockwise about C(2, 0) gives the same image as rotating P by 90° counterclockwise about the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q5/5

SET 8 • HSG-CO.A.4

FORWARD 2

Find the center and angle of a rotation from points and their images • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

A rotation maps B(1, 0) to B′(0, 1) and A(3, 1) to A′(−1, 3). The center is where the perpendicular bisectors of BB′ and AA′ meet. Find the center and the angle of rotation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q1/5

A rotation maps P(4, 1) to P′(2, 3) and Q(4, 3) to Q′(0, 3). Find the center and the angle of rotation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q2/5

True or false?

The center of a rotation is the same distance from every point as from that point's image.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q3/5

A rotation about C maps P to P′. $CP = 5$ and $PP' = 5\sqrt{2}$. Which could be the angle of rotation?

A. 90°

B. 60°

C. 45°

D. 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q4/5

A rotation about the origin maps (5, 0) to (3, 4). What is the counterclockwise angle of rotation, to the nearest degree?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q5/5

Draw the image of a figure under a rotation, reflection, or translation

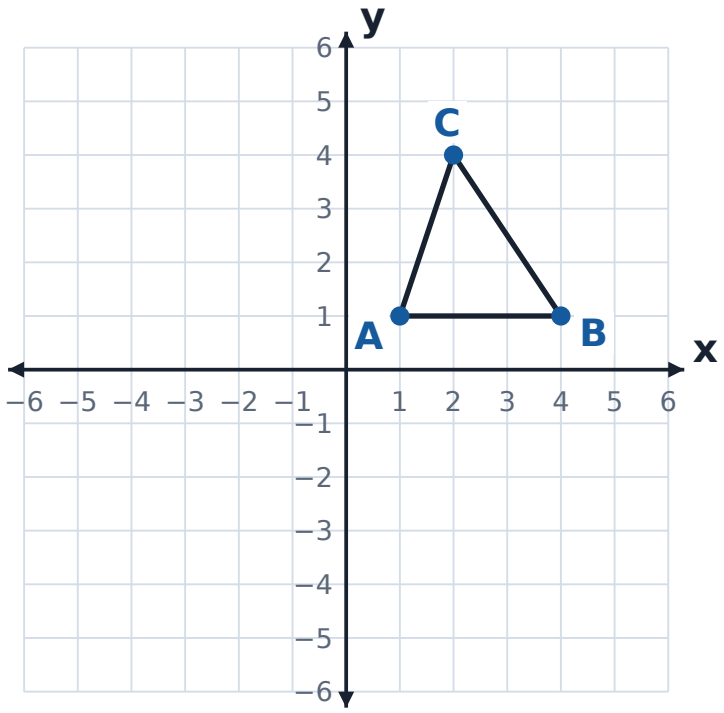
MAIN	Geometry main question	**HSG-CO.A.5**
BACKWARD 1	Plot and read points in all four quadrants	**6.NS.C.6.c**
BACKWARD 2	Find image points with coordinate rules	**8.G.A.3**
BACKWARD 3	Count the distance from a point to a horizontal or vertical line	**6.NS.C.8**
FORWARD 1	Draw the image under a sequence of two transformations	**HSG-CO.A.5**
FORWARD 2	Draw rotations and reflections off the grid with compass, protract…	**HSG-CO.A.5**

SET 9 • HSG-CO.A.5

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Draw the image of triangle ABC under the translation 5 units left and 2 units down. Label the image A′B′C′.

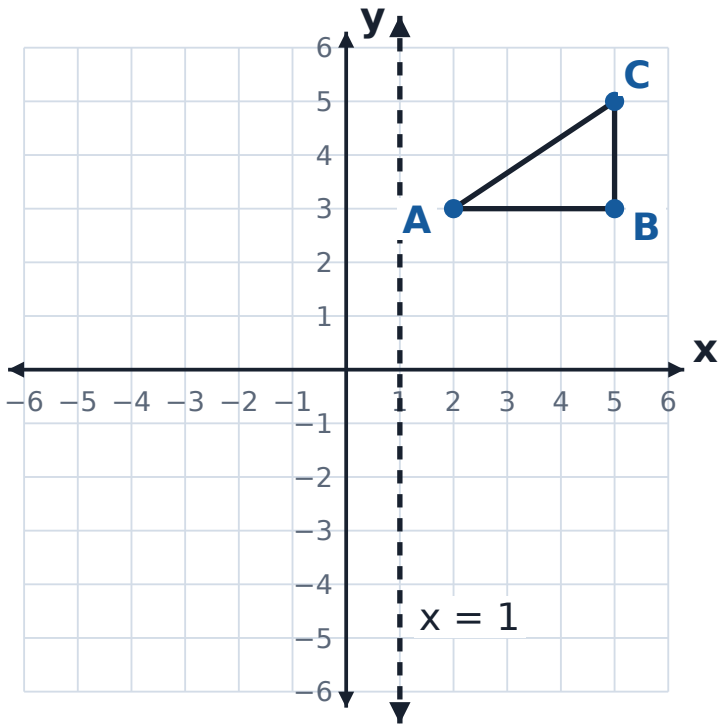


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q1/5

Draw the image of triangle ABC under the reflection across the line x = 1. Label the image A′B′C′.

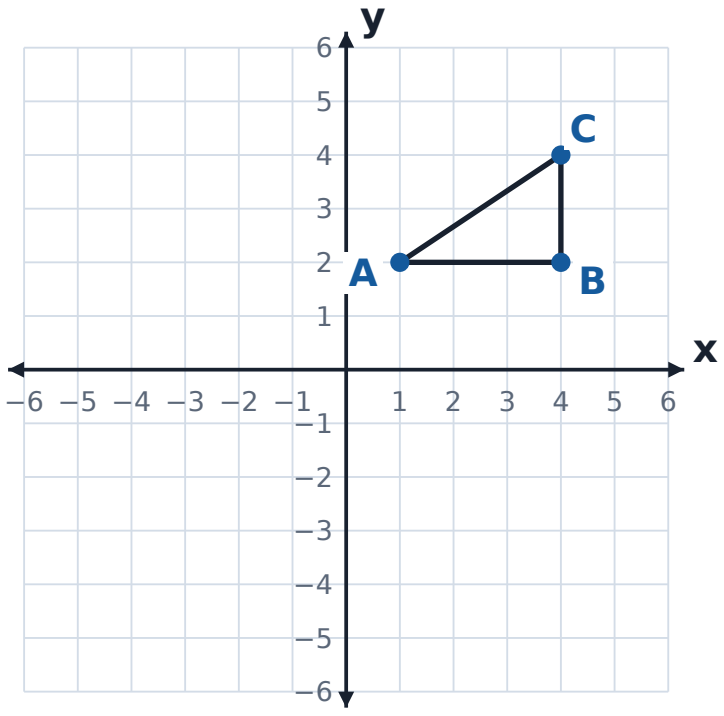


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q2/5

Draw the image of triangle ABC under the rotation of 90° counterclockwise about the origin. Label the image A′B′C′.

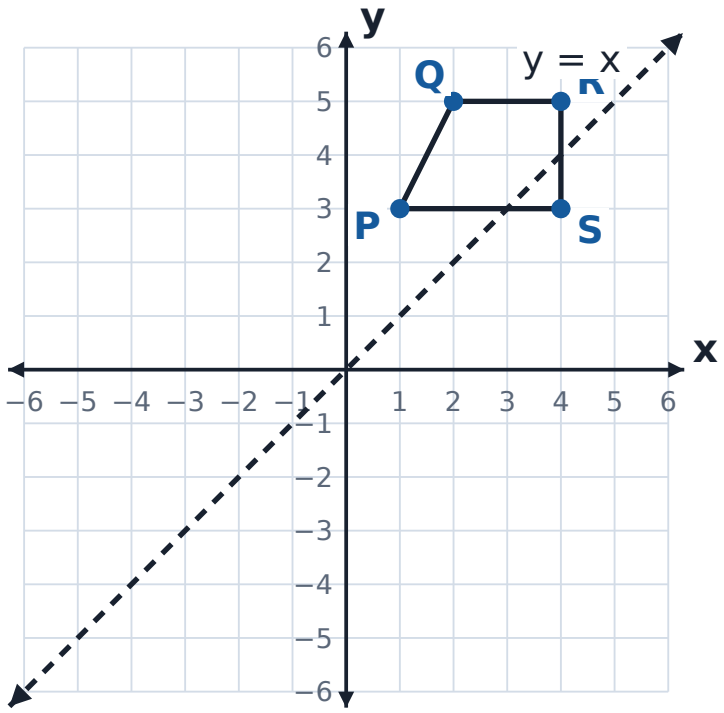


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q3/5

Draw the image of quadrilateral PQRS under the reflection across the line $y = x$. Label the image $P'Q'R'S'$.

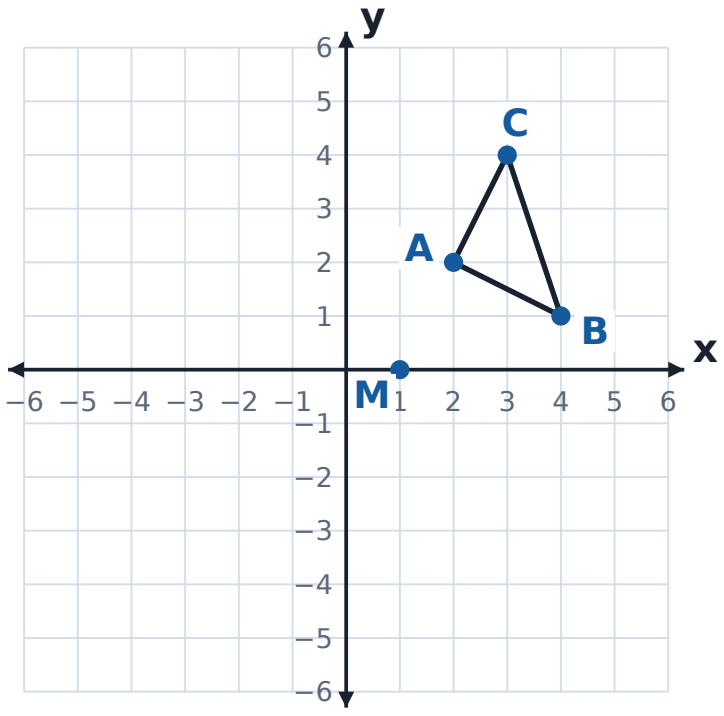


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q4/5

Draw the image of triangle ABC under the rotation of 180° about the point M(1, 0). Label the image A′B′C′.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q5

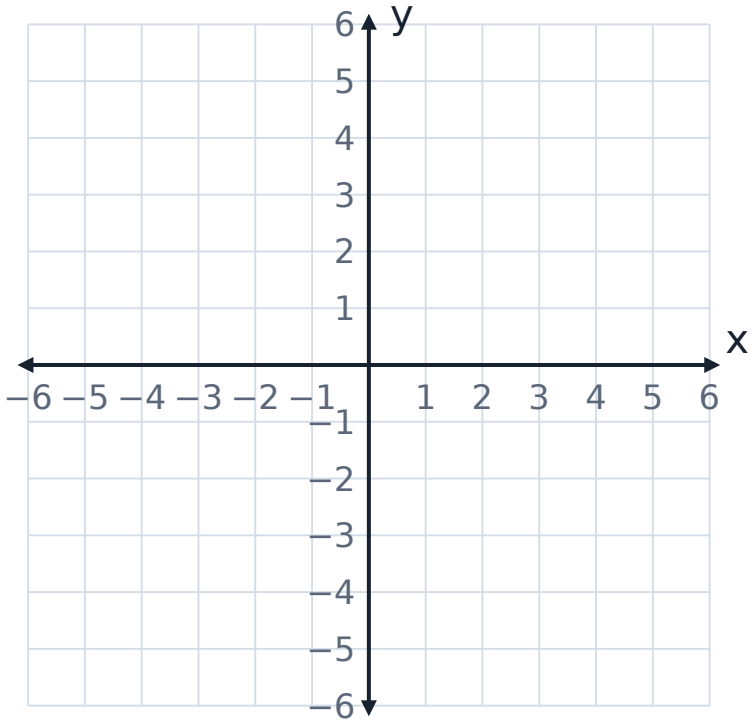
High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q5/5

SET 9 • HSG-CO.A.5

BACKWARD 1

Plot and read points in all four quadrants • Grade 6 • The Number System • 6.NS.C.6.c

Plot and label the points A(−3, 2), B(0, −4), and C(5, 1).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q1/5

A point is 4 units left of the y-axis and 1 unit below the x-axis. What are its coordinates?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q2/5

Which point is 2 units above the x-axis and on the y-axis?

A. (2, 2)

B. (0, 2)

C. (0, −2)

D. (2, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q3/5

True or false?

The point (3, −2) is in Quadrant IV.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q4/5

True or false?

To plot (2, 5), start at the origin and move 2 units up and 5 units right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q5/5

SET 9 • HSG-CO.A.5

BACKWARD 2

Find image points with coordinate rules • Grade 8 • Geometry • 8.G.A.3

Find the image of $(3, -2)$ under a translation 4 units left and 1 unit up.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q1/5

Find the image of $(-2, 5)$ under a reflection across the x-axis.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q2/5

Find the image of (3, 1) under a 90° clockwise rotation about the origin.

A. (−3, −1)

B. (3, −1)

C. (−1, 3)

D. (1, −3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q3/5

True or false?

The image of $(4, -1)$ under a reflection across the y-axis is $(-4, -1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q4/5

True or false?

The image of (2, 2) under a 180° rotation about the origin is (2, −2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q5/5

SET 9 • HSG-CO.A.5

BACKWARD 3

Count the distance from a point to a horizontal or vertical line • Grade 6 • The Number System • 6.NS.C.8

How far is the point (5, 3) from the vertical line x = 1?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q1/5

How far is the point $(-2, -1)$ from the horizontal line $y = 3$?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q2/5

Which point is 3 units from the line x = 2, on its left?

A. (−3, 4)

B. (2, 1)

C. (5, 4)

D. (−1, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q3/5

True or false?

The point (0, 4) is 4 units from the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q4/5

True or false?

The point (6, 2) is 6 units from the line $x = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q5

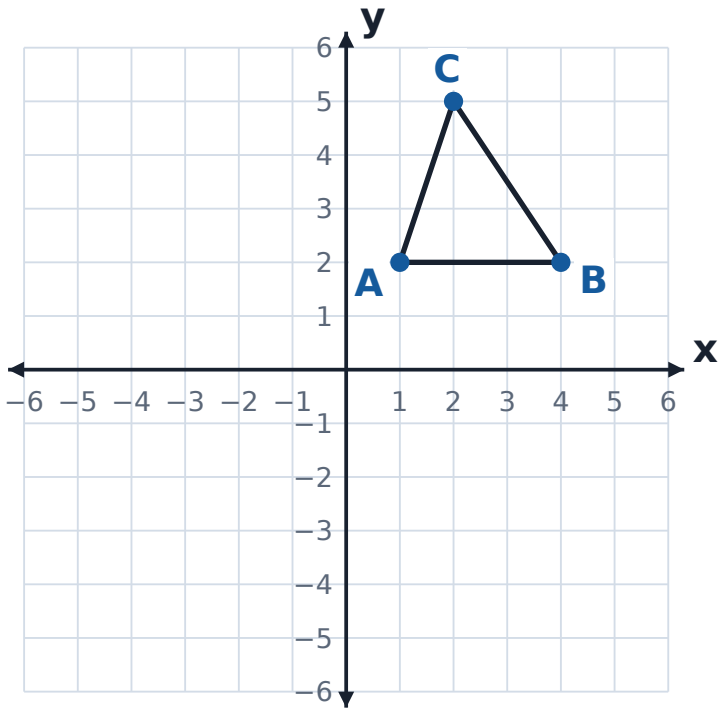
Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q5/5

SET 9 • HSG-CO.A.5

FORWARD 1

Draw the image under a sequence of two transformations • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Reflect triangle ABC across the y-axis, then translate the result 3 units down. Draw and label the final image A″B″C″.

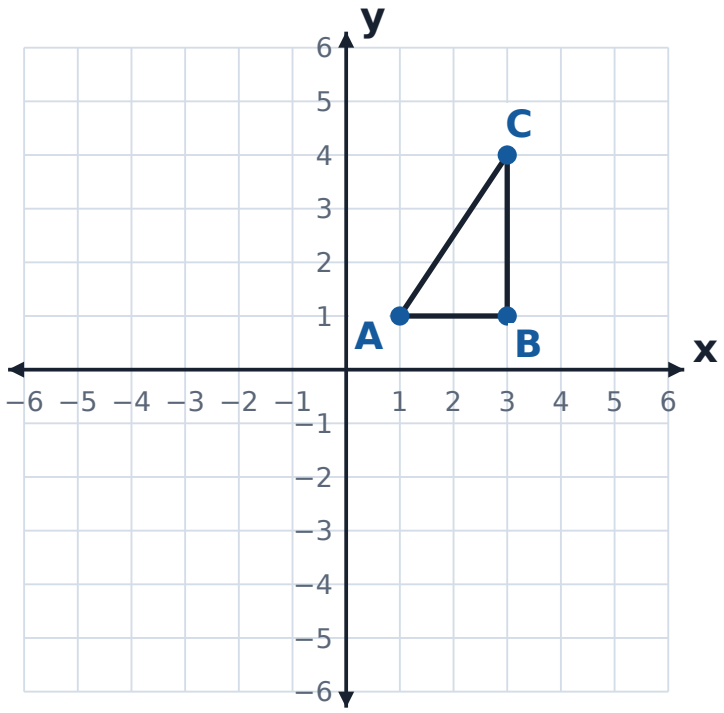


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q1/5

Rotate triangle ABC by 90° counterclockwise about the origin, then reflect the result across the x-axis. Draw and label the final image A″B″C″.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q2/5

Triangle ABC has A(2, 1). It is translated 2 units right and then rotated 180° about the origin. Where does A end up?

A″ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q3/5

True or false?

Reflecting a figure across the x-axis and then across the y-axis gives the same image as rotating it 180° about the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q4/5

Point P(1, 3) is translated 2 units left and then reflected across the line y = x. Where does it end up?

A. (3, 1)

B. (−1, 3)

C. (1, −3)

D. (3, −1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q5/5

SET 9 • HSG-CO.A.5

FORWARD 2

Draw rotations and reflections off the grid with compass, protractor, and straightedge • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Use a compass and protractor. Rotate point P 60° counterclockwise about point O. Mark and label the image P′.

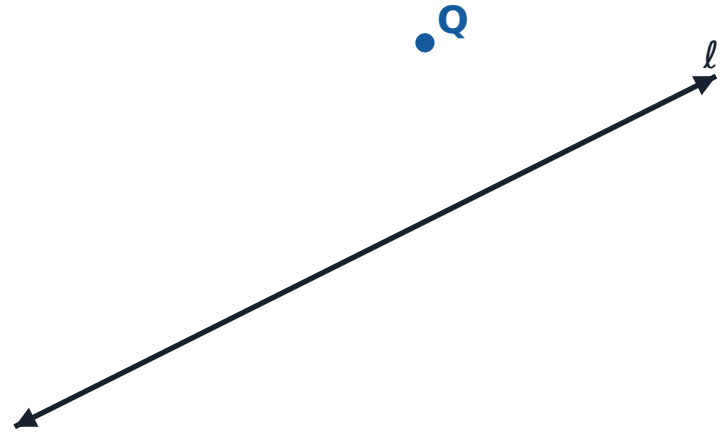


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q1/5

Use a compass and straightedge. Reflect point Q across line ℓ. Mark and label the image Q′.

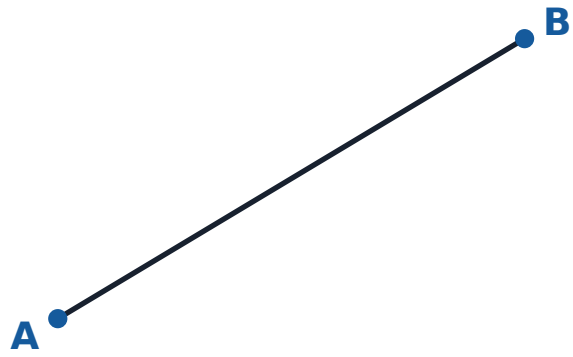


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q2/5

Use a compass and protractor. Rotate segment AB 90° clockwise about point A. Draw and label the image AB′.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q3/5

True or false?

To rotate point P about point O by 45°, it is enough to draw a 45° angle at O; the distance from O does not matter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q4/5

Describe how to reflect point P across a slanted line ℓ using only a compass and straightedge.

Steps: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q5/5

Specify a sequence of transformations that carries one figure onto another

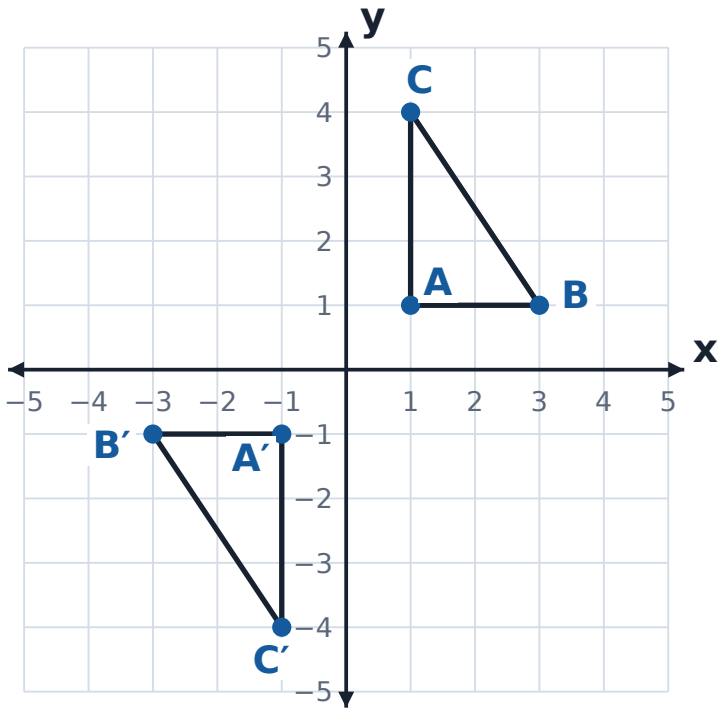
MAIN	Geometry main question	**HSG-CO.A.5**
BACKWARD 1	Understand congruence as the result of one rigid motion	**8.G.A.2**
BACKWARD 2	Find images with coordinate rules for sequences	**8.G.A.3**
BACKWARD 3	Read the orientation (clockwise or counterclockwise order) of a tr…	**8.G.A.1**
FORWARD 1	Use the fewest transformations and justify	**HSG-CO.A.5**
FORWARD 2	Specify a similarity sequence: a dilation and rigid motions	**HSG-SRT.A.2**

SET 10 • HSG-CO.A.5

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Describe a sequence of rigid transformations that carries triangle ABC onto triangle A′B′C′ (A to A′, B to B′, C to C′).



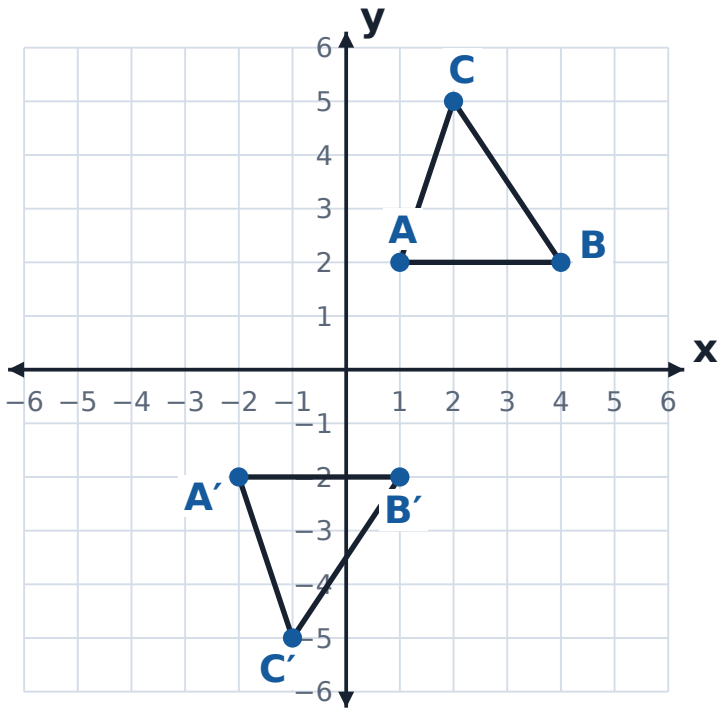
Sequence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q1/5

Describe a sequence of rigid transformations that carries triangle ABC onto triangle A′B′C′.



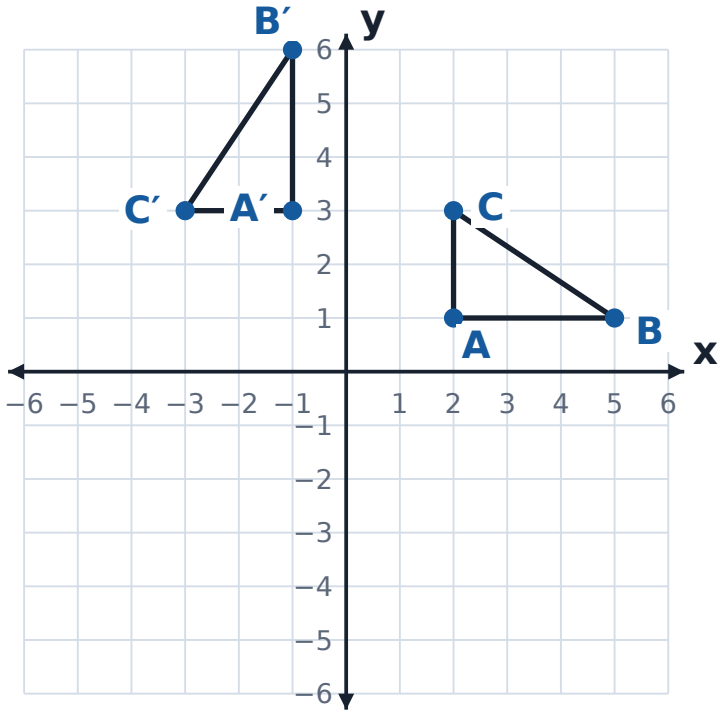
Sequence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q2/5

On the grid, triangle A′B′C′ is congruent to triangle ABC. Describe a sequence of rigid transformations that carries ABC onto A′B′C′, with A → A′, B → B′, C → C′.



Sequence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q3/5

Right triangle ABC has A(0, 0), B(2, 0), C(0, 3). Triangle A′B′C′ has A′(4, 0), B′(4, 2), C′(1, 0). Which sequence carries ABC onto A′B′C′?

A. Reflect across the y-axis, then translate 4 units right

B. Rotate 90° counterclockwise about the origin, then translate 4 units right

C. Rotate 90° clockwise about the origin, then translate 4 units right

D. Translate 4 units right, then rotate 90° counterclockwise about the origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q4/5

True or false?

Triangles PQR and P′Q′R′ are congruent. P, Q, R are in clockwise order and P′, Q′, R′ are in counterclockwise order. Some sequence of rotations and translations alone carries P to P′, Q to Q′, and R to R′.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q5/5

SET 10 • HSG-CO.A.5

BACKWARD 1

Understand congruence as the result of one rigid motion • Grade 8 • Geometry • 8.G.A.2

Triangle 2 is the image of triangle 1 after one rigid motion. Every point moved 3 units right. Which motion was it?

A. A reflection

B. A rotation

C. A translation

D. A dilation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q1/5

A triangle and its image face opposite ways, as in a mirror. Which kind of rigid motion was most likely used?

Motion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q2/5

True or false?

If one figure is the image of another under a sequence of rigid motions, the figures are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q3/5

True or false?

A figure and its image under a rigid motion can have different areas.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q4/5

Segment AB goes from (0, 0) to (2, 0). Its image goes from (0, 0) to (0, 2). Which single rigid motion could do this?

A. A 90° counterclockwise rotation about the origin

B. A translation 2 units up

C. A reflection across the x-axis

D. A 180° rotation about the origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q5/5

SET 10 • HSG-CO.A.5

BACKWARD 2

Find images with coordinate rules for sequences • Grade 8 • Geometry • 8.G.A.3

Point (2, −3) is reflected across the y-axis. Find its image.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q1/5

Point $(-1, 4)$ is rotated 90° counterclockwise about the origin. Find its image.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q2/5

Point (5, 1) is translated 2 units left and 3 units down. Find its image.

A. (7, 4)

B. (3, −2)

C. (7, −2)

D. (3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q3/5

True or false?

Point (0, 3) rotated 180° about the origin is (0, −3).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q4/5

True or false?

Point (4, 1) reflected across the line $y = x$ is $(4, -1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q5/5

SET 10 • HSG-CO.A.5

BACKWARD 3

Read the orientation (clockwise or counterclockwise order) of a triangle • Grade 8 • Geometry • 8.G.A.1

Triangle ABC has A(0, 0), B(4, 0), C(0, 3). Going A → B → C, is the order clockwise or counterclockwise?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q1/5

True or false?

A reflection reverses the orientation of a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q2/5

True or false?

A rotation reverses the orientation of a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q3/5

Triangle PQR is clockwise. After two reflections, its image is

A. clockwise

B. counterclockwise

C. either, depending on the lines

D. not a triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q4/5

Triangle ABC is counterclockwise. It is translated and then reflected. Is the final image clockwise or counterclockwise?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q5/5

SET 10 • HSG-CO.A.5

FORWARD 1

Use the fewest transformations and justify • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Two reflections are done one after the other, across parallel lines that are 3 units apart. Which single transformation has the same result?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q1/5

Two reflections are done one after the other, across lines that meet at point O and form a 40° angle. Which single transformation has the same result?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q2/5

Triangle ABC is clockwise and triangle A′B′C′ is congruent to it and counterclockwise. Explain why no single rotation or translation carries ABC onto A′B′C′.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q3/5

True or false?

Every rigid motion of the plane can be written as a sequence of at most three reflections.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q4/5

Triangle A′B′C′ is the image of triangle ABC under some rigid motion, and both are counterclockwise. What is the fewest number of reflections that can always carry ABC onto A′B′C′?

A. 2

B. 4

C. 3

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q5/5

SET 10 • HSG-CO.A.5

FORWARD 2

Specify a similarity sequence: a dilation and rigid motions • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Triangle ABC has A(0, 0), B(2, 0), C(0, 1). Triangle A′B′C′ has A′(0, 0), B′(0, 6), C′(−3, 0). Describe a sequence (a dilation and a rigid motion) that carries ABC onto A′B′C′.

Sequence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q1/5

Square S has vertices (0, 0), (1, 0), (1, 1), (0, 1). Square T has vertices (2, 2), (4, 2), (4, 4), (2, 4). Describe a similarity sequence carrying S onto T.

Sequence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q2/5

True or false?

Any two figures that a sequence of a dilation and rigid motions carries one onto the other are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q3/5

Segment AB is from (0, 0) to (3, 0). Segment CD is from (1, 1) to (1, 7). Which sequence carries AB onto CD (A to C, B to D)?

A. Rotate 90° counterclockwise about the origin, then translate 1 right and 1 up

B. Dilate by 2 about the origin, then translate 1 right and 1 up

C. Dilate by 2 about the origin, rotate 90° counterclockwise about the origin, then translate 1 right and 1 up

D. Dilate by 3 about the origin, rotate 90° counterclockwise, then translate 1 right and 1 up

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q4/5

Triangle PQR has sides 3, 4, 5. Triangle XYZ has sides 6, 8, 10. What scale factor must the dilation in a similarity sequence from PQR to XYZ have?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q5/5

Predict the effect of a rigid motion on a figure

MAIN	Geometry main question	**HSG-CO.B.6**
BACKWARD 1	Know what rigid motions keep: lines, lengths, angles, and parallel l…	**8.G.A.1**
BACKWARD 2	Match the vertices and sides of a figure with its image	**8.G.A.2**
FORWARD 1	Predict the effect of a sequence of rigid motions	**HSG-CO.B.6**
FORWARD 2	Use a rigid motion to conclude that parts of a figure are congruent	**HSG-CO.B.7**

SET 11 • HSG-CO.B.6

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.6

Triangle ABC has AB = 7, BC = 5, and m∠B = 48°. It is reflected across a line ℓ. What are A′B′ and m∠B′?

A. 14 and 48°

B. 7 and 132°

C. 7 and 48°

D. They depend on where ℓ is

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q1/5

Quadrilateral PQRS has PQ ∥ SR. It is rotated 90° about a point O. What is true about P′Q′ and S′R′? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q2/5

Point A is 4 cm from line m. Triangle ABC is reflected across m. How far is A′ from m, and on which side of m is it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q3/5

True or false?

A translation can reverse the orientation (the clockwise or counterclockwise order) of a triangle's vertices.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q4/5

Segment AB lies on line ℓ. A 180° rotation about a point O on ℓ maps AB to A′B′. Where does A′B′ lie? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q5/5

SET 11 • HSG-CO.B.6

BACKWARD 1

Know what rigid motions keep: lines, lengths, angles, and parallel lines • Grade 8 • Geometry • 8.G.A.1

True or false?

Under a rotation, the image of a line is a line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q1/5

True or false?

A reflection can turn two parallel lines into two lines that meet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q2/5

A 52° angle is translated 6 units to the right. What is the measure of its image?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q3/5

A 9 cm segment is rotated 30° about one of its endpoints. How long is its image?

A. 9 cm

B. 30 cm

C. 4.5 cm

D. 39 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q4/5

Name two properties of a figure that a reflection does NOT change.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q5/5

SET 11 • HSG-CO.B.6

BACKWARD 2

Match the vertices and sides of a figure with its image • Grade 8 • Geometry • 8.G.A.2

Triangle ABC is rotated to triangle A′B′C′. Which side of the image corresponds to side BC?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q1/5

Quadrilateral PQRS is reflected to P′Q′R′S′. Which angle corresponds to ∠Q?

A. ∠R′

B. ∠P′

C. ∠S′

D. ∠Q′

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q2/5

True or false?

If triangle ABC is translated to triangle A′B′C′, then AB = A′B′.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q3/5

Triangle DEF is reflected to D′E′F′. DE = 6 and EF = 9. Find D′E′ and E′F′.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q4/5

True or false?

After a rotation, side A′B′ of the image always lies on top of side AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q5/5

SET 11 • HSG-CO.B.6

FORWARD 1

Predict the effect of a sequence of rigid motions • High School — Geometry • Geometry: Congruence • HSG-CO.B.6

Triangle ABC is reflected across line ℓ and then across line m, where $\ell \parallel$ m. Is the final image clockwise or counterclockwise if ABC is counterclockwise? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q1/5

A figure is rotated 70° about point O and then rotated 50° about the same point O. Which single rigid motion has the same effect?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q2/5

True or false?

A translation that is not the identity moves every point, so it has no fixed points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q3/5

Which rigid motion keeps exactly one point fixed?

A. A translation 3 units right

B. A rotation of 90° about point O

C. A reflection across line ℓ

D. The identity

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q4/5

Line ℓ is reflected across line m, and ℓ is perpendicular to m. What is the image of ℓ? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q5/5

SET 11 • HSG-CO.B.6

FORWARD 2

Use a rigid motion to conclude that parts of a figure are congruent • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

A rotation about O maps triangle OAB onto triangle OCD, with A → C and B → D. OA = 5 and AB = 7. Find OC and CD.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q1/5

A reflection across line ℓ maps triangle PQR onto triangle PSR (Q → S, and P and R are on ℓ). $m\angle PQR = 64°$. Find $m\angle PSR$.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q2/5

True or false?

If a rigid motion maps segment AB onto segment CD, then AB = CD.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q3/5

A translation maps triangle KLM onto triangle K′L′M′. Which conclusion is NOT justified?

A. KL is on the same line as K′L′

B. KL ∥ K′L′ or KL is on line K′L′

C. ∠L ≅ ∠L′

D. KL = K′L′

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q4/5

Explain why the image of an isosceles triangle under any rigid motion is also isosceles.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q5/5

Decide whether two figures are congruent using rigid motions

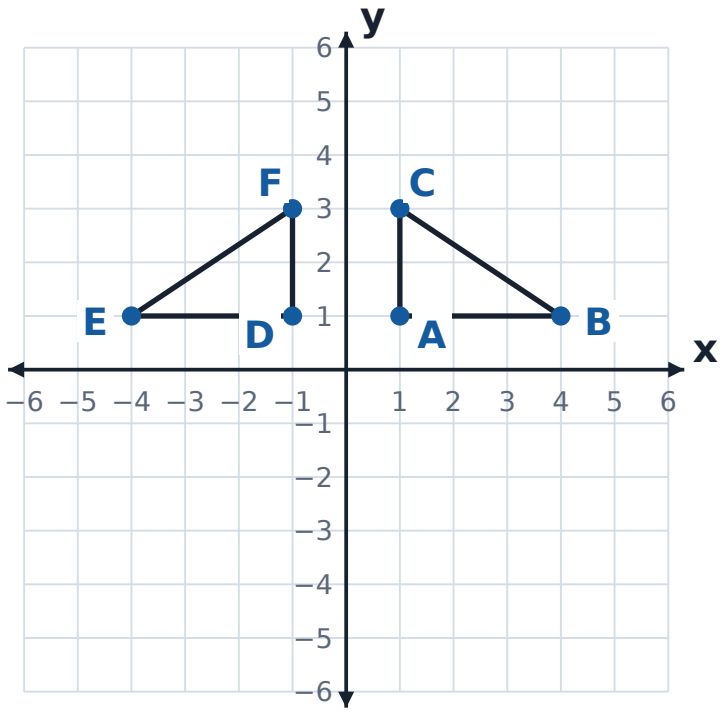
MAIN	Geometry main question	**HSG-CO.B.6**
BACKWARD 1	Recognize congruent figures as images under rigid motions	**8.G.A.2**
BACKWARD 2	Find lengths on a grid with the Pythagorean Theorem	**8.G.B.8**
FORWARD 1	Decide congruence for figures given by measurements, including…	**HSG-CO.B.6**
FORWARD 2	Use an established congruence to find unknown parts	**HSG-CO.B.7**

SET 12 • HSG-CO.B.6

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.6

Is triangle ABC congruent to triangle DEF? If so, give a rigid motion that carries ABC onto DEF. If not, explain why.



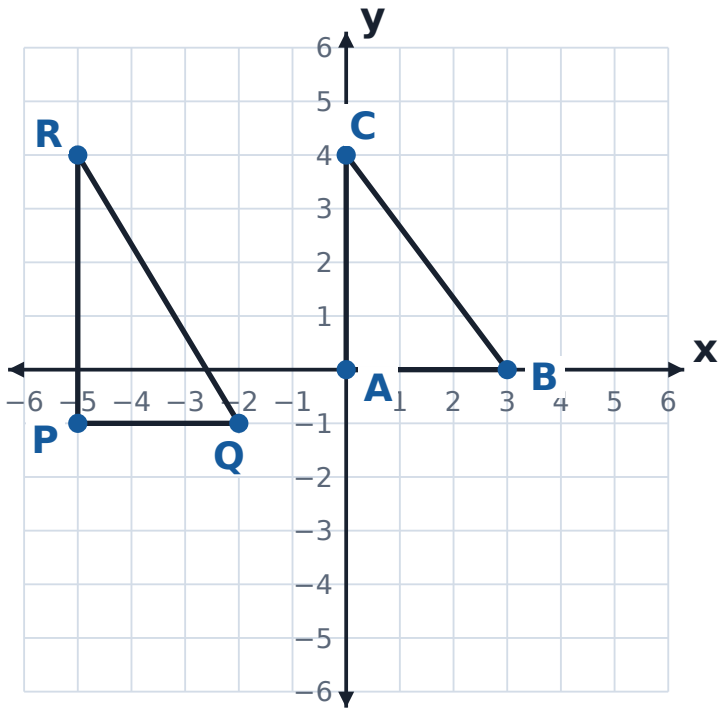
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q1/5

Is triangle ABC congruent to triangle PQR? If so, give a rigid motion that carries one onto the other. If not, explain why.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q2/5

Which fact proves that two figures are NOT congruent?

A. One is above the other

B. One has area 12 and the other has area 15

C. They face opposite directions

D. They are in different quadrants

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q3/5

True or false?

Two figures are congruent if some sequence of rotations, reflections, and translations carries one onto the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q4/5

Rectangle R is 4 cm wide and 6 cm tall. Rectangle S is 6 cm wide and 4 cm tall. Are they congruent? Give a rigid motion or explain why not.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q5/5

SET 12 • HSG-CO.B.6

BACKWARD 1

Recognize congruent figures as images under rigid motions • Grade 8 • Geometry • 8.G.A.2

True or false?

If one figure is a translation of another, the figures are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q1/5

True or false?

Two figures with the same shape but different sizes are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q2/5

Which pair of figures must be congruent?

A. A square and a larger square

B. A circle of radius 2 and a circle of radius 3

C. A triangle and a dilation of it by 2

D. A triangle and its reflection

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q3/5

A triangle has sides 5, 6, and 7. It is rotated 90°. What are the side lengths of its image?

Lengths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q4/5

Is a figure congruent to its image after a reflection followed by a translation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q5/5

SET 12 • HSG-CO.B.6

BACKWARD 2

Find lengths on a grid with the Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.8

Find the length of the segment from (1, 2) to (5, 2).

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q1/5

Find the length of the segment from (0, 0) to (3, 4).

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q2/5

Find the length of the segment from $(-2, 1)$ to $(1, 5)$.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q3/5

True or false?

The segments from (0, 0) to (2, 1) and from (0, 0) to (1, 2) have the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q4/5

True or false?

The segment from (0, 0) to (3, 3) has length 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q5/5

SET 12 • HSG-CO.B.6

FORWARD 1

Decide congruence for figures given by measurements, including curved figures • High School — Geometry • Geometry: Congruence • HSG-CO.B.6

Circle P has radius 3 and circle Q has radius 3. Explain, using a rigid motion, why they are congruent.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q1/5

Two quadrilaterals each have sides 5, 5, 5, 5. One is a square and the other has a 60° angle. Are they congruent? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q2/5

True or false?

Two triangles with the same perimeter must be congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q3/5

Which information is enough to show two segments are congruent?

A. They have the same length

B. They have the same slope

C. They have the same midpoint

D. They both start at the origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q4/5

Angle 1 measures 35° and angle 2 measures 35°. Describe a rigid motion that carries angle 1 onto angle 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q5/5

SET 12 • HSG-CO.B.6

FORWARD 2

Use an established congruence to find unknown parts • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

A rigid motion carries quadrilateral ABCD onto quadrilateral EFGH with A → E, B → F, C → G, D → H. AB = 6, ∠C = 110°, and GH = 4. Find EF, ∠G, and CD.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q1/5

Triangle RST is congruent to triangle XYZ by a rigid motion with R → X, S → Y, T → Z. RS = 2x + 3 and XY = 15. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q2/5

True or false?

If a rigid motion carries triangle ABC onto triangle DEF with A → D, B → E, C → F, then ∠B ≅ ∠F.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q3/5

A reflection carries pentagon PQRST onto pentagon P′Q′R′S′T′. Which pair must be congruent?

A. PQ and RS

B. QR and S′T′

C. ∠S and ∠T′

D. ∠S and ∠S′

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q4/5

Triangle ABC is congruent to triangle CDA by a rigid motion with A → C, B → D, C → A. m∠BAC = 3y° and m∠DCA = 42°. Find y.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q5/5

Congruent triangles: a rigid motion exactly when all corresponding parts are congruent

MAIN	Geometry main question	**HSG-CO.B.7**
BACKWARD 1	Read a congruence statement to match corresponding parts	**8.G.A.2**
BACKWARD 2	Rigid motions keep lengths and angle measures	**8.G.A.1**
BACKWARD 3	Move a point onto another and a ray onto another with rigid motio…	**8.G.A.3**
FORWARD 1	Find which three corresponding parts are enough	**HSG-CO.B.8**
FORWARD 2	Use congruent triangles to prove a relationship in a figure	**HSG-SRT.B.5**

SET 13 • HSG-CO.B.7

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

A rigid motion carries $\triangle ABC$ onto $\triangle DEF$ with $A \to D$, $B \to E$, $C \to F$. Explain why $AB = DE$, $BC = EF$, $CA = FD$, $\angle A \cong \angle D$, $\angle B \cong \angle E$, and $\angle C \cong \angle F$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q1/5

In $\triangle ABC$ and $\triangle DEF$, $AB = DE$, $BC = EF$, $CA = FD$, $\angle A \cong \angle D$, $\angle B \cong \angle E$, and $\angle C \cong \angle F$. Explain how to build a rigid motion that carries $\triangle ABC$ onto $\triangle DEF$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q2/5

A rigid motion maps $\triangle PQR$ onto $\triangle XYZ$ with $P \to X$, $Q \to Y$, $R \to Z$. Which must be true?

A. $QR = YZ$

B. $PQ = YZ$

C. $\angle P \cong \angle Y$

D. $QR = XZ$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q3/5

$\triangle JKL \cong \triangle MNO$ with $J \leftrightarrow M$, $K \leftrightarrow N$, $L \leftrightarrow O$. $JK = 8$, $KL = 11$, and $m\angle L = 37°$. Find MN, NO, and $m\angle O$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q4/5

True or false?

Two triangles whose three pairs of corresponding angles are congruent must be congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q5/5

SET 13 • HSG-CO.B.7

BACKWARD 1

Read a congruence statement to match corresponding parts • Grade 8 • Geometry • 8.G.A.2

△ABC ≅ △XYZ. Which angle corresponds to ∠B?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q1/5

△ABC ≅ △XYZ. Which side corresponds to AC?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q2/5

△PQR ≅ △LMN. Which pair corresponds?

A. QR and LN

B. PQ and MN

C. PR and LM

D. QR and MN

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q3/5

True or false?

$\triangle DEF \cong \triangle GHI$ means that $\angle F \cong \angle I$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q4/5

True or false?

$\triangle DEF \cong \triangle GHI$ means that $DE = HI$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q5/5

SET 13 • HSG-CO.B.7

BACKWARD 2

Rigid motions keep lengths and angle measures • Grade 8 • Geometry • 8.G.A.1

True or false?

A rotation keeps the measure of every angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q1/5

Segment PQ is 12 units long. It is reflected and then translated. How long is the image?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q2/5

A 71° angle is reflected across a line. Its image measures

A. 289°

B. 109°

C. 71°

D. 19°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q3/5

True or false?

A translation can change the length of a segment if it moves the segment far enough.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q4/5

Which three basic rigid motions keep lengths and angle measures?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q5/5

SET 13 • HSG-CO.B.7

BACKWARD 3

Move a point onto another and a ray onto another with rigid motions • Grade 8 • Geometry • 8.G.A.3

Which translation moves (2, 3) onto (5, −1)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q1/5

A ray starts at the origin and points along the positive x-axis. Which rotation about the origin makes it point along the positive y-axis?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q2/5

After a rotation, point C is below line DE but should be above it, and D and E are on the line. Which motion moves C across without moving D or E?

A. A rotation about D

B. A reflection across line DE

C. A translation along DE

D. A rotation about the midpoint of DE

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q3/5

True or false?

Reflecting across a line keeps every point of that line fixed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q4/5

True or false?

Translating a triangle can keep exactly one of its vertices fixed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q5/5

SET 13 • HSG-CO.B.7

FORWARD 1

Find which three corresponding parts are enough • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

Must two triangles be congruent if all three pairs of corresponding sides are congruent? Use the result of the full six-part test.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q1/5

Give an example of two triangles with two pairs of congruent sides and a pair of congruent angles that are NOT congruent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q2/5

True or false?

Knowing that all three pairs of corresponding angles are congruent is enough to prove two triangles congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q3/5

Which set of corresponding parts is NOT always enough for congruence?

A. Two sides and an angle not between them

B. Three sides

C. Two sides and the angle between them

D. Two angles and the side between them

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q4/5

Two triangles have AB = DE, ∠A ≅ ∠D, and ∠B ≅ ∠E. Must ∠C ≅ ∠F? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q5/5

SET 13 • HSG-CO.B.7

FORWARD 2

Use congruent triangles to prove a relationship in a figure • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

In quadrilateral ABCD, AB = CB and AD = CD. Prove that diagonal BD bisects ∠ABC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q1/5

Segments AC and BD bisect each other at E. Prove that AB = CD.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q2/5

In △PQR and △STU, PQ = ST, QR = TU, and PR = SU. The triangles are congruent. What must be true of ∠R and ∠U? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q3/5

True or false?

Once two triangles are proved congruent, every pair of corresponding parts is congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q4/5

To prove that point M is the midpoint of segment XY using congruent triangles, which pair of parts must you show is congruent?

A. ∠M and ∠X

B. ∠X and ∠Y

C. XY and XM

D. XM and YM

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q5/5

Explain SAS congruence with rigid motions

MAIN	Geometry main question	**HSG-CO.B.8**
BACKWARD 1	Identify the included angle and included side	**7.G.A.2**
BACKWARD 2	Rigid motions map rays, lengths, and angles to equal ones	**8.G.A.1**
BACKWARD 3	Choose a translation, rotation, or reflection to move a point or ray…	**8.G.A.2**
FORWARD 1	Prove triangles congruent by SAS inside a larger figure	**HSG-SRT.B.5**
FORWARD 2	Decide whether given information is enough for SAS, and name a…	**HSG-SRT.B.5**

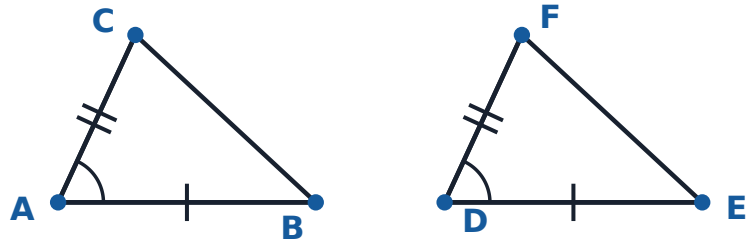
SET 14 • HSG-CO.B.8

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In $\triangle ABC$ and $\triangle DEF$, $AB = DE$, $AC = DF$, and $\angle A \cong \angle D$. Explain, using rigid motions, why $\triangle ABC \cong \triangle DEF$.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q1/5

In the rigid-motion proof of SAS, A has been moved to D and ray AB lies on ray DE, with C on the same side of line DE as F. Why must ray AC lie on ray DF?

A. $\angle A \cong \angle D$, so both rays make the same angle with ray DE on the same side

B. BC = EF

C. The triangles have equal areas

D. AC = DF

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q2/5

In the rigid-motion proof of SAS, ray AC now lies on ray DF. Why does C land exactly on F?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q3/5

Explain why a reflection is sometimes needed in the rigid-motion proof of SAS, and why it does not move D or E.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q4/5

True or false?

The rigid-motion argument for SAS still works if the congruent angle is not between the two congruent sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q5/5

SET 14 • HSG-CO.B.8

BACKWARD 1

Identify the included angle and included side • Grade 7 • Geometry • 7.G.A.2

In △ABC, which angle is included between sides AB and AC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q1/5

In △PQR, which side is included between ∠P and ∠Q?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q2/5

In △XYZ, which angle is NOT included between sides XY and YZ?

A. Both ∠X and ∠Y

B. ∠X

C. ∠Y

D. None of the angles is excluded

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q3/5

True or false?

Two sides of 5 cm and 7 cm with a 40° angle between them determine one triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q4/5

True or false?

Two sides of 5 cm and 7 cm with a 40° angle that is not between them always determine one triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q5/5

SET 14 • HSG-CO.B.8

BACKWARD 2

Rigid motions map rays, lengths, and angles to equal ones • Grade 8 • Geometry • 8.G.A.1

True or false?

Under a rigid motion, the image of a ray is a ray.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q1/5

A 30° angle is rotated about its vertex. What is the measure of the image angle?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q2/5

A point is 5 units from D along a ray. After a rotation about D, the image point is

A. closer to D

B. not on the image ray

C. farther from D

D. 5 units from D on the image ray

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q3/5

True or false?

A reflection across line DE can move point D.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q4/5

Segment AB is translated so that A lands on D. AB = 9. How far is the image of B from D?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q5/5

SET 14 • HSG-CO.B.8

BACKWARD 3

Choose a translation, rotation, or reflection to move a point or ray onto a target • Grade 8 • Geometry • 8.G.A.2

Which kind of rigid motion moves point A onto point D without turning anything?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q1/5

A and D are already together. Which kind of rigid motion turns ray AB onto ray DE without moving D?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q2/5

D and E are fixed, and point C is on the wrong side of line DE. Which motion fixes D and E and moves C across the line?

A. A translation along DE

B. A rotation of 90° about D

C. A reflection across line DE

D. A rotation of 180° about E

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q3/5

True or false?

A rotation about point D keeps D fixed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q4/5

True or false?

A translation that moves A onto D also keeps A fixed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q5/5

SET 14 • HSG-CO.B.8

FORWARD 1

Prove triangles congruent by SAS inside a larger figure • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

In quadrilateral ABCD, AB ∥ DC and AB = DC. Diagonal AC is drawn. Prove that △ABC ≅ △CDA.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q1/5

Point M is the midpoint of segment AB, and line ℓ is perpendicular to AB at M. Point P is on ℓ. Prove that $\triangle PMA \cong \triangle PMB$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q2/5

In the figure, AE = CE and BE = DE, where segments AC and BD cross at E. Name the congruent triangles and the criterion.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q3/5

True or false?

In an SAS proof, a side that the two triangles share can be one of the pairs of congruent sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q4/5

Segments PQ and RS bisect each other at T. Which reason gives the included angles for SAS?

A. Vertical angles are congruent

B. The angles are both right angles

C. Base angles are congruent

D. Corresponding angles are congruent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q5/5

SET 14 • HSG-CO.B.8

FORWARD 2

Decide whether given information is enough for SAS, and name a missing condition • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

In △ABC and △DEF, AB = DE and BC = EF. Name ONE more pair of congruent parts that would prove congruence by SAS.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q1/5

In $\triangle PQR$ and $\triangle XYZ$, $PQ = XY$, $PR = XZ$, and $\angle Q \cong \angle Y$. Is this enough to prove the triangles congruent? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q2/5

True or false?

In $\triangle ABC$ and $\triangle ADC$ (sharing AC), $AB = AD$ and $\angle BAC \cong \angle DAC$. These facts prove $\triangle ABC \cong \triangle ADC$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q3/5

Kites ABCD and EFGH have AB = EF, BC = FG, and ∠B ≅ ∠F. Which triangles are congruent by SAS?

A. △BCD and △FGH

B. △ABD and △EFH

C. △ABC and △EFG

D. △ACD and △EGH

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-F2-Q4**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q4/5

Two triangles have sides 7 and 9 with a 50° angle between them. A student says they need not be congruent because the third sides were not measured. Is the student right? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q5/5

Explain ASA congruence with rigid motions

MAIN	Geometry main question	**HSG-CO.B.8**
BACKWARD 1	Two angles and the included side determine a triangle	**7.G.A.2**
BACKWARD 2	Two different lines meet in at most one point	**8.EE.C.8.a**
BACKWARD 3	Rigid motions keep rays and angle measures	**8.G.A.1**
FORWARD 1	Explain AAS from ASA and the triangle angle sum	**HSG-CO.C.10**
FORWARD 2	Prove congruence with ASA or AAS in overlapping triangles	**HSG-SRT.B.5**

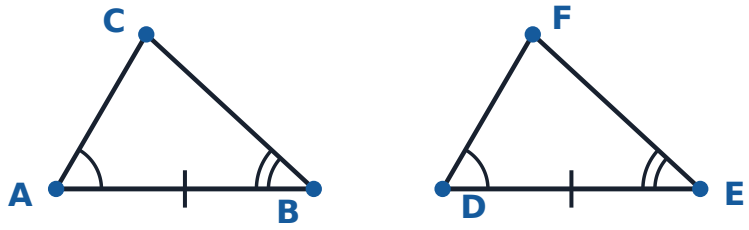
SET 15 • HSG-CO.B.8

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In $\triangle ABC$ and $\triangle DEF$, $\angle A \cong \angle D$, $AB = DE$, and $\angle B \cong \angle E$. Explain, using rigid motions, why $\triangle ABC \cong \triangle DEF$.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q1/5

In the rigid-motion proof of ASA, A is on D, B is on E, and C is on F's side of line DE. Why does C land on F?

A. C is where rays AC and BC meet; those rays now lie on rays DF and EF, which meet only at F

B. C and F are both the highest points

C. BC = EF

D. AC = DF

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q2/5

In the rigid-motion proof of ASA, A is on D, B is on E, and C is on the same side of line DE as F. Explain why ray BC lies on ray EF.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q3/5

In the rigid-motion proof of ASA, why can rays DF and EF meet in only one point?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q4/5

True or false?

In the rigid-motion proof of ASA, a reflection across line DE leaves D and E where they are.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q5/5

SET 15 • HSG-CO.B.8

BACKWARD 1

Two angles and the included side determine a triangle • Grade 7 • Geometry • 7.G.A.2

True or false?

Two angles of 40° and 60° with a 5 cm side between them determine exactly one triangle (up to position).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q1/5

True or false?

Any two angles with a side between them make a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q2/5

A triangle has angles of 50° and 70° at the ends of a 6 cm side. What is the third angle?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q3/5

Which set of conditions determines exactly one triangle?

A. One angle of 45° and nothing else

B. Angles 30°, 60°, and 90°

C. Angles 100° and 90° with a 3 cm side between them

D. Angles 30° and 80° with a 4 cm side between them

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q4/5

Two angles of a triangle are 95° and 90°. Can the triangle exist? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q5/5

SET 15 • HSG-CO.B.8

BACKWARD 2

Two different lines meet in at most one point • Grade 8 • Expressions & Equations • 8.EE.C.8.a

How many points do the lines $y = x + 1$ and $y = -x + 3$ have in common?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q1/5

True or false?

Two different lines can meet in exactly two points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q2/5

Lines $y = 2x + 1$ and $y = 2x - 4$

A. are the same line

B. never meet

C. meet at one point

D. meet at two points

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q3/5

True or false?

If two lines share two points, they are the same line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q4/5

Where do the lines $x = 2$ and $y = -3$ meet?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q5/5

SET 15 • HSG-CO.B.8

BACKWARD 3

Rigid motions keep rays and angle measures • Grade 8 • Geometry • 8.G.A.1

A 64° angle is translated and then reflected. What is the measure of its image?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q1/5

True or false?

A rigid motion maps an angle to an angle with the same measure.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q2/5

A ray from E makes a 35° angle with ray ED above line DE. How many such rays are there?

A. Exactly one

B. Two

C. Infinitely many

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q3/5

True or false?

After a rotation, an angle can become larger if its sides are longer.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q4/5

Segment AB = 4 cm is rotated so that A lands on D. Where is B, relative to D?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q5/5

SET 15 • HSG-CO.B.8

FORWARD 1

Explain AAS from ASA and the triangle angle sum • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In $\triangle ABC$ and $\triangle DEF$, $\angle A \cong \angle D$, $\angle B \cong \angle E$, and $BC = EF$ (the side is NOT between the angles). Prove $\triangle ABC \cong \triangle DEF$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q1/5

Two triangles have angles 40° and 75° and a 9 cm side opposite the 75° angle. Must they be congruent? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q2/5

True or false?

AAS is a valid congruence criterion only because the angles of a triangle add to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q3/5

In $\triangle PQR$ and $\triangle XYZ$, $\angle P \cong \angle X$, $\angle Q \cong \angle Y$, and $PR = XZ$. Which criterion proves congruence?

A. ASA

B. SAS

C. SSS

D. AAS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q4/5

Which corresponding side would make AAS apply if ∠A ≅ ∠D and ∠C ≅ ∠F?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q5/5

SET 15 • HSG-CO.B.8

FORWARD 2

Prove congruence with ASA or AAS in overlapping triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

Segments AD and BC cross at E. AB ∥ CD and E is the midpoint of AD. Prove that △ABE ≅ △DCE.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q1/5

In $\triangle ABC$, the bisector of $\angle A$ is perpendicular to BC at D. Prove that AB = AC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q2/5

In overlapping triangles ABE and ACD, ∠A is shared, AB = AC, and ∠B ≅ ∠C. Which criterion proves △ABE ≅ △ACD?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q3/5

True or false?

An angle shared by two overlapping triangles can be used as one pair of congruent angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q4/5

In △JKL and △JML, ∠KJL ≅ ∠MJL and ∠JLK ≅ ∠JLM. Which side completes ASA?

A. JK = JM

B. JL = JL

C. KL = ML

D. JK = KL

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F2-Q5**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q5/5

Explain SSS congruence with rigid motions

MAIN	Geometry main question	**HSG-CO.B.8**
BACKWARD 1	Three side lengths determine a triangle (triangle inequality)	**7.G.A.2**
BACKWARD 2	Find the points at given distances from two points (two circles)	**7.G.B.4**
BACKWARD 3	Rigid motions keep every length	**8.G.A.1**
FORWARD 1	Explain the Hypotenuse-Leg criterion for right triangles	**HSG-SRT.B.5**
FORWARD 2	Decide which criteria guarantee congruence, with counterexampl…	**HSG-SRT.B.5**

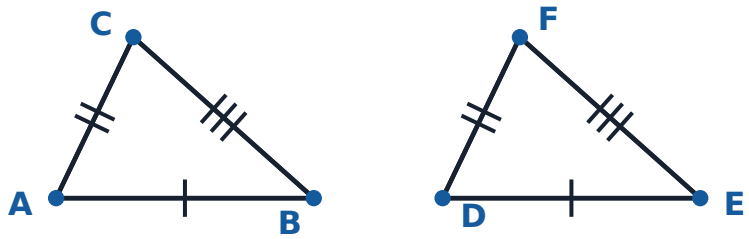
SET 16 • HSG-CO.B.8

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In $\triangle ABC$ and $\triangle DEF$, $AB = DE$, $BC = EF$, and $CA = FD$. Explain, using rigid motions, why $\triangle ABC \cong \triangle DEF$.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q1/5

In the rigid-motion proof of SSS, A is on D, B is on E, and C′ (the image of C) is not F. Why is line DE the perpendicular bisector of C′F?

A. D and E are each the same distance from C′ as from F

B. DE bisects ∠D

C. DE is parallel to C′F

D. C′F is the longest segment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q2/5

In the rigid-motion proof of SSS, line DE is the perpendicular bisector of C′F. Why does the reflection across DE carry C′ onto F, and why is that enough?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q3/5

The rigid-motion proof of SSS uses no angle information. Explain how the three side lengths alone locate the third vertex.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q4/5

True or false?

A triangle with sides 5, 7, and 9 is congruent to a triangle with sides 9, 5, and 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q5/5

SET 16 • HSG-CO.B.8

BACKWARD 1

Three side lengths determine a triangle (triangle inequality) • Grade 7 • Geometry • 7.G.A.2

True or false?

Side lengths 3, 4, and 8 can form a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q1/5

True or false?

Side lengths 5, 6, and 10 form exactly one triangle (up to position).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q2/5

Which three lengths can form a triangle?

A. 5, 5, 11

B. 4, 5, 7

C. 2, 3, 6

D. 1, 1, 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q3/5

Two sides of a triangle are 6 and 10. Give the range of possible lengths for the third side.

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q4/5

Can two different-shaped triangles both have sides 7, 8, and 9? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q5/5

SET 16 • HSG-CO.B.8

BACKWARD 2

Find the points at given distances from two points (two circles) • Grade 7 • Geometry • 7.G.B.4

Point P is 5 units from A and 5 units from B, and A and B are 6 units apart. How many such points P are there in the plane?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q1/5

True or false?

All points 3 cm from a point A form a circle with center A and radius 3 cm.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q2/5

Circles of radius 4 centered at A and B meet at two points. AB = 6. Where are the two points?

A. One on each side of line AB

B. Both on line AB

C. At A and B

D. Both on the same side of AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q3/5

True or false?

Two circles always meet in exactly two points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q4/5

Points A and B are 10 apart. Is there a point 3 units from A and 4 units from B? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q5/5

SET 16 • HSG-CO.B.8

BACKWARD 3

Rigid motions keep every length • Grade 8 • Geometry • 8.G.A.1

Triangle ABC has sides 3, 5, and 7. After a rotation and a reflection, what are the side lengths of the image?

Lengths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q1/5

True or false?

Under a rigid motion, the distance from the image of A to the image of C equals AC.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q2/5

Which can a rigid motion change?

A. The position of a triangle

B. The perimeter

C. The length of a side

D. The area

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q3/5

True or false?

A reflection doubles the distance from a point to the line of reflection.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q4/5

A point C is 6 units from D. After a reflection across a line through D, how far is the image of C from D?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q5/5

SET 16 • HSG-CO.B.8

FORWARD 1

Explain the Hypotenuse-Leg criterion for right triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

Right triangles ABC and DEF have right angles at C and F, hypotenuses AB = DE, and legs AC = DF. Prove that △ABC ≅ △DEF.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q1/5

Two right triangles each have hypotenuse 13 and one leg 5. Are they congruent? What is the other leg?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q2/5

True or false?

Hypotenuse-Leg is a special case of SSA that does guarantee congruence.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q3/5

Which information proves two right triangles congruent?

A. One pair of acute angles is equal

B. Their hypotenuses are equal and one pair of legs is equal

C. Their right angles are equal

D. Their hypotenuses are equal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q4/5

In right triangles PQR and XYZ (right angles at R and Z), PQ = XY = 10 and QR = YZ = 6. Find PR and XZ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q5/5

SET 16 • HSG-CO.B.8

FORWARD 2

Decide which criteria guarantee congruence, with counterexamples • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

Explain with a counterexample why AAA is not a congruence criterion.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q1/5

Explain with a counterexample why SSA is not a congruence criterion.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q2/5

Which list contains only valid triangle congruence criteria?

A. SSS, SSA, ASA, AAS

B. SSA, AAA, SAS, HL

C. SSS, SAS, ASA, AAS

D. SAS, AAA, ASA, SSS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q3/5

True or false?

If two triangles have the same three side lengths, they have the same three angle measures.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q4/5

True or false?

Two triangles with two pairs of congruent sides and equal perimeters must be congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q5/5

SET 17 • HSG-CO.C.9 • CONGRUENCE: PROVING THEOREMS

Prove that vertical angles are congruent

MAIN	Geometry main question	**HSG-CO.C.9**
BACKWARD 1	Angles in a linear pair add to 180°	**7.G.B.5**
BACKWARD 2	Use properties of equality as reasons	**HSA-REI.A.1**
BACKWARD 3	Identify vertical angles and linear pairs in a figure	**7.G.B.5**
FORWARD 1	Use the vertical angle theorem in multistep proofs	**HSG-CO.C.9**
FORWARD 2	Prove statements about the bisectors of angles formed by two lin…	**HSG-CO.C.9**

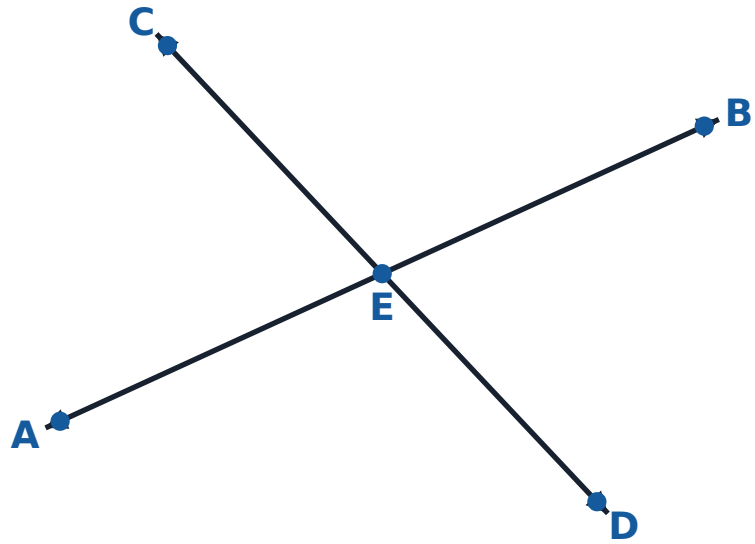
SET 17 • HSG-CO.C.9

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines AB and CD intersect at E. Prove that ∠AEC ≅ ∠BED.

Work space



Conclusion: _____

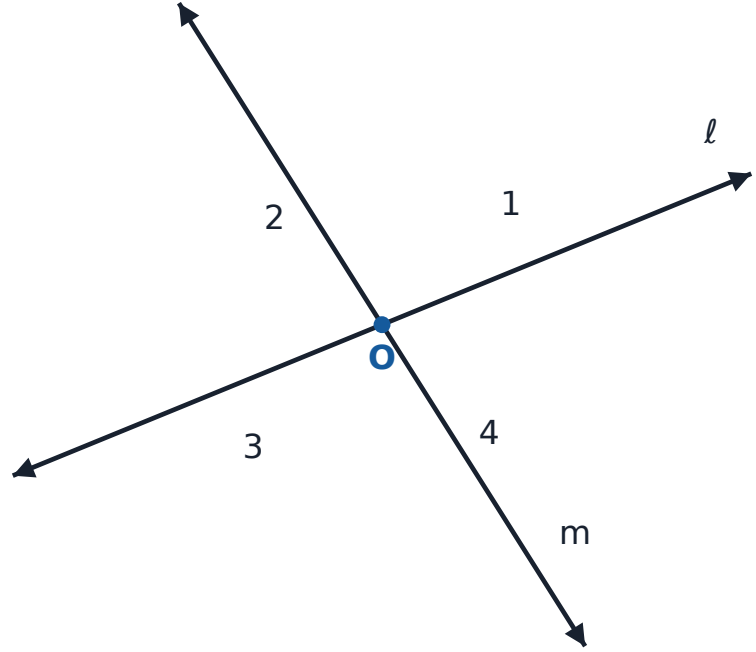
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q1/5

Lines ℓ and m meet at O, forming $\angle 1$, $\angle 2$, $\angle 3$, and $\angle 4$ in order around O. Prove that $\angle 2 \cong \angle 4$.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q2/5

A proof that $\angle 1 \cong \angle 3$ for vertical angles 1 and 3 (with $\angle 2$ next to both):

1. $m\angle 1 + m\angle 2 = 180°$ (linear pair)

2. $m\angle 2 + m\angle 3 = 180°$ (linear pair)

3. $m\angle 1 + m\angle 2 = m\angle 2 + m\angle 3$ (?)

4. $m\angle 1 = m\angle 3$ (?)

Give the reasons for steps 3 and 4.

Reasons: _____

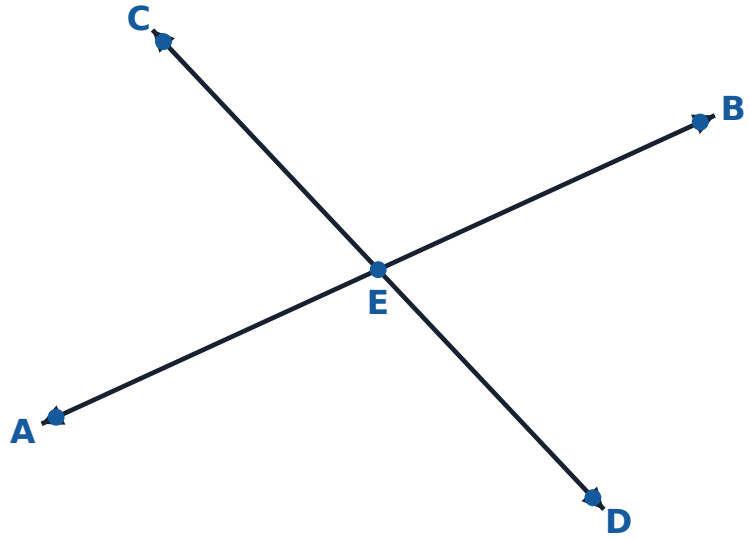
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q3/5

Lines AB and CD intersect at E. Explain, using a rigid motion, why ∠AEC ≅ ∠BED.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q4/5

A student measures two vertical angles in one drawing, finds that both are 47°, and writes "vertical angles are congruent." Why is this not a proof?

A. Protractors are never accurate

B. It checks one example; a proof must show the result for every pair of vertical angles

C. Vertical angles are not always congruent

D. The student should have measured all four angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q5/5

SET 17 • HSG-CO.C.9

BACKWARD 1

Angles in a linear pair add to 180° • Grade 7 • Geometry • 7.G.B.5

Two angles form a linear pair. One measures 128°. Find the other.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q1/5

∠PQR and ∠RQS form a linear pair, and m∠PQR = 3 · m∠RQS. Find m∠RQS.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q2/5

True or false?

The two angles of a linear pair are always supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q3/5

Points A, B, C are on a line with B between A and C, and ray BD is drawn. Which angles form a linear pair?

A. ∠DBC and ∠ABC

B. ∠ABD and ∠DBC

C. ∠BAD and ∠BCD

D. ∠ABD and ∠ABC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q4/5

True or false?

Two angles that add to 180° must form a linear pair.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q5/5

SET 17 • HSG-CO.C.9

BACKWARD 2

Use properties of equality as reasons • High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

From $m\angle 1 + 40° = m\angle 2 + 40°$, which property gives $m\angle 1 = m\angle 2$?

Property: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q1/5

If $x + y = 180$ and $y + z = 180$, which property lets you write $x + y = y + z$?

Property: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q2

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q2/5

Which property justifies: if a = b, then b = a?

A. Reflexive property

B. Distributive property

C. Symmetric property

D. Transitive property

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-B2-Q3**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q3/5

True or false?

The reflexive property says that any quantity equals itself, such as $m\angle A = m\angle A$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q4/5

True or false?

From $2x = 2y$ you may conclude $x = y$ by the subtraction property of equality.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

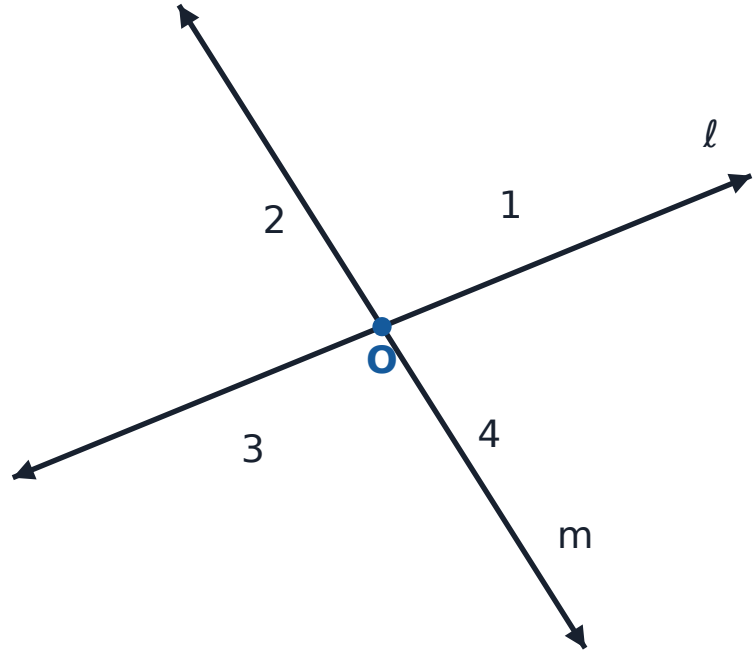
High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q5/5

SET 17 • HSG-CO.C.9

BACKWARD 3

Identify vertical angles and linear pairs in a figure • Grade 7 • Geometry • 7.G.B.5

Lines ℓ and m meet at O, forming $\angle 1$, $\angle 2$, $\angle 3$, $\angle 4$ in order. Which angle is vertical to $\angle 1$?



A. $\angle 3$

B. $\angle 1$

C. $\angle 4$

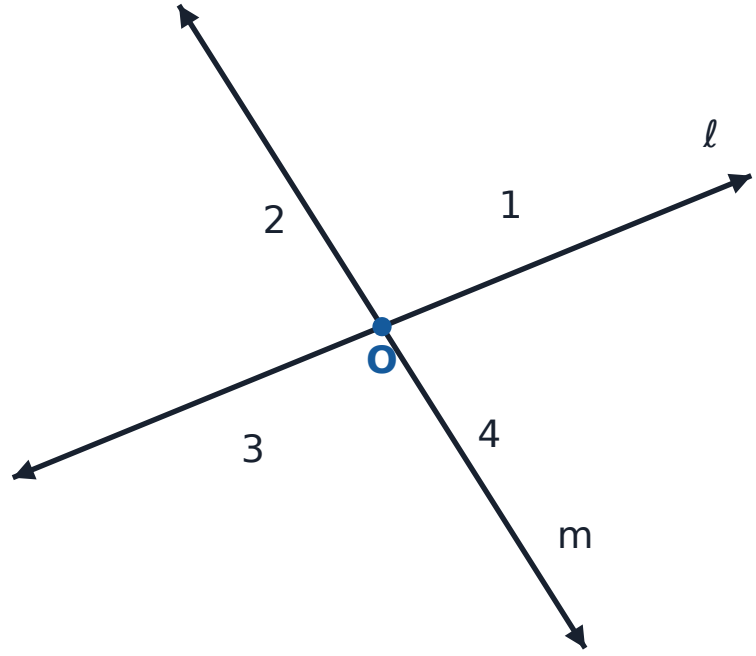
D. $\angle 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q1/5

Lines ℓ and m meet at O, forming $\angle 1$, $\angle 2$, $\angle 3$, $\angle 4$ in order. Name the two angles that form linear pairs with $\angle 2$.



Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q2/5

True or false?

Vertical angles share a side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q3/5

True or false?

When two lines cross, they form two pairs of vertical angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q4/5

Two lines cross. One angle is 63°. Find its vertical angle and the angles next to it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q5/5

SET 17 • HSG-CO.C.9

FORWARD 1

Use the vertical angle theorem in multistep proofs • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines AB and CD intersect at E, and ray EF bisects ∠AEC. Ray EG is the ray opposite ray EF. Prove that ray EG bisects ∠BED.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q1/5

Three lines meet at point O, forming six angles. Two of the angles that are next to each other are 50° and 70°. Prove that the six angles are 50°, 70°, 60°, 50°, 70°, 60° in order.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q2/5

Two lines meet. One angle is (4x + 10)° and its vertical angle is (6x − 30)°. Find x and all four angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q3/5

True or false?

If one angle formed by two crossing lines is a right angle, all four angles are right angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q4/5

Lines AB and CD meet at E with m∠AEC = 2x and m∠CEB = 3x. What is m∠BED?

A. 144°

B. 36°

C. 72°

D. 108°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q5/5

SET 17 • HSG-CO.C.9

FORWARD 2

Prove statements about the bisectors of angles formed by two lines • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

∠ABD and ∠DBC form a linear pair. Ray BE bisects ∠ABD and ray BF bisects ∠DBC. Prove that ∠EBF is a right angle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q1/5

Is the converse of the vertical angle theorem true: "If two angles are congruent, they are vertical angles"? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q2/5

Two lines meet at O. Prove that the bisectors of a pair of vertical angles lie on one line.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q3/5

True or false?

The bisectors of two angles that form a linear pair are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q4/5

Two lines meet, and the bisectors of two adjacent angles are drawn. What angle do the bisectors form?

A. 45°

B. 90°

C. It depends on the lines

D. 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q5/5

SET 18 • HSG-CO.C.9 • CONGRUENCE: PROVING THEOREMS

Prove the angle theorems for parallel lines cut by a transversal

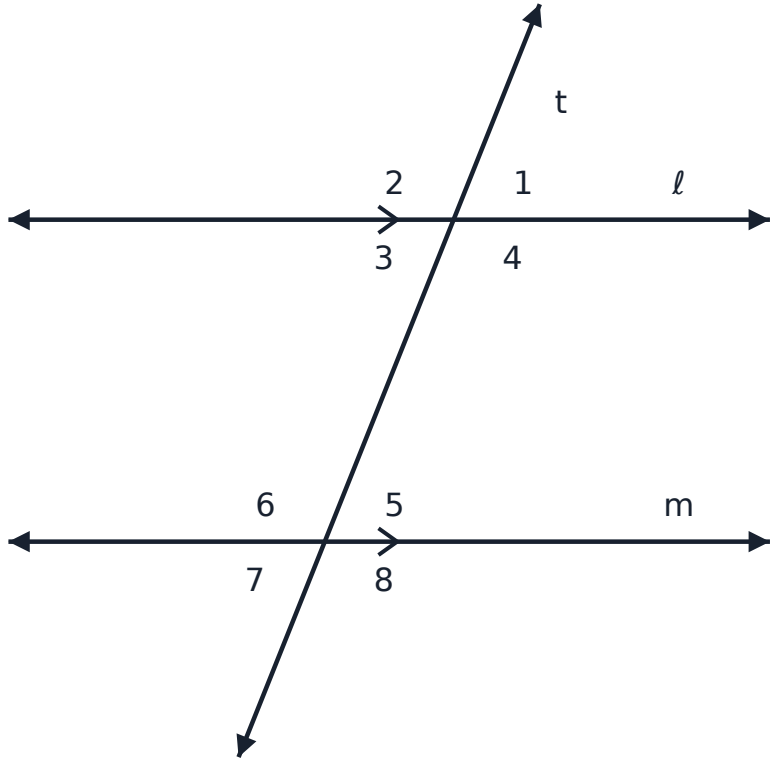
MAIN	Geometry main question	**HSG-CO.C.9**
BACKWARD 1	Name and use angles formed by parallel lines and a transversal	**8.G.A.5**
BACKWARD 2	Vertical angles and linear pairs as reasons	**HSG-CO.C.9**
BACKWARD 3	The parallel postulate and the definition of parallel lines	**HSG-CO.A.1**
FORWARD 1	Prove the converse: congruent alternate interior angles make line...	**HSG-CO.C.9**
FORWARD 2	Use parallel-line theorems to prove the triangle angle sum	**HSG-CO.C.10**

SET 18 • HSG-CO.C.9

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines ℓ and m are parallel and are cut by transversal t. Assume the corresponding angles postulate (corresponding angles formed by a transversal of parallel lines are congruent). Prove that alternate interior angles $\angle 3$ and $\angle 5$ are congruent.



Work space

Conclusion: _____

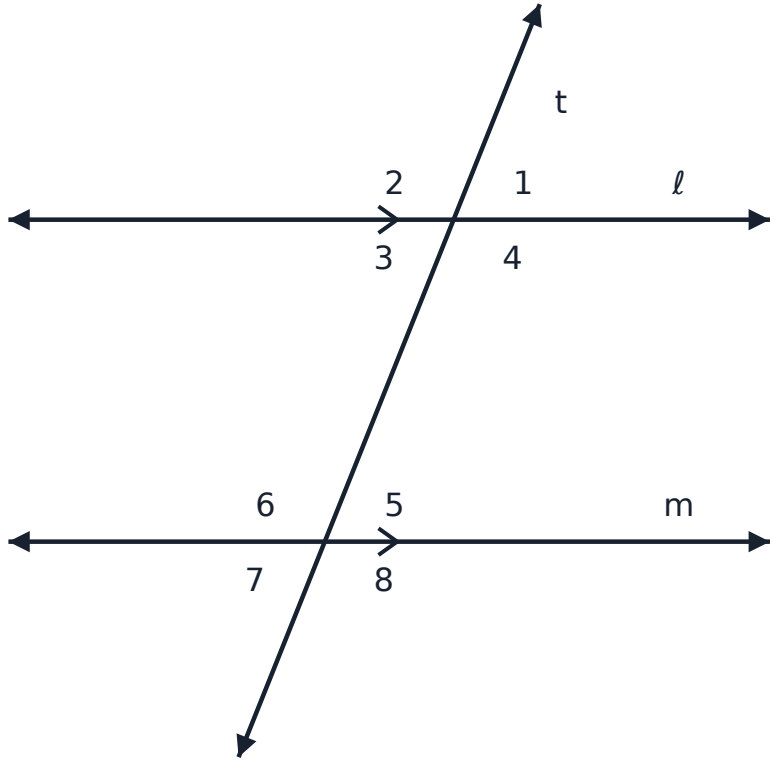
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q1/5

Lines ℓ and m are parallel and are cut by transversal t. Assume that alternate interior angles are congruent. Prove that corresponding angles $\angle 2$ and $\angle 6$ are congruent.

Work space



Conclusion: _____

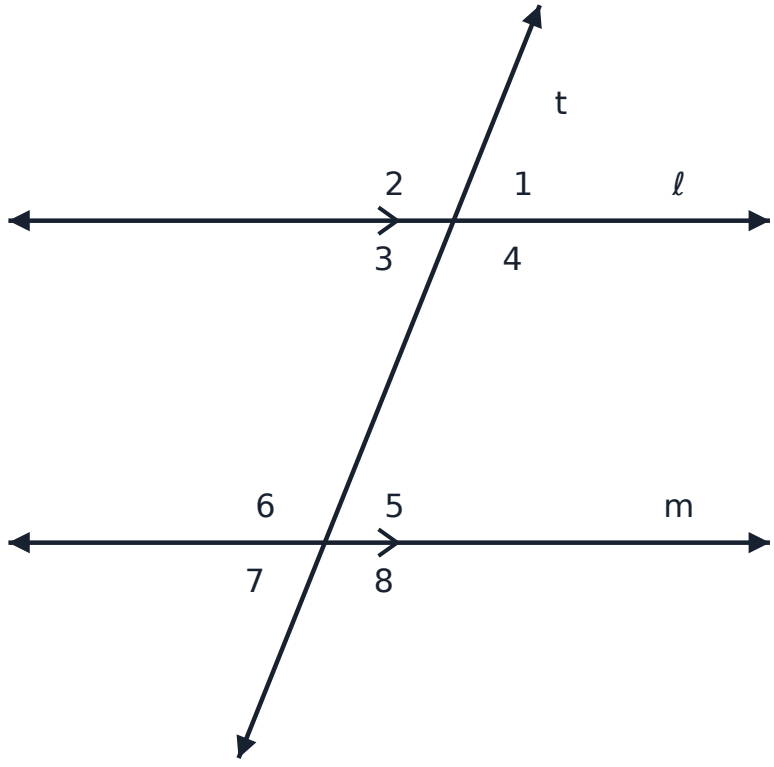
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q2/5

Lines ℓ and m are parallel and are cut by transversal t, crossing ℓ at P and m at Q. Explain, using a translation, why corresponding angles $\angle 1$ and $\angle 5$ are congruent.

Work space



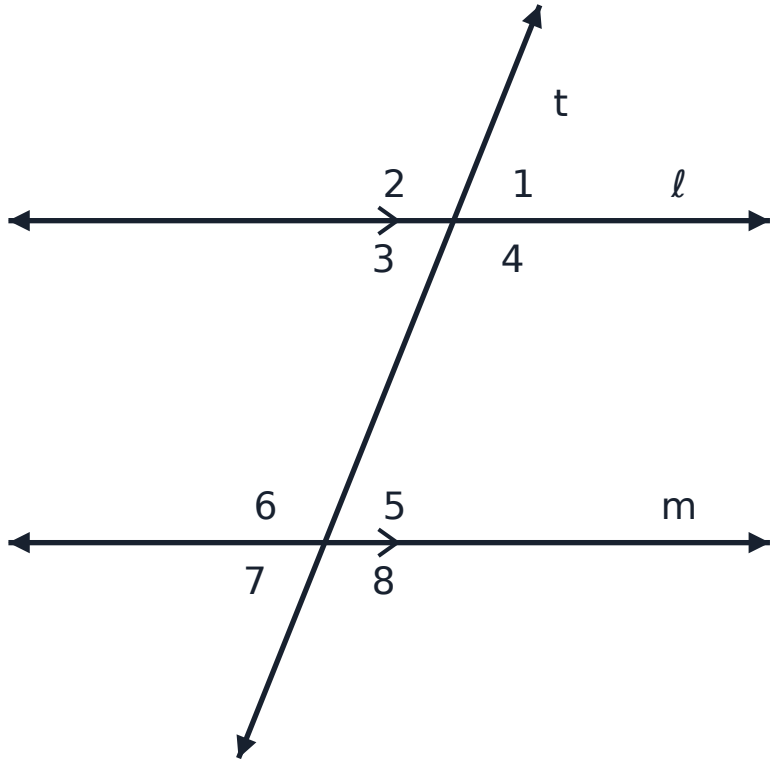
Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q3/5

Lines $\ell \parallel m$ are cut by transversal t, and $m\angle 1 = 115°$. Find $m\angle 5$, $m\angle 3$, and $m\angle 6$. Give a reason for each.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q4/5

A student "proves" that alternate interior angles are congruent by saying: "The lines are parallel because the alternate interior angles are congruent." What is wrong?

A. Nothing; the argument is correct

B. Alternate interior angles are supplementary

C. The lines must be perpendicular

D. The argument is circular: it assumes the conclusion

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q5

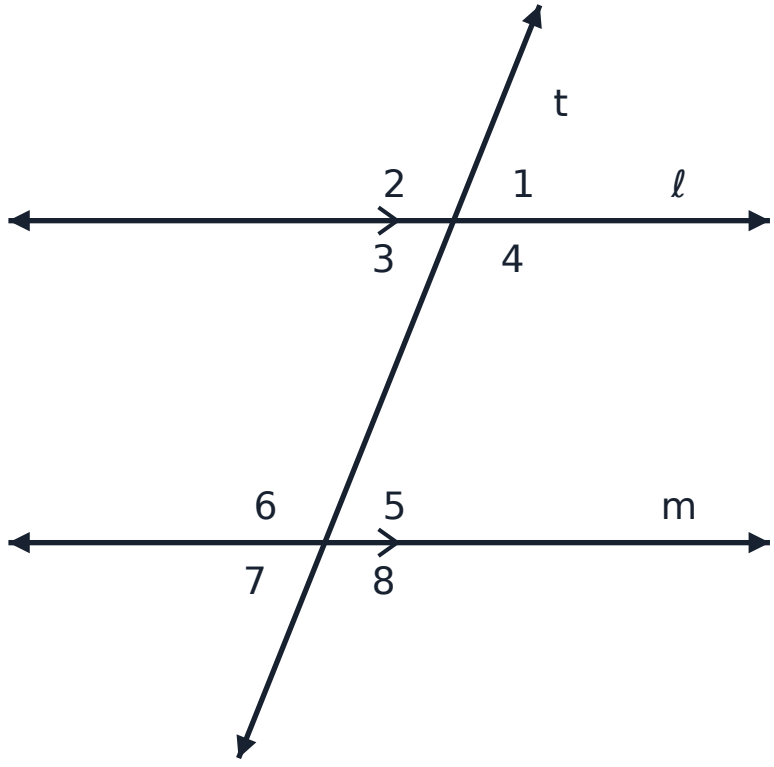
High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q5/5

SET 18 • HSG-CO.C.9

BACKWARD 1

Name and use angles formed by parallel lines and a transversal • Grade 8 • Geometry • 8.G.A.5

Lines $\ell \parallel m$ are cut by transversal t. $\angle 4$ measures $70°$. Find the measure of $\angle 6$.



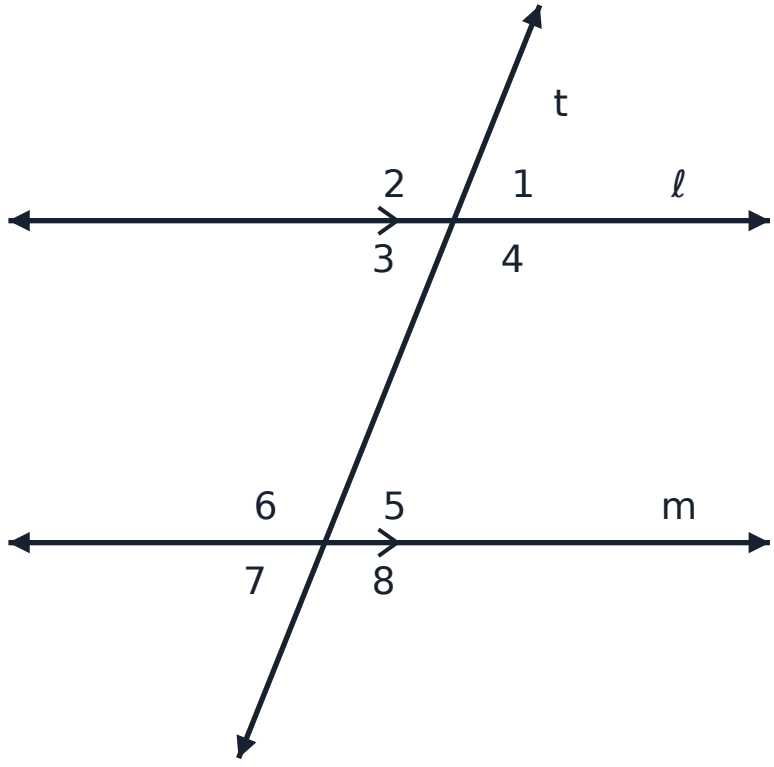
Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q1/5

Lines $\ell \parallel m$ are cut by transversal t. Which angle corresponds to $\angle 2$?



A. $\angle 7$

B. $\angle 8$

C. $\angle 5$

D. $\angle 6$

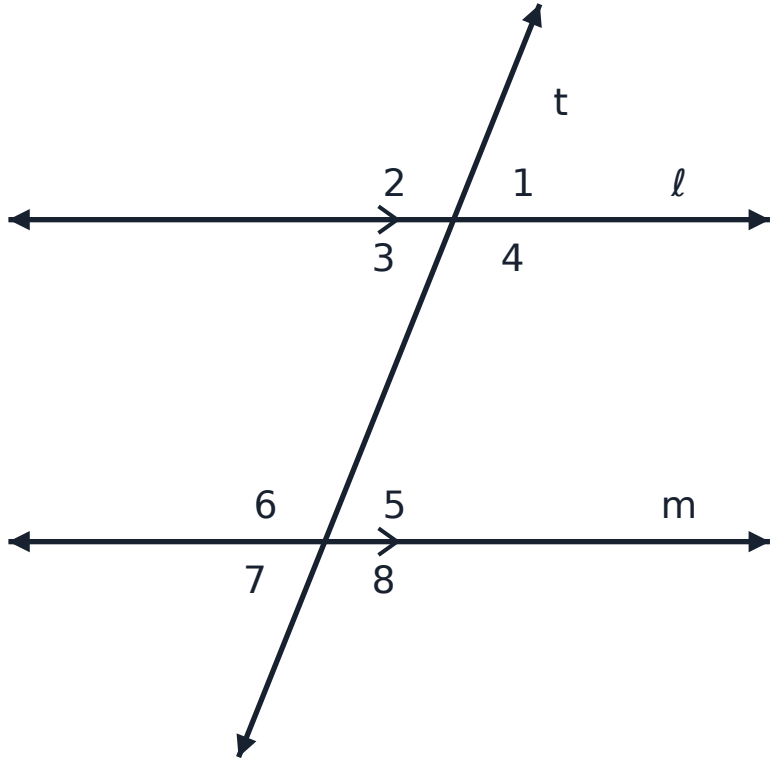
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q2/5

True or false?

Lines $\ell \parallel m$ are cut by transversal t. $\angle 4$ and $\angle 5$ are same-side interior angles, so they add to 180°.



A. True **B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q3/5

True or false?

When two lines are cut by a transversal, corresponding angles are congruent even if the lines are not parallel.

A. True

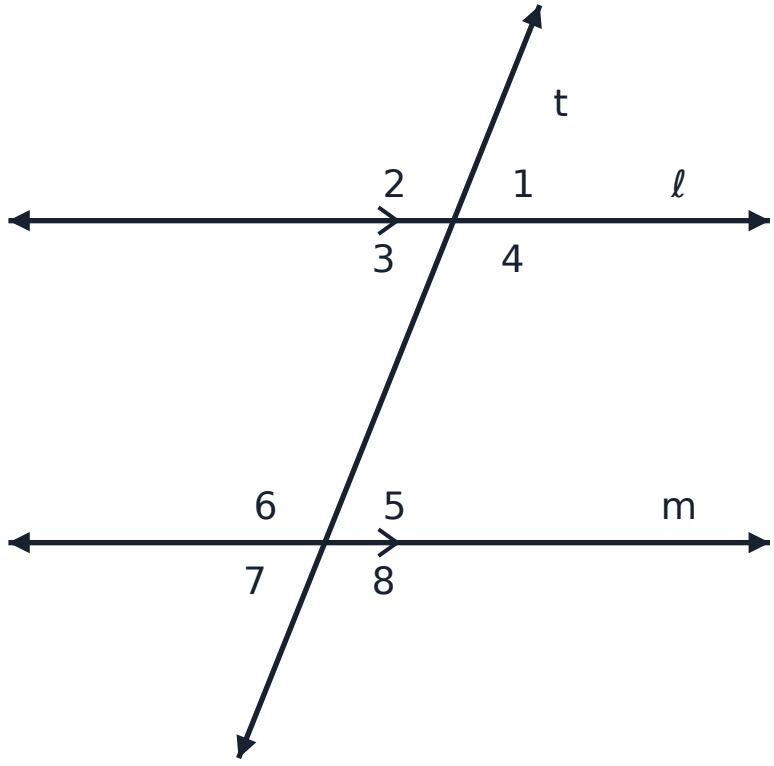
B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q4/5

Lines $\ell \parallel m$ are cut by transversal t. $\angle 8$ measures 112°. Find the measure of $\angle 2$.



Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q5/5

SET 18 • HSG-CO.C.9

BACKWARD 2

Vertical angles and linear pairs as reasons • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Two lines cross. Name the reason two opposite angles are congruent.

Reason: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q1/5

Name the reason two adjacent angles on a line add to 180°.

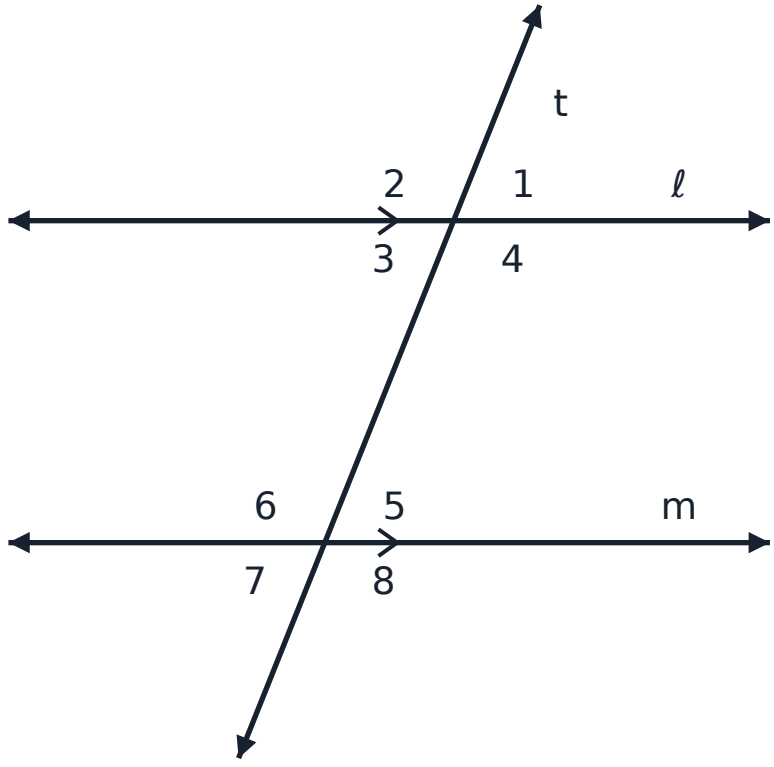
Reason: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q2/5

At the top crossing, $\angle 1 = 110°$. What is $\angle 2$ and why?



A. 110°, linear pair

B. 70°, linear pair

C. 110°, vertical angles

D. 70°, vertical angles

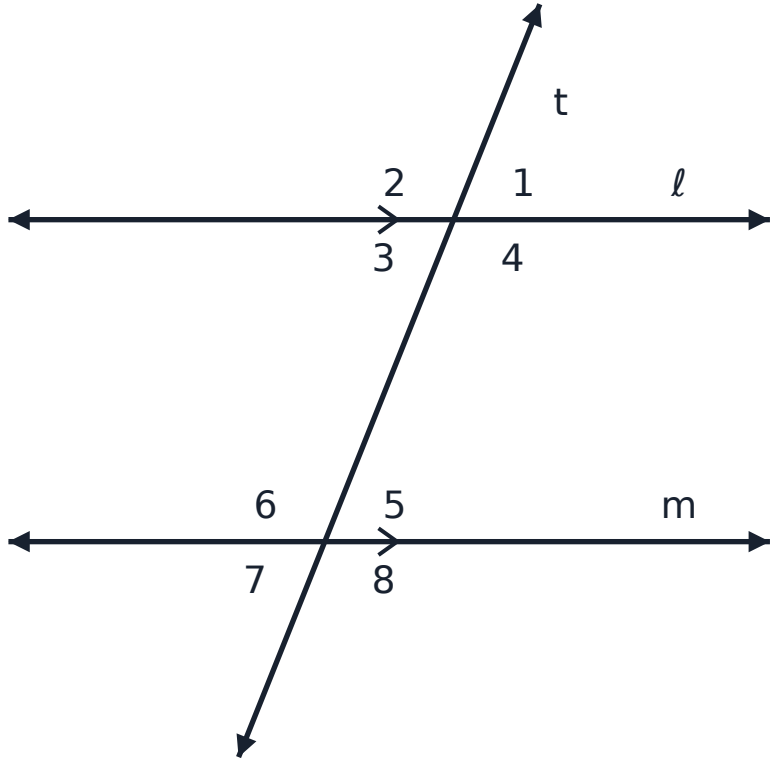
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q3/5

True or false?

Angles 1 and 3 are vertical angles in the figure.



A. True

B. False

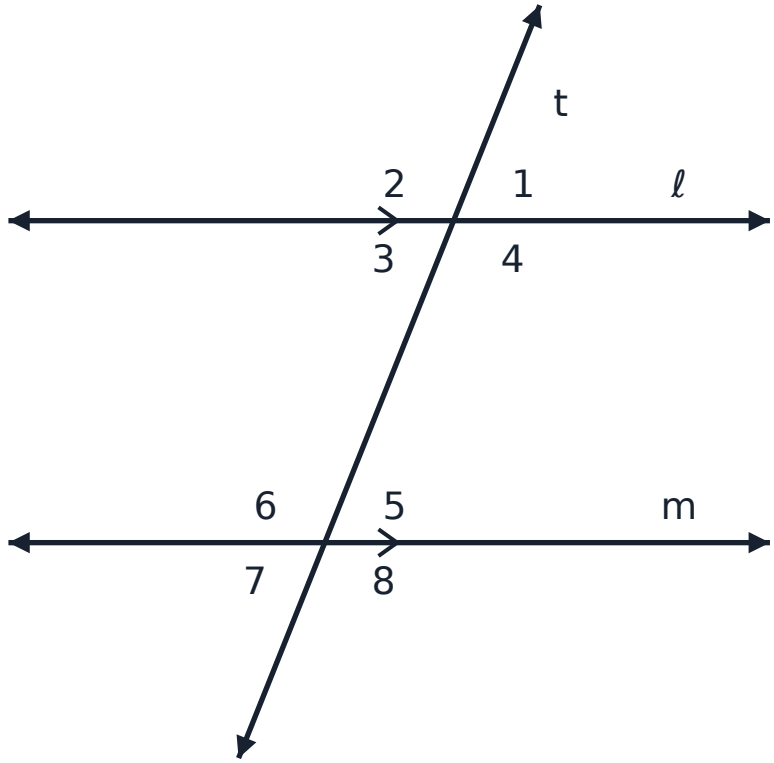
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q4/5

True or false?

Angles 1 and 4 are vertical angles in the figure.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q5/5

SET 18 • HSG-CO.C.9

BACKWARD 3

The parallel postulate and the definition of parallel lines • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

True or false?

Through a point not on a line, there is exactly one line parallel to it (in a plane).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q1/5

Line ℓ is parallel to line m. Line n passes through a point of ℓ and is also parallel to m. What can you conclude about n?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q2/5

Which pair of lines in a plane are parallel?

A. $x = 2$ and $y = 2$

B. $y = x$ and $y = -x$

C. $y = 2x + 1$ and $y = 2x - 5$

D. $y = 2x + 1$ and $y = -2x + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q3/5

True or false?

Two lines in a plane that never meet are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q4/5

Line $\ell \parallel$ line m. A translation maps m to a line through a point of ℓ. Must that image line be parallel to m or the same as m?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q5/5

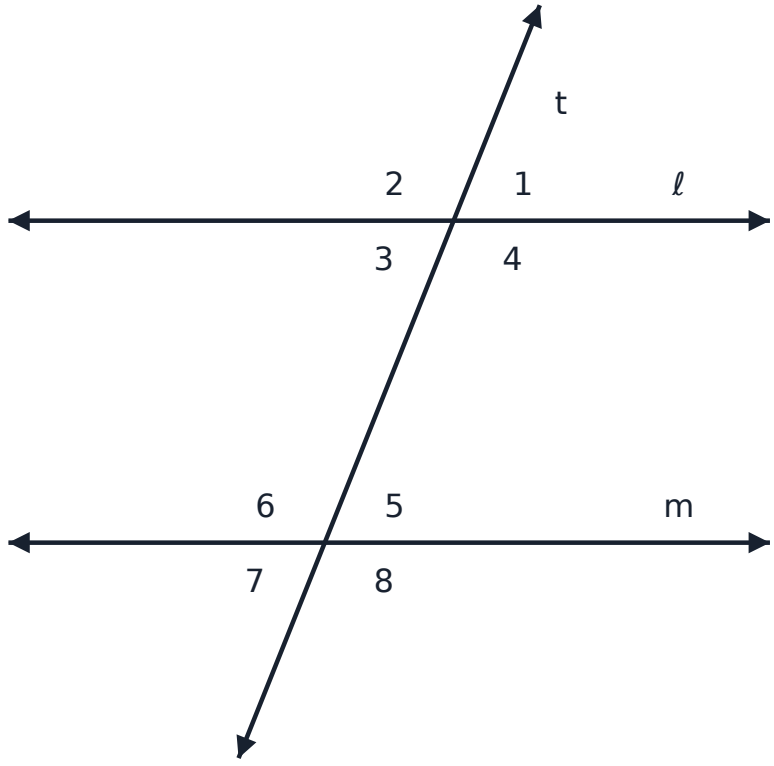
SET 18 • HSG-CO.C.9

FORWARD 1

Prove the converse: congruent alternate interior angles make lines parallel • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines ℓ and m are cut by transversal t, and alternate interior angles $\angle 3$ and $\angle 5$ are congruent. Prove that $\ell \parallel$ m.

Work space



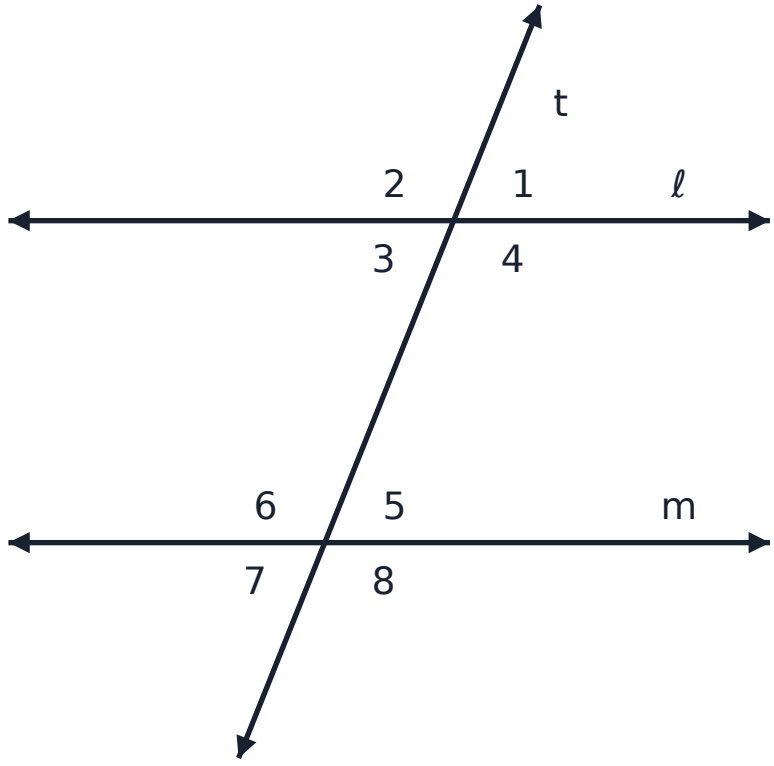
Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q1/5

Lines ℓ and m are cut by transversal t with $m\angle 4 = (5x - 20)°$ and $m\angle 6 = (3x + 30)°$. For what value of x is $\ell \parallel m$? Explain.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q2/5

True or false?

If same-side interior angles are supplementary, the lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q3/5

Which fact proves that lines ℓ and m are parallel?

A. A pair of vertical angles is congruent

B. A linear pair adds to 180°

C. Two angles at the same crossing are equal

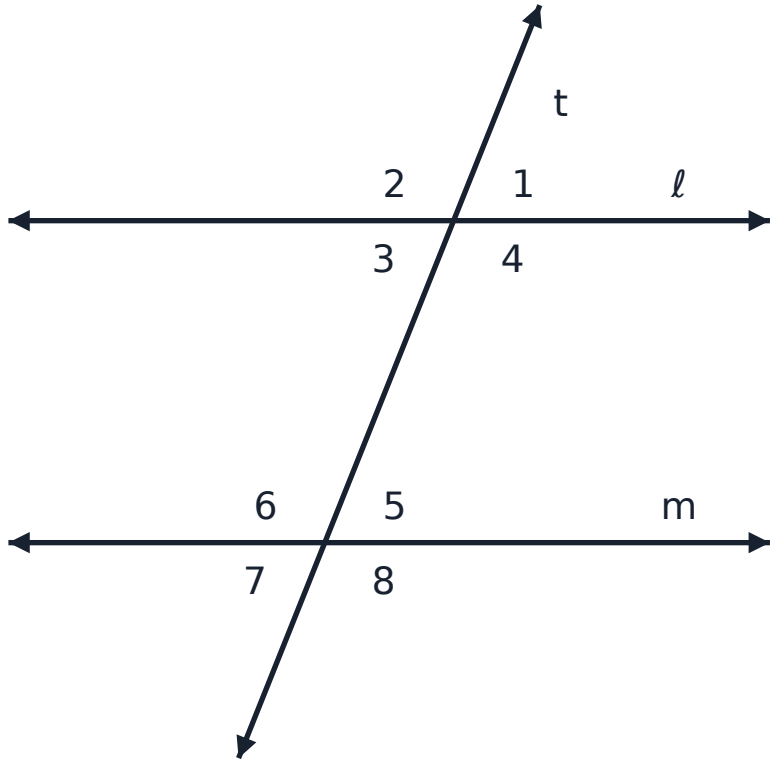
D. A pair of corresponding angles is congruent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q4/5

Lines ℓ and m are cut by a transversal. $m\angle 1 = 72°$ and $m\angle 7 = 108°$. Are ℓ and m parallel? Explain.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q5/5

SET 18 • HSG-CO.C.9

FORWARD 2

Use parallel-line theorems to prove the triangle angle sum • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

Prove that the angles of triangle ABC add to 180°. Begin by drawing the line through B parallel to AC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q1/5

In triangle ABC, a line through B parallel to AC makes a 50° angle with BA (on A's side) and a 65° angle with BC (on C's side). Find all three angles of the triangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q2/5

True or false?

The proof of the triangle angle sum depends on the parallel postulate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q3/5

Which theorem is used to show the new angles at B equal ∠A and ∠C?

A. Isosceles triangle theorem

B. Triangle inequality

C. Alternate interior angles theorem

D. Vertical angles theorem

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q4/5

Explain why the exterior angle at C of triangle ABC equals m∠A + m∠B.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q5/5

Prove the perpendicular bisector theorem and its converse

MAIN	Geometry main question	**HSG-CO.C.9**
BACKWARD 1	Use the definitions of midpoint and perpendicular bisector	**HSG-CO.A.1**
BACKWARD 2	Use SAS and SSS to prove triangles congruent	**HSG-CO.B.8**
BACKWARD 3	Corresponding parts of congruent triangles and right angles from…	**HSG-CO.B.7**
FORWARD 1	Use the perpendicular bisector to find points equidistant from two…	**HSG-GPE.B.5**
FORWARD 2	Prove that the perpendicular bisectors of a triangle meet at one p…	**HSG-C.A.3**

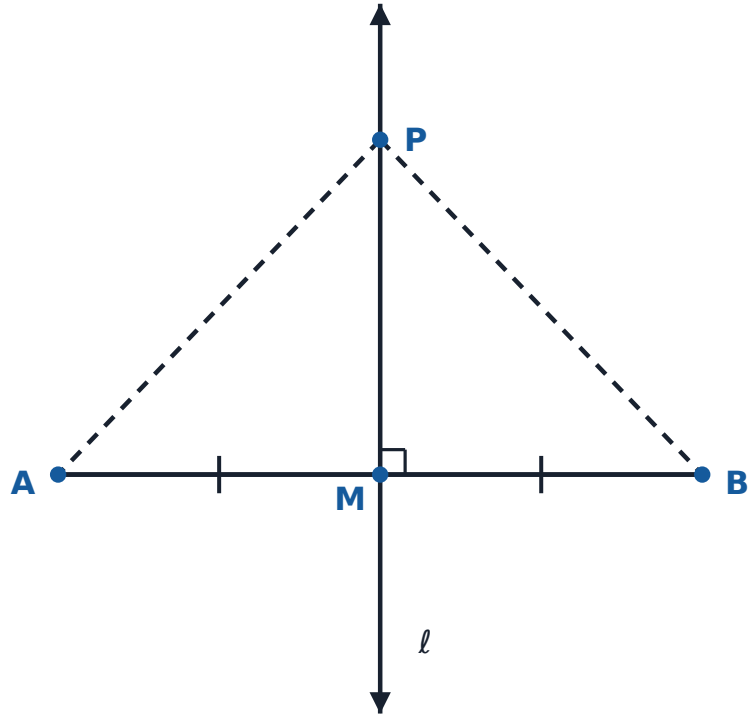
SET 19 • HSG-CO.C.9

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Line ℓ is the perpendicular bisector of segment AB, meeting AB at M. Point P is on ℓ. Prove that PA = PB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q1/5

Point P is not on line AB, and PA = PB. Let M be the midpoint of AB. Prove that line PM is perpendicular to AB, so P is on the perpendicular bisector of AB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q2/5

The theorem says the points on the perpendicular bisector of AB are EXACTLY the points equidistant from A and B. Which two statements must be proved?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q3/5

In a coordinate plane, A(−a, 0) and B(a, 0) with a > 0. Prove that every point P(0, y) on the y-axis is equidistant from A and B, and that every point P(x, y) with PA = PB has x = 0.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q4/5

A student proves only that every point on the perpendicular bisector of AB is equidistant from A and B. What is missing?

A. A proof that the bisector is a line

B. Nothing; one direction proves "exactly"

C. The converse: every point equidistant from A and B is on the perpendicular bisector

D. A measurement of PA and PB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q5/5

SET 19 • HSG-CO.C.9

BACKWARD 1

Use the definitions of midpoint and perpendicular bisector • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

M is the midpoint of AB and AB = 14. Find AM.

AM = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q1/5

True or false?

The perpendicular bisector of AB passes through the midpoint of AB and is perpendicular to AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q2/5

Line ℓ meets AB at M with AM = MB, but ℓ is not perpendicular to AB. What is ℓ?

A. Not related to AB

B. Parallel to AB

C. A bisector of AB that is not the perpendicular bisector

D. The perpendicular bisector of AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q3/5

Line ℓ is perpendicular to AB at M, and AM = 3, MB = 5. Is ℓ the perpendicular bisector of AB? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q4/5

True or false?

A segment has exactly one perpendicular bisector in a plane.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q5/5

SET 19 • HSG-CO.C.9

BACKWARD 2

Use SAS and SSS to prove triangles congruent • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In $\triangle PMA$ and $\triangle PMB$, PM is shared, AM = BM, and $\angle PMA \cong \angle PMB$. Which criterion proves they are congruent?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q1/5

In $\triangle PMA$ and $\triangle PMB$, $PA = PB$, $AM = BM$, and PM is shared. Which criterion proves they are congruent?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q2/5

Two triangles share a side and have two other pairs of congruent sides. Which criterion applies?

A. SAS

B. SSS

C. ASA

D. AAS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q3/5

True or false?

A side shared by two triangles counts as a pair of congruent sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q4/5

True or false?

SSA is a valid congruence criterion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q5/5

SET 19 • HSG-CO.C.9

BACKWARD 3

Corresponding parts of congruent triangles and right angles from a linear pair • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

$\triangle PMA \cong \triangle PMB$. Which angle corresponds to $\angle PMA$?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q1/5

Two congruent angles form a linear pair. What is each angle's measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q2/5

True or false?

If $\triangle PMA \cong \triangle PMB$, then $PA = PB$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q3/5

$\triangle XYZ \cong \triangle XWZ$. Which pair of segments must be congruent?

A. YZ and XW

B. XY and XZ

C. XZ and YW

D. YZ and WZ

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q4/5

True or false?

Two angles in a linear pair are always congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q5/5

SET 19 • HSG-CO.C.9

FORWARD 1

Use the perpendicular bisector to find points equidistant from two points • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Find an equation of the perpendicular bisector of the segment from A(2, 1) to B(6, 5).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q1/5

Find the point on the x-axis that is equidistant from A(2, 1) and B(6, 5).

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q2/5

True or false?

The point (4, 3) is equidistant from (0, 0) and (8, 6).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q3/5

Which point is equidistant from A(0, 0) and B(4, 0)?

A. (3, 1)

B. (4, 4)

C. (2, 7)

D. (0, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q4/5

Points A(−1, 2) and B(3, 2). Describe all points equidistant from A and B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q5/5

SET 19 • HSG-CO.C.9

FORWARD 2

Prove that the perpendicular bisectors of a triangle meet at one point • High School — Geometry • Geometry: Circles • HSG-C.A.3

Prove that the perpendicular bisectors of the sides of triangle ABC meet at one point O, and that O is equidistant from A, B, and C.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q1/5

Triangle ABC has A(0, 0), B(6, 0), C(0, 8). Find the point equidistant from all three vertices.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q2/5

True or false?

The point where the perpendicular bisectors of a triangle meet is always inside the triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q3/5

Three towns are to share a well equally distant from all three. Where should it go?

A. Where the angle bisectors meet

B. At the midpoint of the longest side

C. Where the perpendicular bisectors of the triangle's sides meet

D. Where the medians meet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q4/5

Explain why showing that the perpendicular bisectors of AB and BC meet at O is enough to show that all three meet at O.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q5/5

Prove that the angles of a triangle add to 180°

MAIN	Geometry main question	**HSG-CO.C.10**
BACKWARD 1	Use the angle sum and parallel-line angles informally	**8.G.A.5**
BACKWARD 2	Angles that form a straight angle add to 180°	**4.MD.C.7**
BACKWARD 3	Alternate interior angles of parallel lines are congruent	**HSG-CO.C.9**
FORWARD 1	Prove the exterior angle theorem	**HSG-CO.C.10**
FORWARD 2	Generalize the angle sum to polygons	**HSG-CO.C.10**

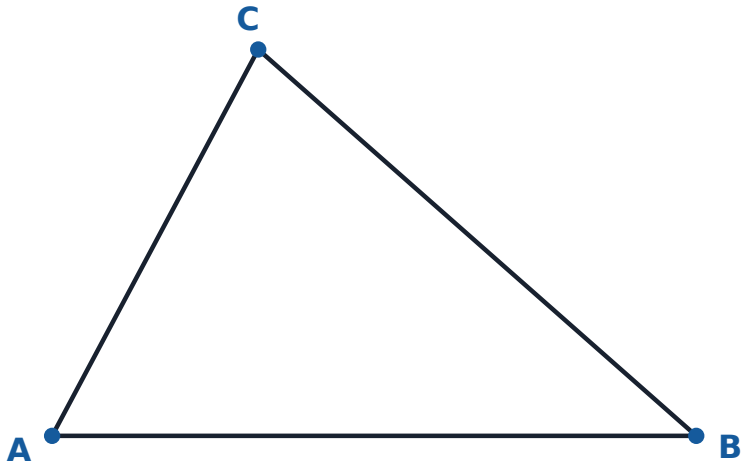
SET 20 • HSG-CO.C.10

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

Prove that
$m\angle A + m\angle B + m\angle C = 180°$ in triangle ABC. Draw the line through C parallel to AB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q1/5

In triangle PQR, draw the line through P parallel to QR. Prove that the angles of triangle PQR add to 180°.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q2/5

In triangle ABC, extend side BC past C to point D, and draw ray CE parallel to BA (E on the same side of line BD as A). Prove that the angles of triangle ABC add to 180°.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q3/5

A proof of the triangle angle sum (line ℓ through C parallel to AB, making $\angle 1$ next to CA and $\angle 2$ next to CB):

1. $m\angle 1 + m\angle ACB + m\angle 2 = 180°$ (?)

2. $\angle 1 \cong \angle A$ and $\angle 2 \cong \angle B$ (?)

3. $m\angle A + m\angle ACB + m\angle B = 180°$ (?)

Give the three reasons.

Reasons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q4/5

A student proves the angle sum theorem using "an exterior angle equals the sum of the two remote interior angles." What is wrong?

A. Exterior angles do not exist in triangles

B. Nothing is wrong

C. The exterior angle theorem is false

D. That theorem is usually proved FROM the angle sum, so the argument is circular

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q5/5

SET 20 • HSG-CO.C.10

BACKWARD 1

Use the angle sum and parallel-line angles informally • Grade 8 • Geometry • 8.G.A.5

Two angles of a triangle are 47° and 68°. Find the third.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q1/5

True or false?

Tearing off the three corners of a paper triangle and lining them up makes a straight angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q2/5

A triangle has angles x, 2x, and 3x. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q3/5

Lines $\ell \parallel m$ are cut by a transversal. An angle at ℓ is 72°. Which angle at m must also be 72°?

A. Its alternate interior angle

B. Its same-side interior angle

C. Every angle at m

D. The angle next to it on the same line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q4/5

True or false?

A triangle can have two obtuse angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q5/5

SET 20 • HSG-CO.C.10

BACKWARD 2

Angles that form a straight angle add to 180° • Grade 4 • Measurement & Data • 4.MD.C.7

Three angles together form a straight angle. Two are 40° and 75°. Find the third.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q1/5

True or false?

The angles around a point on one side of a straight line add to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q2/5

$\angle 1$, $\angle 2$, and $\angle 3$ form a straight angle, with $\angle 1 = \angle 3 = 50°$. Find $\angle 2$.

A. 50°

B. 80°

C. 100°

D. 130°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q3/5

Angle ABC is a straight angle. Ray BD makes ∠ABD = 3x and ∠DBC = 2x. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q4/5

True or false?

Two angles that make up a straight angle must be equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q5/5

SET 20 • HSG-CO.C.10

BACKWARD 3

Alternate interior angles of parallel lines are congruent • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines $\ell \parallel m$ are cut by transversal t. An alternate interior angle of a 58° angle measures

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q1/5

True or false?

Alternate interior angles formed by a transversal of parallel lines are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q2/5

Line ℓ through C is parallel to AB. Which angle at C is congruent to $\angle A$?

A. The angle between CA and CB

B. The angle between ℓ and CB

C. $\angle ACB$

D. The angle between ℓ and CA

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q3/5

Why is it allowed to draw the line through C parallel to AB?

Reason: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q4/5

True or false?

Alternate interior angles are congruent for any two lines cut by a transversal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q5/5

SET 20 • HSG-CO.C.10

FORWARD 1

Prove the exterior angle theorem • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In triangle ABC, side BC is extended past C to D. Prove that $m\angle ACD = m\angle A + m\angle B$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q1/5

An exterior angle of a triangle measures 125°. One remote interior angle is 48°. Find the other remote interior angle and the adjacent interior angle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q2/5

True or false?

An exterior angle of a triangle is always greater than each remote interior angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q3/5

The exterior angle at C is (5x + 10)° and the remote interior angles are (2x + 15)° and (2x + 20)°. Find x.

A. 5

B. 35

C. 25

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q4/5

Prove that the three exterior angles of a triangle (one at each vertex) add to 360°.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q5/5

SET 20 • HSG-CO.C.10

FORWARD 2

Generalize the angle sum to polygons • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

Prove that the interior angles of any convex polygon with n sides add to $(n - 2) \cdot 180°$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q1/5

Find the measure of each interior angle of a regular octagon.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q2/5

Each interior angle of a regular polygon is 156°. How many sides does it have?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q3/5

True or false?

The interior angles of a hexagon add to 720°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q4/5

A pentagon has angles 100°, 110°, 120°, 95°, and x°. Find x.

A. 115

B. 125

C. 135

D. 105

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q5/5

Prove that the base angles of an isosceles triangle are congruent

MAIN	Geometry main question	**HSG-CO.C.10**
BACKWARD 1	Prove triangles congruent with SAS or SSS	**HSG-CO.B.8**
BACKWARD 2	Use the definitions of angle bisector and midpoint	**HSG-CO.A.1**
BACKWARD 3	Classify triangles and name legs, base, and base angles	**4.G.A.2**
FORWARD 1	Prove the converse: congruent base angles give congruent sides	**HSG-CO.C.10**
FORWARD 2	Use isosceles triangle theorems in multistep problems and proofs	**HSG-CO.C.10**

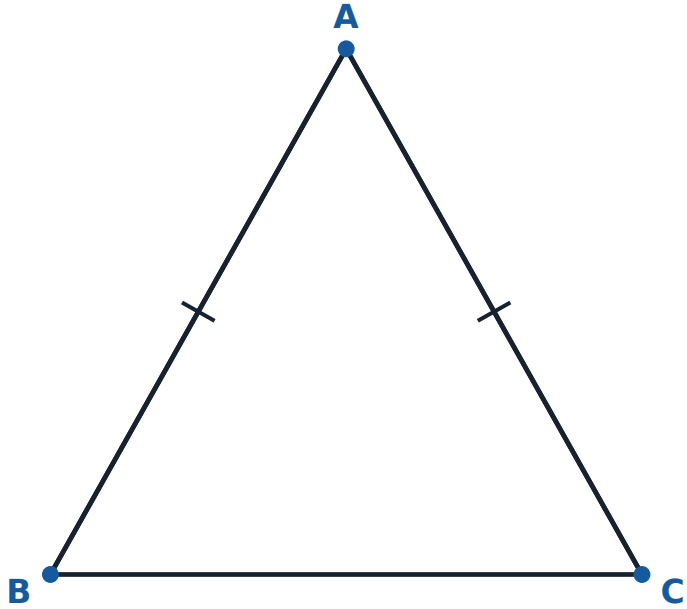
SET 21 • HSG-CO.C.10

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In triangle ABC, AB = AC. Prove that ∠B ≅ ∠C. (Draw the bisector of ∠A, meeting BC at D.)

Work space



Conclusion: _____

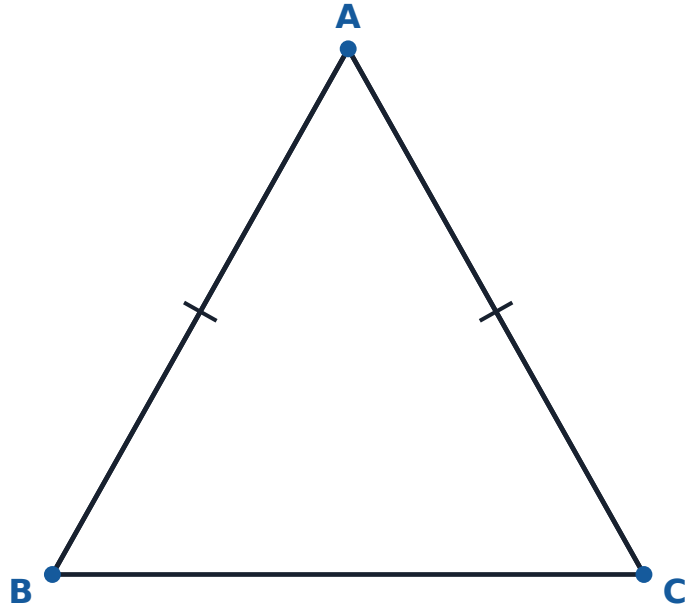
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q1/5

In triangle ABC, AB = AC. Prove that ∠B ≅ ∠C using the midpoint M of BC.

Work space



Conclusion: _____

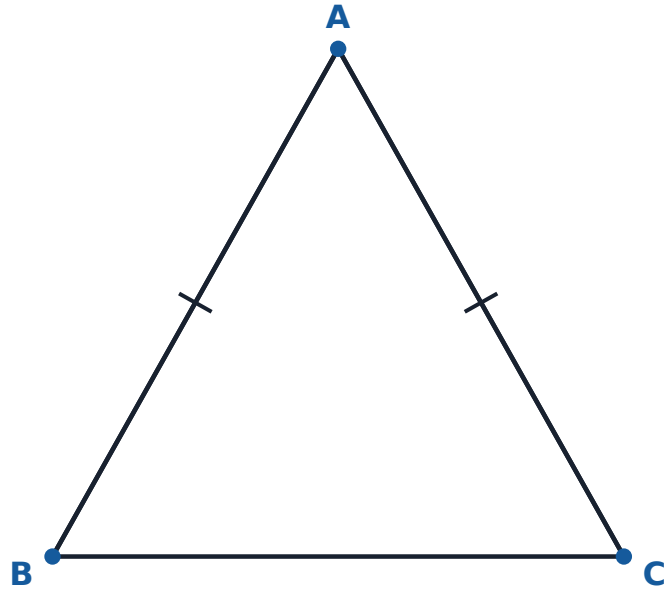
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q2/5

In triangle ABC, AB = AC. Prove that ∠B ≅ ∠C using a rigid motion: the reflection across the bisector of ∠A.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q3/5

Prove that every equilateral triangle is equiangular (all three angles are congruent).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q4/5

In the proof that base angles of isosceles triangle ABC (AB = AC) are congruent, the bisector of ∠A meets BC at D. Which criterion proves △ABD ≅ △ACD?

A. SAS

B. SSS

C. AAS

D. ASA

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q5/5

SET 21 • HSG-CO.C.10

BACKWARD 1

Prove triangles congruent with SAS or SSS • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

Two triangles share side AD, and AB = AC, ∠BAD ≅ ∠CAD. Which criterion shows △ABD ≅ △ACD?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q1/5

Two triangles share side AM, and AB = AC, BM = CM. Which criterion shows $\triangle ABM \cong \triangle ACM$?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q2/5

True or false?

A shared side counts as a pair of congruent sides in a congruence proof.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q3/5

Which information is NOT enough for congruence?

A. Two sides and a non-included angle

B. Three sides

C. Two sides and the included angle

D. Two angles and the included side

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q4/5

True or false?

If $\triangle ABD \cong \triangle ACD$, then $\angle ABD \cong \angle ACD$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • HSG-CO.C.10

BACKWARD 2

Use the definitions of angle bisector and midpoint • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Ray AD bisects ∠BAC, and m∠BAC = 64°. Find m∠BAD.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q1/5

M is the midpoint of BC, and BC = 18. Find BM.

BM = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q2/5

True or false?

An angle bisector divides an angle into two congruent angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q3/5

Ray AD bisects ∠BAC. Which equation is true?

A. m∠BAD = m∠BAC

B. AD = AB

C. BD = DC

D. m∠BAD = m∠DAC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q4/5

True or false?

The bisector of $\angle A$ of triangle ABC always passes through the midpoint of BC.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q5/5

SET 21 • HSG-CO.C.10

BACKWARD 3

Classify triangles and name legs, base, and base angles • Grade 4 • Geometry • 4.G.A.2

Triangle PQR has PQ = PR. Name the base and the base angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q1/5

True or false?

An isosceles triangle has at least two congruent sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q2/5

Which triangle is scalene?

A. Sides 4, 5, 6

B. Sides 3, 3, 3

C. Sides 5, 5, 5

D. Sides 4, 4, 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q3/5

In isosceles triangle XYZ with XY = XZ, which angle is the vertex angle?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q4/5

True or false?

Every equilateral triangle is isosceles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q5/5

SET 21 • HSG-CO.C.10

FORWARD 1

Prove the converse: congruent base angles give congruent sides • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In triangle ABC, $\angle B \cong \angle C$. Prove that AB = AC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q1/5

In triangle PQR, m∠Q = 70° and m∠R = 70°, and PQ = 3x + 2, PR = 5x − 6. Find x and PQ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q2/5

True or false?

A triangle with two congruent angles must have two congruent sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q3/5

Triangle ABC has $m\angle A = 50°$ and $m\angle B = 65°$. Which two sides are congruent?

A. No two sides

B. AB and AC

C. AB and BC

D. AC and BC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q4/5

Prove that every equiangular triangle is equilateral.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q5/5

SET 21 • HSG-CO.C.10

FORWARD 2

Use isosceles triangle theorems in multistep problems and proofs • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In triangle ABC, AB = AC and the vertex angle ∠A measures 38°. Find m∠B.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q1/5

Point D is on side BC of triangle ABC with AD = BD = CD. Find m∠BAC.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q2/5

In triangle ABC, AB = AC, and D and E are points on BC with BD = CE. Prove that AD = AE.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q3/5

True or false?

If the vertex angle of an isosceles triangle is 100°, each base angle is 40°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q4/5

In isosceles triangle PQR with PQ = PR, the exterior angle at Q (on the extension of RQ past Q) is 112°. Find m∠P.

A. 56°

B. 112°

C. 68°

D. 44°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q5/5

SET 22 • HSG-CO.C.10 • CONGRUENCE: PROVING THEOREMS

Prove the midsegment theorem

MAIN	Geometry main question	**HSG-CO.C.10**
BACKWARD 1	Find midpoints of segments in the coordinate plane	**6.G.A.3**
BACKWARD 2	Find and compare slopes to decide whether lines are parallel	**8.EE.B.6**
BACKWARD 3	Find lengths with the distance formula	**8.G.B.8**
BACKWARD 4	Prove lines parallel from congruent alternate interior angles	**HSG-CO.C.9**
FORWARD 1	Prove that the midpoints of a quadrilateral's sides form a parallelo	**HSG-CO.C.11**
FORWARD 2	Midsegment triangles: similarity, scale factor, and area	**HSG-SRT.A.2**

SET 22 • HSG-CO.C.10

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

Triangle ABC has A(0, 0), B(2a, 0), and C(2b, 2c), with a > 0 and c ≠ 0. D is the midpoint of AC and E is the midpoint of BC. Prove that DE ∥ AB and DE = ½AB.

Work space

Conclusion: _____

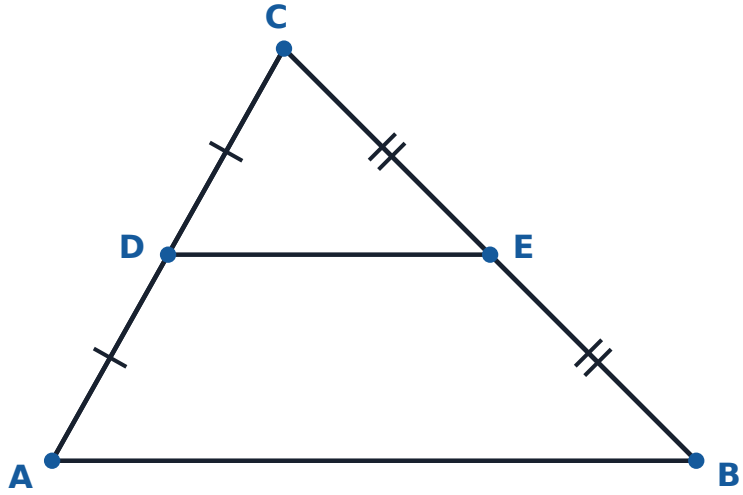
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q1/5

In triangle ABC, D is the midpoint of AC and E is the midpoint of BC. Extend DE past E to F so that EF = DE. Prove that DE ∥ AB and DE = ½AB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q2/5

In the coordinate proof of the midsegment theorem, the vertices are A(0, 0), B(2a, 0), C(2b, 2c). Why are the coordinates written with 2a, 2b, and 2c?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q3/5

A proof of the midsegment theorem with A(0, 0), B(2a, 0), C(2b, 2c):

1. D = (b, c), E = (a + b, c) (?)

2. slope DE = 0 = slope AB (?)

3. DE ∥ AB (?)

4. DE = a and AB = 2a, so DE = ½AB (?)

Give the four reasons.

Reasons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q4/5

A student checks the triangle A(0, 0), B(6, 0), C(2, 4), finds the midsegment is parallel to AB and 3 units long, and concludes the theorem is proved. What is the flaw?

A. One example cannot prove a statement about every triangle

B. The midsegment should be 6 units long

C. Midpoints cannot be found with coordinates

D. The midsegment is not parallel to AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q5/5

SET 22 • HSG-CO.C.10

BACKWARD 1

Find midpoints of segments in the coordinate plane • Grade 6 • Geometry • 6.G.A.3

Find the midpoint of the segment from (2, 6) to (8, 2).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q1/5

Find the midpoint of the segment from $(-3, 1)$ to $(5, -7)$.

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q2/5

Find the midpoint of the segment from (0, 0) to (2a, 2c).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q3/5

True or false?

The midpoint of the segment from (1, 1) to (7, 1) is (4, 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q4/5

True or false?

The midpoint of the segment from (0, 4) to (6, 0) is (6, 4).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q5/5

SET 22 • HSG-CO.C.10

BACKWARD 2

Find and compare slopes to decide whether lines are parallel • Grade 8 • Expressions & Equations • 8.EE.B.6

Find the slope of the line through (1, 2) and (4, 8).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q1/5

Find the slope of the line through (b, c) and (a + b, c).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q2/5

True or false?

Two different nonvertical lines with the same slope are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q3/5

Which segment is parallel to the segment from (0, 0) to (4, 2)?

A. From (2, 2) to (2, 6)

B. From (1, 3) to (3, 7)

C. From (1, 3) to (5, 5)

D. From (0, 0) to (2, −1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q4/5

True or false?

A horizontal segment and a vertical segment have the same slope.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q5/5

SET 22 • HSG-CO.C.10

BACKWARD 3

Find lengths with the distance formula • Grade 8 • Geometry • 8.G.B.8

Find the distance between (b, c) and (a + b, c), with a > 0.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B3-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q1/5

Find the distance between (0, 0) and (6, 8).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B3-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q2/5

Find the length of the segment from $(-2, 5)$ to $(4, 5)$.

A. 3

B. 10

C. 2

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B3-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q3/5

True or false?

The distance from (1, 2) to (4, 6) is 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B3-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q4/5

True or false?

The distance from (0, 0) to (2a, 0) is a, when $a > 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B3-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q5/5

SET 22 • HSG-CO.C.10

BACKWARD 4

Prove lines parallel from congruent alternate interior angles • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines ℓ and m are cut by a transversal, and alternate interior angles are congruent. What can you conclude?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B4-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q1/5

True or false?

If two lines are cut by a transversal and alternate interior angles are congruent, the lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B4-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q2/5

In the synthetic midsegment proof, which fact shows BF ∥ AC?

A. DE = EF

B. ∠DCE ≅ ∠FBE are alternate interior angles

C. BF = DC

D. E is the midpoint of BC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B4-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q3/5

Two lines are cut by a transversal and corresponding angles measure 70° and 70°. Are the lines parallel?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B4-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q4/5

True or false?

Congruent vertical angles prove that two lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B4-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q5/5

SET 22 • HSG-CO.C.10

FORWARD 1

Prove that the midpoints of a quadrilateral's sides form a parallelogram • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ABCD is any quadrilateral, and P, Q, R, S are the midpoints of AB, BC, CD, DA. Prove that PQRS is a parallelogram.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q1/5

Quadrilateral ABCD has diagonals AC = 10 and BD = 14. Find the perimeter of the quadrilateral formed by the midpoints of its sides.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q2/5

True or false?

The midpoint quadrilateral of any quadrilateral is a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q3/5

If the diagonals of ABCD are perpendicular, the midpoint quadrilateral is a

A. trapezoid that is not a parallelogram

B. rhombus

C. rectangle

D. kite that is not a parallelogram

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q4/5

In triangle ABC, D, E, F are the midpoints of AB, BC, CA. AB = 12, BC = 16, CA = 10. Find the perimeter of triangle DEF.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q5/5

SET 22 • HSG-CO.C.10

FORWARD 2

Midsegment triangles: similarity, scale factor, and area • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

D, E, F are the midpoints of the sides of triangle ABC. Explain why triangle DEF is similar to triangle ABC, and give the scale factor.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q1/5

Triangle ABC has area 48. Find the area of the triangle formed by the midpoints of its sides.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q2/5

True or false?

The four triangles formed by drawing all three midsegments of a triangle are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q3/5

The midsegment parallel to AB in triangle ABC is 9. A line segment connects the midpoints of that midsegment's triangle the same way. How long is the new segment parallel to AB?

A. 9

B. 4.5

C. 18

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q4/5

Triangle ABC has vertices A(0, 0), B(8, 0), C(2, 6). Find the vertices of its midsegment triangle and verify the ratio of areas.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q5/5

Prove that the medians of a triangle meet at one point

MAIN	Geometry main question	**HSG-CO.C.10**
BACKWARD 1	Find midpoints, including with variables	**6.G.A.3**
BACKWARD 2	Use the midsegment theorem	**HSG-CO.C.10**
BACKWARD 3	Find the point a given fraction of the way along a segment	**7.RP.A.2**
BACKWARD 4	Write the equation of a line through two points	**8.F.B.4**
FORWARD 1	Locate the centroid and use the 2 : 1 ratio	**HSG-GPE.B.6**
FORWARD 2	Prove that the angle bisectors of a triangle meet at one point	**HSG-C.A.3**

SET 23 • HSG-CO.C.10

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

Triangle ABC has A(0, 0), B(2b, 0), and C(2c, 2d), with b ≠ 0 and d ≠ 0. Prove that the point G((2b + 2c)/3, 2d/3) lies on all three medians.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q1/5

In triangle ABC, medians BE and CF meet at G. Let H and K be the midpoints of BG and CG. Prove that BG = 2GE. Then explain why the third median also passes through G.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q2/5

Explain why proving that the point two-thirds of the way along EACH median is the same point proves that the three medians meet at one point.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q3/5

Triangle PQR has P(0, 0), Q(6, 0), and R(0, 6). Find the three medians' equations and prove that they meet at one point.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q4/5

True or false?

Drawing medians carefully in three different triangles and seeing that they meet proves that the medians of every triangle meet at one point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q5/5

SET 23 • HSG-CO.C.10

BACKWARD 1

Find midpoints, including with variables • Grade 6 • Geometry • 6.G.A.3

Find the midpoint of the segment from (2b, 0) to (2c, 2d).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q1/5

Find the midpoint of the segment from (0, 6) to (6, 0).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q2/5

A median of a triangle joins a vertex to

A. another vertex

B. the midpoint of the opposite side

C. the foot of the perpendicular from that vertex

D. the midpoint of an adjacent side

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q3/5

True or false?

A triangle has exactly three medians.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q4/5

True or false?

A median always bisects the angle at its vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q5/5

SET 23 • HSG-CO.C.10

BACKWARD 2

Use the midsegment theorem • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In triangle ABC, F and E are the midpoints of AB and AC. BC = 14. Find FE.

FE = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q1/5

True or false?

The segment joining the midpoints of two sides of a triangle is parallel to the third side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q2/5

In triangle GBC, H and K are midpoints of GB and GC. Which statement is true?

A. HK ∥ BC and HK = ½BC

B. HK ⊥ BC

C. HK ∥ GB

D. HK = BC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q3/5

A midsegment of a triangle is 11 cm. How long is the side parallel to it?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q4/5

True or false?

A midsegment of a triangle is always half as long as each of the other two sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q5/5

SET 23 • HSG-CO.C.10

BACKWARD 3

Find the point a given fraction of the way along a segment • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

On a number line, what point is ⅔ of the way from 0 to 9?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q1/5

On a number line, what point is ⅔ of the way from 3 to 12?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q2/5

A point is ⅔ of the way from A to D. What is the ratio of its distances from A and from D?

A. 2 : 3

B. 3 : 2

C. 1 : 2

D. 2 : 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q3/5

True or false?

The point ⅔ of the way from (0, 0) to (6, 3) is (4, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q4/5

True or false?

The point ⅔ of the way from (0, 0) to (3, 6) is (2, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q5/5

SET 23 • HSG-CO.C.10

BACKWARD 4

Write the equation of a line through two points • Grade 8 • Functions • 8.F.B.4

Write an equation of the line through (0, 0) and (3, 3).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B4-Q1

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q1/5

Write an equation of the line through (6, 0) and (0, 3).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B4-Q2

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q2/5

Which point is on the line $y = -2x + 6$?

A. (2, 2)

B. (1, 2)

C. (3, 3)

D. (0, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B4-Q3

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q3/5

True or false?

The line through (0, 6) and (3, 0) has slope −2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B4-Q4

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q4/5

True or false?

The line $y = x$ passes through $(2, 3)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B4-Q5

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q5/5

SET 23 • HSG-CO.C.10

FORWARD 1

Locate the centroid and use the 2 : 1 ratio • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6

Find the point where the medians of the triangle with vertices (1, 2), (7, 4), and (4, 9) meet.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q1/5

Median AD of a triangle is 18 cm long. How far is the centroid from A and from D?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q2/5

True or false?

The centroid divides each median so that the part from the vertex is twice the part from the midpoint.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q3/5

G is the centroid of triangle ABC, and GE = 5 on median BE. Find BE.

A. 15

B. 20

C. 10

D. 7.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q4/5

Explain why the centroid of a triangle is the average of its three vertices.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q5/5

SET 23 • HSG-CO.C.10

FORWARD 2

Prove that the angle bisectors of a triangle meet at one point • High School — Geometry • Geometry: Circles • HSG-C.A.3

Prove that the bisectors of the three angles of triangle ABC meet at one point that is equidistant from the three sides.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q1/5

Why must the bisectors of ∠A and ∠B of a triangle meet inside the triangle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q2/5

True or false?

A point on the bisector of an angle is equidistant from the two sides of the angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q3/5

The point where the angle bisectors of a triangle meet is the center of the

A. circumscribed circle

B. nine-point circle

C. inscribed circle

D. largest square inside the triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q4/5

True or false?

The angle bisectors and the medians of every triangle meet at the same point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q5/5

Prove that opposite sides and opposite angles of a parallelogram are congruent

MAIN	Geometry main question	**HSG-CO.C.11**
BACKWARD 1	Alternate interior and same-side interior angles of parallel lines	**HSG-CO.C.9**
BACKWARD 2	Prove triangles congruent with ASA	**HSG-CO.B.8**
BACKWARD 3	Know the definition of a parallelogram	**5.G.B.3**
FORWARD 1	Prove that the diagonals of a rhombus are perpendicular	**HSG-CO.C.11**
FORWARD 2	Prove the parallelogram properties with coordinates	**HSG-GPE.B.4**

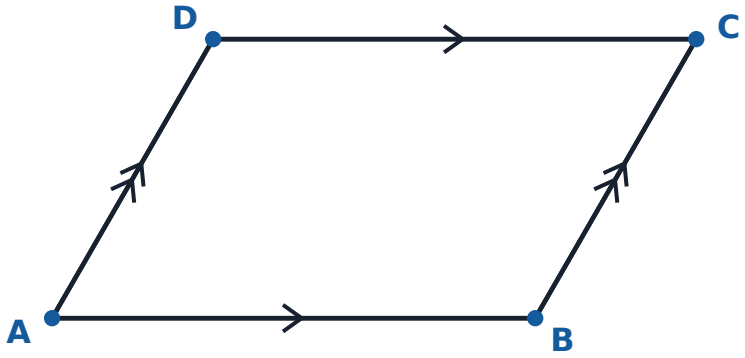
SET 24 • HSG-CO.C.11

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ABCD is a parallelogram (AB ∥ DC and AD ∥ BC). Prove that AB = CD and AD = CB. (Draw diagonal AC.)

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

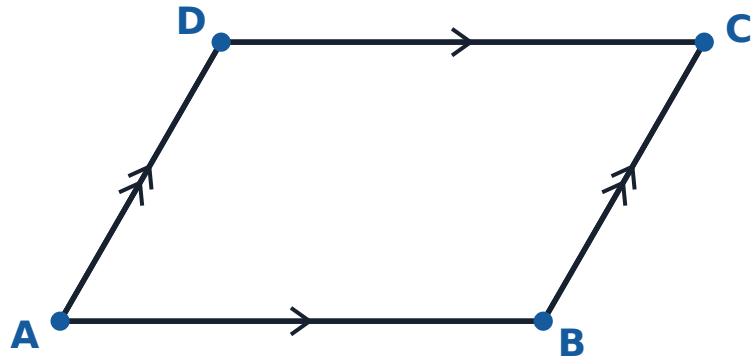
ID S24-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q1/5

ABCD is a parallelogram.

Prove that $\angle B \cong \angle D$.

Work space



Conclusion: _____

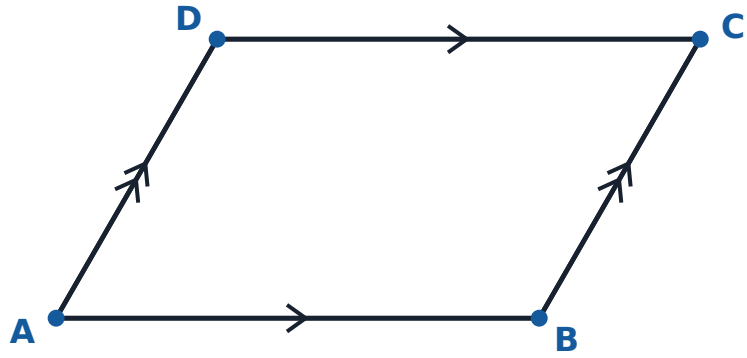
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q2/5

ABCD is a parallelogram. Prove that ∠A ≅ ∠C using same-side interior angles (no diagonal).

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q3/5

A proof that opposite sides of parallelogram ABCD are congruent (diagonal AC drawn):

1. $\angle BAC \cong \angle DCA$ (?)

2. $\angle BCA \cong \angle DAC$ (?)

3. $AC = CA$ (?)

4. $\triangle ABC \cong \triangle CDA$ (?)

5. $AB = CD$ and $BC = DA$ (?)

Give the five reasons.

Reasons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q4/5

In the proof that opposite sides of parallelogram ABCD are congruent, which congruence criterion is used for triangles ABC and CDA?

A. SSA

B. SAS

C. SSS

D. ASA

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q5/5

SET 24 • HSG-CO.C.11

BACKWARD 1

Alternate interior and same-side interior angles of parallel lines • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

AB ∥ DC and diagonal AC is drawn. Which angle at C is congruent to ∠BAC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q1/5

True or false?

Same-side interior angles formed by a transversal of parallel lines are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q2/5

In parallelogram ABCD, side AB is a transversal of AD and BC. What is m∠A + m∠B?

A. It depends on the figure

B. 90°

C. 360°

D. 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q3/5

In parallelogram ABCD, m∠A = 65°. Find m∠B.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q4/5

True or false?

Alternate interior angles of parallel lines are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q5/5

SET 24 • HSG-CO.C.11

BACKWARD 2

Prove triangles congruent with ASA • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

Two triangles have two pairs of congruent angles and the congruent sides between them. Which criterion applies?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q1/5

True or false?

A diagonal shared by two triangles can be the included side in ASA.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q2/5

In $\triangle ABC$ and $\triangle CDA$, $\angle BAC \cong \angle DCA$, $\angle BCA \cong \angle DAC$, and AC is shared. Which correspondence is correct?

A. A ↔ C, B ↔ D, C ↔ A

B. A ↔ D, B ↔ C, C ↔ A

C. A ↔ A, B ↔ D, C ↔ C

D. A ↔ C, B ↔ A, C ↔ D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q3/5

After proving $\triangle ABC \cong \triangle CDA$, which side corresponds to AB?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q4/5

True or false?

ASA requires the side to be between the two angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q5/5

SET 24 • HSG-CO.C.11

BACKWARD 3

Know the definition of a parallelogram • Grade 5 • Geometry • 5.G.B.3

True or false?

A parallelogram is a quadrilateral with both pairs of opposite sides parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B3-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q1/5

Which quadrilateral is always a parallelogram?

A. A rectangle

B. A trapezoid

C. A kite

D. Any quadrilateral with one right angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B3-Q2

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q2/5

Name the two pairs of opposite sides of parallelogram PQRS.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B3-Q3

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q3/5

True or false?

In parallelogram ABCD, sides AB and BC are opposite sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B3-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q4/5

Name the two pairs of opposite angles of parallelogram WXYZ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B3-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q5/5

SET 24 • HSG-CO.C.11

FORWARD 1

Prove that the diagonals of a rhombus are perpendicular • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ABCD is a rhombus (a parallelogram with all four sides congruent), with diagonals meeting at E. Prove that AC ⊥ BD.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q1/5

In rhombus ABCD, diagonals AC = 16 and BD = 12. Find the side length.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q2/5

True or false?

Each diagonal of a rhombus bisects two of its angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q3/5

Which property does every rhombus have but not every parallelogram?

A. Perpendicular diagonals

B. Opposite sides parallel

C. Opposite angles congruent

D. Diagonals that bisect each other

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q4/5

In rhombus PQRS, m∠PQR = 110°. Find m∠PQS, where QS is a diagonal.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q5/5

SET 24 • HSG-CO.C.11

FORWARD 2

Prove the parallelogram properties with coordinates • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

A parallelogram has vertices A(0, 0), B(a, 0), C(a + b, c), D(b, c), with a > 0 and c > 0. Prove that AB = DC and AD = BC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q1/5

Explain why the vertices A(0, 0), B(a, 0), C(a + b, c), D(b, c) describe a general parallelogram.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q2/5

True or false?

With vertices A(0, 0), B(a, 0), C(a + b, c), D(b, c), slopes show AD ∥ BC.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q3/5

For the parallelogram A(0, 0), B(6, 0), C(8, 4), D(2, 4), what is AD?

A. $\sqrt{20} + 2$

B. 4

C. 6

D. $2\sqrt{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q4/5

Parallelogram ABCD has A(1, 1), B(5, 1), and D(2, 4). Find C.

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q5/5

SET 25 • HSG-CO.C.11 • CONGRUENCE: PROVING THEOREMS

Prove that the diagonals of a parallelogram bisect each other

MAIN	Geometry main question	**HSG-CO.C.11**
BACKWARD 1	Opposite sides of a parallelogram are congruent	**HSG-CO.C.11**
BACKWARD 2	Vertical and alternate interior angles at the crossing of the diago…	**HSG-CO.C.9**
BACKWARD 3	Find the midpoint of a segment and use the definition of bisect	**6.G.A.3**
FORWARD 1	Prove that a 180° rotation carries a parallelogram onto itself	**HSG-CO.A.3**
FORWARD 2	Use diagonal bisection with coordinates and algebra	**HSG-GPE.B.4**

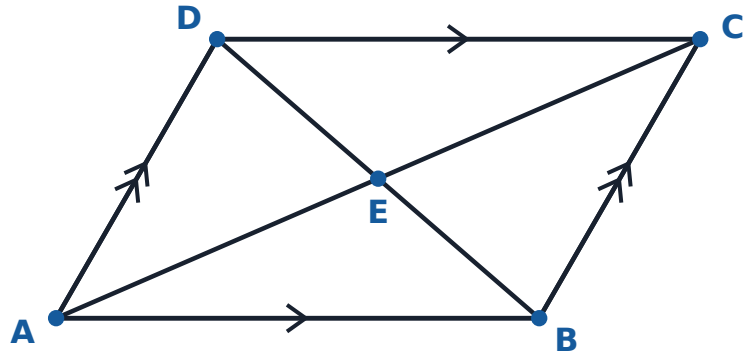
SET 25 • HSG-CO.C.11

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ABCD is a parallelogram with diagonals AC and BD meeting at E. Prove that AE = CE and BE = DE.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q1/5

PQRS is a parallelogram with diagonals PR and QS meeting at T. Prove that T is the midpoint of both diagonals.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q2/5

A parallelogram has vertices A(0, 0), B(a, 0), C(a + b, c), D(b, c). Prove that its diagonals bisect each other.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q3/5

A proof that the diagonals of parallelogram ABCD bisect each other (they meet at E):

1. AB = CD (?)

2. ∠BAE ≅ ∠DCE (?)

3. ∠AEB ≅ ∠CED (?)

4. △ABE ≅ △CDE (?)

5. AE = CE, BE = DE (?)

Give the five reasons.

Reasons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q4/5

Which previously proved fact does the proof that a parallelogram's diagonals bisect each other use?

A. Consecutive sides are congruent

B. Opposite sides of a parallelogram are congruent

C. Diagonals are perpendicular

D. Diagonals of a parallelogram are congruent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q5/5

SET 25 • HSG-CO.C.11

BACKWARD 1

Opposite sides of a parallelogram are congruent • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

In parallelogram ABCD, AB = 9 and BC = 5. Find CD and DA.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q1/5

True or false?

Opposite sides of every parallelogram are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q2/5

In parallelogram WXYZ, WX = 2x + 1 and YZ = x + 6. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q3/5

In parallelogram PQRS, which side is congruent to QR?

A. PQ

B. SP

C. PR

D. RS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q4/5

True or false?

Consecutive sides of every parallelogram are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q5/5

SET 25 • HSG-CO.C.11

BACKWARD 2

Vertical and alternate interior angles at the crossing of the diagonals • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Diagonals of parallelogram ABCD meet at E. Which angle is vertical to ∠AEB?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q1/5

AB ∥ DC with transversal AC. Which angle is congruent to ∠BAC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q2/5

True or false?

Vertical angles are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q3/5

AB ∥ DC with transversal BD. Which angle is congruent to ∠ABD?

A. ∠ABC

B. ∠CDB

C. ∠ADB

D. ∠CBD

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q4/5

True or false?

$\angle AEB$ and $\angle BEC$ are vertical angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q5/5

SET 25 • HSG-CO.C.11

BACKWARD 3

Find the midpoint of a segment and use the definition of bisect • Grade 6 • Geometry • 6.G.A.3

Find the midpoint of the segment from (0, 0) to (a + b, c).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q1/5

Find the midpoint of the segment from (a, 0) to (b, c).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q2/5

True or false?

A segment bisects another segment if it passes through its midpoint.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q3/5

Segment AC = 18 is bisected at E. Find AE.

AE = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q4/5

True or false?

If two segments have the same midpoint, each bisects the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q5/5

SET 25 • HSG-CO.C.11

FORWARD 1

Prove that a 180° rotation carries a parallelogram onto itself • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

ABCD is a parallelogram whose diagonals meet at E. Prove that the 180° rotation about E maps ABCD onto itself.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q1/5

Parallelogram ABCD has diagonals meeting at E(2, 3), and A(−1, 1). Find C.

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q2/5

True or false?

A quadrilateral whose diagonals bisect each other has 180° rotation symmetry about their crossing point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q3/5

The 180° rotation about the center of parallelogram PQRS maps side PQ onto

A. side SP

B. side QR

C. diagonal PR

D. side RS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q4/5

Explain why the result implies that opposite angles of a parallelogram are congruent.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q5/5

SET 25 • HSG-CO.C.11

FORWARD 2

Use diagonal bisection with coordinates and algebra • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Parallelogram ABCD has A(1, 2), B(7, 2), C(9, 6). Find D and the point where the diagonals meet.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q1/5

In parallelogram ABCD, diagonals meet at E.
AE = 3x − 2 and CE = x + 8. Find AC.

AC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q2/5

True or false?

The diagonals of every parallelogram are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q3/5

Three vertices of a parallelogram are (0, 0), (4, 0), (1, 3). Which could be the fourth vertex?

A. (5, 3)

B. (4, 3)

C. (3, 3)

D. (1, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q4/5

In parallelogram PQRS, diagonals meet at T. PT = y + 4, TR = 2y − 1, QT = 3y. Find QS.

QS = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q5/5

SET 26 • HSG-CO.C.11 • CONGRUENCE: PROVING THEOREMS

Prove the converses: conditions that make a quadrilateral a parallelogram

MAIN	Geometry main question	**HSG-CO.C.11**
BACKWARD 1	Converses of the parallel-line angle theorems	**HSG-CO.C.9**
BACKWARD 2	Prove triangles congruent with SSS and SAS	**HSG-CO.B.8**
BACKWARD 3	Angle sum of a quadrilateral	**8.G.A.5**
FORWARD 1	Prove that a quadrilateral is a parallelogram using coordinates	**HSG-GPE.B.4**
FORWARD 2	Decide which conditions are enough to guarantee a parallelogram	**HSG-CO.C.11**

SET 26 • HSG-CO.C.11

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

In quadrilateral ABCD, AB = CD and AD = CB. Prove that ABCD is a parallelogram. (Draw diagonal AC.)

Work space

Conclusion: _____

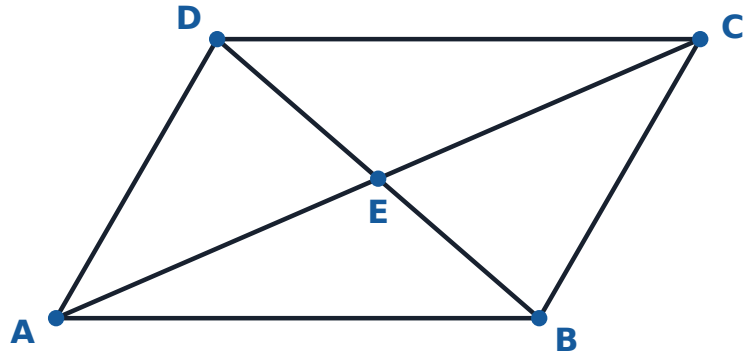
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q1/5

In quadrilateral ABCD, diagonals AC and BD bisect each other at E. Prove that ABCD is a parallelogram.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q2/5

In quadrilateral ABCD, AB ∥ DC and AB = DC. Prove that ABCD is a parallelogram.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q3/5

In quadrilateral ABCD, $\angle A \cong \angle C$ and $\angle B \cong \angle D$. Prove that ABCD is a parallelogram.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q4/5

Quadrilateral ABCD has AB ∥ DC and AD = BC. Which conclusion is justified?

A. ABCD may not be a parallelogram (it could be an isosceles trapezoid)

B. ABCD must be a rhombus

C. ABCD must be a rectangle

D. ABCD must be a parallelogram

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q5/5

SET 26 • HSG-CO.C.11

BACKWARD 1

Converses of the parallel-line angle theorems • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Two lines are cut by a transversal so that alternate interior angles are congruent. What can you conclude?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q1/5

True or false?

If same-side interior angles are supplementary, the lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q2/5

Which fact proves two lines are parallel?

A. A linear pair

B. Two acute angles

C. Congruent corresponding angles

D. Congruent vertical angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q3/5

Same-side interior angles measure 104° and 76°. Are the lines parallel?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q4/5

True or false?

If alternate interior angles are 40° and 50°, the lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q5/5

SET 26 • HSG-CO.C.11

BACKWARD 2

Prove triangles congruent with SSS and SAS • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

△ABC and △CDA share AC, and AB = CD, BC = DA. Which criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q1/5

△AEB and △CED: AE = CE, BE = DE, and ∠AEB ≅ ∠CED. Which criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q2/5

True or false?

Vertical angles can serve as the included angle in SAS.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q3/5

After $\triangle ABC \cong \triangle CDA$, which angle is congruent to $\angle BAC$?

A. $\angle DCA$

B. $\angle BCA$

C. $\angle ABC$

D. $\angle DAC$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q4/5

True or false?

SSA proves two triangles congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q5/5

SET 26 • HSG-CO.C.11

BACKWARD 3

Angle sum of a quadrilateral • Grade 8 • Geometry • 8.G.A.5

Three angles of a quadrilateral are 80°, 95°, and 110°. Find the fourth.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q1/5

True or false?

The angles of every quadrilateral add to 360°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B3-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q2/5

A quadrilateral has angles x, x, 2x, 2x. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q3/5

Opposite angles of a quadrilateral are 70° and 70°, and another is 110°. Find the fourth.

A. 110°

B. 90°

C. 70°

D. 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q4/5

True or false?

A quadrilateral can have four acute angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q5/5

SET 26 • HSG-CO.C.11

FORWARD 1

Prove that a quadrilateral is a parallelogram using coordinates • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Prove or disprove that A(1, 1), B(6, 2), C(8, 6), D(3, 5) form a parallelogram.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q1/5

Prove or disprove that P(0, 0), Q(4, 1), R(5, 4), S(0, 3) form a parallelogram.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q2/5

True or false?

If the midpoints of both diagonals of a quadrilateral are the same point, the quadrilateral is a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q3/5

Which test alone proves that ABCD is a parallelogram?

A. AC and BD have equal lengths

B. AB $\perp$ BC

C. AB and DC have equal slopes and equal lengths

D. AB and DC have equal lengths

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q4/5

A(0, 0), B(5, 0), C(7, 3), D(x, 3) form a parallelogram ABCD. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q5/5

SET 26 • HSG-CO.C.11

FORWARD 2

Decide which conditions are enough to guarantee a parallelogram • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

Is a quadrilateral with one pair of opposite sides congruent and the other pair parallel always a parallelogram? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q1/5

Is a quadrilateral with both pairs of opposite angles congruent always a parallelogram? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q2/5

True or false?

A quadrilateral with one diagonal that bisects the other must be a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q3/5

Which condition is NOT enough to prove a quadrilateral is a parallelogram?

A. Both pairs of opposite sides congruent

B. One pair of opposite sides congruent

C. Diagonals bisect each other

D. One pair of opposite sides parallel and congruent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q4/5

Is a quadrilateral with perpendicular diagonals always a parallelogram? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q5/5

SET 27 • HSG-CO.C.11 • CONGRUENCE: PROVING THEOREMS

Prove that rectangles are parallelograms with congruent diagonals

MAIN	Geometry main question	**HSG-CO.C.11**
BACKWARD 1	Parallelogram properties: opposite sides and consecutive angles	**HSG-CO.C.11**
BACKWARD 2	SAS and SSS with overlapping triangles	**HSG-CO.B.8**
BACKWARD 3	Use the Pythagorean Theorem for diagonals	**8.G.B.7**
FORWARD 1	Decide whether four points form a rectangle	**HSG-GPE.B.4**
FORWARD 2	Prove diagonal properties of rhombuses and squares	**HSG-CO.C.11**

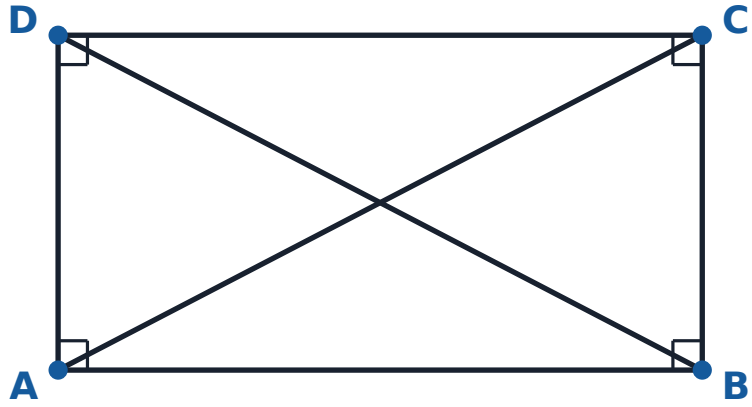
SET 27 • HSG-CO.C.11

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ABCD is a rectangle. Prove that AC = BD.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q1/5

ABCD is a parallelogram with AC = BD. Prove that ABCD is a rectangle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q2/5

Rectangle ABCD has A(0, 0), B(a, 0), C(a, b), D(0, b), with a, b > 0. Prove that its diagonals are congruent.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q3/5

Explain why a rectangle is a parallelogram.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q4/5

Which statement is the converse proved in this set?

A. Every rectangle has congruent diagonals

B. A parallelogram with congruent diagonals is a rectangle

C. Every quadrilateral with congruent diagonals is a rectangle

D. A rectangle has perpendicular diagonals

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q5/5

SET 27 • HSG-CO.C.11

BACKWARD 1

Parallelogram properties: opposite sides and consecutive angles • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

In parallelogram ABCD, m∠A = 90°. Find the other three angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q1/5

True or false?

Consecutive angles of a parallelogram are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q2/5

In parallelogram PQRS, m∠P = m∠Q. Find m∠P.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q3/5

In parallelogram ABCD, which side is congruent to AD?

A. BC

B. AC

C. CD

D. AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q4/5

True or false?

In every parallelogram, all angles are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q5/5

SET 27 • HSG-CO.C.11

BACKWARD 2

SAS and SSS with overlapping triangles • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

△ABC and △BAD overlap and share AB. Which corresponding angle is at A in △BAD for ∠B in △ABC?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q1/5

True or false?

Overlapping triangles can share a side used in a congruence proof.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q2/5

△ABC and △BAD: AB shared, BC = AD, AC = BD. Which criterion?

A. ASA

B. SAS

C. AAS

D. SSS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q3/5

△ABC and △BAD: AB shared, BC = AD, ∠ABC ≅ ∠BAD. Which criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q4/5

True or false?

In $\triangle ABC \cong \triangle BAD$, vertex C corresponds to vertex C.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q5/5

SET 27 • HSG-CO.C.11

BACKWARD 3

Use the Pythagorean Theorem for diagonals • Grade 8 • Geometry • 8.G.B.7

A rectangle is 6 by 8. Find the length of a diagonal.

Diagonal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q1/5

A rectangle is 5 by 12. Find the length of a diagonal.

Diagonal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q2/5

True or false?

A rectangle with sides a and b has diagonal $\sqrt{(a^2 + b^2)}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q3/5

A square has side 4. Its diagonal is

A. 8

B. $4\sqrt{2}$

C. $2\sqrt{2}$

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q4/5

True or false?

The two diagonals of a 3-by-7 rectangle have different lengths.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q5/5

SET 27 • HSG-CO.C.11

FORWARD 1

Decide whether four points form a rectangle • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Prove or disprove that A(1, 1), B(5, 3), C(4, 5), D(0, 3) form a rectangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q1/5

Prove or disprove that P(0, 0), Q(4, 0), R(5, 3), S(1, 3) form a rectangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q2/5

True or false?

A quadrilateral whose diagonals are congruent and bisect each other is a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q3/5

Which set of facts proves ABCD is a rectangle?

A. Diagonals bisect each other and are congruent

B. Diagonals are congruent

C. One angle is a right angle

D. Opposite sides are congruent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q4/5

A(0, 0), B(6, 0), and D(0, 4) are three vertices of rectangle ABCD. Find C and verify the diagonals are congruent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q5/5

SET 27 • HSG-CO.C.11

FORWARD 2

Prove diagonal properties of rhombuses and squares • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

Prove that a parallelogram whose diagonals are perpendicular is a rhombus.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q1/5

Which property must a quadrilateral have to be a square, using diagonals only?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q2/5

True or false?

A rectangle with perpendicular diagonals is a square.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q3/5

A parallelogram has congruent, perpendicular diagonals. It is a

A. rectangle that is not a square

B. square

C. rhombus that is not a square

D. trapezoid

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q4/5

A square has diagonal 10. Find its side length (exact) and its area.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q5/5

Construct a copy of a segment

MAIN	Geometry main question	**HSG-CO.D.12**
BACKWARD 1	Compare and measure lengths	**2.MD.A.1**
BACKWARD 2	A compass draws all points at one distance from its center	**7.G.B.4**
BACKWARD 3	Know rays, segments, and their endpoints	**HSG-CO.A.1**
FORWARD 1	Construct a triangle from three given segments	**HSG-CO.D.12**
FORWARD 2	Decide whether segments can form a triangle, and construct sums	**HSG-CO.D.12**

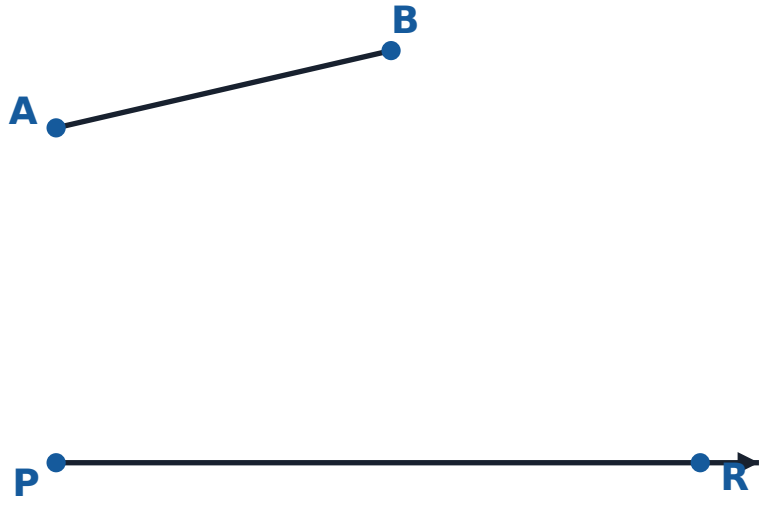
SET 28 • HSG-CO.D.12

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct a point Q on ray PR so that PQ = AB.



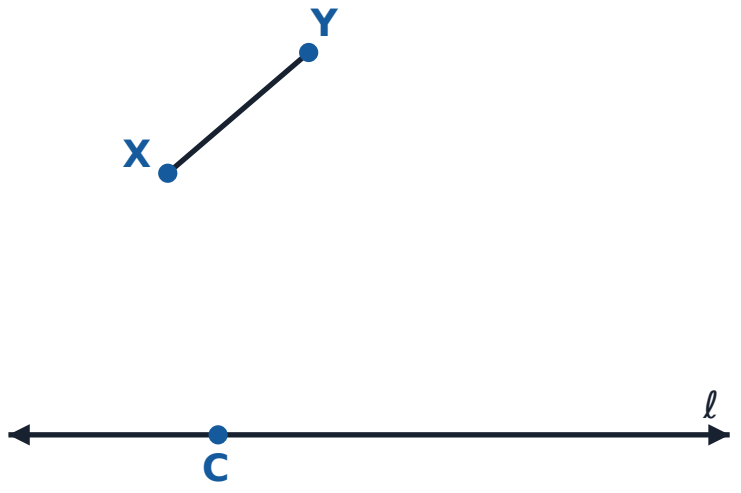
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

On line ℓ, starting at point C, construct a segment CD with CD = XY.



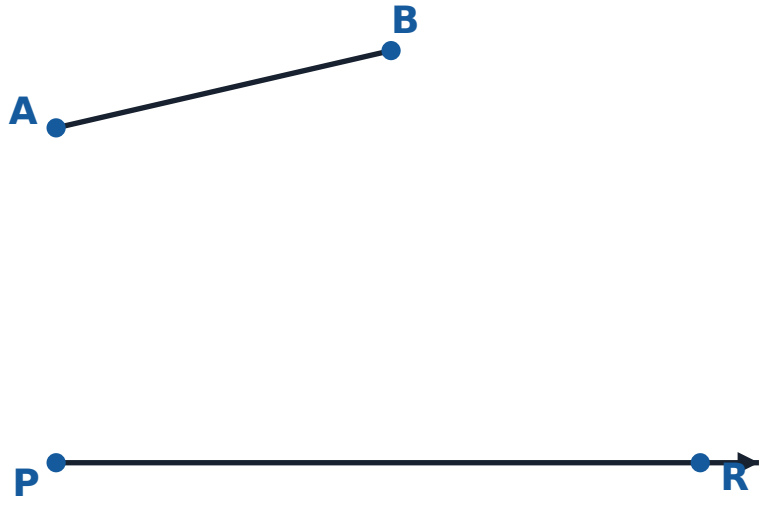
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct a point Q on ray PR so that PQ = 2 · AB.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q3/5

Write the steps for copying segment AB onto ray PR, starting at P, with a compass and straightedge.

Steps: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q4/5

You have no compass, only a straightedge and a strip of paper. Describe how to copy segment AB onto line ℓ starting at point C.

Method: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q5/5

SET 28 • HSG-CO.D.12

BACKWARD 1

Compare and measure lengths • Grade 2 • Measurement & Data • 2.MD.A.1

Segment AB is 6 cm and segment CD is 6 cm. Are they congruent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q1

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q1/5

True or false?

Two segments are congruent if they have the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q2

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q2/5

Which tool transfers a length without reading numbers?

A. A calculator

B. A protractor

C. A compass

D. A thermometer

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q3

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q3/5

A segment 4.5 cm long is copied twice end to end on a line. How long is the result?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q4

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q4/5

True or false?

A segment drawn longer on the page than another is congruent to it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q5

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q5/5

SET 28 • HSG-CO.D.12

BACKWARD 2

A compass draws all points at one distance from its center • Grade 7 • Geometry • 7.G.B.4

A compass is opened to 5 cm with its point on P. How far from P is every point on the arc it draws?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q1/5

True or false?

Changing the compass opening between steps of a construction keeps the copy accurate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q2/5

An arc centered at P crosses a ray from P at Q. What is PQ?

A. It cannot be known

B. Half the opening

C. Twice the opening

D. The compass opening

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q3/5

True or false?

Every point on a circle is the same distance from the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q4/5

An arc centered at C with radius 3 cm crosses a line through C. How many crossing points are there?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q5/5

SET 28 • HSG-CO.D.12

BACKWARD 3

Know rays, segments, and their endpoints • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

True or false?

Ray PR starts at P and continues through R without end.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q1/5

Q is on ray PR with PQ = 5. Must Q be between P and R?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q2/5

Segment AB has endpoints

A. none

B. A and B

C. only B

D. only A

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q3/5

True or false?

A point on ray PR can be on the opposite side of P from R.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q4/5

Name the endpoint of ray PR.

Endpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q5/5

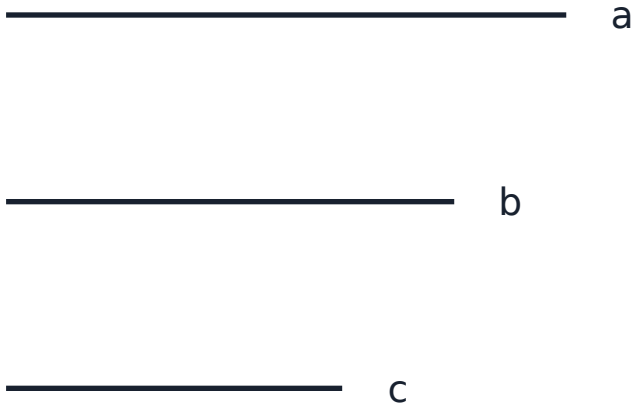
SET 28 • HSG-CO.D.12

FORWARD 1

Construct a triangle from three given segments • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct a triangle whose sides are congruent to the three segments a, b, and c.



Explain why the two arcs in the triangle construction locate the third vertex.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q2/5

True or false?

Any three segments can be used to construct a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q3/5

In the construction, the arcs centered at the ends of the base cross at two points. The two triangles formed are

A. similar but not congruent

B. different in size

C. congruent (mirror images)

D. not triangles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q4/5

Segments of length 3, 4, and 6 are given. Can a triangle be constructed? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q5/5

SET 28 • HSG-CO.D.12

FORWARD 2

Decide whether segments can form a triangle, and construct sums and differences • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Segments of length 2, 3, and 6 are given. Can a triangle be constructed? Explain what happens to the arcs.

Answer: _____

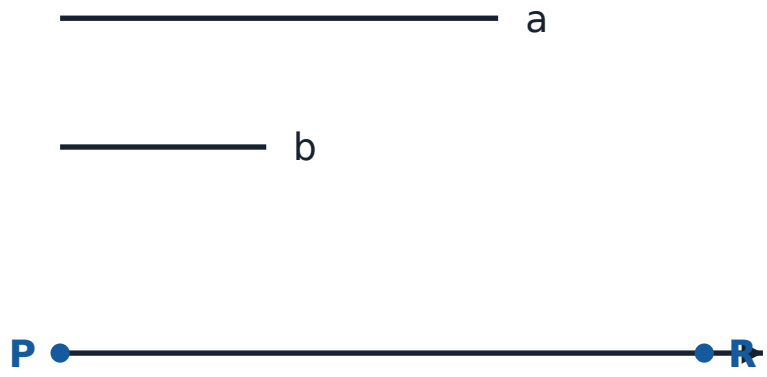
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q1/5

Compass and straightedge only; leave all arcs.

On the ray from P, construct a segment of length a − b (a is the longer segment).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q2/5

True or false?

If the arcs in a triangle construction just touch at one point on the base, the "triangle" is flat.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q3/5

Which lengths CANNOT form a triangle?

A. 3, 4, 5

B. 1, 2, 3

C. 4, 4, 7

D. 2, 3, 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q4/5

Two sides of a constructed triangle are 5 and 8. For what third lengths will the arcs meet in two points?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q5/5

SET 29 • HSG-CO.D.12 • CONGRUENCE: CONSTRUCTIONS

Construct a copy of an angle

MAIN	Geometry main question	**HSG-CO.D.12**
BACKWARD 1	Measure and draw angles with a protractor	**4.MD.C.6**
BACKWARD 2	Copy a segment with a compass	**HSG-CO.D.12**
BACKWARD 3	Arcs and circles in constructions	**7.G.B.4**
FORWARD 1	Justify why the angle-copy construction works	**HSG-CO.B.8**
FORWARD 2	Use angle copying to construct a parallel line	**HSG-CO.D.12**

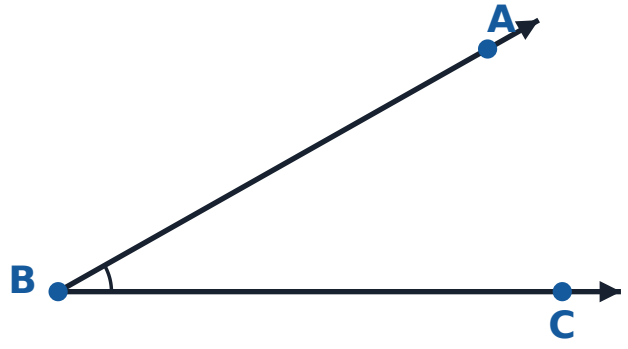
SET 29 • HSG-CO.D.12

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct an angle with vertex P and one side ray PR that is congruent to ∠ABC.





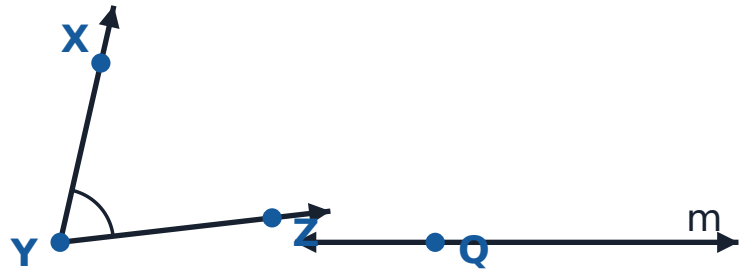
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct an angle at point Q on line m congruent to ∠XYZ, opening upward.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q2/5

Write the steps to copy ∠ABC so that its vertex is at P and one side is ray PR.

Steps: _____

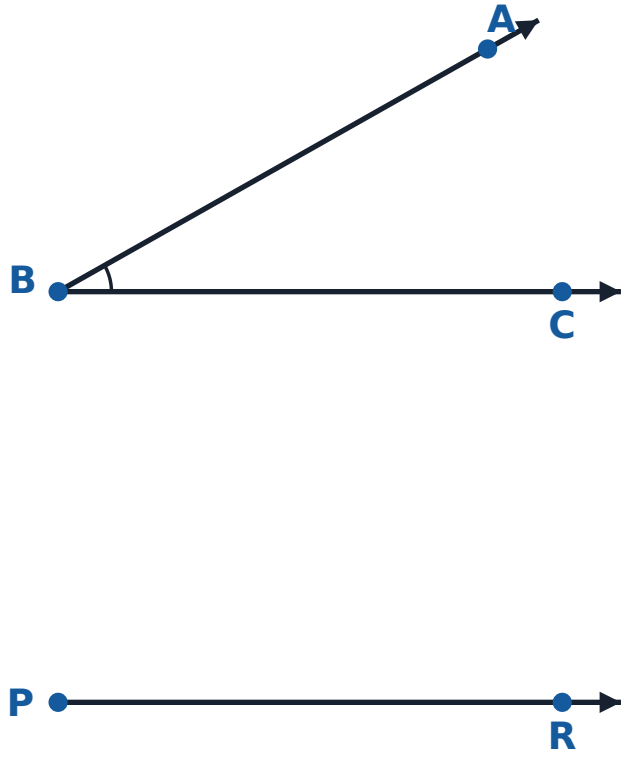
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

Construct an angle at P on ray PR whose measure is twice m∠ABC.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q4/5

True or false?

When copying an angle, the first arcs at B and at P must have the same radius.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q5/5

SET 29 • HSG-CO.D.12

BACKWARD 1

Measure and draw angles with a protractor • Grade 4 • Measurement & Data • 4.MD.C.6

A protractor shows a ray at 0° and the other ray at 65°. What is the angle?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q1/5

True or false?

An angle's measure does not change when its sides are drawn longer.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q2/5

Which angle is obtuse?

A. 90°

B. 45°

C. 20°

D. 130°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q3/5

Two angles of 35° and 35° are placed side by side. What is the combined angle?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q4/5

True or false?

Two angles with sides of different lengths cannot be congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q5/5

SET 29 • HSG-CO.D.12

BACKWARD 2

Copy a segment with a compass • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

In copying an angle, which length is transferred with the compass in the fourth step?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q1/5

True or false?

To copy a length, set the compass to the length and do not change it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q2/5

An arc centered at F with radius DE crosses another arc at G. What is FG?

A. Half of DE

B. DE

C. BD

D. PF

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q3/5

Arcs centered at B and at P are both drawn with radius 3 cm. How far is F (on the arc at P) from P?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q4/5

True or false?

A segment copy can be checked only by measuring with a ruler.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q5/5

SET 29 • HSG-CO.D.12

BACKWARD 3

Arcs and circles in constructions • Grade 7 • Geometry • 7.G.B.4

True or false?

Two arcs centered at different points can cross at two points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q1/5

Points D and E are both on an arc centered at B with radius 4. Find BD and BE.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q2/5

Which describes an arc drawn with a compass?

A. A ray

B. A straight segment

C. Part of a circle

D. A polygon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q3/5

True or false?

An arc centered at P passes through P.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q4/5

What does the point of the compass mark in a construction?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q5/5

SET 29 • HSG-CO.D.12

FORWARD 1

Justify why the angle-copy construction works • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In copying ∠ABC to ∠FPG, BD = BE = PF = PG (same arc radius) and DE = FG (same compass setting). Prove that ∠DBE ≅ ∠FPG.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q1/5

Which congruence criterion explains the angle-copy construction?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q2/5

True or false?

The angle copy would fail if the second arc used a different radius than the first.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q3/5

Which three pairs of segments are congruent in the angle-copy construction?

A. BA and PR, BC and PG, AC and RG

B. Only DE and FG

C. BD and DE, PF and FG, BE and PG

D. BD and PF, BE and PG, DE and FG

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q4/5

Why does the construction not need to measure the angle with a protractor?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q5/5

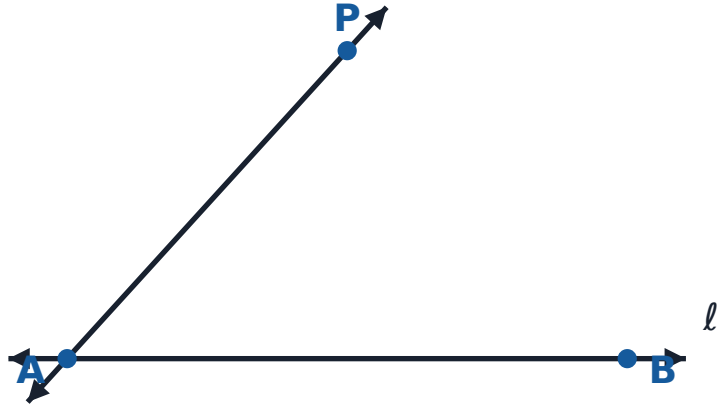
SET 29 • HSG-CO.D.12

FORWARD 2

Use angle copying to construct a parallel line • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct the line through P parallel to line ℓ by copying an angle.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q1/5

Why does copying a corresponding angle at P give a line parallel to ℓ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q2/5

True or false?

Copying the angle into the alternate interior position also produces a parallel line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q3/5

In the parallel-line construction, which angle is copied?

A. The angle between ℓ and itself

B. A right angle

C. The angle the transversal makes with ℓ

D. Any 45° angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q4/5

Describe a different way to construct a parallel through P using two perpendicular constructions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q5/5

SET 30 • HSG-CO.D.12 • CONGRUENCE: CONSTRUCTIONS

Construct the perpendicular bisector of a segment (bisect a segment)

MAIN	Geometry main question	**HSG-CO.D.12**
BACKWARD 1	Points equidistant from A and B lie on the perpendicular bisector	**HSG-CO.C.9**
BACKWARD 2	Know midpoint and perpendicular bisector definitions	**HSG-CO.A.1**
BACKWARD 3	Arcs with one radius from two centers	**7.G.B.4**
FORWARD 1	Justify the perpendicular bisector construction	**HSG-CO.C.9**
FORWARD 2	Construct the circumcenter and circumscribed circle of a triangle	**HSG-C.A.3**

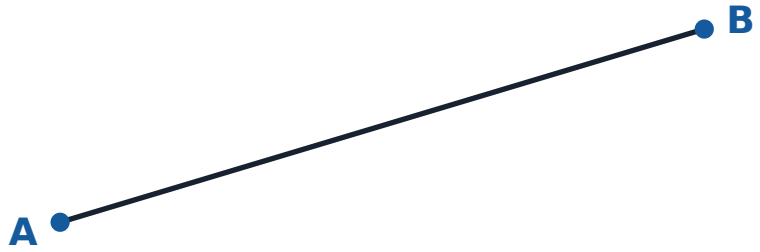
SET 30 • HSG-CO.D.12

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct the perpendicular bisector of segment AB. Label the midpoint M.



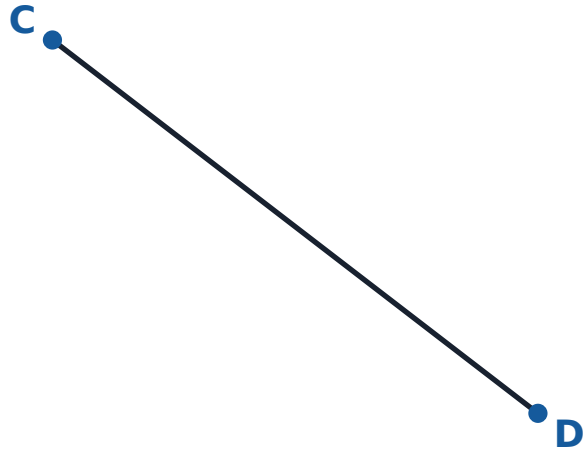
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct the midpoint M of segment CD.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct the point P on line ℓ that is the same distance from A as from B.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q3/5

Write the steps for constructing the perpendicular bisector of segment AB.

Steps: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q4/5

Describe how to find the perpendicular bisector of segment AB by folding paper.

Method: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q5/5

SET 30 • HSG-CO.D.12

BACKWARD 1

Points equidistant from A and B lie on the perpendicular bisector • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

A point equidistant from A and B lies on the perpendicular bisector of AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q1/5

Points P and Q are each equidistant from A and B. What line do P and Q determine?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q2/5

In the construction, the arc crossings are equidistant from A and B because

A. the crossings are on AB

B. the compass is opened to AB

C. the arcs have the same radius

D. the arcs are drawn above AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q3/5

True or false?

Every point on the perpendicular bisector of AB is equidistant from A and B.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q4/5

True or false?

Two points determine more than one line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q5/5

SET 30 • HSG-CO.D.12

BACKWARD 2

Know midpoint and perpendicular bisector definitions • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

M is the midpoint of AB, and AM = 3.5. Find AB.

AB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q1/5

True or false?

The perpendicular bisector of a segment passes through its midpoint.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q2/5

A line crosses AB at its midpoint at a 60° angle. It is

A. the perpendicular bisector

B. not a bisector

C. parallel to AB

D. a bisector but not the perpendicular bisector

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q3/5

Line k is perpendicular to AB and passes through its midpoint. What is k called?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q4/5

True or false?

A segment has many perpendicular bisectors in one plane.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q5/5

SET 30 • HSG-CO.D.12

BACKWARD 3

Arcs with one radius from two centers • Grade 7 • Geometry • 7.G.B.4

Two arcs of radius 5 are centered at A and B, and AB = 6. Do they cross?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q1/5

True or false?

If the compass opening is less than half of AB, the arcs from A and B do not meet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q2/5

Why must the opening be more than half of AB?

A. It does not matter

B. So the midpoint is on the arc

C. So the line is longer

D. So the arcs from A and B cross

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q3/5

A point X is on both arcs (radius r) centered at A and B. Find XA and XB.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q4/5

True or false?

The two arc crossings are on the same side of AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q5/5

SET 30 • HSG-CO.D.12

FORWARD 1

Justify the perpendicular bisector construction • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

In the construction, arcs of radius r centered at A and B meet at P and Q. Prove that line PQ is the perpendicular bisector of AB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q1/5

Explain why the construction would still work with a larger compass opening.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q2/5

True or false?

The construction works only if the arcs are drawn with radius exactly AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q3/5

Which theorem justifies that the arc crossings lie on the perpendicular bisector?

A. Converse of the perpendicular bisector theorem

B. Midsegment theorem

C. Triangle angle sum

D. Vertical angles theorem

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q4/5

Why does folding A onto B produce the perpendicular bisector?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q5/5

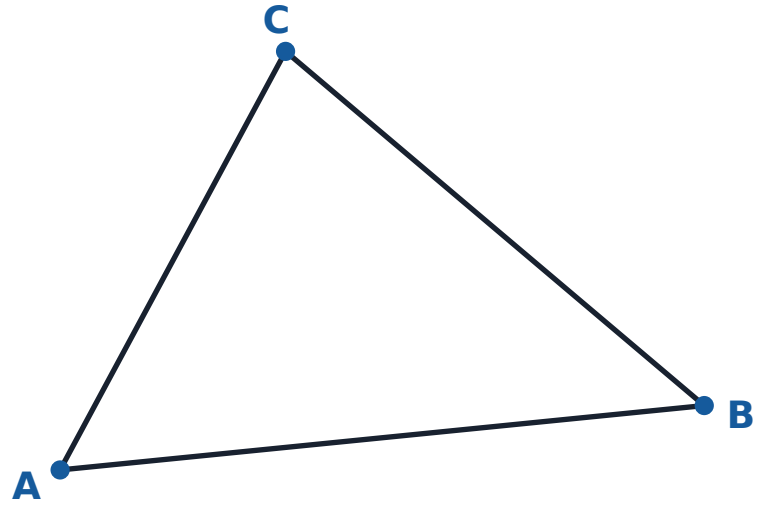
SET 30 • HSG-CO.D.12

FORWARD 2

Construct the circumcenter and circumscribed circle of a triangle • High School — Geometry • Geometry: Circles • HSG-C.A.3

Compass and straightedge only; leave all arcs.

Construct the point O equidistant from A, B, and C, and draw the circle through A, B, and C.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q1/5

Why is it enough to construct only two perpendicular bisectors to find the circumcenter?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q2/5

True or false?

The circumcenter of an obtuse triangle lies outside the triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q3/5

Where is the circumcenter of a right triangle?

A. Inside, near the centroid

B. At the midpoint of the hypotenuse

C. Outside the triangle

D. At the right-angle vertex

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q4/5

A triangle has A(0, 0), B(8, 0), C(0, 6). Where is its circumcenter, and what is the radius?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q5/5

SET 31 • HSG-CO.D.12 • CONGRUENCE: CONSTRUCTIONS

Construct the bisector of an angle

MAIN	Geometry main question	**HSG-CO.D.12**
BACKWARD 1	Know the definition of an angle bisector	**HSG-CO.A.1**
BACKWARD 2	Prove triangles congruent with SSS	**HSG-CO.B.8**
BACKWARD 3	Measure angles with a protractor	**4.MD.C.6**
FORWARD 1	Justify the angle bisector construction	**HSG-CO.B.8**
FORWARD 2	Construct special angles and the incenter by bisecting	**HSG-CO.D.12**

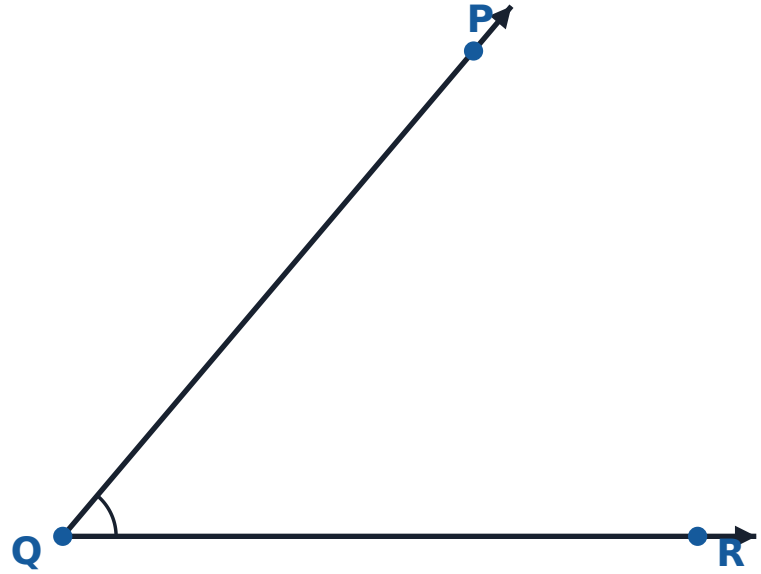
SET 31 • HSG-CO.D.12

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct the bisector of $\angle PQR$.



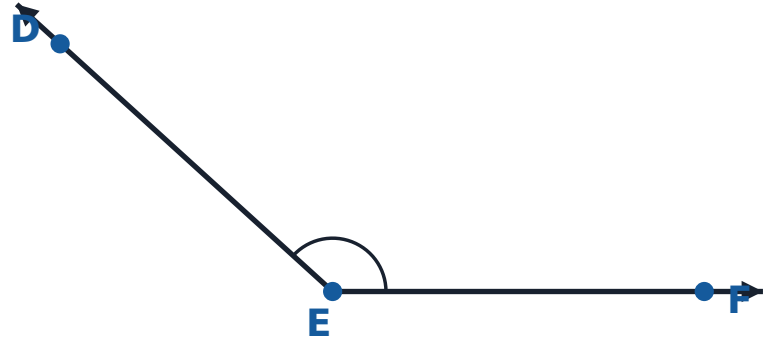
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct the bisector of obtuse angle ∠DEF.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct a 45° angle with vertex A on line ℓ.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q3/5

Write the steps for bisecting ∠ABC with compass and straightedge.

Steps: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q4/5

Describe how to bisect an angle drawn on paper by folding.

Method: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q5/5

SET 31 • HSG-CO.D.12

BACKWARD 1

Know the definition of an angle bisector • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Ray BD bisects ∠ABC, and m∠ABD = 27°. Find m∠ABC.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q1/5

True or false?

An angle bisector divides the angle into two congruent angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q2/5

Ray BD bisects ∠ABC with m∠ABC = 3x + 6 and m∠ABD = 24. Find x.

A. 6

B. 14

C. 10

D. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q3/5

A 90° angle is bisected. What is each part?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q4/5

True or false?

An angle bisector is a segment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q5/5

SET 31 • HSG-CO.D.12

BACKWARD 2

Prove triangles congruent with SSS • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

Two triangles share a side and have two more pairs of congruent sides. Which criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q1/5

True or false?

SSS guarantees that corresponding angles are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q2/5

In the bisector construction, which triangles are congruent?

A. △ABC and △XYZ

B. △BXZ and △BYZ

C. △BXY and △XYZ

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q3/5

After $\triangle BXZ \cong \triangle BYZ$, which angle is congruent to $\angle XBZ$?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q4/5

True or false?

SSS requires knowing one angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q5/5

SET 31 • HSG-CO.D.12

BACKWARD 3

Measure angles with a protractor • Grade 4 • Measurement & Data • 4.MD.C.6

A protractor reads 0° on one side and 112° on the other. Half of this angle is

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q1/5

True or false?

Measuring with a protractor is a construction.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q2/5

An angle of 150° is bisected. Each part is

A. 60°

B. 150°

C. 30°

D. 75°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q3/5

After bisecting, a student measures 38° and 38°. What was the original angle?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q4/5

True or false?

A protractor can check a construction.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q5/5

SET 31 • HSG-CO.D.12

FORWARD 1

Justify the angle bisector construction • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In bisecting ∠ABC, an arc centered at B crosses the sides at X and Y, and equal arcs centered at X and Y meet at Z. Prove that ray BZ bisects ∠ABC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q1/5

Why must the arcs from X and Y have the same radius?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q2/5

True or false?

The bisector construction works for obtuse angles too.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q3/5

Which criterion justifies the angle bisector construction?

A. AAS

B. ASA

C. SSS

D. SAS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q4/5

Explain why folding one side onto the other gives the bisector.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q5/5

SET 31 • HSG-CO.D.12

FORWARD 2

Construct special angles and the incenter by bisecting • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct a 30° angle. (Hint: start from an equilateral triangle.)



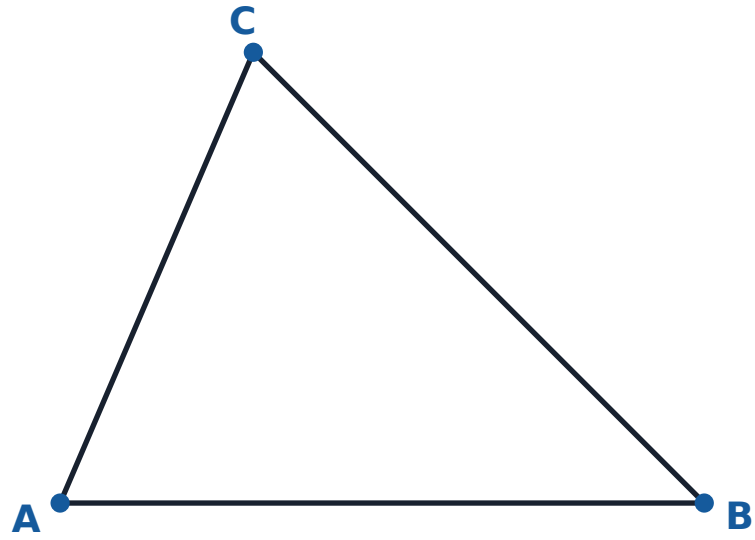
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q1/5

Compass and straightedge only; leave all arcs.

Construct the point inside triangle ABC that is equidistant from its three sides.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q2/5

True or false?

Bisecting a 45° angle gives a 22.5° angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q3/5

Starting from 60° and 90° angles, which angle CANNOT be constructed by bisecting angles and adding angles together?

A. 15°

B. 20°

C. 75°

D. 22.5°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q4/5

Explain how to construct a 135° angle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q5/5

SET 32 • HSG-CO.D.12 • CONGRUENCE: CONSTRUCTIONS

Construct a perpendicular to a line through a point on or off the line

MAIN	Geometry main question	**HSG-CO.D.12**
BACKWARD 1	Know the definition of perpendicular lines	**HSG-CO.A.1**
BACKWARD 2	Construct the perpendicular bisector of a segment	**HSG-CO.D.12**
BACKWARD 3	Recognize perpendicular lines in figures	**4.G.A.1**
FORWARD 1	Construct an altitude of a triangle and use it for area	**HSG-CO.D.12**
FORWARD 2	Construct a square on a given segment	**HSG-CO.D.12**

SET 32 • HSG-CO.D.12

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct the line through P perpendicular to line ℓ. (P is on ℓ.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct the line through P perpendicular to line ℓ. (P is not on ℓ.)



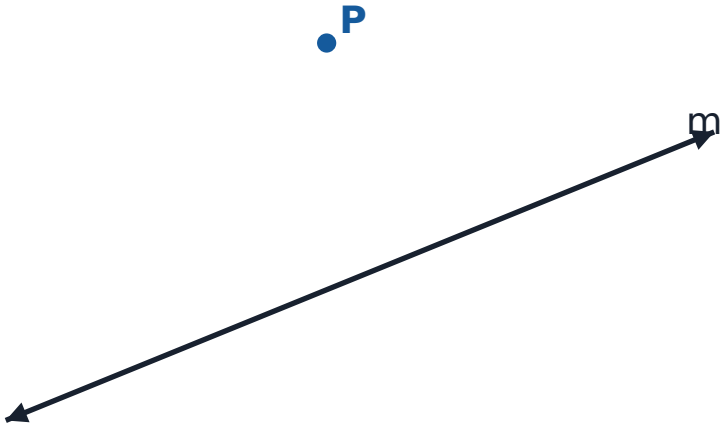
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct the perpendicular from point P to the slanted line m.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q3/5

Write the steps for constructing the perpendicular from a point P not on line ℓ.

Steps: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q4/5

Explain why the perpendicular construction from a point P not on ℓ is really a perpendicular bisector construction.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q5/5

SET 32 • HSG-CO.D.12

BACKWARD 1

Know the definition of perpendicular lines • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

True or false?

Perpendicular lines meet at a right angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q1/5

Line k $\perp$ line ℓ. What is the measure of each angle they form?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q2/5

Which pair of lines is perpendicular?

A. $y = 3$ and $y = -3$

B. $y = 2x$ and $y = -\frac{1}{2}x$

C. $y = 2x$ and $y = 2x + 1$

D. $y = x$ and $y = 2x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q3/5

True or false?

Through a point, there are many lines perpendicular to a given line in a plane.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q4/5

The shortest path from a point to a line follows which kind of line?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q5/5

SET 32 • HSG-CO.D.12

BACKWARD 2

Construct the perpendicular bisector of a segment • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

After marking A and B on ℓ, which construction finishes the perpendicular?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q1/5

True or false?

The perpendicular bisector construction needs arcs of equal radius from both endpoints.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q2/5

The arc crossings of the perpendicular bisector construction are

A. on segment AB

B. on line ℓ

C. at A and B

D. equidistant from A and B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q3/5

P is on ℓ, and A and B are marked on ℓ with PA = PB. Where does the perpendicular bisector of AB cross ℓ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q4/5

True or false?

The perpendicular bisector of AB passes through A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q5/5

SET 32 • HSG-CO.D.12

BACKWARD 3

Recognize perpendicular lines in figures • Grade 4 • Geometry • 4.G.A.1

True or false?

A square's adjacent sides are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q1/5

Which letter is made of perpendicular segments?

A. V

B. O

C. S

D. T

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q2/5

How many right angles do two perpendicular lines make?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q3/5

True or false?

Two lines that look almost perpendicular must be perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q4/5

Name a pair of perpendicular edges of a rectangular door.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q5/5

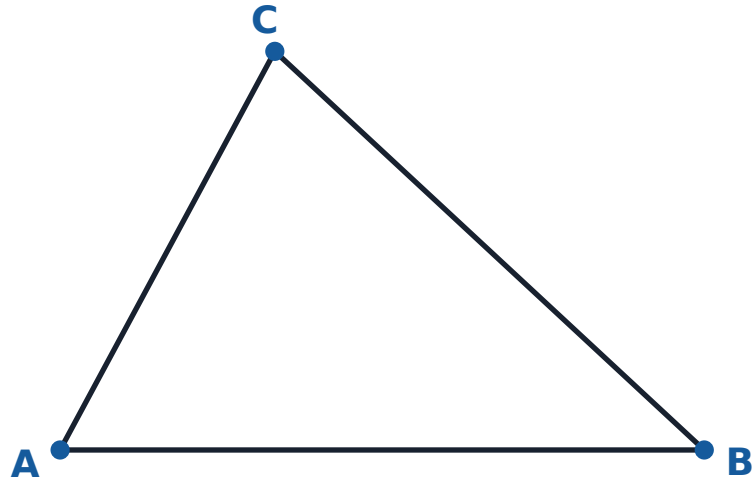
SET 32 • HSG-CO.D.12

FORWARD 1

Construct an altitude of a triangle and use it for area • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct the altitude from C to side AB of triangle ABC.



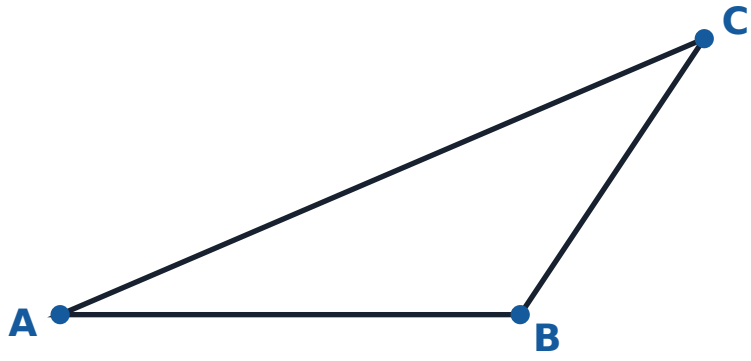
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q1/5

Compass and straightedge only; leave all arcs.

Construct the altitude from C in obtuse triangle ABC (the foot is outside AB). Extend AB as needed.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q2/5

A triangle has base 10 and its constructed altitude measures 6. Find its area.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q3/5

True or false?

Every triangle has three altitudes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q4/5

In a right triangle, the altitude from the right-angle vertex to the hypotenuse is

A. the hypotenuse

B. outside the triangle

C. a leg

D. inside the triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q5/5

SET 32 • HSG-CO.D.12

FORWARD 2

Construct a square on a given segment • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct a square with side AB.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q1/5

Explain why the figure constructed this way is a square.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q2/5

True or false?

A quadrilateral with four congruent sides is always a square.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q3/5

To construct a rectangle 2AB long and AB wide, which additional step is needed?

A. Bisect AB

B. Construct a 60° angle

C. Draw a circle through A and B

D. Copy AB twice along the base

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q4/5

A square is constructed on a segment of length 5. What is the length of its diagonal?

Diagonal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q5/5

SET 33 • HSG-CO.D.12 • CONGRUENCE: CONSTRUCTIONS

Construct a line parallel to a given line through a point not on it

MAIN	Geometry main question	**HSG-CO.D.12**
BACKWARD 1	Copy an angle with compass and straightedge	**HSG-CO.D.12**
BACKWARD 2	Converse: congruent corresponding angles give parallel lines	**HSG-CO.C.9**
BACKWARD 3	Recognize parallel lines	**4.G.A.1**
FORWARD 1	Justify the parallel-line constructions	**HSG-CO.C.9**
FORWARD 2	Divide a segment into equal parts with parallel lines	**HSG-SRT.B.4**

SET 33 • HSG-CO.D.12

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct the line through P parallel to line ℓ.



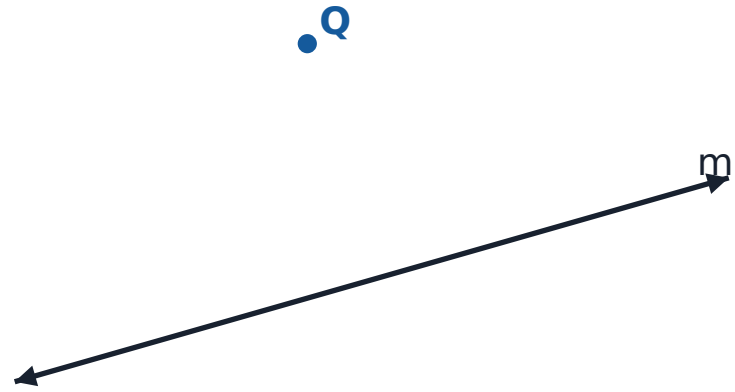
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct the line through Q parallel to the slanted line m.



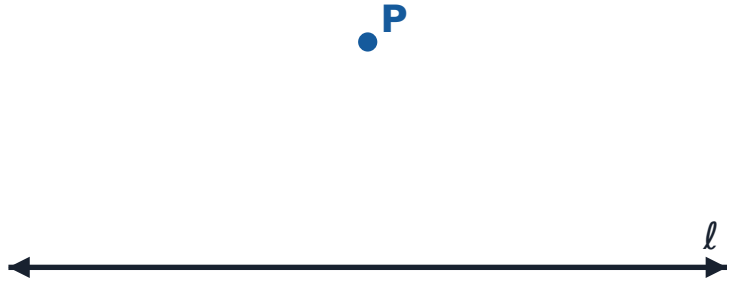
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct the line through P parallel to ℓ using two perpendiculars.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q3/5

Write the steps for constructing the line through P parallel to line ℓ by copying an angle.

Steps: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q4/5

Why is the line constructed by copying the corresponding angle parallel to ℓ?

A. It is perpendicular to the transversal

B. It passes through P

C. It looks parallel

D. Congruent corresponding angles make the lines parallel

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q5/5

SET 33 • HSG-CO.D.12

BACKWARD 1

Copy an angle with compass and straightedge • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

In copying an angle, which three lengths are transferred?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q1/5

True or false?

An angle copy uses arcs of the same radius at both vertices.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q2/5

Which step completes an angle copy?

A. Draw a circle

B. Measure with a protractor

C. Draw the ray from the new vertex through the last arc crossing

D. Bisect the angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q3/5

Why does the angle copy work?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q4/5

True or false?

An angle copy needs a protractor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q5/5

SET 33 • HSG-CO.D.12

BACKWARD 2

Converse: congruent corresponding angles give parallel lines • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

Congruent corresponding angles mean the lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q1/5

A transversal makes 62° angles with two lines in corresponding positions. Are the lines parallel?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q2/5

Which also proves the lines parallel?

A. Congruent vertical angles

B. Supplementary linear pairs

C. Two acute angles

D. Congruent alternate interior angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q3/5

True or false?

Same-side interior angles of 70° and 70° prove the lines parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q4/5

Corresponding angles are (2x + 10)° and 50°. For which x are the lines parallel?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q5/5

SET 33 • HSG-CO.D.12

BACKWARD 3

Recognize parallel lines • Grade 4 • Geometry • 4.G.A.1

True or false?

The opposite edges of a ruler are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q1/5

Which letter has parallel segments?

A. V

B. A

C. H

D. X

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q2/5

Two railroad rails stay the same distance apart. What are they modeling?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q3/5

True or false?

Lines that cross are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q4/5

How many lines through a point are parallel to a given line (in a plane)?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q5/5

SET 33 • HSG-CO.D.12

FORWARD 1

Justify the parallel-line constructions • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

In the construction, the angle at P is a copy of the angle at A in the corresponding position along transversal AP. Prove that the new line is parallel to ℓ.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q1/5

In the two-perpendicular construction, k ⊥ ℓ and n ⊥ k at P. Prove that n ∥ ℓ.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q2/5

True or false?

Copying the alternate interior angle at P also gives a parallel line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q3/5

Which construction gives a line through P parallel to ℓ?

A. Bisect the angle at P

B. Draw a circle centered at P

C. Copy, at P, the angle that a transversal through P makes with ℓ, in the corresponding position

D. Draw the perpendicular bisector of the transversal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q4/5

By the parallel postulate, how many lines could the construction produce?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q5/5

SET 33 • HSG-CO.D.12

FORWARD 2

Divide a segment into equal parts with parallel lines • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

Compass and straightedge only; leave all arcs.

Divide segment AB into three congruent parts.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q1/5

Why do the parallel lines divide AB into equal parts?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q2/5

True or false?

The same method can divide a segment into 5 equal parts.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q3/5

To divide AB in the ratio 2 : 3, how many equal steps are marked on the auxiliary ray?

A. 2

B. 5

C. 3

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q4/5

A segment of length 12 is divided into 4 parts this way. How long is each part?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q5/5

SET 34 • HSG-CO.D.13 • CONGRUENCE: CONSTRUCTIONS

Construct an equilateral triangle inscribed in a circle

MAIN	Geometry main question	**HSG-CO.D.13**
BACKWARD 1	Central angles of a regular polygon	**HSG-CO.A.3**
BACKWARD 2	Copy a length with a compass	**HSG-CO.D.12**
BACKWARD 3	Circle vocabulary: center, radius, chord, diameter	**7.G.B.4**
FORWARD 1	Justify that the inscribed triangle is equilateral	**HSG-C.A.2**
FORWARD 2	Construct a regular dodecagon by bisecting arcs	**HSG-CO.D.13**

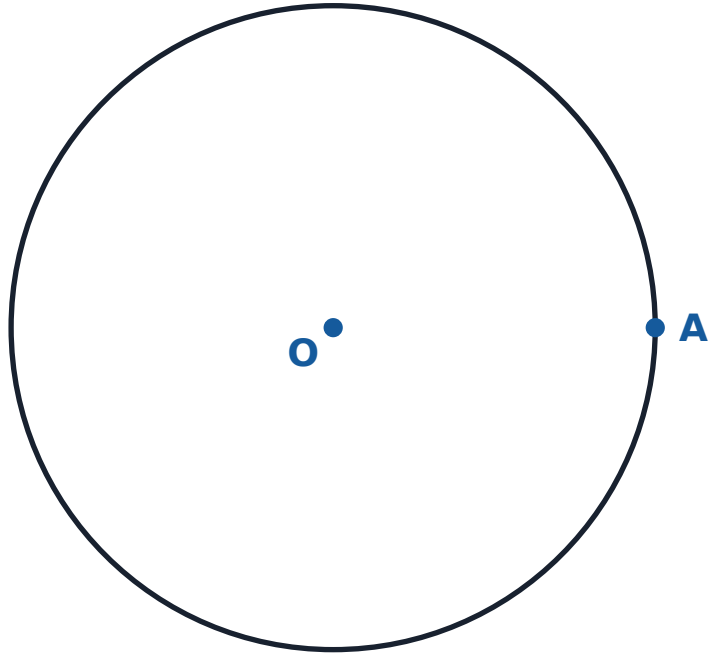
SET 34 • HSG-CO.D.13

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

Compass and straightedge only; leave all arcs.

Construct an equilateral triangle inscribed in the circle, with one vertex at A.



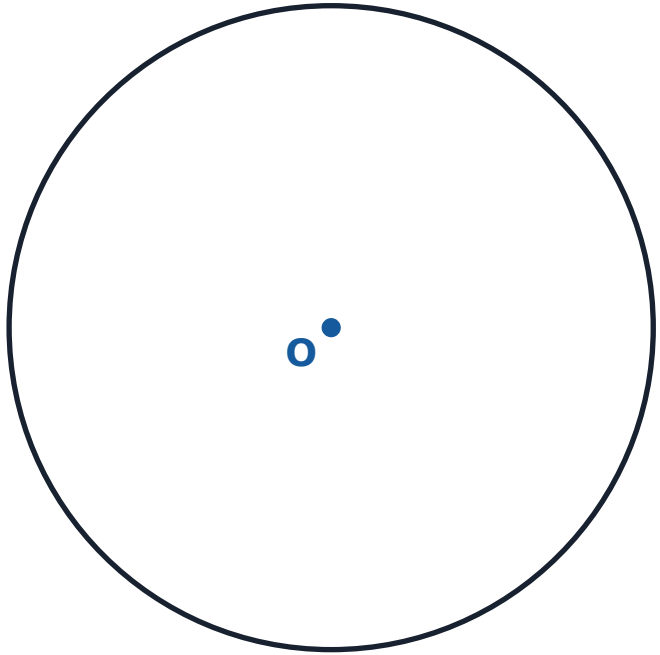
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct an equilateral triangle inscribed in the circle (choose any starting vertex).



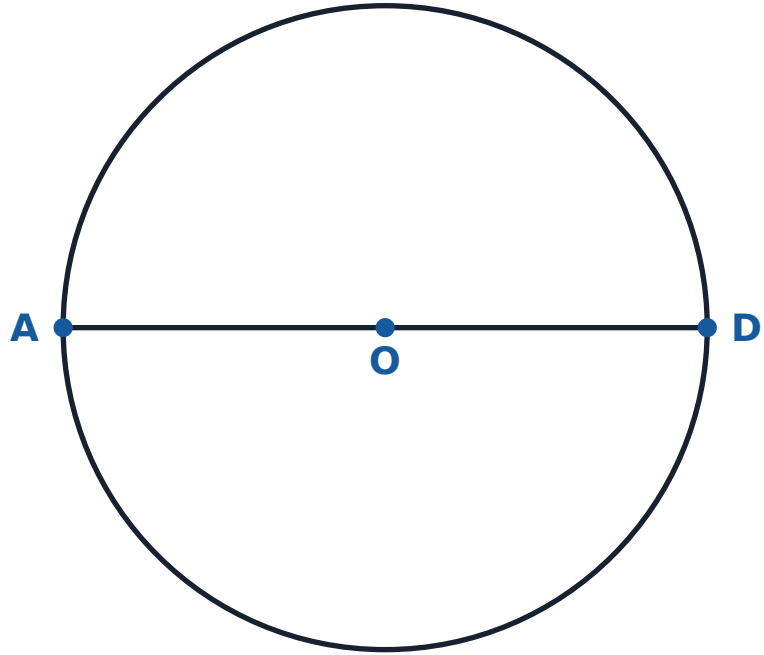
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct an equilateral triangle inscribed in the circle using a diameter: draw diameter AD, then an arc centered at D with radius DO.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q3/5

Write the steps for inscribing an equilateral triangle in a circle with center O.

Steps: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q4/5

In the stepping method, why does the radius fit exactly six times around the circle?

A. Any length fits six times

B. Each step makes an equilateral triangle with the center, a 60° central angle

C. The compass is 6 cm

D. The circle has six parts

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q5/5

SET 34 • HSG-CO.D.13

BACKWARD 1

Central angles of a regular polygon • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

What rotation carries an equilateral triangle onto itself, about its center (smallest positive)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q1/5

Into how many equal central angles does a regular hexagon divide 360°?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q2/5

True or false?

The vertices of an inscribed equilateral triangle are 120° apart around the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q3/5

Joining every other vertex of a regular hexagon gives

A. a square

B. a rectangle

C. a pentagon

D. an equilateral triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q4/5

True or false?

The central angles of an inscribed equilateral triangle are 60°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q5/5

SET 34 • HSG-CO.D.13

BACKWARD 2

Copy a length with a compass • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

The compass is set to the radius of a circle. What length does each step around the circle mark?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q1/5

True or false?

Keeping the compass opening fixed is required when stepping around the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q2/5

Each radius-length chord and the two radii to its ends form

A. an equilateral triangle

B. a square

C. a right triangle

D. an isosceles triangle with a 90° angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q3/5

A circle has radius 4 cm. How long is each chord marked by stepping the radius?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q4/5

True or false?

Stepping a compass set to the diameter marks six points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q5/5

SET 34 • HSG-CO.D.13

BACKWARD 3

Circle vocabulary: center, radius, chord, diameter • Grade 7 • Geometry • 7.G.B.4

What is a chord of a circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q1/5

True or false?

A diameter is a chord through the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q2/5

A circle has radius 6. Its diameter is

A. 3

B. 12

C. 36

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q3/5

What does "inscribed in a circle" mean for a polygon?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q4/5

True or false?

An inscribed triangle can have a vertex inside the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q5/5

SET 34 • HSG-CO.D.13

FORWARD 1

Justify that the inscribed triangle is equilateral • High School — Geometry • Geometry: Circles • HSG-C.A.2

Points A, B, C, D, E, F are marked around a circle with center O by stepping the radius. Prove that triangle ACE is equilateral.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q1/5

An equilateral triangle is inscribed in a circle of radius 6. Find its side length.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q2/5

True or false?

Congruent central angles cut off congruent chords.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q3/5

Each angle of the inscribed triangle ACE intercepts an arc (not containing its vertex) of

A. 120°

B. 240°

C. 180°

D. 60°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q4/5

Find the area of an equilateral triangle inscribed in a circle of radius 2.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q5/5

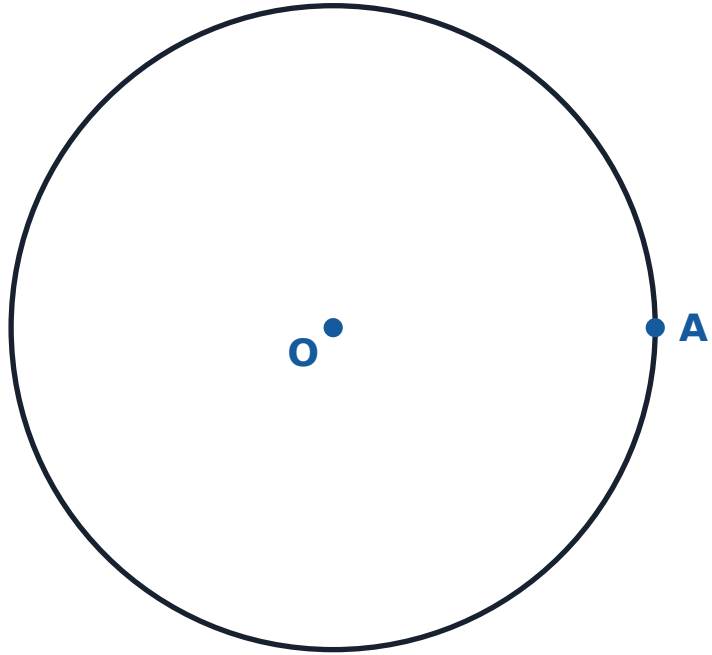
SET 34 • HSG-CO.D.13

FORWARD 2

Construct a regular dodecagon by bisecting arcs • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

Compass and straightedge only; leave all arcs.

Construct a regular 12-sided polygon inscribed in the circle.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q1/5

Each side of the inscribed regular hexagon subtends 60° at the center. What does each side of the dodecagon subtend?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q2/5

True or false?

Bisecting the arcs of an inscribed regular n-gon gives the vertices of a regular 2n-gon.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q3/5

Starting from an inscribed square, bisecting arcs gives a regular

A. pentagon

B. octagon

C. hexagon

D. decagon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q4/5

How many sides does the polygon have after bisecting the arcs of an inscribed equilateral triangle twice?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q5/5

Construct a square inscribed in a circle

MAIN	Geometry main question	**HSG-CO.D.13**
BACKWARD 1	Construct a perpendicular bisector	**HSG-CO.D.12**
BACKWARD 2	Diagonals of a square	**HSG-CO.C.11**
BACKWARD 3	Diameters and the center of a circle	**7.G.B.4**
FORWARD 1	Justify that the inscribed quadrilateral is a square	**HSG-CO.C.11**
FORWARD 2	Construct a regular octagon inscribed in a circle	**HSG-CO.D.13**

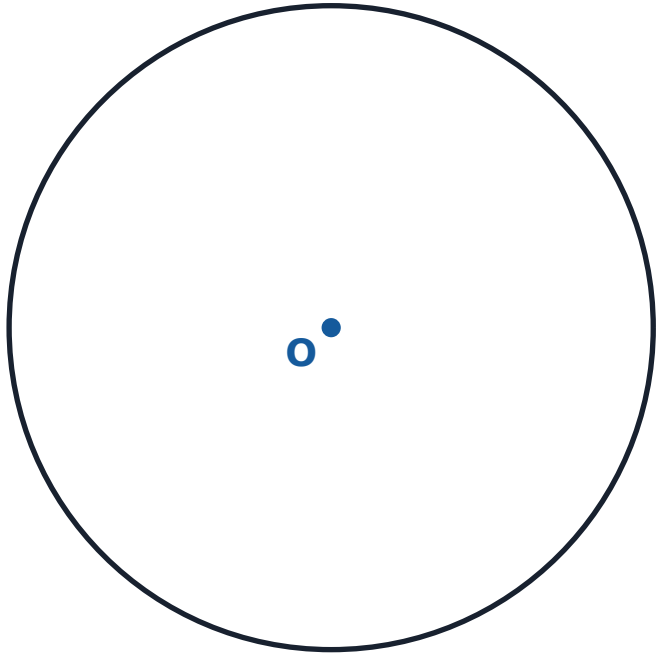
SET 35 • HSG-CO.D.13

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

Compass and straightedge only; leave all arcs.

Construct a square inscribed in the circle.



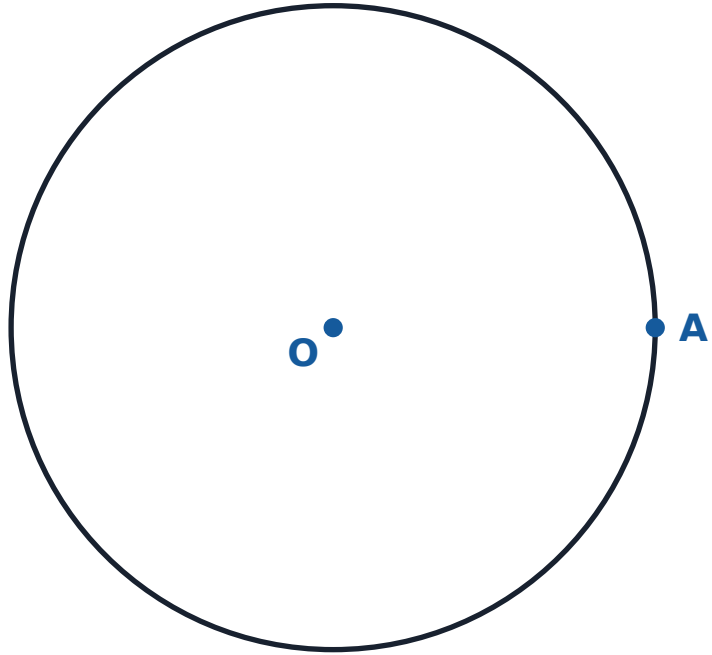
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct a square inscribed in the circle with one vertex at A.



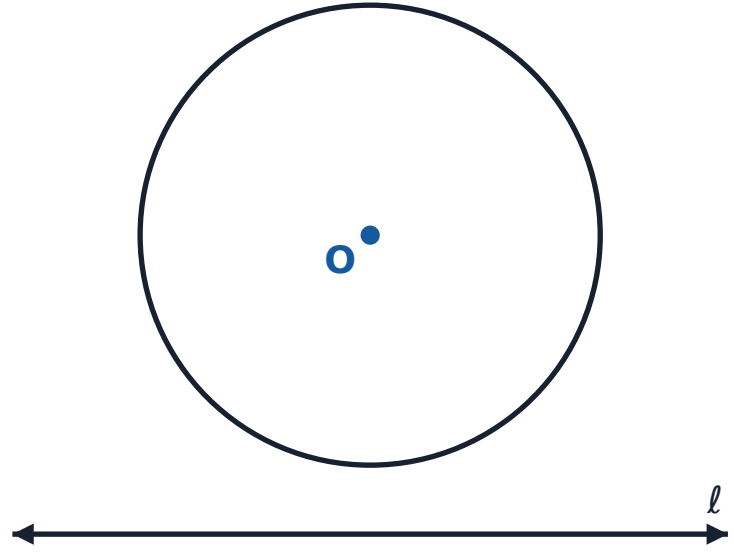
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct a square inscribed in the circle with one side parallel to the given line ℓ.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q3/5

Write the steps for inscribing a square in a circle with center O.

Steps: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q4/5

Which property of a square's diagonals does the construction use?

A. They are congruent, perpendicular, and bisect each other

B. They are parallel

C. They are equal to the sides

D. They meet outside the square

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q5/5

SET 35 • HSG-CO.D.13

BACKWARD 1

Construct a perpendicular bisector • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Which construction turns one diameter into two perpendicular diameters?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q1/5

True or false?

The perpendicular bisector of a diameter passes through the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q2/5

Two perpendicular diameters cut the circle into

A. six 60° arcs

B. four 45° arcs

C. four 90° arcs

D. two 90° arcs

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q3/5

A diameter is 10 cm. Where does its perpendicular bisector cross it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q4/5

True or false?

A perpendicular bisector can be drawn by measuring with a ruler only.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q5/5

SET 35 • HSG-CO.D.13

BACKWARD 2

Diagonals of a square • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

True or false?

The diagonals of a square are congruent and perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q1/5

A quadrilateral's diagonals are congruent, perpendicular, and bisect each other. It is a

A. square

B. rhombus that is not a square

C. trapezoid

D. kite

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q2/5

A square has diagonal 8. Find its side.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q3/5

True or false?

A rectangle always has perpendicular diagonals.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q4/5

The diagonals of a square meet at O. Name the angle between them.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q5/5

SET 35 • HSG-CO.D.13

BACKWARD 3

Diameters and the center of a circle • Grade 7 • Geometry • 7.G.B.4

A circle has radius 3. How long is any diameter?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q1/5

True or false?

Every diameter passes through the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q2/5

Where do two different diameters of a circle meet?

A. At the midpoint of a radius

B. On the circle

C. Nowhere

D. At the center

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q3/5

What is the longest chord of a circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q4/5

True or false?

All diameters of a circle have different lengths.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q5/5

SET 35 • HSG-CO.D.13

FORWARD 1

Justify that the inscribed quadrilateral is a square • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

Diameters AC and BD of a circle with center O are perpendicular. Prove that ABCD is a square.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q1/5

A square is inscribed in a circle of radius 5. Find its side length and area.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q2/5

True or false?

The side of a square inscribed in a circle of radius r is $r\sqrt{2}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q3/5

The area of a square inscribed in a circle of radius r is

A. πr^2

B. $4r^2$

C. r^2

D. $2r^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q4/5

What fraction of the circle's area does the inscribed square cover? Give the exact value.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q5/5

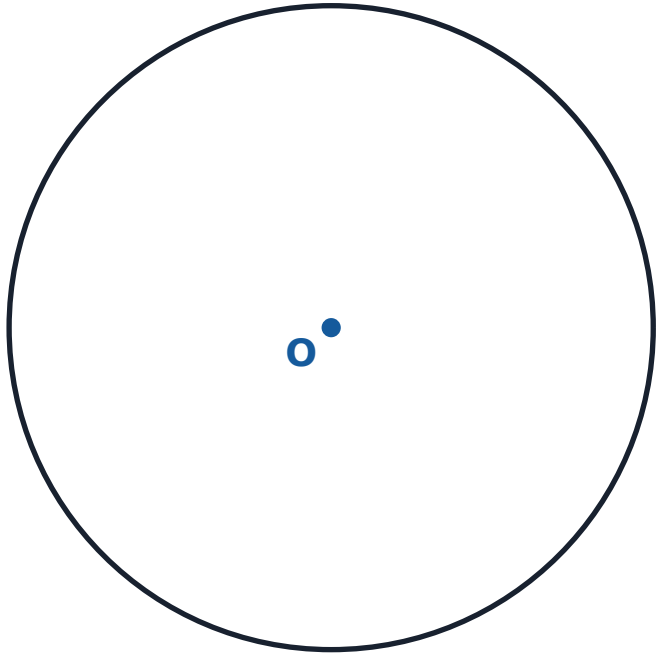
SET 35 • HSG-CO.D.13

FORWARD 2

Construct a regular octagon inscribed in a circle • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

Compass and straightedge only; leave all arcs.

Construct a regular octagon inscribed in the circle.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q1/5

What is the central angle for each side of a regular octagon?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q2/5

True or false?

A regular octagon has interior angles of 135°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q3/5

Bisecting the arcs of an inscribed regular octagon gives a regular

A. hexagon

B. 12-gon

C. square

D. 16-gon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q4/5

Explain why the eight points are equally spaced.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q5/5

SET 36 • HSG-CO.D.13 • CONGRUENCE: CONSTRUCTIONS

Construct a regular hexagon inscribed in a circle

MAIN	Geometry main question	**HSG-CO.D.13**
BACKWARD 1	Construct an inscribed equilateral triangle	**HSG-CO.D.13**
BACKWARD 2	An equilateral triangle has equal sides and 60° angles	**4.G.A.2**
BACKWARD 3	Circle facts: radius and chord	**7.G.B.4**
FORWARD 1	Justify that the inscribed hexagon is regular	**HSG-C.A.2**
FORWARD 2	Use inscribed hexagons to estimate π	**HSG-GMD.A.1**

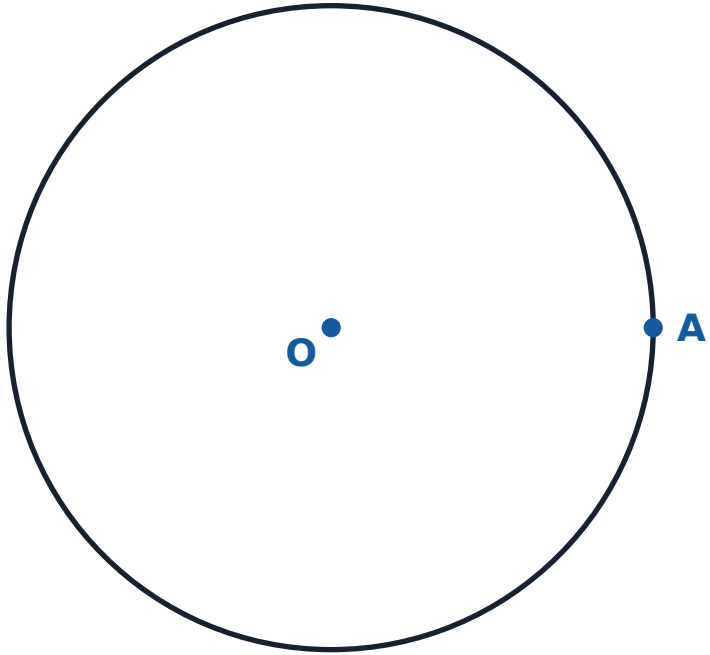
SET 36 • HSG-CO.D.13

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

Compass and straightedge only; leave all arcs.

Construct a regular hexagon inscribed in the circle, with one vertex at A.



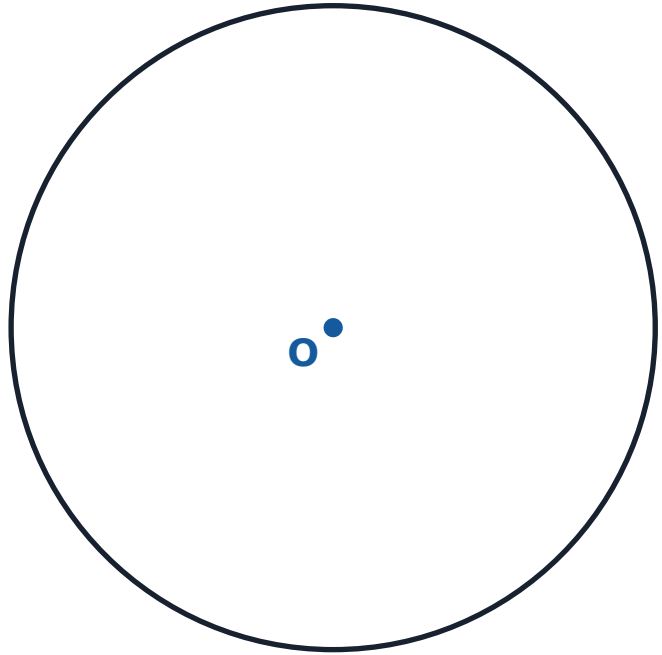
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct a regular hexagon inscribed in the circle (choose any starting point).



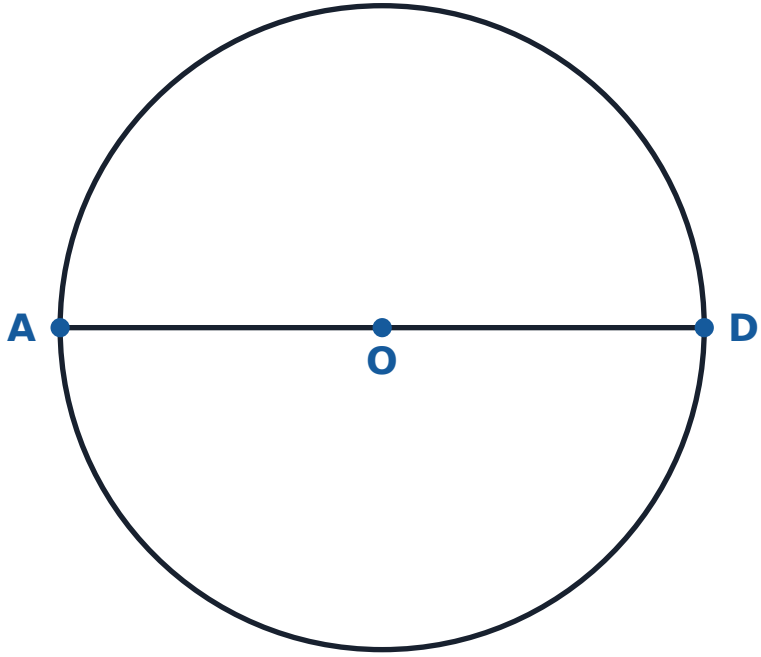
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct a regular hexagon inscribed in the circle using diameter AD and arcs centered at A and D with radius equal to the circle's radius.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q3/5

Write the steps for inscribing a regular hexagon in a circle.

Steps: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q4/5

True or false?

In the hexagon construction, the compass is set to the diameter of the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q5/5

SET 36 • HSG-CO.D.13

BACKWARD 1

Construct an inscribed equilateral triangle • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

How are the inscribed equilateral triangle and the inscribed regular hexagon related?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q1/5

True or false?

Both constructions start by stepping the radius around the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q2/5

How many points does stepping the radius mark?

A. 12

B. 4

C. 6

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q3/5

An equilateral triangle is inscribed. How many more points must be added to make a regular hexagon?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q4/5

True or false?

The inscribed equilateral triangle and hexagon have the same side length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q5/5

SET 36 • HSG-CO.D.13

BACKWARD 2

An equilateral triangle has equal sides and 60° angles • Grade 4 • Geometry • 4.G.A.2

True or false?

Each angle of an equilateral triangle measures 60°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q1/5

A triangle has three sides equal to the radius of a circle. What kind of triangle is it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q2/5

Six equilateral triangles meeting at a point fill an angle of

A. 360°

B. 300°

C. 60°

D. 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q3/5

A regular hexagon has side 5. How long are the segments from its center to its vertices?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q4/5

True or false?

A regular hexagon splits into four equilateral triangles from its center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q5/5

SET 36 • HSG-CO.D.13

BACKWARD 3

Circle facts: radius and chord • Grade 7 • Geometry • 7.G.B.4

A chord equals the radius. What central angle does it subtend?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q1/5

True or false?

A chord equal to the radius forms an equilateral triangle with the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q2/5

A circle has radius 7. A regular inscribed hexagon has perimeter

A. 42

B. 21

C. 14

D. 49

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q3/5

A circle has circumference 31.4 (π ≈ 3.14). Find its radius.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q4/5

True or false?

Every chord of a circle is as long as the radius.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q5/5

SET 36 • HSG-CO.D.13

FORWARD 1

Justify that the inscribed hexagon is regular • High School — Geometry • Geometry: Circles • HSG-C.A.2

Six points are marked around a circle with center O by stepping the radius. Prove that the hexagon they form is regular.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q1/5

Why does the sixth step land exactly back on the starting point?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q2/5

True or false?

Each interior angle of a regular hexagon is 120°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q3/5

A regular hexagon inscribed in a circle of radius 4 has perimeter

A. 24

B. 12

C. 48

D. 8π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q4/5

Find the area of a regular hexagon inscribed in a circle of radius 2.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q5/5

SET 36 • HSG-CO.D.13

FORWARD 2

Use inscribed hexagons to estimate π • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

A regular hexagon is inscribed in a circle of radius r. Compare its perimeter with the circumference. What estimate of π does this give?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q1/5

A regular 12-gon inscribed in a circle of radius 1 has perimeter about 6.21. What estimate of π does it give?

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q2/5

True or false?

Doubling the number of sides of an inscribed regular polygon brings its perimeter closer to the circumference.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q3/5

Which polygon's perimeter is closest to the circumference of its circle?

A. Inscribed regular hexagon

B. Inscribed square

C. Inscribed triangle

D. Inscribed regular 96-gon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q4/5

Explain why an inscribed polygon's perimeter is always less than the circumference.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q5/5

Verify that a dilation maps a line to a parallel line (or to itself through the center)

MAIN	Geometry main question	**HSG-SRT.A.1.a**
BACKWARD 1	Dilate points with coordinates	**8.G.A.3**
BACKWARD 2	Compare slopes to test whether lines are parallel	**8.EE.B.6**
BACKWARD 3	Write the equation of a line through two points	**8.F.B.4**
FORWARD 1	Dilate lines about a center other than the origin	**HSG-SRT.A.1.a**
FORWARD 2	Prove algebraically that a dilation maps a line to a parallel line	**HSG-SRT.A.1.a**

SET 37 • HSG-SRT.A.1.a

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a

Line ℓ is $y = 2x + 1$. Dilate it about the origin with scale factor 3: find the images of (0, 1) and (1, 3), and write an equation of the image line. Is it parallel to ℓ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q1/5

Line m is $y = -x$, which passes through the origin. Dilate it about the origin with scale factor 2: find the images of $(1, -1)$ and $(3, -3)$. What is the image line?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q2/5

The horizontal line y = 4 is dilated about the origin with scale factor ½. Find an equation of the image and compare it with the original.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q3/5

A dilation with center C maps line ℓ, which does not pass through C, to line ℓ'. Which is true?

A. ℓ' passes through C

B. ℓ' is the same line as ℓ

C. $\ell' \parallel \ell$

D. $\ell' \perp \ell$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q4/5

True or false?

A dilation about point C maps every line through C onto itself.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q5/5

SET 37 • HSG-SRT.A.1.a

BACKWARD 1

Dilate points with coordinates • Grade 8 • Geometry • 8.G.A.3

Dilate $(2, -3)$ about the origin with scale factor 4.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q1/5

Dilate (6, 10) about the origin with scale factor ½.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q2/5

A dilation about the origin sends (3, 1) to (9, 3). What is the scale factor?

A. 6

B. 9

C. 3

D. $\frac{1}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q3/5

True or false?

A dilation about the origin with scale factor 1 leaves every point where it is.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q4/5

True or false?

A dilation with scale factor 2 moves the center to a new point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q5/5

SET 37 • HSG-SRT.A.1.a

BACKWARD 2

Compare slopes to test whether lines are parallel • Grade 8 • Expressions & Equations • 8.EE.B.6

Find the slope of the line through (0, 3) and (3, 9).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q1/5

True or false?

$y = 2x + 1$ and $y = 2x + 3$ are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q2/5

Which line is parallel to $y = -3x + 2$?

A. $y = \frac{1}{3}x$

B. $y = -3$

C. $y = 3x + 2$

D. $y = -3x - 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q3/5

Are y = 5 and y = 2 parallel?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q4/5

True or false?

Lines with slopes 2 and −2 are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q5/5

SET 37 • HSG-SRT.A.1.a

BACKWARD 3

Write the equation of a line through two points • Grade 8 • Functions • 8.F.B.4

Write an equation of the line through (0, 3) and (3, 9).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q1

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q1/5

Write an equation of the line through (2, −2) and (6, −6).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q2

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q2/5

Which point is on $y = 2x + 3$?

A. (1, 5)

B. (5, 1)

C. (2, 3)

D. (0, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q3

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q3/5

True or false?

The line through (0, 0) and (4, 8) is $y = 2x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q4

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q4/5

True or false?

The line through (1, 1) and (2, 4) has slope 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q5

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q5/5

SET 37 • HSG-SRT.A.1.a

FORWARD 1

Dilate lines about a center other than the origin • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a

Dilate the line $y = x + 2$ about the center $(2, 0)$ with scale factor 2. Find the images of $(0, 2)$ and $(2, 4)$ and the image line.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q1/5

Does the line $y = x - 2$ pass through the center $(2, 0)$? What happens to it under any dilation about $(2, 0)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q2/5

True or false?

Under a dilation about (1, 1) with scale factor 3, the point (2, 1) goes to (4, 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q3/5

A dilation about (0, 2) with scale factor ½ maps (4, 6) to

A. (4, 4)

B. (8, 10)

C. (2, 3)

D. (2, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q4/5

Line ℓ: y = 3x. A dilation about (1, 0) with scale factor 2 maps ℓ to ℓ'. Find ℓ'.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q5/5

SET 37 • HSG-SRT.A.1.a

FORWARD 2

Prove algebraically that a dilation maps a line to a parallel line • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a

Prove that the dilation about the origin with scale factor k ($k > 0, k \neq 1$) maps the line $y = mx + b$ to the line $y = mx + kb$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q1/5

From the general result, when does the image of $y = mx + b$ coincide with the original line?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q2/5

True or false?

The dilation $(x, y) \to (kx, ky)$ maps vertical lines $x = c$ to vertical lines $x = kc$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q3/5

Under the dilation about the origin with scale factor 4, the line $y = -2x + 5$ maps to

A. $y = -8x + 5$

B. $y = -2x + 20$

C. $y = -2x + 5$

D. $y = -8x + 20$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q4/5

Explain why measuring a few dilated lines is not a proof of the property.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q5/5

Verify that a dilation multiplies lengths by the scale factor

MAIN	Geometry main question	**HSG-SRT.A.1.b**
BACKWARD 1	Find lengths with the distance formula	**8.G.B.8**
BACKWARD 2	Scale factor in scale drawings	**7.G.A.1**
BACKWARD 3	Dilate points about the origin	**8.G.A.3**
FORWARD 1	Find the center or scale factor from a segment and its image	**HSG-SRT.A.1.b**
FORWARD 2	Perimeter and area of dilated figures	**HSG-SRT.A.2**

SET 38 • HSG-SRT.A.1.b

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b

Segment AB has A(1, 2) and B(4, 6). Dilate it about the origin with scale factor 3. Find A′B′ and AB, and compare.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q1/5

Segment PQ has P(−4, 2) and Q(8, −6). Dilate it about the origin with scale factor ½. Find PQ and P′Q′.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q2/5

Segment CD has C(2, 1) and D(5, 5). Dilate it about the center (2, 1) with scale factor 2. Find C′D′.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q3/5

A segment 8 cm long is dilated with scale factor 0.75. How long is the image?

A. 8.75 cm

B. 10.67 cm

C. 7.25 cm

D. 6 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q4/5

True or false?

A dilation with scale factor 2.5 makes every segment 2.5 units longer.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q5/5

SET 38 • HSG-SRT.A.1.b

BACKWARD 1

Find lengths with the distance formula • Grade 8 • Geometry • 8.G.B.8

Find the distance between (3, 6) and (12, 18).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q1/5

Find the distance between $(-2, 1)$ and $(4, -3)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q2/5

Find the distance between (0, 0) and (5, 12).

A. $\sqrt{17}$

B. 7

C. 17

D. 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q3/5

True or false?

The distance between (1, 2) and (4, 6) is 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q4/5

True or false?

The distance between (0, 0) and (3, 3) is 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q5/5

SET 38 • HSG-SRT.A.1.b

BACKWARD 2

Scale factor in scale drawings • Grade 7 • Geometry • 7.G.A.1

A drawing uses scale factor 4. A 3 cm segment in the original is how long in the drawing?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q1/5

A 15 m wall is drawn 5 cm long. What scale factor (drawing ÷ actual, in the same units) was used?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q2/5

True or false?

A scale factor less than 1 makes a figure smaller.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q3/5

A rectangle 4 by 6 is enlarged by a scale factor of 1.5. Its new dimensions are

A. 5.5 by 7.5

B. 6 by 9

C. 4 by 9

D. 8 by 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q4/5

True or false?

Enlarging by a scale factor of 2 doubles the area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q5/5

SET 38 • HSG-SRT.A.1.b

BACKWARD 3

Dilate points about the origin • Grade 8 • Geometry • 8.G.A.3

Dilate (4, 6) about the origin with scale factor 3.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q1/5

Dilate (−4, 2) about the origin with scale factor ½.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q2/5

Dilation about the origin sends (2, 5) to (5, 12.5). Scale factor?

A. 2.5

B. 3

C. 0.4

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q3/5

True or false?

Dilating about the origin multiplies both coordinates by the scale factor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q4/5

True or false?

Dilating (3, 0) by 2 about the origin gives (5, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q5/5

SET 38 • HSG-SRT.A.1.b

FORWARD 1

Find the center or scale factor from a segment and its image • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b

A dilation about the origin maps a segment of length 6 to a segment of length 15. What is the scale factor?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q1/5

A dilation maps A(1, 1) to A′(3, 3) and B(2, 1) to B′(5, 3). Find the scale factor and the center.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q2/5

True or false?

Lines through each point and its image under a dilation all pass through the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q3/5

A dilation maps P(4, 0) to P′(1, 0) and its center is the origin. Scale factor?

A. 4

B. −3

C. 3

D. ¼

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q4/5

A segment of length 9 is dilated by an unknown factor k, and the image is 3 units shorter than the original. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q5/5

SET 38 • HSG-SRT.A.1.b

FORWARD 2

Perimeter and area of dilated figures • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

A triangle with perimeter 18 and area 12 is dilated with scale factor 3. Find the new perimeter and area.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q1/5

A square of area 50 is dilated so its area becomes 8. Find the scale factor.

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q2/5

True or false?

If lengths are multiplied by k, areas are multiplied by k^2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q3/5

A figure is dilated with scale factor ½. Its area is multiplied by

A. ½

B. 4

C. 2

D. ¼

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q4/5

Explain why a dilation with factor k multiplies the perimeter by k.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q5/5

Decide whether two figures are similar using similarity transformations

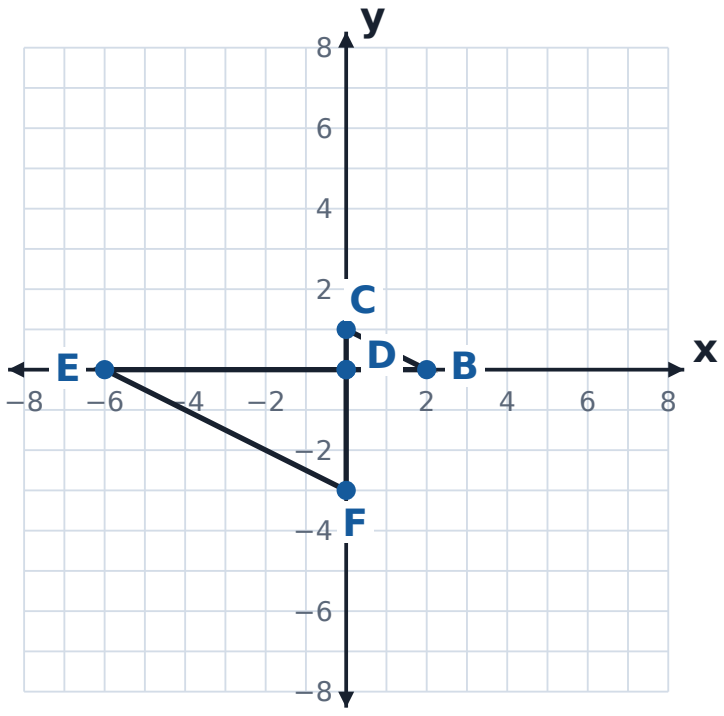
MAIN	Geometry main question	**HSG-SRT.A.2**
BACKWARD 1	Similar figures as a sequence of transformations including a dilati…	**8.G.A.4**
BACKWARD 2	Specify a sequence of rigid motions	**HSG-CO.A.5**
BACKWARD 3	Dilate figures about the origin	**8.G.A.3**
FORWARD 1	Find the scale factor and correspondence when the image is turne	**HSG-SRT.A.2**
FORWARD 2	Prove that all circles are similar	**HSG-C.A.1**

SET 39 • HSG-SRT.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Is triangle ABC similar to triangle DEF? If so, describe a similarity transformation that maps ABC onto DEF; if not, explain.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q1/5

Is rectangle R (4 by 6) similar to rectangle S (6 by 8)? Explain using similarity transformations.

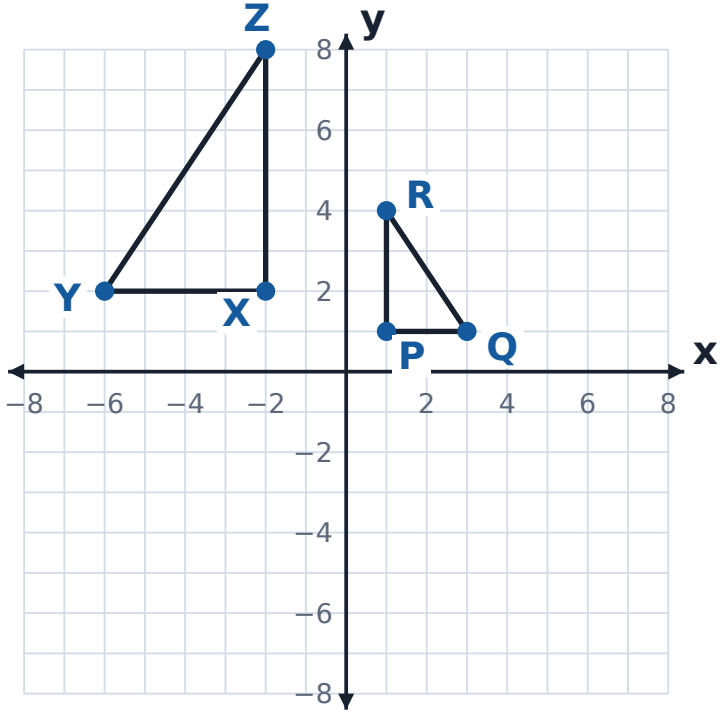
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q2/5

Is triangle PQR similar to triangle XYZ? If so, describe a similarity transformation; if not, explain.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q3/5

Which pair of figures must be similar?

A. Any two isosceles triangles

B. Any two rectangles

C. Any two rhombuses

D. Any two squares

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q4/5

True or false?

If a sequence of rigid motions and one dilation carries figure F onto figure G, then F and G are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q5/5

SET 39 • HSG-SRT.A.2

BACKWARD 1

Similar figures as a sequence of transformations including a dilation • Grade 8 • Geometry • 8.G.A.4

True or false?

A figure and its image after a dilation and a translation are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q1

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q1/5

Two figures are similar. What may be different?

A. Their shape

B. Their angle measures

C. Their size

D. The ratios of their sides

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q2

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q2/5

A triangle is dilated by 2 and then rotated. Its sides were 3, 4, 5. What are the image's sides?

Lengths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q3

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q3/5

True or false?

Two figures with the same angles but different side ratios are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q4

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q4/5

Name the transformation that changes size.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q5

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q5/5

SET 39 • HSG-SRT.A.2

BACKWARD 2

Specify a sequence of rigid motions • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Which rigid motion maps (x, y) to (−x, y)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q1/5

Which rigid motion maps (x, y) to (−x, −y)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q2/5

Which rigid motion maps (2, 3) to (−3, 2)?

A. Translation 5 left

B. Rotation 180°

C. Rotation 90° counterclockwise about the origin

D. Reflection across the x-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q3/5

True or false?

A translation then a reflection is a valid sequence of rigid motions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q4/5

True or false?

A dilation is a rigid motion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q5/5

SET 39 • HSG-SRT.A.2

BACKWARD 3

Dilate figures about the origin • Grade 8 • Geometry • 8.G.A.3

Dilate triangle (0, 0), (2, 0), (0, 1) by 3 about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q1/5

True or false?

A dilation about the origin with factor 3 triples each coordinate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q2/5

(4, −2) dilated by ½ about the origin is

A. (8, −4)

B. (4.5, −1.5)

C. (2, −1)

D. (2, −2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q3/5

A dilation about the origin maps (5, 5) to (2, 2). What is the factor?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q4/5

True or false?

A dilation can change the angles of a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q5/5

SET 39 • HSG-SRT.A.2

FORWARD 1

Find the scale factor and correspondence when the image is turned or flipped • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Triangle ABC has sides AB = 4, BC = 6, CA = 8. Triangle PQR has PQ = 9, QR = 12, RP = 6. Are they similar? If so, give the correspondence and the scale factor.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q1/5

A 3-4-5 triangle and a triangle with sides 7.5, 10, 12.5: similar? Scale factor?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q2/5

True or false?

Matching the shortest side with the shortest side, middle with middle, and longest with longest gives the correspondence for similar triangles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q3/5

Triangles with sides 5, 6, 7 and 10, 12, 15: are they similar?

A. Cannot tell

B. Yes, factor 2.5

C. Yes, factor 2

D. No, 15/7 ≠ 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q4/5

Triangle ABC ~ triangle DEF with a factor of 4 from ABC to DEF. DE = 10. Find AB.

AB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q5/5

SET 39 • HSG-SRT.A.2

FORWARD 2

Prove that all circles are similar • High School — Geometry • Geometry: Circles • HSG-C.A.1

Circle C_1 has center A and radius r; circle C_2 has center B and radius s. Prove that C_1 and C_2 are similar.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q1/5

Circle P has radius 4 and circle Q has radius 10. What scale factor maps P onto Q (after a translation)?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q2/5

True or false?

Any two circles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q3/5

Which pair of figures is NOT always similar?

A. Two equilateral triangles

B. Two squares

C. Two circles

D. Two ellipses

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q4/5

Explain why all equilateral triangles are similar.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q5/5

Explain similarity of triangles as equal angles and proportional sides

MAIN	Geometry main question	**HSG-SRT.A.2**
BACKWARD 1	Read correspondence from a similarity or congruence statement	**HSG-CO.B.7**
BACKWARD 2	Solve proportions	**7.RP.A.2**
BACKWARD 3	Scale factor between figures	**7.G.A.1**
FORWARD 1	Use the AA criterion to recognize similar triangles	**HSG-SRT.A.3**
FORWARD 2	Use similarity to find unknown lengths in problems	**HSG-SRT.B.5**

SET 40 • HSG-SRT.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

A similarity transformation (a dilation with scale factor k followed by rigid motions) maps $\triangle ABC$ onto $\triangle DEF$ with $A \to D$, $B \to E$, $C \to F$. Explain why $\angle A \cong \angle D$, $\angle B \cong \angle E$, $\angle C \cong \angle F$ and $DE/AB = EF/BC = FD/CA = k$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q1/5

$\triangle ABC \sim \triangle DEF$ with $A \leftrightarrow D$, $B \leftrightarrow E$, $C \leftrightarrow F$. $AB = 6$, $BC = 9$, $DE = 10$. Find EF and the scale factor from ABC to DEF.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q2/5

$\triangle PQR \sim \triangle STU$ with $P \leftrightarrow S$, $Q \leftrightarrow T$, $R \leftrightarrow U$. $m\angle P = 42°$ and $m\angle Q = 71°$. Find $m\angle U$.

$m\angle U =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q3/5

Which information shows that △ABC is NOT similar to △XYZ (A ↔ X, B ↔ Y, C ↔ Z)?

A. ∠A ≅ ∠X

B. AB/XY = BC/YZ = CA/ZX = 2

C. AB/XY = 2 but BC/YZ = 3

D. All angles of both are 60°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q4/5

True or false?

If $\triangle ABC \sim \triangle DEF$, then $AB/DE = BC/EF$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q5/5

SET 40 • HSG-SRT.A.2

BACKWARD 1

Read correspondence from a similarity or congruence statement • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

△ABC ~ △XYZ. Which angle corresponds to ∠C?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q1/5

△ABC ~ △XYZ. Which side corresponds to BC?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q2/5

△KLM ~ △RST. Which ratio equals KL/RS?

A. LM/RT

B. RS/LM

C. LM/ST

D. KM/ST

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q3/5

True or false?

$\triangle DEF \sim \triangle GHI$ means $\angle E \cong \angle H$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q4/5

True or false?

$\triangle DEF \sim \triangle GHI$ means DF corresponds to GH.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q5/5

SET 40 • HSG-SRT.A.2

BACKWARD 2

Solve proportions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

Solve 6/9 = 10/x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q1/5

Solve x/4 = 15/6.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q2/5

Which ratio is equal to 3 : 5?

A. 9 : 12

B. 6 : 8

C. 5 : 3

D. 12 : 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q3/5

True or false?

$2/3 = 8/12$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q4/5

True or false?

$4/6 = 6/9 = 10/14$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q5/5

SET 40 • HSG-SRT.A.2

BACKWARD 3

Scale factor between figures • Grade 7 • Geometry • 7.G.A.1

A figure is enlarged so that a 4 cm side becomes 10 cm. Scale factor?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q1/5

True or false?

All lengths of a scale drawing are multiplied by the same number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q2/5

With scale factor 3, a 7 cm side becomes

A. 21 cm

B. 7 cm

C. 10 cm

D. 73 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q3/5

A scale drawing's 12 cm side represents 60 cm. Scale factor (drawing to actual)?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q4/5

True or false?

A scale drawing changes the angles of a figure.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q5/5

SET 40 • HSG-SRT.A.2

FORWARD 1

Use the AA criterion to recognize similar triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

Two triangles have angles of 35° and 80°. Are they similar? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q1/5

Triangle ABC has m∠A = 50°, m∠B = 60°. Triangle DEF has m∠D = 60°, m∠F = 70°. Are they similar? Give the correspondence.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q2/5

True or false?

Two right triangles with one pair of congruent acute angles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q3/5

Which pair is NOT necessarily similar?

A. Two triangles with angles 30° and 60°

B. Two isosceles triangles

C. Two right isosceles triangles

D. Two equilateral triangles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q4/5

In a figure, DE ∥ BC with D on AB and E on AC. Explain why △ADE ~ △ABC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q5/5

SET 40 • HSG-SRT.A.2

FORWARD 2

Use similarity to find unknown lengths in problems • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

A 1.8 m person casts a 2.4 m shadow when a tree casts a 16 m shadow. How tall is the tree?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q1/5

In $\triangle ABC$, $DE \parallel BC$ with $AD = 4$, $DB = 6$, and $DE = 5$. Find BC.

$BC =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q2/5

True or false?

In similar triangles with scale factor 3, a 7 cm side corresponds to a 21 cm side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q3/5

A 6 ft pole casts a 4 ft shadow. At the same time, a building casts a 30 ft shadow. How tall is the building?

A. 36 ft

B. 20 ft

C. 45 ft

D. 24 ft

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F2-Q4**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q4/5

Two similar triangles have perimeters 24 and 36. A side of the smaller is 10. Find the corresponding side of the larger.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q5/5

Establish the AA criterion with similarity transformations

MAIN	Geometry main question	**HSG-SRT.A.3**
BACKWARD 1	Prove triangles congruent with ASA	**HSG-CO.B.8**
BACKWARD 2	Dilations multiply lengths and keep angles	**HSG-SRT.A.1.b**
BACKWARD 3	Use the angle sum to find a third angle	**8.G.A.5**
FORWARD 1	Establish the SAS and SSS similarity criteria	**HSG-SRT.A.3**
FORWARD 2	Use AA to set up and solve problems	**HSG-SRT.B.5**

SET 41 • HSG-SRT.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

In △ABC and △DEF, ∠A ≅ ∠D and ∠B ≅ ∠E. Prove that △ABC ~ △DEF using similarity transformations.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q1/5

In the AA proof, why is the scale factor chosen as DE/AB?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q2/5

In the AA proof, after dilating $\triangle ABC$ to $\triangle A'B'C'$, which criterion shows $\triangle A'B'C' \cong \triangle DEF$, and which parts does it use?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q3/5

Why is it enough to know TWO pairs of congruent angles?

A. The third angle can be different

B. Two angles fix the side lengths

C. The angle sum makes the third pair congruent too

D. Any two triangles are similar

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q4/5

True or false?

The AA argument uses the fact that dilations preserve angle measure.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q5/5

SET 41 • HSG-SRT.A.3

BACKWARD 1

Prove triangles congruent with ASA • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

Two triangles have $\angle A \cong \angle D$, $AB = DE$, $\angle B \cong \angle E$. Which criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q1/5

True or false?

ASA needs the side between the two angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q2/5

Which set of parts gives ASA?

A. Two angles and the side between them

B. Three angles

C. Three sides

D. Two sides and the angle between them

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q3/5

After proving $\triangle PQR \cong \triangle XYZ$, are their corresponding sides equal?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q4/5

True or false?

ASA works only for right triangles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q5/5

SET 41 • HSG-SRT.A.3

BACKWARD 2

Dilations multiply lengths and keep angles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b

A 5 cm segment is dilated by 1.6. How long is the image?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q1/5

True or false?

A dilation preserves angle measures.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q2/5

A dilation maps AB = 4 to A′B′ = 10. Factor?

A. 6

B. 0.4

C. 14

D. 2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q3/5

What scale factor makes a 6 cm segment 6 cm long?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q4/5

True or false?

A dilation by 2 doubles the angles of a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q5/5

SET 41 • HSG-SRT.A.3

BACKWARD 3

Use the angle sum to find a third angle • Grade 8 • Geometry • 8.G.A.5

A triangle has angles 47° and 88°. Find the third.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q1/5

True or false?

If two angles of one triangle equal two angles of another, the third angles are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B3-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q2/5

Triangles with angles 30°, 70°, x and 70°, 80°, y. Which is true?

A. x = 80, y = 30; not similar

B. x = 80, y = 30; the triangles are similar

C. x = 70, y = 30

D. x = 30, y = 80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q3/5

Two angles of a triangle are equal and the third is 40°. Find the equal angles.

Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q4/5

True or false?

Knowing one angle of each triangle is enough for similarity.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q5/5

SET 41 • HSG-SRT.A.3

FORWARD 1

Establish the SAS and SSS similarity criteria • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

In $\triangle ABC$ and $\triangle DEF$, $DE/AB = DF/AC = k$ and $\angle A \cong \angle D$. Prove that $\triangle ABC \sim \triangle DEF$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q1/5

Triangles with sides 4, 6, 8 and 6, 9, 12. Similar? Which criterion?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q2/5

True or false?

Two triangles with two pairs of proportional sides and congruent included angles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q3/5

Which is NOT a similarity criterion?

A. SAS

B. SSS

C. SSA

D. AA

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q4/5

$\triangle PQR$: $PQ = 5$, $PR = 7$, $\angle P = 40°$. $\triangle XYZ$: $XY = 10$, $XZ = 14$, $\angle X = 40°$. Similar? Factor?

Answer: _____

SET 41 • HSG-SRT.A.3

FORWARD 2

Use AA to set up and solve problems • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

Two vertical poles are 3 m and 7 m tall. A wire from the top of each to the bottom of the other crosses at a point P. How high above the ground is P? (Use similar triangles.)

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q1/5

In $\triangle ABC$, D is on AB and E is on AC with $\angle ADE \cong \angle ACB$. AD = 4, AE = 6, AB = 12. Find AC.

AC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q2/5

True or false?

A mirror on the ground can be used to find a building's height with similar triangles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q3/5

Diagonals of trapezoid ABCD (AB ∥ DC) meet at E. AB = 12, DC = 4, AE = 9. Find EC.

A. 3

B. 6

C. 27

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q4/5

A 2 m stick casts a 1.5 m shadow. A flagpole casts a 9 m shadow. How tall is the flagpole?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q5/5

Prove the triangle proportionality theorem

MAIN	Geometry main question	**HSG-SRT.B.4**
BACKWARD 1	Use AA to show triangles are similar	**HSG-SRT.A.3**
BACKWARD 2	Corresponding angles of parallel lines	**HSG-CO.C.9**
BACKWARD 3	Rewrite and solve proportions algebraically	**7.RP.A.2**
FORWARD 1	Prove that parallel lines cut transversals proportionally	**HSG-SRT.B.4**
FORWARD 2	Prove the angle bisector theorem	**HSG-SRT.B.5**

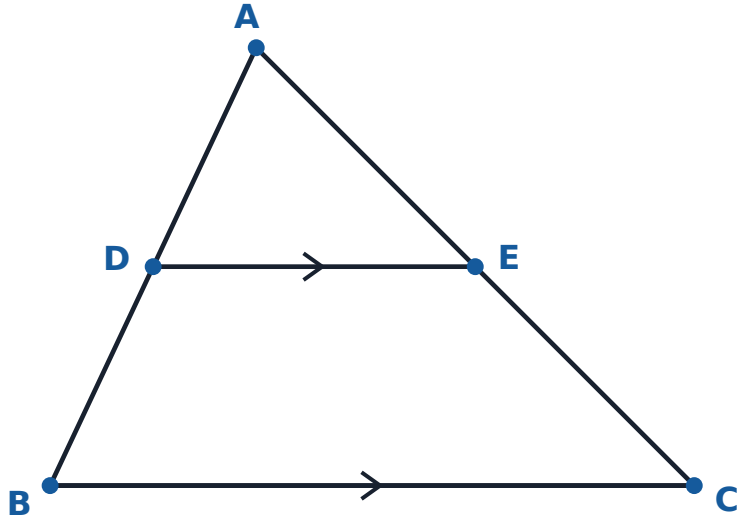
SET 42 • HSG-SRT.B.4

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

In △ABC, D is on AB and E is on AC with DE ∥ BC. Prove that AD/DB = AE/EC.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q1/5

In the proof of the triangle proportionality theorem, the similarity gives AD/AB = AE/AC. Show the algebra that turns this into AD/DB = AE/EC.

Work: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q2/5

In △PQR, a line parallel to QR meets PQ at S and PR at T. Prove that PS/SQ = PT/TR.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q3/5

A proof of the triangle proportionality theorem:

1. $\angle ADE \cong \angle ABC$ (?)

2. $\angle A \cong \angle A$ (?)

3. $\triangle ADE \sim \triangle ABC$ (?)

4. $AD/AB = AE/AC$ (?)

Give the reasons.

Reasons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q4/5

Which fact does the proof use first?

A. AB = AC

B. D and E are midpoints

C. DE = ½BC

D. DE ∥ BC makes corresponding angles congruent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q5/5

SET 42 • HSG-SRT.B.4

BACKWARD 1

Use AA to show triangles are similar • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

True or false?

Two pairs of congruent angles make two triangles similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q1/5

DE ∥ BC in △ABC. Name two pairs of congruent angles in △ADE and △ABC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q2/5

△ADE ~ △ABC. Which ratio equals AD/AB?

A. DB/AB

B. AE/AC

C. DE/AB

D. AE/EC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q3/5

$\triangle ADE \sim \triangle ABC$ with $AD = 3$, $AB = 9$, $AE = 4$. Find AC.

$AC =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q4/5

True or false?

$\triangle ADE \sim \triangle ABC$ means $AD/DB = DE/BC$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q5/5

SET 42 • HSG-SRT.B.4

BACKWARD 2

Corresponding angles of parallel lines • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

When a transversal crosses parallel lines, corresponding angles are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q1/5

DE ∥ BC and AB is a transversal. Which angle corresponds to ∠ADE?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q2/5

DE ∥ BC, AC a transversal. ∠AED corresponds to

A. ∠BAC

B. ∠ABC

C. ∠DEC

D. ∠ACB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q3/5

$\angle ABC = 65°$ and $DE \parallel BC$. Find $\angle ADE$.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q4/5

True or false?

Corresponding angles are always supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q5/5

SET 42 • HSG-SRT.B.4

BACKWARD 3

Rewrite and solve proportions algebraically • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

If a/b = c/d, write the equation you get by taking reciprocals.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q1/5

Solve 4/(4 + x) = 6/15.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q2/5

If $(a + b)/a = (c + d)/c$, then

A. $b = d$

B. $a = c$

C. $b/a = d/c$

D. $a/b = c/d$ is false

True or false?

3/5 = 6/10 implies 5/3 = 10/6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q4/5

True or false?

$(2 + 3)/2 = 5/2$ implies $3/2 = 5/2$.

A. True

B. False

SET 42 • HSG-SRT.B.4

FORWARD 1

Prove that parallel lines cut transversals proportionally • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

Three parallel lines cut transversal t at A, B, C and transversal u at D, E, F (in order). Prove that AB/BC = DE/EF.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q1/5

Three parallel lines cut one transversal into segments 4 and 6 and another into segments 5 and x. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q2/5

True or false?

Parallel lines that cut equal segments on one transversal cut equal segments on every transversal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q3/5

In $\triangle ABC$, $DE \parallel BC$, $AD = 5$, $DB = 3$, $AE = 7.5$. Find EC.

A. 12.5

B. 6

C. 5

D. 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F1-Q4**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q4/5

Lots between parallel streets have frontages 30 m and 45 m on one avenue. On another avenue the first frontage is 40 m. Find the second.

Frontage: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q5/5

SET 42 • HSG-SRT.B.4

FORWARD 2

Prove the angle bisector theorem • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

In $\triangle ABC$, the bisector of $\angle A$ meets BC at D. Prove that $BD/DC = AB/AC$. (Draw the line through C parallel to AD, meeting line AB at E.)

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q1/5

In $\triangle ABC$, the bisector of $\angle A$ meets BC at D. AB = 9, AC = 6, BC = 10. Find BD and DC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q2/5

True or false?

The bisector of an angle of a triangle always meets the opposite side at its midpoint.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q3/5

The bisector of $\angle P$ in $\triangle PQR$ divides QR into 3 and 5. PQ = 6. Find PR.

A. 15

B. 10

C. 3.6

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q4/5

Explain why the angle bisector theorem is a relationship that similarity or proportionality, not congruence, proves.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q5/5

Prove the converse of the triangle proportionality theorem

MAIN	Geometry main question	**HSG-SRT.B.4**
BACKWARD 1	Use the triangle proportionality theorem	**HSG-SRT.B.4**
BACKWARD 2	SAS similarity	**HSG-SRT.A.3**
BACKWARD 3	Converse of the corresponding angles theorem	**HSG-CO.C.9**
FORWARD 1	Use the converse to decide parallelism in figures	**HSG-SRT.B.4**
FORWARD 2	Prove the trapezoid midsegment theorem	**HSG-SRT.B.5**

SET 43 • HSG-SRT.B.4

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

In △ABC, D is on AB and E is on AC with AD/DB = AE/EC. Prove that DE ∥ BC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q1/5

In $\triangle PQR$, S is on PQ and T is on PR with PS = 4, SQ = 6, PT = 6, TR = 9. Prove that $ST \parallel QR$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q2/5

Show the algebra that turns AD/DB = AE/EC into AD/AB = AE/AC.

Work: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q3/5

In $\triangle ABC$, AD = 3, DB = 5, AE = 4.5, EC = 7.5 (D on AB, E on AC). Is DE ∥ BC?

A. Yes, because 3/5 = 4.5/7.5

B. No, because 3 ≠ 4.5

C. Yes, because D and E are midpoints

D. No, because 5 ≠ 7.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q4/5

True or false?

In $\triangle ABC$, D is on AB and E is on AC with $AD = 2$, $DB = 3$, $AE = 3$, $EC = 4$. Then $DE \parallel BC$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q5/5

SET 43 • HSG-SRT.B.4

BACKWARD 1

Use the triangle proportionality theorem • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

In $\triangle ABC$, $DE \parallel BC$, $AD = 6$, $DB = 4$, $AE = 9$. Find EC.

$EC =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q1/5

True or false?

If DE ∥ BC in △ABC, then AD/DB = AE/EC.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q2/5

DE ∥ BC, AD = 2, AB = 8, AC = 12. Find AE.

A. 9

B. 4

C. 6

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q3/5

State the converse of "If DE ∥ BC, then AD/DB = AE/EC."

Converse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q4/5

True or false?

A theorem and its converse are always both true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q5/5

SET 43 • HSG-SRT.B.4

BACKWARD 2

SAS similarity • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

True or false?

Two pairs of proportional sides with congruent included angles make triangles similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q1/5

△ADE and △ABC share ∠A, AD/AB = AE/AC = 0.4. Are they similar? Criterion?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q2/5

After $\triangle ADE \sim \triangle ABC$, which angle is congruent to $\angle ADE$?

A. $\angle ABC$

B. $\angle ACB$

C. $\angle BAC$

D. $\angle DEA$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q3/5

Ratios 3/6 and 4/8 with a shared included angle: similar? Factor?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q4/5

True or false?

SAS similarity needs the angle to be between the two pairs of sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q5/5

SET 43 • HSG-SRT.B.4

BACKWARD 3

Converse of the corresponding angles theorem • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

Congruent corresponding angles make two lines parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q1/5

∠ADE ≅ ∠ABC with transversal AB. What can you conclude about DE and BC?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q2/5

Which angles, if congruent, prove DE ∥ BC?

A. ∠A and ∠B

B. ∠B and ∠C

C. ∠AED and ∠ACB

D. ∠ADE and ∠AED

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q3/5

Corresponding angles measure 58° and 62°. Are the lines parallel?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q4/5

True or false?

Vertical angles prove lines parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q5/5

SET 43 • HSG-SRT.B.4

FORWARD 1

Use the converse to decide parallelism in figures • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

In $\triangle ABC$, D is on AB with AD = 5, DB = 10, and E is on AC with AE = 4. How long must EC be for $DE \parallel BC$?

EC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q1/5

In $\triangle PQR$, S on PQ, T on PR: PS = x, SQ = 6, PT = 4, TR = 8. For which x is ST ∥ QR?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q2/5

True or false?

The midsegment theorem's parallel part is a special case of the converse (ratio 1 : 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q3/5

Points divide two sides in ratios 2 : 3 and 4 : 6. The segment joining them is

A. perpendicular to the third side

B. half the third side

C. parallel to the third side

D. not parallel to the third side

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q4/5

In $\triangle ABC$, D is on AB with AD : DB = 1 : 2 and DE $\parallel$ BC with E on AC. Find DE : BC.

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q5/5

SET 43 • HSG-SRT.B.4

FORWARD 2

Prove the trapezoid midsegment theorem • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

Trapezoid ABCD has AB ∥ DC. M and N are the midpoints of legs AD and BC. Prove that MN ∥ AB and MN = ½(AB + DC).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q1/5

A trapezoid has bases 10 and 16. Find the length of the segment joining the midpoints of the legs.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q2/5

True or false?

The midsegment of a trapezoid is parallel to both bases.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q3/5

A trapezoid's midsegment is 12 and one base is 9. Find the other base.

A. 3

B. 6

C. 15

D. 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q4/5

Explain why M, P, N are on one line in the trapezoid midsegment proof.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q5/5

Prove the Pythagorean Theorem using similar triangles

MAIN	Geometry main question	**HSG-SRT.B.4**
BACKWARD 1	Show triangles similar with AA	**HSG-SRT.A.3**
BACKWARD 2	Cross-multiply proportions	**7.RP.A.2**
BACKWARD 3	Explain a proof of the Pythagorean Theorem informally	**8.G.B.6**
FORWARD 1	Use the geometric mean relationships in a right triangle	**HSG-SRT.B.5**
FORWARD 2	The Law of Cosines as a generalization of the Pythagorean Theor	**HSG-SRT.D.10**

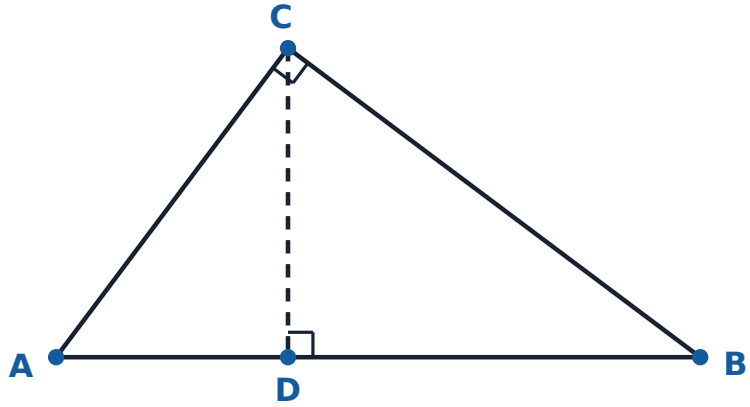
SET 44 • HSG-SRT.B.4

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

Right triangle ABC has the right angle at C, legs a = BC, b = AC, and hypotenuse c = AB. The altitude from C meets AB at D. Prove that $a^2 + b^2 = c^2$.

Work space



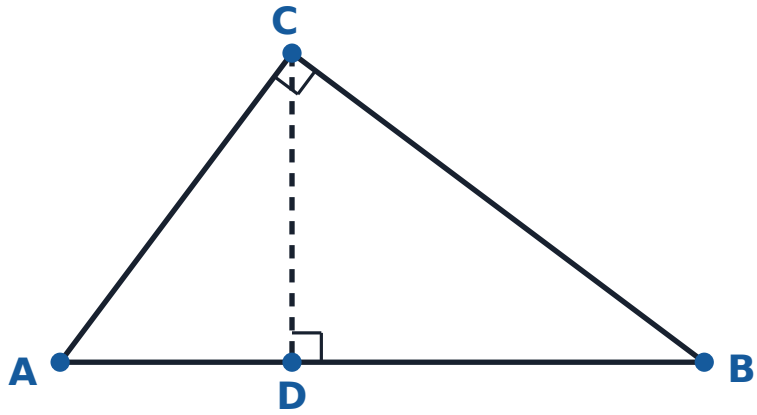
Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q1/5

In the similarity proof of the Pythagorean Theorem, why is $\triangle ADC \sim \triangle ACB$?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q2/5

In the proof, $b^2 = c \cdot AD$ and $a^2 = c \cdot BD$. Show how these give $a^2 + b^2 = c^2$.

Work: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q3/5

A right triangle has legs 6 and 8, and the altitude to the hypotenuse divides it into parts x and y. Use similar triangles (not the Pythagorean Theorem) to find x and y, and show that the hypotenuse is 10.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q4/5

Which pair of triangles is NOT similar in the figure of a right triangle with the altitude to the hypotenuse?

A. $\triangle ACD$ and $\triangle ABC$

B. $\triangle BCD$ and $\triangle ABC$

C. $\triangle ACD$ and $\triangle CBD$

D. None; all three triangles are similar

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q5/5

SET 44 • HSG-SRT.B.4

BACKWARD 1

Show triangles similar with AA • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

True or false?

Two right triangles that share an acute angle are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q1/5

△ADC and △ACB: name two pairs of congruent angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q2/5

△ADC ~ △ACB. Which ratio equals AD/AC?

A. DC/AB

B. AD/AB

C. AC/AB

D. CB/AC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q3/5

If $\triangle ADC \sim \triangle ACB$ with $AD = 4$ and $AB = 9$, find AC.

$AC =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q4/5

True or false?

Similar triangles must be the same size.

A. True

B. False

SET 44 • HSG-SRT.B.4

BACKWARD 2

Cross-multiply proportions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

From AD/AC = AC/AB, write an equation without fractions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q1/5

Solve $4/x = x/9$ for $x > 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q2/5

If $p/q = q/r$, then

A. $p = r$

B. $q^2 = pr$

C. $r^2 = pq$

D. $p^2 = qr$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q3/5

True or false?

From $a/b = b/c$ we get $b^2 = ac$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q4/5

True or false?

From $x/5 = 5/x$ we get $x = 5$ only.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q5/5

SET 44 • HSG-SRT.B.4

BACKWARD 3

Explain a proof of the Pythagorean Theorem informally • Grade 8 • Geometry • 8.G.B.6

True or false?

The Pythagorean Theorem says $a^2 + b^2 = c^2$ for every right triangle with legs a, b and hypotenuse c.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q1/5

Name a way to show the theorem by rearranging area pieces.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q2/5

Which triangle satisfies the theorem?

A. 6, 7, 9

B. 5, 6, 7

C. 2, 3, 4

D. 5, 12, 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q3/5

Legs 9 and 12. Hypotenuse?

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q4/5

True or false?

Checking the theorem with three triangles proves it for all right triangles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q5/5

SET 44 • HSG-SRT.B.4

FORWARD 1

Use the geometric mean relationships in a right triangle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

In right triangle ABC with altitude CD to hypotenuse AB, AD = 4 and DB = 9. Find CD.

CD = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q1/5

In right triangle ABC with right angle at C and altitude CD, prove that $CD^2 = AD \cdot DB$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q2/5

True or false?

The altitude to the hypotenuse is the geometric mean of the two parts of the hypotenuse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q3/5

AD = 2 and AB = 8 in the altitude figure. Find AC.

A. $2\sqrt{2}$

B. 6

C. 16

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q4/5

A right triangle's altitude to the hypotenuse is 12 and one part of the hypotenuse is 9. Find the other part.

Part: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q5/5

SET 44 • HSG-SRT.B.4

FORWARD 2

The Law of Cosines as a generalization of the Pythagorean Theorem • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

For a triangle with sides a, b, c and angle C opposite c, the Law of Cosines says $c^2 = a^2 + b^2 - 2ab\cos C$. What does it become when $C = 90°$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q1/5

A triangle has sides 5 and 8 with a 60° angle between them. Find the third side.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q2/5

True or false?

If $a^2 + b^2 > c^2$, the angle opposite c is acute.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q3/5

A triangle has sides 4, 5, 7. The largest angle is

A. right

B. obtuse

C. impossible to tell

D. acute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q4/5

Use the converse idea: is a triangle with sides 7, 24, 25 right, acute, or obtuse?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q5/5

SET 45 • HSG-SRT.B.5 • SIMILARITY, RIGHT TRIANGLES, AND TRIGONOMETRY

Use similarity criteria to solve problems in geometric figures

MAIN	Geometry main question	**HSG-SRT.B.5**
BACKWARD 1	Similarity criteria: AA, SAS, SSS	**HSG-SRT.A.3**
BACKWARD 2	Solve proportions	**7.RP.A.2**
BACKWARD 3	Identify corresponding sides in overlapping triangles	**HSG-SRT.A.2**
FORWARD 1	Solve multistep problems with two or more pairs of similar triangl...	**HSG-SRT.B.5**
FORWARD 2	Solve similarity problems with algebra (unknowns on both sides)	**HSG-SRT.B.5**

SET 45 • HSG-SRT.B.5

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

In $\triangle ABC$, D is on AB and E is on AC with $DE \parallel BC$. AD = 6, DB = 9, DE = 8. Find BC. Name the criterion you used for similar triangles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q1/5

A lamppost lights a student who is 1.6 m tall and stands 3 m from the post. The student's shadow is 2 m long, measured away from the post. How tall is the lamppost? Name the similarity criterion you used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q2/5

Two triangles: $\triangle PQR$ with $PQ = 8$, $QR = 12$, $\angle Q = 50°$; $\triangle STU$ with $ST = 6$, $TU = 9$, $\angle T = 50°$. Are they similar? If so, find the ratio of their perimeters.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q3/5

A surveyor finds a river's width w with similar right triangles: legs 12 m and 18 m on the small triangle, 12 m corresponding to 40 m on the large one. What is w (the side matching 18 m)?

A. 26.7 m

B. 46 m

C. 60 m

D. 54 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q4/5

True or false?

If two triangles have sides 3, 5, 7 and 6, 10, 15, they are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q5/5

SET 45 • HSG-SRT.B.5

BACKWARD 1

Similarity criteria: AA, SAS, SSS • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

Which criterion uses two pairs of congruent angles?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q1/5

Sides 2, 3, 4 and 4, 6, 8. Criterion?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q2/5

Which information is enough for similarity?

A. One pair of congruent angles

B. One pair of proportional sides

C. Two pairs of congruent angles

D. Two sides of each triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q3/5

True or false?

SAS similarity requires the congruent angle to be between the proportional sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q4/5

True or false?

Any two right triangles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q5/5

SET 45 • HSG-SRT.B.5

BACKWARD 2

Solve proportions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

Solve 6/15 = 8/x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q1/5

Solve 1.6/2 = h/5.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q2/5

Solve 12/40 = 18/x.

A. 26.7

B. 60

C. 54

D. 46

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q3/5

True or false?

$3/4 = 9/12$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q4/5

True or false?

$5/8 = 10/14$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q5/5

SET 45 • HSG-SRT.B.5

BACKWARD 3

Identify corresponding sides in overlapping triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

△ADE ~ △ABC. Which side of △ABC corresponds to DE?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q1/5

True or false?

In $\triangle ADE \sim \triangle ABC$, side AD corresponds to side AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q2/5

In the DE ∥ BC figure, the side of the big triangle matching AD is

A. AB

B. DB

C. AC

D. BC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q3/5

AD = 6, DB = 9. What is AB?

AB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q4/5

True or false?

In $\triangle ADE \sim \triangle ABC$, DB corresponds to DE.

A. True

B. False

SET 45 • HSG-SRT.B.5

FORWARD 1

Solve multistep problems with two or more pairs of similar triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

Trapezoid ABCD has AB ∥ DC, AB = 15, DC = 10. The diagonals meet at E, and AC = 20. Find AE and EC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q1/5

A pinhole camera: an object 2 m tall is 8 m from the pinhole; the image is 5 cm from the pinhole on the screen. How tall is the image?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q2/5

True or false?

In a figure with two pairs of similar triangles, one ratio found in the first pair can be reused in the second.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q3/5

In $\triangle ABC$, $DE \parallel BC$ with $AD = 2$, $DB = 6$; and $FG \parallel BC$ farther down with $AF = 5$. If $BC = 16$, find FG.

A. 12

B. 4

C. 10

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q4/5

Using a mirror on the ground, Ana (eyes 1.5 m high) stands 2 m from the mirror and sees the top of a tree 12 m beyond the mirror. How tall is the tree?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q5/5

SET 45 • HSG-SRT.B.5

FORWARD 2

Solve similarity problems with algebra (unknowns on both sides) • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

In $\triangle ABC$, $DE \parallel BC$ with $AD = x$, $DB = 6$, $AE = 4$, $EC = x + 2$. Find x.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q1/5

Similar triangles have corresponding sides $x + 1$ and $2x - 3$ with scale factor 1.5 (larger to smaller is $2x - 3$ to $x + 1$). Find x.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q2/5

True or false?

A proportion with the unknown in two places can lead to a quadratic equation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q3/5

In the altitude figure of a right triangle, AD = x, DB = x + 5, CD = 6. Find x.

A. 3

B. 6

C. 4

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q4/5

Two similar rectangles: the first is 4 by x, the second is x by 9. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q5/5

Use congruence criteria to prove relationships in figures

MAIN	Geometry main question	**HSG-SRT.B.5**
BACKWARD 1	Choose and apply SSS, SAS, ASA, AAS	**HSG-CO.B.8**
BACKWARD 2	Find hidden parts: shared sides, vertical angles, right angles	**HSG-CO.C.9**
BACKWARD 3	Use corresponding parts to finish a proof	**HSG-CO.B.7**
FORWARD 1	Prove relationships that need two congruences in a row	**HSG-SRT.B.5**
FORWARD 2	Decide which given data are sufficient for a proof, and supply a m…	**HSG-SRT.B.5**

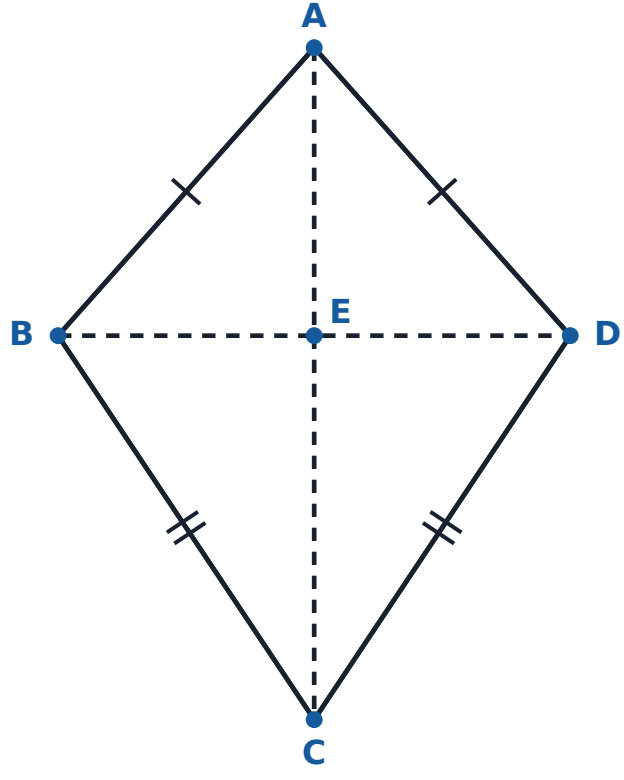
SET 46 • HSG-SRT.B.5

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

Kite ABCD has AB = AD and CB = CD. Its diagonals meet at E. Prove that AC ⊥ BD.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q1/5

Segments AB and CD bisect each other at M. Prove that AC ∥ DB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q2/5

Point P is on the bisector of ∠ABC. PD ⊥ BA at D and PE ⊥ BC at E. Prove that PD = PE.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q3/5

In $\triangle ABC$, $AB = AC$. M is the midpoint of BC. Prove that AM bisects $\angle BAC$ and $AM \perp BC$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q4/5

To prove that two segments in a figure are congruent, the most common first step is to

A. measure them carefully

B. prove two triangles containing them are congruent

C. show the figure is symmetric by looking

D. assume they are congruent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q5/5

SET 46 • HSG-SRT.B.5

BACKWARD 1

Choose and apply SSS, SAS, ASA, AAS • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

Two triangles share a side and have two more pairs of congruent sides. Criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q1/5

Two angles and a non-included side match. Criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q2/5

Which pair of facts plus a shared side gives SAS?

A. Two pairs of congruent sides not including the angle

B. Nothing more is needed

C. A pair of congruent sides and the congruent included angles

D. Two pairs of congruent angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q3/5

True or false?

A shared side is a valid pair for any criterion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q4/5

True or false?

AAA proves congruence.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q5/5

SET 46 • HSG-SRT.B.5

BACKWARD 2

Find hidden parts: shared sides, vertical angles, right angles • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Segments AB and CD cross at M. Which angles are vertical to ∠AMC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q1/5

True or false?

Perpendicular lines give congruent right angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q2/5

Which is a hidden fact in figures where two triangles share a side?

A. All angles are equal

B. The triangles are right

C. Nothing

D. The shared side is congruent to itself

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q3/5

A bisector divides ∠ABC into which two congruent angles?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q4/5

True or false?

Vertical angles are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q5/5

SET 46 • HSG-SRT.B.5

BACKWARD 3

Use corresponding parts to finish a proof • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

$\triangle ABC \cong \triangle ADC$. Which angle corresponds to $\angle BAC$?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q1/5

True or false?

Once triangles are congruent, all corresponding parts are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q2/5

△PBD ≅ △PBE. Which conclusion follows?

A. PD = PE

B. PB = PD

C. BD = PE

D. ∠D ≅ ∠B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q3/5

Two congruent angles form a linear pair. Each measures

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q4/5

True or false?

Corresponding parts can be used before the triangles are proved congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q5/5

SET 46 • HSG-SRT.B.5

FORWARD 1

Prove relationships that need two congruences in a row • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

In quadrilateral ABCD, AB = CD and AD = CB. Diagonals AC and BD meet at E. Prove that E is the midpoint of AC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q1/5

In the figure, AB = AC and BD = CE, with D on AB and E on AC. Prove that BE = CD.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q2/5

True or false?

Some proofs need one congruence to find a part used in a second congruence.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q3/5

In a two-step proof, the first congruence gives $\angle 1 \cong \angle 2$. This is used to

A. measure

B. end the proof

C. restart

D. supply a pair of parts for the second congruence

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q4/5

Explain why you cannot use corresponding parts of the second pair of triangles before proving them congruent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q5/5

SET 46 • HSG-SRT.B.5

FORWARD 2

Decide which given data are sufficient for a proof, and supply a missing condition • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

In $\triangle PQR$ and $\triangle PSR$ (sharing PR), $PQ = PS$. What one more fact would prove $\angle QPR \cong \angle SPR$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q1/5

A student claims a kite's diagonals bisect each other. Prove or disprove.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q2/5

True or false?

Given AB = DE, BC = EF, and ∠C ≅ ∠F, the triangles must be congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q3/5

Given $\angle A \cong \angle D$ and $AB = DE$ in $\triangle ABC$ and $\triangle DEF$, which extra fact gives ASA?

A. $\angle B \cong \angle E$

B. $\angle C \cong \angle F$

C. $BC = EF$

D. $AC = DF$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F2-Q4**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q4/5

Given AB = DE and AC = DF, name an angle condition that guarantees congruence.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q5/5

Use similarity criteria to prove relationships in figures

MAIN	Geometry main question	**HSG-SRT.B.5**
BACKWARD 1	Prove triangles similar with AA	**HSG-SRT.A.3**
BACKWARD 2	Turn a proportion into a product	**7.RP.A.2**
BACKWARD 3	Match corresponding sides of similar triangles	**HSG-SRT.A.2**
FORWARD 1	Use proved product relationships to compute	**HSG-SRT.B.5**
FORWARD 2	Prove relationships in circles with similar triangles	**HSG-C.A.2**

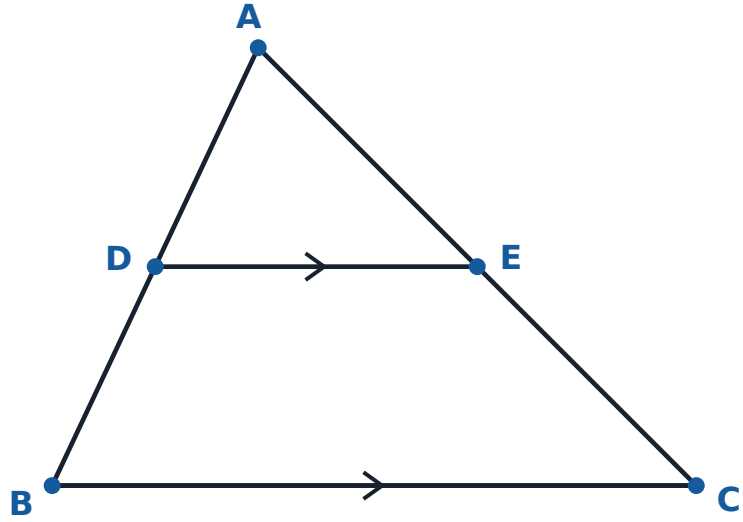
SET 47 • HSG-SRT.B.5

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

In △ABC, DE ∥ BC with D on AB and E on AC. Prove that AD · AC = AE · AB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q1/5

Chords AB and CD of a circle meet at P, and ∠BAD ≅ ∠BCD (given). Prove that PA · PB = PC · PD.

Work space

Conclusion: _____

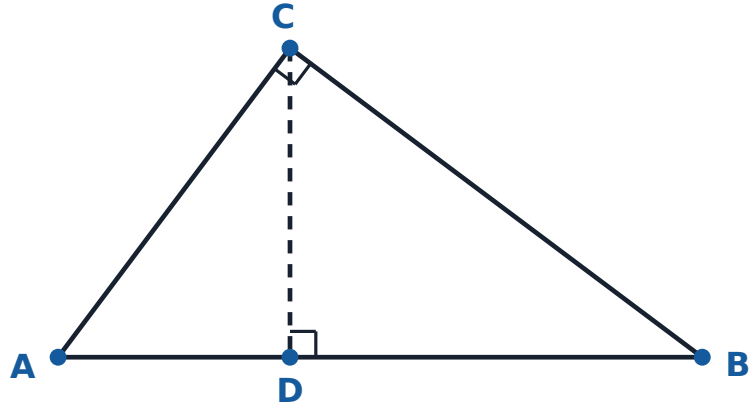
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q2/5

In right triangle ABC (right angle at C), CD is the altitude to AB. Prove that AC · BC = AB · CD.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q3/5

In trapezoid ABCD with AB ∥ DC, the diagonals AC and BD meet at E. Prove that AE/EC = AB/DC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q4/5

To prove a product relationship such as $PA \cdot PB = PC \cdot PD$, which is the usual strategy?

A. Use the Pythagorean Theorem only

B. Measure the four lengths

C. Find two similar triangles whose sides appear in the products

D. Assume the product is true

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q5/5

SET 47 • HSG-SRT.B.5

BACKWARD 1

Prove triangles similar with AA • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

True or false?

Vertical angles can be one pair of congruent angles in an AA proof.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q1/5

Two triangles share ∠A and have another pair of congruent angles. Criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q2/5

Parallel lines in a figure usually supply

A. equal side lengths

B. a scale factor

C. congruent corresponding or alternate interior angles

D. right angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q3/5

△PAD ~ △PCB. Which side of △PCB corresponds to PA?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q4/5

True or false?

Similarity requires all three pairs of sides to be equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q5/5

SET 47 • HSG-SRT.B.5

BACKWARD 2

Turn a proportion into a product • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

From a/b = c/d, write the product relationship.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q1/5

From PA/PC = PD/PB, write the product relationship.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q2/5

If $x/3 = 4/6$, then

A. $6x = 12$

B. $x = 8$

C. $3x = 24$

D. $4x = 18$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q3/5

True or false?

$a/b = c/d$ implies $ab = cd$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q4/5

True or false?

$2/5 = 6/15$ implies $2 \cdot 15 = 5 \cdot 6$.

A. True

B. False

SET 47 • HSG-SRT.B.5

BACKWARD 3

Match corresponding sides of similar triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

△ADE ~ △ABC. Write the proportion with AD, AB, AE, AC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q1/5

True or false?

In $\triangle ABE \sim \triangle CDE$, AB corresponds to CD.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q2/5

△ACB ~ △ADC. Which side corresponds to CB?

A. AD

B. AB

C. DC

D. AC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q3/5

△XYZ ~ △UVW. Write all three equal ratios.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q4/5

True or false?

Any two sides of similar triangles are proportional.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q5/5

SET 47 • HSG-SRT.B.5

FORWARD 1

Use proved product relationships to compute • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

Chords AB and CD meet at P with PA = 4, PB = 9, PC = 6. Find PD.

PD = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q1/5

In right triangle ABC (right angle at C), AC = 6, BC = 8, AB = 10. Find the altitude CD to AB.

CD = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q2/5

True or false?

If two chords meet inside a circle, the products of their segments are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q3/5

Trapezoid diagonals: AB = 18, DC = 12, AE = 9. Find EC.

A. 8

B. 6

C. 4

D. 13.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q4/5

DE ∥ BC in △ABC with AD = 3, AB = 9, AC = 12. Find AE.

AE = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q5/5

SET 47 • HSG-SRT.B.5

FORWARD 2

Prove relationships in circles with similar triangles • High School — Geometry • Geometry: Circles • HSG-C.A.2

From point P outside a circle, a tangent touches the circle at T and a secant meets it at A and B (A between P and B). Given $\angle PTA \cong \angle PBT$, prove that $PT^2 = PA \cdot PB$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q1/5

A tangent segment from P is 8 and the secant from P has outer part 4. Find the whole secant length.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q2/5

True or false?

Two secants from an outside point satisfy $PA \cdot PB = PC \cdot PD$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q3/5

From P, secants meet a circle at A, B and at C, D with PA = 3, PB = 12, PC = 4. Find PD.

A. 7

B. 1

C. 16

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q4/5

Explain why the angle facts used in these circle proofs depend on inscribed angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q5/5

Understand trigonometric ratios as properties of an angle, by similarity

MAIN	Geometry main question	**HSG-SRT.C.6**
BACKWARD 1	Similar triangles have equal side ratios within each triangle	**HSG-SRT.A.2**
BACKWARD 2	Name the opposite side, adjacent side, and hypotenuse for an ang	**HSG-SRT.A.3**
BACKWARD 3	Write and simplify ratios	**6.RP.A.1**
FORWARD 1	Find trig ratios from one ratio, using a similar right triangle	**HSG-SRT.C.6**
FORWARD 2	The unit circle extends the ratios beyond acute angles	**HSF-TF.A.2**

SET 48 • HSG-SRT.C.6

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

Right triangles ABC and DEF have right angles at C and F and $\angle A \cong \angle D$. Explain why $BC/AB = EF/DE$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q1/5

Two right triangles each have a 30° angle. In one, the side opposite 30° is 4 and the hypotenuse is 8. In the other, the hypotenuse is 15. Find the side opposite 30°, and explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q2/5

In right triangle PQR, the right angle is at R, PR = 5, QR = 12, PQ = 13. Write sin P, cos P, and tan P.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q3/5

Why is sin 40° the same in every right triangle with a 40° angle?

A. Every hypotenuse is 1

B. The sides are always 3, 4, 5

C. All such triangles are similar, so the opposite-to-hypotenuse ratio is the same

D. Sine depends on the triangle's size

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q4/5

True or false?

Doubling every side of a right triangle doubles the tangent of its acute angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q5/5

SET 48 • HSG-SRT.C.6

BACKWARD 1

Similar triangles have equal side ratios within each triangle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Triangles with sides 3, 4, 5 and 6, 8, 10 are similar. Compare 3/5 and 6/10.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q1/5

True or false?

If $\triangle ABC \sim \triangle DEF$, then $AB/AC = DE/DF$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q2/5

$\triangle ABC \sim \triangle DEF$ with $BC/AB = 0.8$. Find EF/DE.

A. 0.8

B. Cannot tell

C. 1.25

D. 0.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q3/5

A triangle has legs 2 and 3. A similar one has shorter leg 8. Find the longer leg.

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q4/5

True or false?

Similar triangles have equal side lengths.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q5/5

SET 48 • HSG-SRT.C.6

BACKWARD 2

Name the opposite side, adjacent side, and hypotenuse for an angle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

In right triangle ABC with right angle at C, which side is opposite ∠A?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q1/5

In right triangle ABC with right angle at C, which side is adjacent to ∠A (not the hypotenuse)?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q2/5

The hypotenuse is always

A. next to the right angle

B. the shortest side

C. opposite the smallest angle

D. opposite the right angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q3/5

True or false?

The side opposite $\angle A$ is adjacent to $\angle B$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q4/5

True or false?

The opposite side of an angle depends on the triangle's size.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q5/5

SET 48 • HSG-SRT.C.6

BACKWARD 3

Write and simplify ratios • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1

Write 12 : 13 as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q1/5

Simplify 8/10.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q2/5

Which equals 0.75?

A. 3/4

B. 7/5

C. 3/7

D. 4/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q3/5

True or false?

$5/13 \approx 0.385$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q4/5

True or false?

A ratio changes when both quantities are tripled.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q5/5

SET 48 • HSG-SRT.C.6

FORWARD 1

Find trig ratios from one ratio, using a similar right triangle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

For an acute angle θ, $\sin\theta = 3/5$. Find $\cos\theta$ and $\tan\theta$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q1/5

$\tan \theta = 5/12$ for an acute angle θ. Find $\sin \theta$.

$\sin \theta =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q2/5

True or false?

For any acute angle, $(\sin \theta)^2 + (\cos \theta)^2 = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q3/5

$\cos\theta = 0.6$. Which is $\sin\theta$ (θ acute)?

A. 1.6

B. 0.8

C. 0.4

D. 0.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q4/5

Explain why $\sin\theta < 1$ for every acute angle θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q5/5

SET 48 • HSG-SRT.C.6

FORWARD 2

The unit circle extends the ratios beyond acute angles • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2

On a unit circle, the point for angle θ is (cos θ, sin θ). Find the point for 90°.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q1/5

Find the coordinates of the point on the unit circle at 30°.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q2/5

True or false?

For acute angles, the unit-circle definition agrees with the right-triangle definition.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q3/5

sin 150° on the unit circle equals

A. $-\frac{1}{2}$

B. $\frac{1}{2}$

C. $-\sqrt{3}/2$

D. $\sqrt{3}/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F2-Q4**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q4/5

Explain why the unit circle lets us define sine for 120°, when no right triangle has a 120° acute angle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q5/5

Explain and use the relationship between the sine and cosine of complementary angles

MAIN	Geometry main question	**HSG-SRT.C.7**
BACKWARD 1	Complementary angles add to 90°	**7.G.B.5**
BACKWARD 2	Opposite and adjacent sides depend on the chosen angle	**HSG-SRT.C.6**
BACKWARD 3	Angle sum in a right triangle	**8.G.A.5**
FORWARD 1	Relationships between tangents of complementary angles	**HSG-SRT.C.7**
FORWARD 2	The Pythagorean identity	**HSF-TF.C.8**

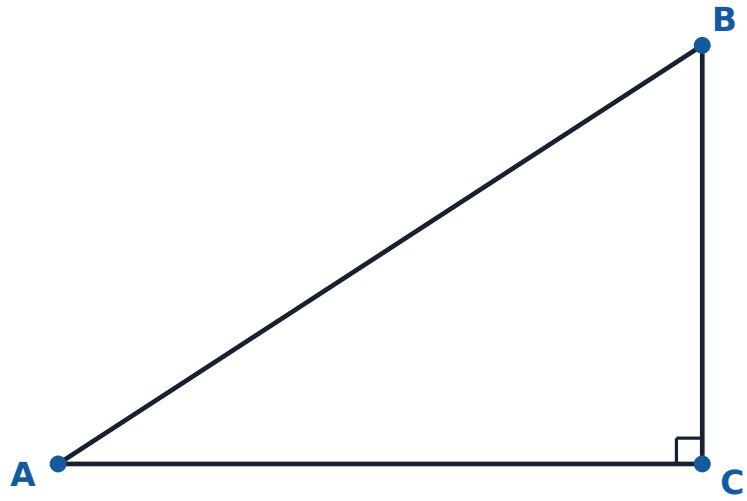
SET 49 • HSG-SRT.C.7

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7

Right triangle ABC has its right angle at C. Explain why sin A = cos B.

Work space



Not drawn to scale.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q1/5

$\sin 25° = \cos x°$, where $0 < x < 90$. Find x.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q2/5

θ is acute and $\cos\theta = 0.6$. Find $\sin(90° - \theta)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q3/5

Both angles are acute and $\sin(2x + 10)° = \cos(3x)°$. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q4/5

Which expression equals cos 72°?

A. cos 18°

B. sin 72°

C. tan 18°

D. sin 18°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q5/5

SET 49 • HSG-SRT.C.7

BACKWARD 1

Complementary angles add to 90° • Grade 7 • Geometry • 7.G.B.5

Find the complement of 37°.

Complement: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q1/5

True or false?

The two acute angles of a right triangle are complementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q2/5

Angles x and 2x are complementary. x =

A. 90°

B. 30°

C. 45°

D. 60°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q3/5

The complement of θ is $90° - \theta$. If $\theta = 64°$, find it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q4/5

True or false?

Complementary angles must be adjacent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q5/5

SET 49 • HSG-SRT.C.7

BACKWARD 2

Opposite and adjacent sides depend on the chosen angle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

In right triangle ABC (right angle at C), name the side adjacent to ∠B.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q1/5

True or false?

The side opposite $\angle A$ is the side adjacent to $\angle B$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q2/5

cos B equals

A. AC/AB

B. AC/BC

C. BC/AB

D. BC/AC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q3/5

Write sin A using the sides of right triangle ABC (right angle at C).

sin A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q4/5

True or false?

sin A and cos A are always equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q5/5

SET 49 • HSG-SRT.C.7

BACKWARD 3

Angle sum in a right triangle • Grade 8 • Geometry • 8.G.A.5

A right triangle has an acute angle of 28°. Find the other acute angle.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q1/5

True or false?

A right triangle can have two right angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q2/5

In a right triangle, the acute angles are x and x + 20. x =

A. 35°

B. 70°

C. 45°

D. 55°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q3/5

Why do the acute angles of a right triangle add to 90°?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q4/5

True or false?

The two acute angles of an isosceles right triangle are 45° each.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q5/5

SET 49 • HSG-SRT.C.7

FORWARD 1

Relationships between tangents of complementary angles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7

Explain why $\tan\theta \cdot \tan(90° - \theta) = 1$ for an acute angle θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q1/5

tan 20° ≈ 0.364. Estimate tan 70°.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q2/5

True or false?

$\sin\theta = \cos\theta$ exactly when $\theta = 45°$ (for acute θ).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q3/5

Which equals 1 for every acute θ?

A. $\tan\theta + \tan(90° - \theta)$

B. $\sin\theta \cdot \cos\theta$

C. $\tan\theta \cdot \tan(90° - \theta)$

D. $\sin\theta + \cos\theta$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q4/5

Show that $\sin\theta / \cos(90° - \theta) = 1$ for acute θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q5/5

SET 49 • HSG-SRT.C.7

FORWARD 2

The Pythagorean identity • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8

Prove that $(\sin \theta)^2 + (\cos \theta)^2 = 1$ for every acute angle θ.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q1/5

$\sin\theta = 0.28$ for acute θ. Find $\cos\theta$.

$\cos\theta =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q2/5

True or false?

$(\sin 35°)^2 + (\cos 35°)^2 = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q3/5

$\cos\theta = 12/13$ (θ acute). $\sin\theta =$

A. 5/13

B. 1/13

C. 12/5

D. 13/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-F2-Q4**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q4/5

Use complementary angles to show $(\sin 20°)^2 + (\sin 70°)^2 = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q5/5

Find unknown sides of right triangles with trigonometric ratios in applied problems

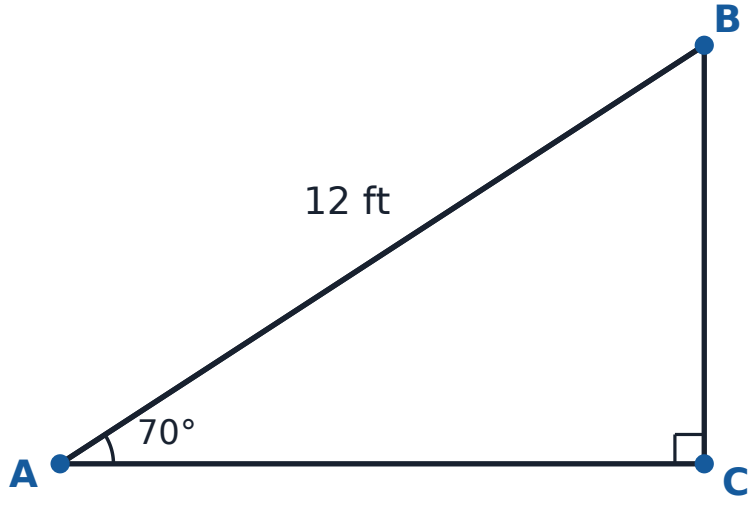
MAIN	Geometry main question	**HSG-SRT.C.8**
BACKWARD 1	Choose the trigonometric ratio that links the known and unknown.	**HSG-SRT.C.6**
BACKWARD 2	Solve one-variable equations with a multiplication or division	**HSA-REI.B.3**
BACKWARD 3	Round to a stated precision and keep track of units	**HSN-Q.A.3**
FORWARD 1	Find a height from two angle-of-elevation observations	**HSG-SRT.C.8**
FORWARD 2	Find a feasible location under a clearance constraint	**HSG-MG.A.3**

SET 50 • HSG-SRT.C.8

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

A 12 ft ladder leans against a wall, making a 70° angle with the level ground. How high up the wall does it reach, to the nearest tenth of a foot?



Not drawn to scale.

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q1/5

A wheelchair ramp rises 2.5 ft and makes an 8° angle with the ground. How long is the ramp surface, to the nearest tenth of a foot?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q2/5

From a point 25 m from the base of a tree, on level ground, the angle of elevation to the top is 38°. How tall is the tree, to the nearest tenth of a meter?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q3/5

A kite string 60 m long makes a 50° angle with the level ground. About how high is the kite above the point where the string is held (nearest meter)?

A. 78 m

B. 46 m

C. 39 m

D. 72 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q4/5

A right triangle has hypotenuse 20 and a 32° angle. Find the leg adjacent to the 32° angle, to the nearest tenth.

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q5/5

SET 50 • HSG-SRT.C.8

BACKWARD 1

Choose the trigonometric ratio that links the known and unknown sides • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

You know the hypotenuse and an angle and want the opposite side. Which ratio?

A. none

B. sine

C. cosine

D. tangent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q1/5

You know the adjacent side and an angle and want the opposite side. Which ratio?

A. tangent

B. none

C. cosine

D. sine

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q2/5

Which ratio uses the adjacent side and the hypotenuse?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q3/5

True or false?

To find the hypotenuse from the opposite side and the angle, use sine.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q4/5

True or false?

Tangent uses the hypotenuse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q5/5

SET 50 • HSG-SRT.C.8

BACKWARD 2

Solve one-variable equations with a multiplication or division • High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve $x/12 = 0.94$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q1/5

Solve 0.14 = 2.5/x.

x ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q2

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q2/5

If $\sin 30° = h/10$, then $h =$

A. $10 \div \sin 30°$

B. $\sin 30° \div 10$

C. $10 \sin 30°$

D. $30 \div 10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-B2-Q3**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q3/5

True or false?

From $0.5 = 4/x$ we get $x = 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q4/5

True or false?

From $\tan 40° = 20/x$ we get $x = 20 \tan 40°$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q5/5

SET 50 • HSG-SRT.C.8

BACKWARD 3

Round to a stated precision and keep track of units • High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3

Round 11.2763 to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q1

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q1/5

A calculation gives 19.532 m. Report it to the nearest tenth of a meter.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q2

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q2/5

True or false?

Rounding in the middle of a calculation can change the final answer.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q3

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q3/5

A height is measured in feet and the angle in degrees. The answer's unit is

A. degrees

B. feet

C. feet per degree

D. no unit

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-B3-Q4**

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q4/5

True or false?

17.96 rounded to the nearest tenth is 17.9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q5

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q5/5

SET 50 • HSG-SRT.C.8

FORWARD 1

Find a height from two angle-of-elevation observations • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

From point A on level ground the angle of elevation to the top of a tower is 25°. From point B, 40 m closer, it is 40°. Find the tower's height to the nearest tenth of a meter.

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q1/5

In the same situation, how far is point B from the base of the tower, to the nearest tenth of a meter?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q2/5

True or false?

With only one angle of elevation and no distance, the height cannot be found.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q3/5

Which equations describe two observations 40 m apart (angles 25° and 40°, distance x from the closer point)?

A. $h = x \tan 40°$ and $h = (x + 40) \tan 25°$

B. $h = 40 \tan 25°$

C. $h = x \tan 25°$ and $h = (x + 40) \tan 40°$

D. $h = x \sin 40°$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q4/5

Explain why the closer observation point has the larger angle of elevation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q5/5

SET 50 • HSG-SRT.C.8

FORWARD 2

Find a feasible location under a clearance constraint • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

A ramp must rise 3 ft, and its angle with the ground may be at most 4.8°. What is the shortest allowed horizontal run, to the nearest foot?

Run: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q1/5

A camera on a 6 m pole must see a point on the ground at an angle of depression of at least 20°. How far from the pole can that point be, at most, to the nearest tenth of a meter?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q2/5

True or false?

Rounding down the run of a ramp with a maximum angle can make the ramp too steep.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q3/5

A ladder must reach 10 ft up and make an angle of at most 75° with the ground. The shortest ladder that works is about

A. 10.4 ft

B. 10.0 ft

C. 9.7 ft

D. 2.7 ft

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q4/5

A building code allows stairs with an angle of at most 37°. A staircase must rise 9 ft. What is the shortest horizontal run, to the nearest tenth of a foot?

Run: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q5/5

Find unknown angles with inverse trigonometric ratios in applied problems

MAIN	Geometry main question	**HSG-SRT.C.8**
BACKWARD 1	Inverse trig undoes a trig ratio	**HSG-SRT.C.6**
BACKWARD 2	Choose the ratio from the two known sides	**HSG-SRT.C.6**
BACKWARD 3	Use degree mode and round angles	**HSN-Q.A.3**
FORWARD 1	Solve a right triangle completely from two sides	**HSG-SRT.C.8**
FORWARD 2	Find directions (bearings) with inverse trigonometry	**HSG-SRT.C.8**

SET 51 • HSG-SRT.C.8

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

A ramp rises 1 m over a horizontal distance of 12 m. What angle does it make with the ground, to the nearest tenth of a degree?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q1/5

A 6 m ladder reaches 5.5 m up a vertical wall. What angle does it make with the level ground, to the nearest tenth of a degree?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q2/5

A 10 m flagpole casts a 14 m shadow on level ground. Find the angle of elevation of the sun, to the nearest tenth of a degree.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q3/5

A right triangle has legs 7 and 24. What is its smaller acute angle, to the nearest tenth of a degree?

A. 73.7°

B. 32.5°

C. 16.3°

D. 26.3°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q4/5

A road rises 6 m for every 100 m of horizontal distance (a 6% grade). What angle does it make with the horizontal, to the nearest tenth of a degree?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q5/5

SET 51 • HSG-SRT.C.8

BACKWARD 1

Inverse trig undoes a trig ratio • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

$\sin \theta = 0.5$ for an acute angle θ. Find θ.

$\theta =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q1/5

$\tan \theta = 1$ for an acute angle θ. Find θ.

$\theta =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q2/5

Which gives an angle from a ratio?

A. cos

B. $\sin^{-1}$

C. tan

D. sin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q3/5

True or false?

$\cos^{-1}(0.5) = 60°$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q4/5

True or false?

$\sin^{-1}(x)$ means $1 \div \sin(x)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q5/5

SET 51 • HSG-SRT.C.8

BACKWARD 2

Choose the ratio from the two known sides • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

You know the opposite side and the hypotenuse. Which inverse function finds the angle?

A. none

B. $\sin^{-1}$

C. $\cos^{-1}$

D. $\tan^{-1}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q1/5

You know both legs. Which inverse function finds an acute angle?

A. $\sin^{-1}$

B. $\cos^{-1}$

C. none

D. $\tan^{-1}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q2/5

Rise 5 and run 12. Write the ratio for the angle with the ground.

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q3/5

True or false?

Adjacent side and hypotenuse give the angle by $\cos^{-1}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q4/5

True or false?

Opposite side and hypotenuse give the angle by $\tan^{-1}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q5/5

SET 51 • HSG-SRT.C.8

BACKWARD 3

Use degree mode and round angles • High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3

Round 4.7636° to the nearest tenth of a degree.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q1

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q1/5

True or false?

A calculator in radian mode gives $\sin^{-1}(0.5) \approx 0.524$, not 30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q2

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q2/5

An acute angle of a right triangle must be

A. more than 90°

B. between 0° and 90°

C. any angle

D. exactly 45°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-B3-Q3**

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q3/5

A computed angle is 112°. Can it be an acute angle of a right triangle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q4

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q4/5

True or false?

35.537° to the nearest tenth is 35.6°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-B3-Q5**

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q5/5

SET 51 • HSG-SRT.C.8

FORWARD 1

Solve a right triangle completely from two sides • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

A right triangle has legs 9 and 14. Find the hypotenuse and both acute angles (nearest tenth).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q1/5

A right triangle has hypotenuse 17 and one leg 8. Find the other leg and both acute angles (nearest tenth of a degree).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q2/5

True or false?

Once one acute angle is known, the other is 90° minus it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q3/5

Legs 5 and 5. Angles?

A. 45° and 45°

B. 50° and 40°

C. 30° and 60°

D. 5° and 85°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q4/5

Explain how to check a solved right triangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q5/5

SET 51 • HSG-SRT.C.8

FORWARD 2

Find directions (bearings) with inverse trigonometry • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

A boat travels 8 km east and then 5 km north. Find the angle north of east of its straight-line direction from the start, to the nearest tenth of a degree.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q1/5

A hiker walks 3 km north and 4 km west. What is the bearing angle west of north from the start, to the nearest tenth of a degree?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q2/5

True or false?

Bearing problems become right-triangle problems when the motions are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q3/5

A drone flies 120 m horizontally and rises 50 m. Its angle of climb is about

A. 24.6°

B. 22.6°

C. 67.4°

D. 50°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-F2-Q4**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q4/5

A ship is 10 km east and 10 km south of port. Find the straight-line distance and the angle south of east.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q5/5

Solve right triangles in applied problems with trigonometry and the Pythagorean Theorem

MAIN	Geometry main question	**HSG-SRT.C.8**
BACKWARD 1	Use the Pythagorean Theorem to find a side	**8.G.B.7**
BACKWARD 2	Interpret angles of elevation and depression	**HSG-SRT.C.6**
BACKWARD 3	Choose sine, cosine, or tangent	**HSG-SRT.C.6**
FORWARD 1	Solve problems that need two right triangles	**HSG-SRT.C.8**
FORWARD 2	Solve non-right triangles with the Law of Sines	**HSG-SRT.D.11**

SET 52 • HSG-SRT.C.8

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

From the top of an 80 m cliff, the angle of depression to a boat is 25°. How far is the boat from the base of the cliff, to the nearest tenth of a meter? How far is it from the observer?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q1/5

A plane flying at an altitude of 2,000 m is seen at an angle of elevation of 14°. Find the horizontal ground distance and the line-of-sight distance to the plane, to the nearest meter.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q2/5

A ramp is 15 ft long and rises 2 ft. Find its horizontal run (nearest tenth of a foot) and its angle with the ground (nearest tenth of a degree).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q3/5

A 25 ft guy wire from the top of a pole meets the level ground 9 ft from the pole. How tall is the pole and what angle does the wire make with the ground (nearest tenth)?

A. 23.3 ft; 68.9°

B. 26.6 ft; 68.9°

C. 16.0 ft; 21.1°

D. 23.3 ft; 21.1°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q4/5

A right triangle has hypotenuse 25 and a 40° angle. Find both legs and the other acute angle (legs to the nearest tenth).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q5/5

SET 52 • HSG-SRT.C.8

BACKWARD 1

Use the Pythagorean Theorem to find a side • Grade 8 • Geometry • 8.G.B.7

A right triangle has hypotenuse 25 and a leg 9. Find the other leg, to the nearest tenth.

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q1/5

Legs 8 and 15. Find the hypotenuse.

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q2/5

True or false?

In a right triangle, the hypotenuse is the longest side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q3/5

Hypotenuse 13, leg 5. Other leg?

A. 12

B. √194

C. 8

D. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q4/5

True or false?

$a^2 + b^2 = c^2$ works for every triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q5/5

SET 52 • HSG-SRT.C.8

BACKWARD 2

Interpret angles of elevation and depression • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

True or false?

An angle of depression is measured down from the horizontal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q1/5

The angle of depression from a cliff top to a boat is 25°. What is the angle of elevation from the boat to the cliff top?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q2/5

Which side is horizontal in an angle-of-elevation triangle?

A. The hypotenuse

B. The leg along the ground

C. The vertical leg

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q3/5

An observer looks up at 30° to a bird. Is that an angle of elevation or depression?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q4/5

True or false?

The angle of depression is measured from the vertical line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q5/5

SET 52 • HSG-SRT.C.8

BACKWARD 3

Choose sine, cosine, or tangent • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

Known: opposite side and angle. Wanted: hypotenuse. Ratio?

A. none

B. tangent

C. cosine

D. sine

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q1/5

Known: adjacent side and angle. Wanted: opposite side. Ratio?

A. tangent

B. cosine

C. none

D. sine

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-B3-Q2**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q2/5

Known: hypotenuse and angle. Wanted: adjacent side. Ratio?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q3/5

True or false?

Two different ratios can lead to the same correct side if used correctly.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q4/5

True or false?

Tangent relates the hypotenuse and a leg.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q5/5

SET 52 • HSG-SRT.C.8

FORWARD 1

Solve problems that need two right triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

From a window 20 m up, the angle of elevation to the top of a building across the street is 30° and the angle of depression to its base is 40°. How tall is the building, to the nearest tenth of a meter?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q1/5

Two observers 100 m apart on opposite sides of a tower see its top at 35° and 50°. Find the tower's height, to the nearest tenth of a meter.

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q2/5

True or false?

A shared side links two right triangles in such problems.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q3/5

In the two-observer problem, which equation is correct?

A. $h = 100 \tan 85°$

B. $h/\sin 35° + h/\sin 50° = 100$

C. $h/\tan 35° + h/\tan 50° = 100$

D. $h \tan 35° + h \tan 50° = 100$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q4/5

Explain why the two-observer problem cannot be solved with one triangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q5/5

SET 52 • HSG-SRT.C.8

FORWARD 2

Solve non-right triangles with the Law of Sines • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

In $\triangle ABC$, $A = 40°$, $B = 65°$, and $a = 12$. Find b to the nearest tenth.

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q1/5

In $\triangle ABC$, $A = 52°$, $C = 70°$, and $c = 20$. Find a to the nearest tenth.

$a =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q2/5

True or false?

The Law of Sines can solve a triangle with no right angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q3/5

Which data allow the Law of Sines directly?

A. Two angles and any side

B. Two sides and the included angle

C. Three sides

D. Three angles only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q4/5

Explain how dropping an altitude turns a non-right triangle into two right triangles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q5/5

SET 53 • HSG-C.A.1 • CIRCLES

Prove that all circles are similar

MAIN	Geometry main question	**HSG-C.A.1**
BACKWARD 1	A dilation multiplies distances by the scale factor	**HSG-SRT.A.1.b**
BACKWARD 2	Translate a point to another point	**8.G.A.3**
BACKWARD 3	A circle is all points at one distance from its center	**HSG-CO.A.1**
FORWARD 1	Find a single dilation that maps one circle onto another	**HSG-C.A.1**
FORWARD 2	Arcs of similar circles scale with the radius	**HSG-C.B.5**

SET 53 • HSG-C.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.1

Circle C has center A and radius r, and circle D has center B and radius s. Prove that the two circles are similar.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q1/5

Circle C_1 has center (1, 2) and radius 3. Circle C_2 has center (−4, 0) and radius 6. Describe a similarity transformation that maps C_1 onto C_2 and explain why it works.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q2/5

In the proof that all circles are similar, why must the scale factor be the ratio of the radii (s/r)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q3/5

Explain why two circles with the same center but different radii, r and s, are similar using only ONE transformation.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q4/5

True or false?

A circle can be similar to an ellipse that is not a circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q5/5

SET 53 • HSG-C.A.1

BACKWARD 1

A dilation multiplies distances by the scale factor • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b

A dilation with factor 2.5 is applied to a segment of length 4. How long is the image?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q1/5

True or false?

A dilation about point O keeps O fixed and multiplies each distance from O by the factor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q2/5

Which factor maps a circle of radius 4 about its center to one of radius 10?

A. 0.4

B. 2.5

C. 6

D. 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q3/5

A circle of radius 9 is dilated about its center by ⅓. New radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q4/5

True or false?

A dilation can turn a circle into a square.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q5/5

SET 53 • HSG-C.A.1

BACKWARD 2

Translate a point to another point • Grade 8 • Geometry • 8.G.A.3

Which translation moves (1, 2) onto (−4, 0)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q1/5

True or false?

A translation keeps the radius of a circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q2/5

A circle centered at (3, 3) is translated by (x, y) → (x − 3, y + 1). New center?

A. (0, 4)

B. (6, 2)

C. (0, 3)

D. (3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q3/5

A translation moves (2, −1) to (7, 4). Describe it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q4/5

True or false?

A translation changes the size of a figure.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q5/5

SET 53 • HSG-C.A.1

BACKWARD 3

A circle is all points at one distance from its center • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

True or false?

Every point of a circle is the same distance from its center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q1/5

A circle has center (0, 0) and passes through (3, 4). Its radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q2/5

A circle is completely determined by

A. its color

B. its center and radius

C. one point on it

D. its center only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q3/5

Two circles have the same center and the same radius. What can you say?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q4/5

True or false?

A circle's radius can be different in different directions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q5/5

SET 53 • HSG-C.A.1

FORWARD 1

Find a single dilation that maps one circle onto another • High School — Geometry • Geometry: Circles • HSG-C.A.1

Circle P: center (0, 0), radius 2. Circle Q: center (6, 0), radius 4. Find the center of the dilation with a positive factor that maps P onto Q.

Center: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q1/5

For the same circles, what is the scale factor?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q2/5

True or false?

Two circles with different radii can always be mapped onto each other by one dilation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q3/5

Circles with equal radii and different centers are related by

A. nothing

B. a translation (no dilation needed)

C. a dilation with factor 2

D. a reflection only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q4/5

Circle A has radius 3 and circle B has radius 12, both centered at the origin. Which dilation maps A onto B?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q5/5

SET 53 • HSG-C.A.1

FORWARD 2

Arcs of similar circles scale with the radius • High School — Geometry • Geometry: Circles • HSG-C.B.5

Two circles have radii 3 and 7. A 60° central angle cuts off an arc in each. What is the ratio of the arc lengths?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q1/5

A 90° arc of a circle with radius 4 has length 2π. How long is a 90° arc of a circle with radius 10?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q2/5

True or false?

For a fixed central angle, arc length divided by radius is the same in every circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q3/5

An arc of length 6 in a circle of radius 2 corresponds, by similarity, to what arc length in a circle of radius 5 (same angle)?

A. 15

B. 2.4

C. 30

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q4/5

Explain how the similarity of circles leads to radian measure.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q5/5

SET 54 • HSG-C.A.2 • CIRCLES

Relate inscribed angles and central angles

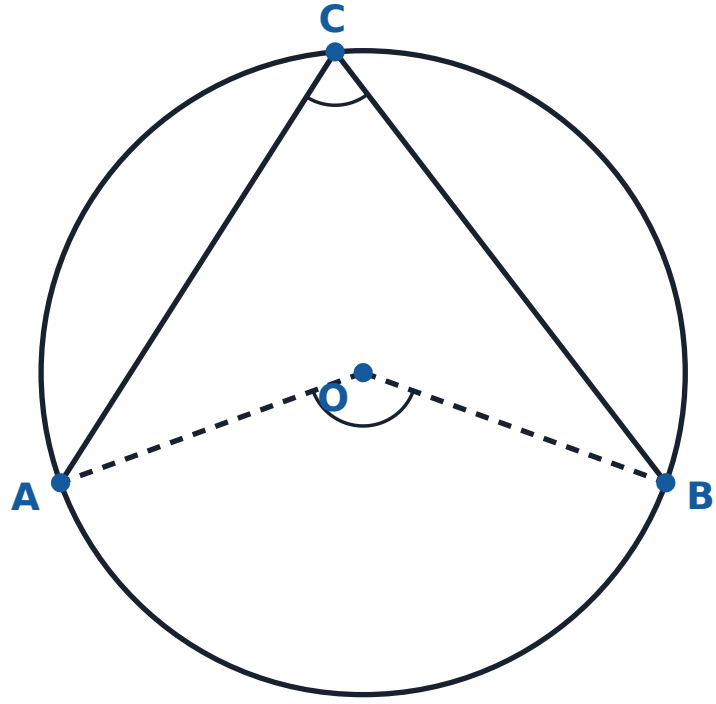
MAIN	Geometry main question	**HSG-C.A.2**
BACKWARD 1	Base angles of an isosceles triangle are congruent	**HSG-CO.C.10**
BACKWARD 2	The exterior angle theorem	**HSG-CO.C.10**
BACKWARD 3	A central angle and its arc have the same measure	**4.MD.C.5.a**
FORWARD 1	Prove the inscribed angle theorem when the center is inside or out…	**HSG-C.A.2**
FORWARD 2	Inscribed angles on the same arc, and arc applications	**HSG-C.A.2**

SET 54 • HSG-C.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.2

In circle O, central angle AOB measures 110°. Inscribed angle ACB intercepts the same arc AB. Find m∠ACB.



m∠ACB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q1/5

In circle O, inscribed angle PQR measures 38°. Find the measure of arc PR (the arc it intercepts) and of central angle POR.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q2/5

In circle O, AC is a diameter and B is another point on the circle. Prove that m∠BAC = ½ m∠BOC (the inscribed angle theorem when one side of the angle is a diameter).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q3/5

Inscribed angle XYZ and central angle XOZ intercept the same arc XZ, and m∠XYZ = 47°. What is m∠XOZ?

A. 23.5°

B. 94°

C. 47°

D. 133°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q4/5

True or false?

An inscribed angle and a central angle that intercept the same arc have equal measures.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q5/5

SET 54 • HSG-C.A.2

BACKWARD 1

Base angles of an isosceles triangle are congruent • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In $\triangle AOB$, $OA = OB$ and $m\angle OAB = 35°$. Find $m\angle OBA$ and $m\angle AOB$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q1/5

True or false?

Two radii of a circle and the chord joining their ends form an isosceles triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q2/5

Isosceles triangle with vertex angle 80°. Each base angle?

A. 100°

B. 80°

C. 50°

D. 40°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q3/5

In $\triangle OPQ$ with $OP = OQ$, $m\angle P = 2x$ and $m\angle Q = 50°$. Find x.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q4/5

True or false?

In an isosceles triangle, the vertex angle always equals a base angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q5/5

SET 54 • HSG-C.A.2

BACKWARD 2

The exterior angle theorem • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

A triangle has remote interior angles 30° and 30°. Find the exterior angle.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q1/5

True or false?

An exterior angle equals the sum of the two remote interior angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q2/5

Exterior angle 100°, one remote interior angle 45°. The other?

A. 45°

B. 55°

C. 80°

D. 135°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q3/5

If both remote interior angles equal x, the exterior angle is

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q4/5

True or false?

An exterior angle equals the adjacent interior angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q5/5

SET 54 • HSG-C.A.2

BACKWARD 3

A central angle and its arc have the same measure • Grade 4 • Measurement & Data • 4.MD.C.5.a

A central angle cuts off ¼ of a circle. Its measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q1/5

True or false?

The measure of a minor arc equals its central angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q2/5

An arc measures 70°. Its central angle measures

A. 70°

B. 140°

C. 290°

D. 35°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q3/5

A central angle of 150°. What fraction of the circle is its arc?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q4/5

True or false?

All arcs of 60° have the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q5/5

SET 54 • HSG-C.A.2

FORWARD 1

Prove the inscribed angle theorem when the center is inside or outside the angle • High School — Geometry • Geometry: Circles • HSG-C.A.2

Inscribed angle ABC has the center O inside it. Prove that m∠ABC = ½ m∠AOC. (Draw diameter BD.)

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q1/5

Inscribed angle ABC has the center O outside it. Prove that m∠ABC = ½ m∠AOC. (Draw diameter BD.)

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q2/5

True or false?

The inscribed angle theorem needs three cases for a complete proof.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q3/5

In the "center inside" case, the two diameter-case results are

A. averaged

B. multiplied

C. added

D. subtracted

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q4/5

Why can't the proof stop after the diameter case?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q5/5

SET 54 • HSG-C.A.2

FORWARD 2

Inscribed angles on the same arc, and arc applications • High School — Geometry • Geometry: Circles • HSG-C.A.2

Inscribed angles ∠ADB and ∠ACB intercept the same arc AB, and m∠ADB = 41°. Find m∠ACB.

m∠ACB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q1/5

Prove that two inscribed angles that intercept the same arc are congruent.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q2/5

True or false?

An inscribed angle that intercepts a 200° arc measures 100°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q3/5

Points A, B, C, D are on a circle with arc AB = 80° and arc CD = 50° (arcs not containing the other points). Chords AC and BD meet at E inside. m∠AEB?

A. 30°

B. 15°

C. 65°

D. 130°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q4/5

An inscribed angle measures 3x + 10 and its intercepted arc measures 110°. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q5/5

SET 55 • HSG-C.A.2 • CIRCLES

Inscribed angles on a diameter are right angles

MAIN	Geometry main question	**HSG-C.A.2**
BACKWARD 1	Use the inscribed angle theorem	**HSG-C.A.2**
BACKWARD 2	Isosceles triangles from radii	**HSG-CO.C.10**
BACKWARD 3	Triangle angle sum	**8.G.A.5**
FORWARD 1	Use the converse to find a circle through a right triangle	**HSG-C.A.2**
FORWARD 2	Use the diameter property to construct tangents from an external…	**HSG-C.A.4**

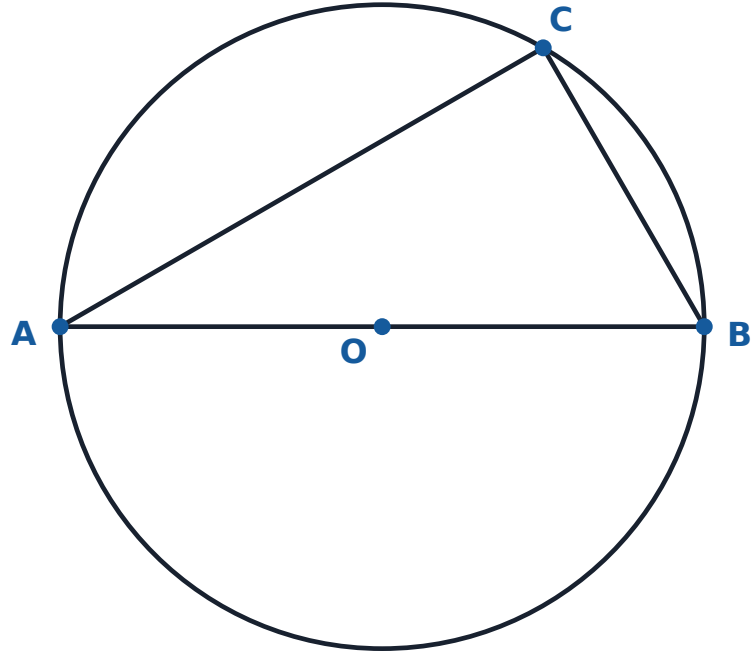
SET 55 • HSG-C.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.2

AB is a diameter of circle O and C is any other point on the circle. Prove that ∠ACB is a right angle.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q1/5

AB is a diameter of a circle and C is on the circle. AC = 6 and BC = 8. Find AB and the radius.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q2/5

AB is a diameter of circle O and C is on the circle with $m\angle CAB = 34°$. Find $m\angle CBA$.

$m\angle CBA =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q3/5

Triangle PQR is inscribed in a circle with PR a diameter. Which statement must be true?

A. $PQ = QR$

B. $\angle Q = 90°$

C. $PR = PQ$

D. $\angle P = 90°$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q4/5

True or false?

Every triangle inscribed in a circle is a right triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q5/5

SET 55 • HSG-C.A.2

BACKWARD 1

Use the inscribed angle theorem • High School — Geometry • Geometry: Circles • HSG-C.A.2

An inscribed angle intercepts an arc of 140°. Its measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q1/5

True or false?

An inscribed angle is half its intercepted arc.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q2/5

A semicircle measures

A. 180°

B. 360°

C. 90°

D. 120°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q3/5

An inscribed angle measures 25°. Its arc?

Arc: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q4/5

True or false?

An inscribed angle equals its intercepted arc.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q5/5

SET 55 • HSG-C.A.2

BACKWARD 2

Isosceles triangles from radii • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

OA = OC (radii) and m∠OAC = 40°. Find m∠OCA.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q1/5

True or false?

Any two radii and the chord between them form an isosceles triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q2/5

In $\triangle OBC$ with $OB = OC$, $m\angle BOC = 100°$. $m\angle OBC =$

A. 80°

B. 100°

C. 50°

D. 40°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q3/5

Why are ∠OAC and ∠OCA congruent when OA = OC?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q4/5

True or false?

A radius is always longer than a chord.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q5/5

SET 55 • HSG-C.A.2

BACKWARD 3

Triangle angle sum • Grade 8 • Geometry • 8.G.A.5

Two angles of a triangle are 90° and 34°. The third?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q1/5

True or false?

The acute angles of a right triangle add to 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q2/5

$2x + 2y = 180$. Then $x + y =$

A. 90

B. 45

C. 360

D. 180

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q3/5

A triangle has angles x, x, and 2x. Find 2x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q4/5

True or false?

A triangle can have angles 90°, 60°, and 40°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q5/5

SET 55 • HSG-C.A.2

FORWARD 1

Use the converse to find a circle through a right triangle • High School — Geometry • Geometry: Circles • HSG-C.A.2

Right triangle ABC has its right angle at C, with A(0, 0) and B(8, 6). Where is the center of the circle through A, B, and C, and what is its radius?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q1/5

In $\triangle ABC$, $m\angle C = 90°$. Prove that C lies on the circle with diameter AB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q2/5

True or false?

The hypotenuse of a right triangle is a diameter of its circumscribed circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q3/5

A right triangle has legs 5 and 12. Radius of its circumscribed circle?

A. 8.5

B. 6

C. 6.5

D. 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q4/5

A carpenter's square (a right angle) is placed with its corner touching a circular board and its edges crossing the edge at P and Q. What is segment PQ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q5/5

SET 55 • HSG-C.A.2

FORWARD 2

Use the diameter property to construct tangents from an external point • High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4

Compass and straightedge only; leave all arcs.

Construct the two tangent lines from point P to circle O.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q1/5

Why are the lines PT_1 and PT_2 tangent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q2/5

True or false?

From a point outside a circle there are exactly two tangent lines.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q3/5

In the construction, the circle with diameter OP is used because

A. it passes through the center only

B. every angle inscribed in it on OP is a right angle

C. it has the same radius as circle O

D. it is tangent to circle O

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q4/5

O is the center, OP = 10, and the radius is 6. How long is each tangent segment?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q5/5

SET 56 • HSG-C.A.2 • CIRCLES

A radius is perpendicular to the tangent at the point of tangency

MAIN	Geometry main question	**HSG-C.A.2**
BACKWARD 1	Use the Pythagorean Theorem and its converse	**8.G.B.7**
BACKWARD 2	The perpendicular is the shortest segment from a point to a line	**7.G.A.2**
BACKWARD 3	A tangent touches a circle at exactly one point	**HSG-CO.A.1**
FORWARD 1	Tangent segments from an external point are congruent	**HSG-C.A.2**
FORWARD 2	Write the equation of a tangent line in the coordinate plane	**HSG-GPE.B.5**

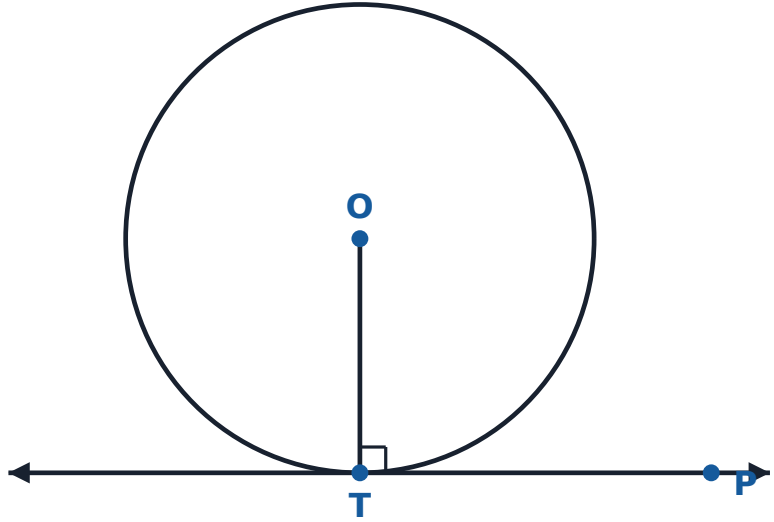
SET 56 • HSG-C.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.2

Line ℓ is tangent to circle O at T. Prove that OT $\perp \ell$. (Use an indirect argument.)

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q1/5

Line PT is tangent to circle O at T. OT = 5 and PT = 12. Find OP.

OP = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q2/5

A tangent from P touches circle O at T. OP = 17 and the radius is 8. Find PT.

PT = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q3/5

Segment PT touches circle O at T. OT = 7, PT = 24, OP = 26. Is PT tangent to the circle?

A. Yes, because $26 > 24$

B. Yes, because it touches at T

C. No, because $7 < 24$

D. No, because $7^2 + 24^2 \neq 26^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q4/5

True or false?

A line perpendicular to a radius at its endpoint on the circle is tangent to the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q5/5

SET 56 • HSG-C.A.2

BACKWARD 1

Use the Pythagorean Theorem and its converse • Grade 8 • Geometry • 8.G.B.7

Legs 5 and 12. Hypotenuse?

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q1/5

Hypotenuse 17, leg 8. Other leg?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q2/5

True or false?

A triangle with sides 7, 24, 25 is a right triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q3/5

Sides 6, 8, 11. Right triangle?

A. No

B. Yes

C. Only if 11 is a leg

D. Cannot tell

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q4/5

True or false?

The Pythagorean Theorem applies to any triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q5/5

SET 56 • HSG-C.A.2

BACKWARD 2

The perpendicular is the shortest segment from a point to a line • Grade 7 • Geometry • 7.G.A.2

True or false?

The shortest segment from a point to a line is perpendicular to the line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q1/5

In a right triangle, which side is longest?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q2/5

O is 4 cm from line ℓ (perpendicular distance). Another segment from O to ℓ could be

A. 4 cm only

B. 2 cm

C. 5 cm

D. 3 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q3/5

Why is the perpendicular segment shorter than any slanted one?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q4/5

True or false?

The distance from a point to a line is measured along any segment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q5/5

SET 56 • HSG-C.A.2

BACKWARD 3

A tangent touches a circle at exactly one point • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

True or false?

A tangent line meets the circle at exactly one point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q1/5

A line meets a circle at two points. It is a

A. radius

B. diameter

C. tangent

D. secant

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q2/5

Point Q on a line is inside the circle. Can the line be tangent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q3/5

What is the point where a tangent touches the circle called?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q4/5

True or false?

A tangent can pass through the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q5/5

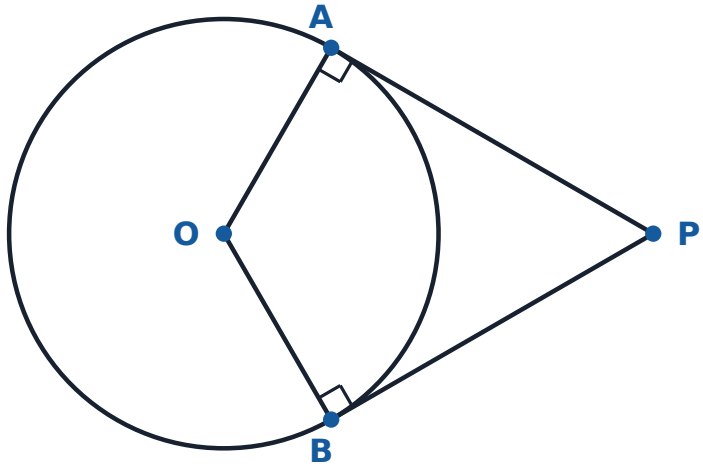
SET 56 • HSG-C.A.2

FORWARD 1

Tangent segments from an external point are congruent • High School — Geometry • Geometry: Circles • HSG-C.A.2

From point P outside circle O, tangent segments PA and PB touch the circle at A and B. Prove that PA = PB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q1/5

Tangent segments from P to a circle are 3x + 4 and 25. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q2/5

True or false?

The segment from P to the center bisects the angle between the two tangents.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q3/5

A circle is inscribed in a triangle with sides tangent to it. From one vertex, the two tangent segments are

A. parallel

B. unequal

C. equal

D. perpendicular

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q4/5

A circle is tangent to all sides of a quadrilateral ABCD. AB = 9, BC = 11, CD = 13. Find DA.

DA = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q5/5

SET 56 • HSG-C.A.2

FORWARD 2

Write the equation of a tangent line in the coordinate plane • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Find an equation of the line tangent to $x^2 + y^2 = 25$ at $(3, 4)$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q1/5

Find an equation of the tangent to $x^2 + y^2 = 13$ at $(2, -3)$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q2/5

True or false?

A tangent line's slope is the negative reciprocal of the radius's slope (when neither is vertical or horizontal).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q3/5

The tangent to $x^2 + y^2 = 16$ at $(0, 4)$ is

A. $x = 0$

B. $y = x + 4$

C. $x = 4$

D. $y = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q4/5

Is y = 7 tangent to the circle with center (2, 3) and radius 4? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q5/5

SET 57 • HSG-C.A.2 • CIRCLES

Relate circumscribed angles and central angles

MAIN	Geometry main question	**HSG-C.A.2**
BACKWARD 1	A tangent is perpendicular to the radius at the point of tangency	**HSG-C.A.2**
BACKWARD 2	The angles of a quadrilateral add to 360°	**8.G.A.5**
BACKWARD 3	Supplementary angles add to 180°	**7.G.B.5**
FORWARD 1	Use the circumscribed angle with arcs, including major arcs	**HSG-C.A.2**
FORWARD 2	Angles formed by secants and tangents outside a circle	**HSG-C.A.2**

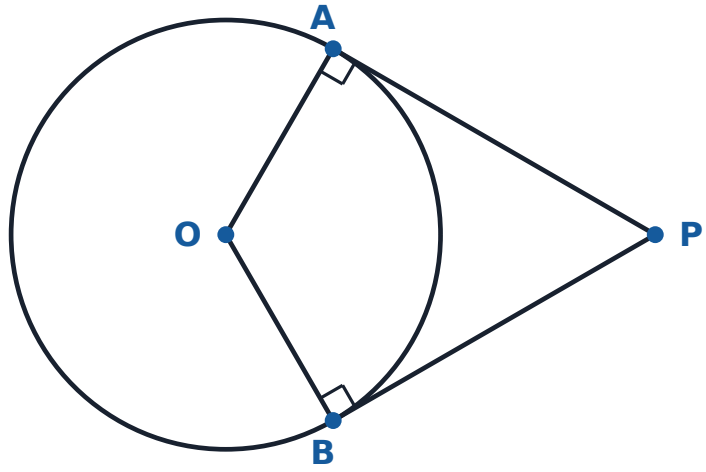
SET 57 • HSG-C.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.2

Tangents from P touch circle O at A and B. Prove that $m\angle APB + m\angle AOB = 180°$.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q1/5

The circumscribed angle APB measures 52°. Find the central angle AOB and the minor arc AB.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q2/5

Minor arc AB measures 100°. Find the circumscribed angle formed by the tangents at A and B.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q3/5

A circumscribed angle measures 90°. The quadrilateral formed by the center, the two tangent points, and the vertex is a

A. kite that is not a square

B. square

C. trapezoid

D. rhombus that is not a square

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q4/5

True or false?

A circumscribed angle and its central angle are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q5/5

SET 57 • HSG-C.A.2

BACKWARD 1

A tangent is perpendicular to the radius at the point of tangency • High School — Geometry • Geometry: Circles • HSG-C.A.2

True or false?

The radius to the point of tangency is perpendicular to the tangent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q1/5

What is m∠OAP when PA is tangent at A?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q2/5

In quadrilateral OAPB with tangents PA and PB, how many right angles are certain?

A. 4

B. 0

C. 2

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q3/5

Tangent PA, radius OA = 9, PA = 12. OP?

OP = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q4/5

True or false?

The angle between a tangent and a radius can be 60°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q5/5

SET 57 • HSG-C.A.2

BACKWARD 2

The angles of a quadrilateral add to 360° • Grade 8 • Geometry • 8.G.A.5

A quadrilateral has angles 90°, 90°, and 70°. The fourth?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q1/5

True or false?

A diagonal splits a quadrilateral into two triangles, so its angles add to 360°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q2/5

Two angles of a quadrilateral are right angles. The other two must

A. each be 90°

B. add to 360°

C. add to 90°

D. add to 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q3/5

Angles x, 90°, 90°, 2x in a quadrilateral. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q4/5

True or false?

The angles of a quadrilateral add to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q5/5

SET 57 • HSG-C.A.2

BACKWARD 3

Supplementary angles add to 180° • Grade 7 • Geometry • 7.G.B.5

Find the supplement of 52°.

Supplement: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q1/5

True or false?

Supplementary angles add to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q2/5

Angles x and 3x are supplementary. x =

A. 60°

B. 90°

C. 45°

D. 30°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q3/5

If two angles are supplementary and one is obtuse, the other is

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q4/5

True or false?

Supplementary angles must form a linear pair.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q5/5

SET 57 • HSG-C.A.2

FORWARD 1

Use the circumscribed angle with arcs, including major arcs • High School — Geometry • Geometry: Circles • HSG-C.A.2

The circumscribed angle at P is 40°. Find the minor arc AB and the major arc AB.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q1/5

Show that a circumscribed angle equals ½(major arc − minor arc).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q2/5

True or false?

A circumscribed angle can measure more than 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q3/5

Tangents from P make a 70° angle. An inscribed angle intercepting the minor arc AB measures

A. 110°

B. 55°

C. 70°

D. 35°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q4/5

The minor arc between two tangent points is x°, and the circumscribed angle is (x − 20)°. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q5/5

SET 57 • HSG-C.A.2

FORWARD 2

Angles formed by secants and tangents outside a circle • High School — Geometry • Geometry: Circles • HSG-C.A.2

Two secants from P meet a circle at A, B and at C, D (A between P and B, C between P and D). Prove that m∠P = ½(arc BD − arc AC).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q1/5

Two secants from P intercept arcs of 110° and 30°. Find m∠P.

m∠P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q2/5

True or false?

The angle outside a circle formed by a tangent and a secant is half the difference of the intercepted arcs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q3/5

An angle outside a circle intercepts arcs of 160° and 50°. Its measure?

A. 110°

B. 55°

C. 210°

D. 105°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q4/5

Explain why the formula for two tangents is a special case of the secant formula.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q5/5

SET 58 • HSG-C.A.2 • CIRCLES

Relationships among radii and chords

MAIN	Geometry main question	**HSG-C.A.2**
BACKWARD 1	Right triangles: Hypotenuse-Leg and the Pythagorean Theorem	**HSG-SRT.B.5**
BACKWARD 2	The perpendicular bisector theorem	**HSG-CO.C.9**
BACKWARD 3	Radii of a circle are equal	**HSG-CO.A.1**
FORWARD 1	Locate the center of a circle from chords	**HSG-C.A.2**
FORWARD 2	Find chord lengths and central angles with trigonometry	**HSG-SRT.C.8**

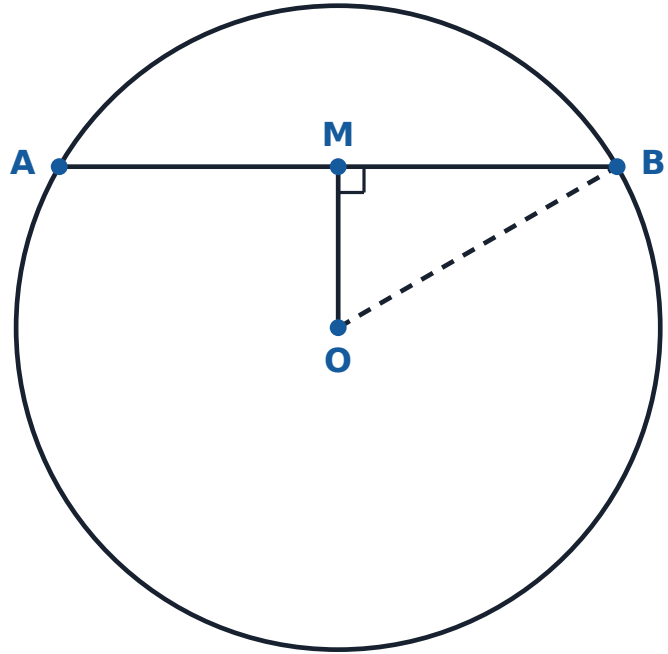
SET 58 • HSG-C.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.2

In circle O, OM ⊥ chord AB at M. Prove that M is the midpoint of AB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q1/5

A circle has radius 13. A chord is 5 units from the center. How long is the chord?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q2/5

A chord of length 16 is in a circle of radius 10. How far is the chord from the center?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q3/5

Two chords of a circle are each 7 units from the center. Which is true?

A. The chords are perpendicular

B. The chords are congruent

C. The chords are parallel

D. The chords meet at the center

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q4/5

True or false?

The perpendicular bisector of any chord passes through the center of the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q5/5

SET 58 • HSG-C.A.2

BACKWARD 1

Right triangles: Hypotenuse-Leg and the Pythagorean Theorem • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

Right triangles with equal hypotenuses and one pair of equal legs are congruent by

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q1/5

Hypotenuse 10, leg 6. Other leg?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q2/5

True or false?

Two radii are congruent hypotenuses in the chord figure.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q3/5

OM ⊥ AB in circle O. △OMA and △OMB share

A. OM

B. nothing

C. AB

D. OA

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q4/5

True or false?

HL works for any triangles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q5/5

SET 58 • HSG-C.A.2

BACKWARD 2

The perpendicular bisector theorem • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

A point equidistant from A and B lies on the perpendicular bisector of AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q1/5

The center O of a circle is equidistant from the endpoints of chord AB. Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q2/5

So the center lies on

A. a tangent

B. the circle

C. every chord

D. the perpendicular bisector of every chord

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q3/5

Two chords' perpendicular bisectors meet at P. What is P?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q4/5

True or false?

The perpendicular bisector of a chord can miss the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q5/5

SET 58 • HSG-C.A.2

BACKWARD 3

Radii of a circle are equal • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

A circle has center O. OA = 7. Find OB if B is on the circle.

OB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q1/5

True or false?

All radii of one circle have the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q2/5

Chord AB with OA = 5. OB =

A. unknown

B. 10

C. 5

D. 2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q3/5

What is the longest chord in a circle of radius 6?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q4/5

True or false?

A chord always passes through the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q5/5

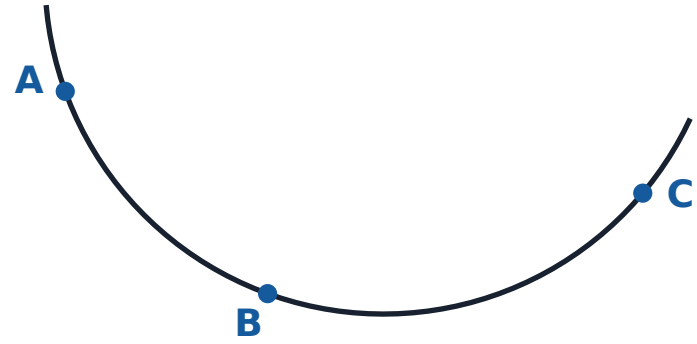
SET 58 • HSG-C.A.2

FORWARD 1

Locate the center of a circle from chords • High School — Geometry • Geometry: Circles • HSG-C.A.2

Compass and straightedge only; leave all arcs.

The arc below is part of a circle. Construct its center.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q1/5

A circle passes through (0, 0), (8, 0), and (0, 6). Find its center.

Center: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q2/5

True or false?

Three points not on one line determine exactly one circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q3/5

Why can't three points on one line lie on a circle?

A. Three points on a line always lie on a circle

B. Their perpendicular bisectors are parallel, so no center exists

C. Only two points can lie on a circle

D. They are too close

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q4/5

A circular plate is broken. Describe how to find its original radius from the piece.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q5/5

SET 58 • HSG-C.A.2

FORWARD 2

Find chord lengths and central angles with trigonometry • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

A chord subtends a central angle of 80° in a circle of radius 10. Find its length to the nearest tenth.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q1/5

A chord of length 12 is in a circle of radius 10. Find the central angle to the nearest tenth of a degree.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q2/5

True or false?

A chord's length is $2r \sin(\theta/2)$ for central angle θ.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q3/5

Central angle 60°, radius 9. Chord length?

A. 4.5

B. 18

C. 9

D. $9\sqrt{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q4/5

A chord equals the radius. Find its central angle.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q5/5

SET 59 • HSG-C.A.3 • CIRCLES

Construct the inscribed circle of a triangle

MAIN	Geometry main question	**HSG-C.A.3**
BACKWARD 1	Construct an angle bisector	**HSG-CO.D.12**
BACKWARD 2	Construct a perpendicular from a point to a line	**HSG-CO.D.12**
BACKWARD 3	Points on an angle bisector are equidistant from the sides	**HSG-SRT.B.5**
FORWARD 1	Prove that the incenter is equidistant from the sides	**HSG-C.A.3**
FORWARD 2	Locate a point equidistant from three lines in a design	**HSG-MG.A.3**

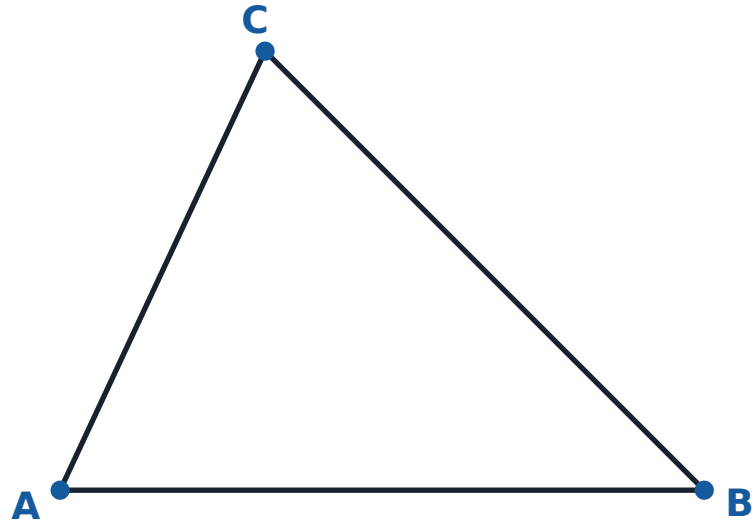
SET 59 • HSG-C.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.3

Compass and straightedge only; leave all arcs.

Construct the inscribed circle of triangle ABC.



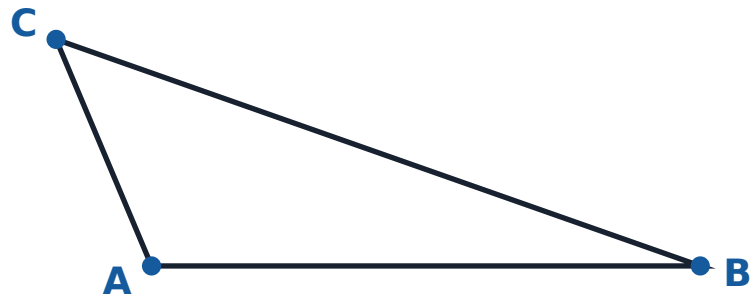
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct the inscribed circle of obtuse triangle ABC.



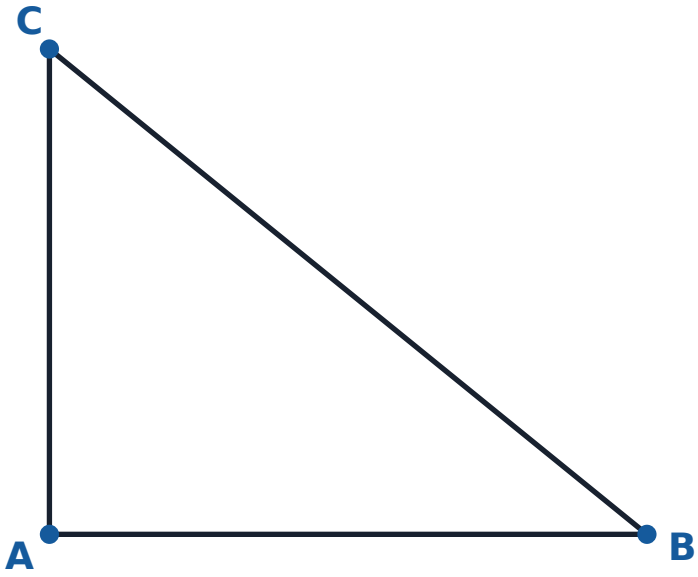
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct the inscribed circle of right triangle ABC.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q3/5

Write the steps for constructing the inscribed circle of a triangle.

Steps: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q4/5

Why is the radius of the inscribed circle measured along a perpendicular from the incenter to a side?

A. The radius goes to a vertex

B. The circle must be tangent to the side, and a tangent is perpendicular to the radius

C. Any segment from I works

D. The perpendicular is the longest segment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q5/5

SET 59 • HSG-C.A.3

BACKWARD 1

Construct an angle bisector • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

How many arcs are needed to bisect an angle with compass and straightedge?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q1/5

True or false?

An angle bisector construction ends with a ray from the vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q2/5

The incenter is where the

A. medians meet

B. perpendicular bisectors meet

C. altitudes meet

D. angle bisectors meet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q3/5

Why construct only two angle bisectors to find the incenter?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q4/5

True or false?

Bisecting the sides locates the incenter.

A. True

B. False

SET 59 • HSG-C.A.3

BACKWARD 2

Construct a perpendicular from a point to a line • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Which construction finds the point where the incircle touches a side?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q1/5

True or false?

The perpendicular from a point to a line is the shortest segment to the line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q2/5

From point I, an arc meets side AB at two points. Next step?

A. Measure AB

B. Draw a circle at A

C. Bisect ∠I

D. Arcs of equal radius from those two points, crossing on the other side

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q3/5

Why not measure from I to a vertex for the radius?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q4/5

True or false?

The incircle passes through the vertices.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q5/5

SET 59 • HSG-C.A.3

BACKWARD 3

Points on an angle bisector are equidistant from the sides • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

True or false?

A point on the bisector of $\angle A$ is equidistant from the two sides of $\angle A$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q1/5

The incenter is 3 cm from side AB. How far is it from side BC?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q2/5

Distance from a point to a side is measured

A. to the nearest vertex

B. along the perpendicular

C. along the bisector

D. along any segment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q3/5

Why is the incenter the same distance from all three sides?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q4/5

True or false?

The incenter is equidistant from the three vertices.

A. True

B. False

SET 59 • HSG-C.A.3

FORWARD 1

Prove that the incenter is equidistant from the sides • High School — Geometry • Geometry: Circles • HSG-C.A.3

The bisectors of ∠A and ∠B of △ABC meet at I. Prove that I is equidistant from all three sides and lies on the bisector of ∠C.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q1/5

A triangle has area 30 and perimeter 30. Find the radius of its inscribed circle.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q2/5

True or false?

The incenter is always inside the triangle.

A. True

B. False

A 3-4-5 right triangle has inradius

A. 2.5

B. 1

C. 1.5

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q4/5

Explain why area = r · s, where s is half the perimeter.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q5/5

SET 59 • HSG-C.A.3

FORWARD 2

Locate a point equidistant from three lines in a design • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

A circular fountain must touch three straight paths that form a triangle with sides 13, 14, and 15 m (area 84 m²). What is the largest possible radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q1/5

Explain why the incircle gives the largest circle that fits inside a triangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q2/5

True or false?

A sprinkler equidistant from three fences (triangle) should be at the incenter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q3/5

To be equidistant from three towns (points), use the

A. circumcenter

B. orthocenter

C. centroid

D. incenter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q4/5

An equilateral triangle has side 6. Find its inradius exactly.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q5/5

SET 60 • HSG-C.A.3 • CIRCLES

Construct the circumscribed circle of a triangle

MAIN	Geometry main question	**HSG-C.A.3**
BACKWARD 1	Construct the perpendicular bisector of a segment	**HSG-CO.D.12**
BACKWARD 2	Points on the perpendicular bisector are equidistant from the end…	**HSG-CO.C.9**
BACKWARD 3	Triangle types: acute, right, obtuse	**4.G.A.2**
FORWARD 1	Find the circumcenter and radius with coordinates	**HSG-GPE.B.5**
FORWARD 2	Find the circumradius with trigonometry (extended Law of Sines)	**HSG-SRT.D.10**

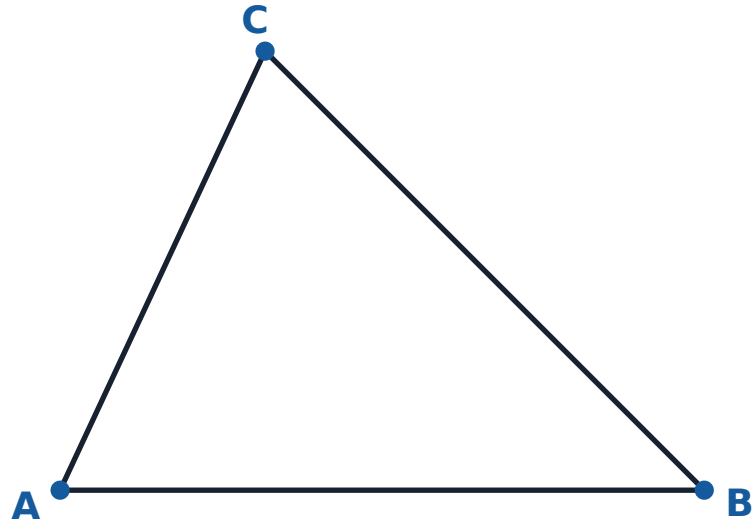
SET 60 • HSG-C.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.3

Compass and straightedge only; leave all arcs.

Construct the circumscribed circle of triangle ABC.



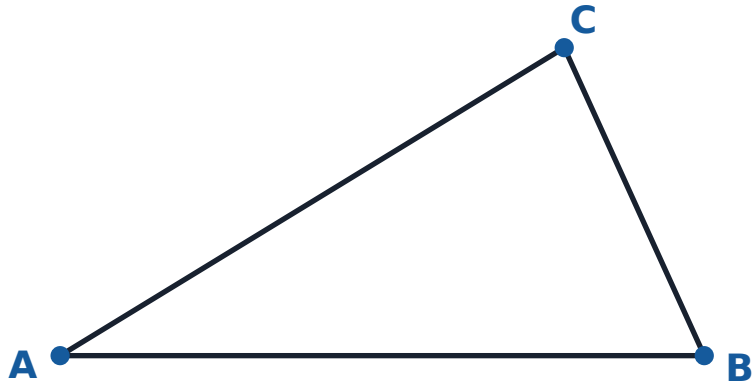
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct the circumscribed circle of obtuse triangle ABC.



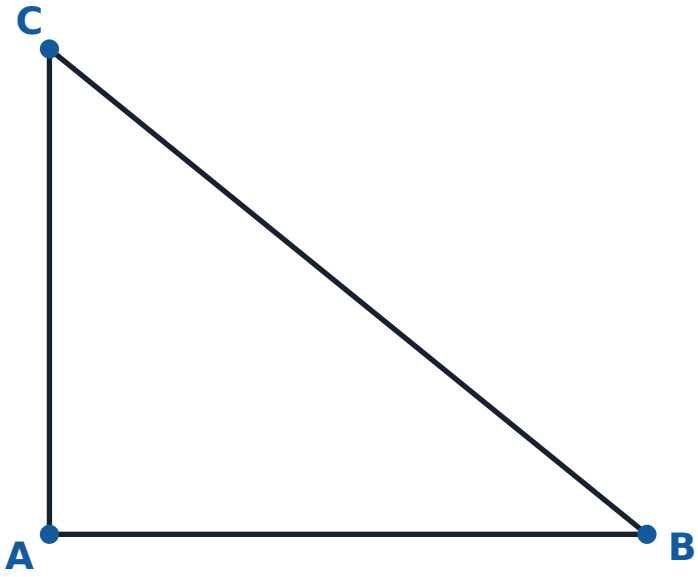
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct the circumscribed circle of right triangle ABC.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q3/5

Write the steps for constructing the circumscribed circle of a triangle.

Steps: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q4/5

True or false?

The circumcenter of an obtuse triangle lies inside the triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q5/5

SET 60 • HSG-C.A.3

BACKWARD 1

Construct the perpendicular bisector of a segment • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Which construction gives all points equidistant from A and B?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q1/5

True or false?

The perpendicular bisector construction uses equal arcs from both endpoints.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q2/5

The circumcenter is where the

A. angle bisectors meet

B. sides meet

C. perpendicular bisectors of the sides meet

D. medians meet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q3/5

After constructing two perpendicular bisectors, why is the third unnecessary?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q4/5

True or false?

The arcs in a perpendicular bisector construction can have any two different radii.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q5/5

SET 60 • HSG-C.A.3

BACKWARD 2

Points on the perpendicular bisector are equidistant from the endpoints • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

Every point on the perpendicular bisector of AB is equidistant from A and B.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q1/5

O is on the perpendicular bisectors of AB and BC. Compare OA, OB, OC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q2/5

The circumcircle's radius is

A. the longest side

B. the distance from O to any vertex

C. the distance from O to a side

D. half a side

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q3/5

A circle passes through A, B, C. Where must its center lie relative to AB?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q4/5

True or false?

The circumcenter is equidistant from the sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q5/5

SET 60 • HSG-C.A.3

BACKWARD 3

Triangle types: acute, right, obtuse • Grade 4 • Geometry • 4.G.A.2

A triangle has angles 30°, 45°, 105°. Classify it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q1/5

True or false?

A right triangle has one 90° angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q2/5

Where is the circumcenter of a right triangle?

A. Inside

B. Outside

C. On the hypotenuse (its midpoint)

D. At the right angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q3/5

A triangle has angles 50°, 60°, 70°. Classify it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q4/5

True or false?

An acute triangle can have an obtuse angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q5/5

SET 60 • HSG-C.A.3

FORWARD 1

Find the circumcenter and radius with coordinates • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Triangle with A(0, 0), B(6, 0), C(2, 4). Find the circumcenter and radius.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q1/5

Triangle with A(−2, 0), B(2, 0), C(0, 2). Find the circumcenter.

Center: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q2/5

True or false?

The circumcenter is the intersection of two perpendicular bisector equations.

A. True

B. False

A right triangle has vertices (0, 0), (8, 0), (0, 6). Circumradius?

A. 10

B. 4

C. 5

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q4/5

Explain how to check a computed circumcenter.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q5/5

SET 60 • HSG-C.A.3

FORWARD 2

Find the circumradius with trigonometry (extended Law of Sines) • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

In a triangle, $a = 10$ and $A = 30°$. Find the circumradius R using $a/\sin A = 2R$.

$R =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q1/5

Equilateral triangle with side 6. Find the circumradius exactly.

R = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q2/5

True or false?

In any triangle, a/sin A equals the diameter of the circumcircle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q3/5

A triangle with a 90° angle opposite side 14 has circumradius

A. 3.5

B. 7

C. 14

D. 28

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F2-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q4/5

Explain why $a/\sin A = 2R$ using an inscribed angle on a diameter.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q5/5

SET 61 • HSG-C.A.3 • CIRCLES

Prove that opposite angles of an inscribed quadrilateral are supplementary

MAIN	Geometry main question	**HSG-C.A.3**
BACKWARD 1	The inscribed angle theorem	**HSG-C.A.2**
BACKWARD 2	Supplementary angles	**7.G.B.5**
BACKWARD 3	Arcs of a circle add to 360°	**4.MD.C.7**
FORWARD 1	Use the converse: decide whether a quadrilateral can be inscribed	**HSG-C.A.3**
FORWARD 2	Exterior angle of a cyclic quadrilateral	**HSG-C.A.3**

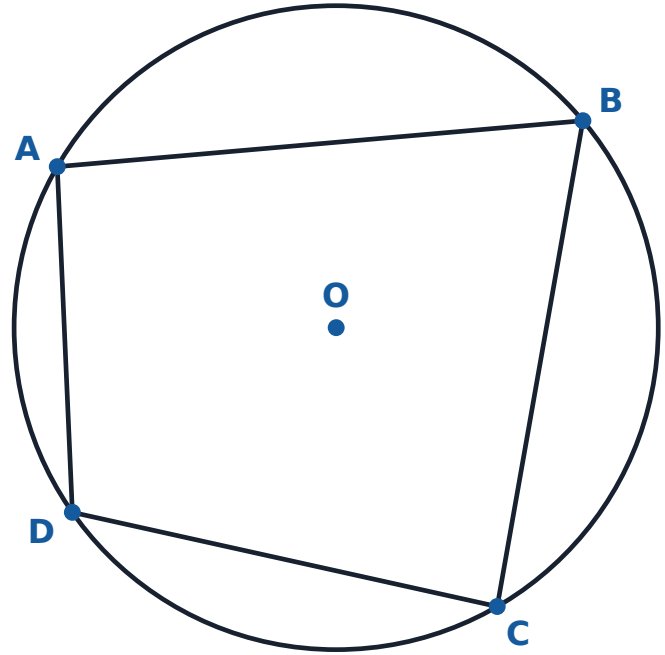
SET 61 • HSG-C.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.3

Quadrilateral ABCD is inscribed in circle O. Prove that $m\angle A + m\angle C = 180°$.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q1/5

Quadrilateral PQRS is inscribed in a circle. Prove that ∠Q and ∠S are supplementary.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q2/5

Quadrilateral ABCD is inscribed in a circle with $m\angle A = 72°$ and $m\angle B = 105°$. Find $m\angle C$ and $m\angle D$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q3/5

Which quadrilateral can always be inscribed in a circle?

A. Any rhombus

B. Any trapezoid

C. Any rectangle

D. Any parallelogram

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q4/5

True or false?

A parallelogram that is not a rectangle can be inscribed in a circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q5/5

SET 61 • HSG-C.A.3

BACKWARD 1

The inscribed angle theorem • High School — Geometry • Geometry: Circles • HSG-C.A.2

An inscribed angle intercepts a 220° arc. Its measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q1/5

True or false?

An inscribed angle can intercept an arc greater than 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q2/5

An inscribed angle of 70° intercepts

A. 70°

B. 290°

C. 35°

D. 140°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q3/5

Two arcs together make a whole circle. Their sum?

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q4/5

True or false?

An inscribed angle equals half the central angle only when it is acute.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q5/5

SET 61 • HSG-C.A.3

BACKWARD 2

Supplementary angles • Grade 7 • Geometry • 7.G.B.5

Supplement of 72°?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q1/5

True or false?

Two angles whose measures add to 180° are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q2/5

Angles 3x and 2x are supplementary. x =

A. 30°

B. 45°

C. 36°

D. 60°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q3/5

If $\angle A = \angle C$ and they are supplementary, find each.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q4/5

True or false?

Supplementary angles are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q5/5

SET 61 • HSG-C.A.3

BACKWARD 3

Arcs of a circle add to 360° • Grade 4 • Measurement & Data • 4.MD.C.7

A circle is split into arcs of 100°, 80°, and x°. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q1/5

True or false?

The arcs between four points on a circle add to 360°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q2/5

Two arcs make a whole circle; one is 150°. The other is

A. 150°

B. 210°

C. 30°

D. 360°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q3/5

Half of 360° is

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q4/5

True or false?

The arcs of a circle add to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q5/5

SET 61 • HSG-C.A.3

FORWARD 1

Use the converse: decide whether a quadrilateral can be inscribed • High School — Geometry • Geometry: Circles • HSG-C.A.3

Quadrilateral ABCD has angles A = 85°, B = 70°, C = 95°, D = 110°. Can it be inscribed in a circle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q1/5

An isosceles trapezoid has base angles of 64°. Can it be inscribed in a circle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q2/5

True or false?

Every kite can be inscribed in a circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q3/5

Inscribed quadrilateral with angles (2x)°, (3x + 5)°, (2x + 20)°, and y°. If 2x and 2x + 20 are opposite, x =

A. 50

B. 40

C. 45

D. 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q4/5

Inscribed quadrilateral: $\angle A = (5y - 10)°$, $\angle C = (3y + 30)°$, A and C opposite. Find y and $\angle A$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q5/5

SET 61 • HSG-C.A.3

FORWARD 2

Exterior angle of a cyclic quadrilateral • High School — Geometry • Geometry: Circles • HSG-C.A.3

ABCD is inscribed in a circle and side DC is extended past C to E. Prove that m∠BCE = m∠A.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q1/5

In inscribed ABCD, $\angle A = 77°$. Side DC is extended past C to E. Find $\angle BCE$.

$\angle BCE =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q2/5

True or false?

An exterior angle of a cyclic quadrilateral equals the interior angle opposite to it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q3/5

Inscribed ABCD with $\angle B = 100°$. The exterior angle at D (extending CD past D) is

A. 50°

B. 180°

C. 80°

D. 100°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q4/5

Why is the exterior angle at C equal to ∠A, not ∠B?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q5/5

SET 62 • HSG-C.B.5 • CIRCLES

Derive arc length from similarity and define radian measure

MAIN	Geometry main question	**HSG-C.B.5**
BACKWARD 1	Similar figures: lengths scale by the scale factor	**HSG-SRT.A.2**
BACKWARD 2	Circumference of a circle	**7.G.B.4**
BACKWARD 3	Find a fraction of a whole	**7.RP.A.3**
FORWARD 1	Convert between degrees and radians	**HSG-C.B.5**
FORWARD 2	Radian measure for angles of rotation and the unit circle	**HSF-TF.A.1**

SET 62 • HSG-C.B.5

MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.B.5

A central angle θ cuts off arc length s_1 on a circle of radius r_1 and arc length s_2 on a circle of radius r_2 (same center). Explain, using similarity, why $s_1/r_1 = s_2/r_2$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q1/5

A central angle cuts off an arc of length 12 cm on a circle of radius 4 cm. What is the angle in radians?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q2/5

Find the length of the arc cut off by a central angle of 2.5 radians in a circle of radius 6 m.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q3/5

Find the exact length of the arc cut off by a 120° central angle in a circle of radius 9.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q4/5

Why is a full circle 2π radians?

A. The circumference is 2πr, so s/r = 2π for the whole circle

B. There are 360 degrees

C. Every circle has radius 1

D. π ≈ 3.14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q5/5

SET 62 • HSG-C.B.5

BACKWARD 1

Similar figures: lengths scale by the scale factor • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

True or false?

Under a dilation with factor k, every length (straight or curved) is multiplied by k.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q1/5

Two similar circles have radii 3 and 12. Scale factor?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q2/5

An arc of length 5 on the small circle (radius 3) matches which arc length on the large one (radius 12)?

A. 15

B. 20

C. 8

D. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q3/5

Are all circles similar?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q4/5

True or false?

Arc lengths do not change under dilation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q5/5

SET 62 • HSG-C.B.5

BACKWARD 2

Circumference of a circle • Grade 7 • Geometry • 7.G.B.4

Find the circumference of a circle with radius 5 (exact).

Circumference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q1/5

True or false?

$C = 2\pi r$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q2/5

A circle has diameter 8. Its circumference is

A. 8π

B. 64π

C. 4π

D. 16π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q3/5

A circle has circumference 18π. Radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q4/5

True or false?

$C = \pi r^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q5/5

SET 62 • HSG-C.B.5

BACKWARD 3

Find a fraction of a whole • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

What fraction of a circle is a 90° arc?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q1/5

Find ⅓ of 18π.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q2/5

An arc of 45° is what fraction of the circumference?

A. 45/100

B. ⅛

C. 1/45

D. ¼

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q3/5

True or false?

A 120° arc is ⅓ of the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q4/5

True or false?

A 60° arc is 1/60 of the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q5/5

SET 62 • HSG-C.B.5

FORWARD 1

Convert between degrees and radians • High School — Geometry • Geometry: Circles • HSG-C.B.5

Convert 150° to radians (exact).

Radians: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q1/5

Convert 3π/4 radians to degrees.

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q2/5

True or false?

1 radian is about 57.3°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q3/5

Which equals π/3 radians?

A. 120°

B. 30°

C. 180°

D. 60°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q4/5

An arc in a circle of radius 10 has length 5π. Find the central angle in degrees.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q5/5

SET 62 • HSG-C.B.5

FORWARD 2

Radian measure for angles of rotation and the unit circle • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1

On the unit circle, an angle of 2 radians cuts off an arc of what length?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q1/5

A wheel of radius 0.3 m turns through 10 radians. How far does a point on its rim travel?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q2/5

True or false?

An angle of 7 radians is more than one full turn.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q3/5

A point on the unit circle travels an arc of length π. The angle is

A. 1 radian

B. 2π radians

C. π radians (180°)

D. π°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-F2-Q4**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q4/5

Explain why the radian measure of an angle does not depend on the radius of the circle used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q5/5

SET 63 • HSG-C.B.5 • CIRCLES

Derive and use the formula for the area of a sector

MAIN	Geometry main question	**HSG-C.B.5**
BACKWARD 1	Area of a circle	**7.G.B.4**
BACKWARD 2	The radius is half the diameter	**7.G.B.4**
BACKWARD 3	Fractions of a circle in degrees and radians	**HSG-C.B.5**
FORWARD 1	Solve inverse sector problems	**HSG-C.B.5**
FORWARD 2	Areas of segments and composite circular regions	**HSG-MG.A.1**

SET 63 • HSG-C.B.5

MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.B.5

Derive a formula for the area of a sector with central angle θ (in degrees) in a circle of radius r. Then rewrite it for θ in radians.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q1/5

Find the exact area of a sector with a 60° central angle in a circle of radius 6.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q2/5

Find the area of a sector with central angle 1.2 radians in a circle of radius 5.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q3/5

A pizza with diameter 14 in. is cut into 8 equal slices. Find the area of one slice to the nearest tenth of a square inch.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q4/5

A sector has area 12π in a circle of radius 6. What is its central angle?

A. 240°

B. 12°

C. 60°

D. 120°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q5/5

SET 63 • HSG-C.B.5

BACKWARD 1

Area of a circle • Grade 7 • Geometry • 7.G.B.4

Find the exact area of a circle with radius 6.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q1/5

A circle has diameter 10. Exact area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q2/5

True or false?

$A = \pi r^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q3/5

A circle has area 49π. Radius?

A. 14

B. 24.5

C. 7

D. 49

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q4/5

True or false?

Doubling the radius doubles the area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q5/5

SET 63 • HSG-C.B.5

BACKWARD 2

The radius is half the diameter • Grade 7 • Geometry • 7.G.B.4

A circle has diameter 14 in. Radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q1/5

True or false?

Using the diameter in $A = \pi r^2$ gives an area four times too large.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q2/5

A sector of a circle with diameter 20 uses r =

A. 20

B. 5

C. 40

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q3/5

A circle's radius is 2.5 cm. Diameter?

Diameter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q4/5

True or false?

The radius and diameter can be used interchangeably in area formulas.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q5/5

SET 63 • HSG-C.B.5

BACKWARD 3

Fractions of a circle in degrees and radians • High School — Geometry • Geometry: Circles • HSG-C.B.5

What fraction of a circle is a sector with central angle 45°?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q1/5

What fraction of a circle is a sector with central angle π/2 radians?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q2/5

A full circle measures

A. 180° or π radians

B. 360 radians

C. 2π°

D. 360° or 2π radians

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q3/5

True or false?

A central angle of 1 radian is 1/(2π) of the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q4/5

True or false?

A 90° sector is ⅓ of the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q5/5

SET 63 • HSG-C.B.5

FORWARD 1

Solve inverse sector problems • High School — Geometry • Geometry: Circles • HSG-C.B.5

A sector with a 40° angle has area 4π. Find the radius.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q1/5

A sector of a circle with radius 10 has area 30. Find the central angle in radians.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q2/5

True or false?

If the radius is tripled and the angle stays the same, the sector area is multiplied by 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q3/5

A sector has area 50 and arc length 10. Radius?

A. 20

B. 5

C. 10

D. 2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q4/5

Show that the area of a sector is A = ½rs, where s is the arc length.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q5/5

SET 63 • HSG-C.B.5

FORWARD 2

Areas of segments and composite circular regions • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

Find the exact area of the region between a 90° arc and its chord in a circle of radius 4 (a circular segment).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q1/5

A sprinkler waters a 120° sector of radius 9 m. Find the watered area to the nearest square meter.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q2/5

True or false?

The area of a circular segment equals the sector area minus the area of the triangle formed by the two radii and the chord.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q3/5

An annulus (ring) has inner radius 3 and outer radius 5. Its area is

A. 8π

B. 4π

C. 25π

D. 16π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q4/5

A windshield wiper blade covers the part of its arm from 20 cm to 60 cm from the pivot and sweeps through 120°. Find the area it wipes to the nearest square centimeter.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q5/5